HL7 FHIR R4 — Chapter 14: Clinical Reasoning (sections 14 – 14.18.9)

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14 Clinical Reasoning

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Work Group Clinical Decision Support & Clinical Quality Information | Standards Status: Informative

14.0 Clinical Reasoning

14.0.1 Introduction

The Clinical Reasoning module provides resources and operations to enable the representation, distribution, and evaluation of clinical knowledge artifacts such as clinical decision support rules, quality measures, public health indicators, order sets, and clinical protocols. In addition, the module describes how expression languages can be used throughout the specification to provide dynamic capabilities.

Clinical Reasoning involves the ability to represent and encode clinical knowledge in a very broad sense so that it can be integrated into clinical systems. This encoding may be as simple as controlling whether or not a particular section of an order set appears based on the conditions that a patient has, or it may be as complex as representing the care pathway for a patient with multiple conditions.

14.0.1.1 Scope

The Clinical Reasoning module focuses on enabling two primary use cases:

• Sharing - The ability to represent clinical knowledge artifacts such as decision support rules, order sets, clinical protocols, and quality measures, and to do so in a way that enables those artifacts to be shared across all organizations involved in care management and delivery.

• Evaluation - The ability to evaluate clinical knowledge artifacts in the context of a specific patient or population, including the ability to request decision support guidance, impact clinical workflow, and retrospectively assess clinical quality metrics.

To enable these use cases, the module defines several components that can each be used independently, or combined to enable more complex functionality. These components are:

• Expression Logic - the representation of logic using languages such as FHIRPath and Clinical Quality Language (CQL) .

• Definitional Resources - resources that are not defined on any specific patient, but are used to define the actions to be performed as part of a clinical knowledge artifact such as an order set or decision support rule. These resources can be used directly, or with profiles to provide intended structure for specific types of resources.

• Knowledge Artifacts - representation of clinical knowledge such as decision support rules and clinical quality measures.

These basic components can then be used to enable a broad variety of clinical decision support and quality measurement use cases, including knowledge sharing, decision support services, and clinical quality assessment and reporting. The topics below provide more detailed discussion on each of these components and uses:

Topic | Description
Overview and Background | If you are interested in the background and development of the FHIR Clinical Reasoning module, this topic covers where it came from and why it exists. See also the general FHIR introductions for clinicians, developers or architects
Using Expressions | If you want to see how to add dynamic capabilities to FHIR resources using expressions, start here.
Definitional Resources | If you want to see how to describe definitional resources using the ActivityDefinition resource, start here.
Representing Knowledge Artifacts | If you want to represent knowledge artifacts such as Event-Condition-Action rules, Order Sets, or Clinical Protocols, start here.
Sharing Knowledge Artifacts | If you want to share and distribute knowledge artifacts, start here.
Clinical Decision Support Service | If you want to use the Clinical Reasoning module to provide or use Clinical Decision Support services, start here.
Quality Reporting | If you want to define or report clinical quality measures, start here.

14.0.1.2 Audience

From the perspective of a Knowledge Author, this module describes an approach to representing knowledge artifacts within FHIR.

From the perspective of a Knowledge Content Provider, this module defines search functionality for using a FHIR server as a knowledge artifact repository.

From the perspective of a Knowledge Evaluation Service Provider, this module defines operations and profiles in support of evaluating quality measures, and defining and using CDS Hooks services.

And finally, from the perspective of a Knowledge Evaluation Service Consumer, this module defines the expected available operations and behavior of a knowledge evaluation service.

14.0.2 Index

14.0.2.1 Resources

Resource | Description
ActivityDefinition | A resource to represent definitional resources.
DataRequirement | A data type that represents a general data requirement for a knowledge asset such as a decision support rule or quality measure.
GuidanceResponse | Represents the result from invoking a decision support service.
Library | Provides a container for knowledge artifacts that includes logic libraries, model definitions, and asset collections.
Measure | Represents a clinical quality measure and provides evaluation through the $evaluate-measure operation.
MeasureReport | Represents the response to a specific measure evaluation request returned by the $evaluate-measure operation.
PlanDefinition | Represents the description of a plan for accomplishing a particular goal. This resource is used to represent a broad variety of clinical knowledge artifacts including decision support rules, order sets, and protocols.
RequestGroup | Represents a group of options for a particular subject that can be used to accomplish a particular goal. This resource is often, but not always, the result of applying a PlanDefinition to a particular patient.

14.0.2.2 Extensions

Extension | Description
cdsHooksEndpoint | An extension applied to a PlanDefinition to indicate that it provides the behavior for a CDS Hooks service endpoint.
expression | A general purpose extension that supports the use of languages such as FHIRPath and Clinical Quality Language within FHIR.
library | A general purpose extension that supports the declaration of dependencies that can be accessed by expression logic.
measureInfo | An extension that can be applied to resources to indicate the measure criteria they satisfy. Used in evaluated resource bundles as part of reporting measure results for a patient to identify resources that contributed to the patient's membership in a particular population criteria.
qualityOfEvidence | An extension that can be applied to indicate the quality of evidence in support of a particular artifact or recommendation.
relativeDateTime | An extension that can be applied to define a date/time value relative to another event.
strengthOfRecommendation | An extension that can be applied to indicate the strength of a recommendation.

14.0.2.3 Profiles

Profile | Description
Shareable ActivityDefinition | Enforces the minimum information set for the activity definition metadata required by HL7 and other organizations that share and publish activity definitions.
CQF-Questionnaire | Defines extensions to the base Questionnaire that allow it to be used as a DocumentationTemplate with behavior specified via logic in CQL libraries.
CDS Hooks GuidanceResponse | Defines a GuidanceResponse that represents the response container for a CDS Hooks response
Shareable Library | Enforces the minimum information set for the library metadata required by HL7 and other organizations that share and publish libraries
CQL Library | Represents a CQL logic library
Shareable Measure | Enforces the minimum information set for the measure metadata required by HL7 and other organizations that share and publish measures
Shareable PlanDefinition | Enforces the minimum information set for the plan definition metadata required by HL7 and other organizations that share and publish plan definitions
Computable PlanDefinition | Defines a computable PlanDefinition that specifies a single library and requires all expressions referenced from the PlanDefinition to be definitions in that single library
CDS Hooks PlanDefinition | Defines a PlanDefinition that implements the behavior for a CDS Hooks service
CDS Hooks RequestGroup | Defines a RequestGroup that can represent a CDS Hooks response

14.0.2.4 Services

Service | Description
Knowledge Repository | Defines minimum service capabilities for a knowledge repository.
Measure Processor | Defines minimum service capabilities for a measure processor.

14.0.3 Security and Privacy

Because Knowledge Artifacts are typically patient-independent, many of the resources in the clinical reasoning module have no patient security and privacy concerns beyond the normal sensitivity that should be paid in any electronic healthcare system environment. However, the evaluation use case, including decision support guidance request/response, as well as quality measure evaluation have significant patient security and privacy concerns.

For the clinical decision support evaluation use case, as with any patient-specific information, care should be taken to ensure that the request and response are properly secured both at rest and in-motion, and that all access to the patient's information is done via a properly authenticated and authorized mechanism. This is particularly true of decision support artifacts where the logic is ingested as part of the definition of the

artifact. In this scenario, the evaluation engine must ensure that data access within the ingested logic is subject to the same authentication and authorization requirements as any other access.

For guidance services that receive patient information, ensure that logging and auditing trails do not inadvertently compromise patient privacy and security by logging potentially sensitive information in an unencrypted way. In addition, guidance and recommendations returned from the service must ensure that content that contains patient information is clearly indicated so that consuming clients can take the appropriate care in integrating and displaying the resulting guidance.

For quality measure evaluation, individual and patient-list reports have the potential to contain large amounts of patient information. As with the decision support guidance responses, care must be taken to ensure the patient information is only accessible to properly authenticated and authorized agents, and that inadvertent breaches are minimized by following appropriate logging and auditing protocols.

In particular, because expression languages, depending on their power and scope, can provide the ability to access large amounts of data, as well as the potential for infinite recursion or looping, care should be taken to ensure that implementations adequately safeguard against Denial-of-Service-style attacks that leverage these capabilities to compromise systems by overloading capacity.

For more general considerations, see the Security and Privacy module.

14.0.4 Common use Cases

Use Case | Description
Providing a dynamic value for a resource element | Using expressions to define the value for an element of a FHIR resource.
Defining a CQL library | Using the Library resource to incorporate a Clinical Quality Language library for use in FHIR resources.
Defining a Model Definition artifact | Using the Library resource to incorporate the definition of an information model for use with expressions in FHIR.
Defining an Event Condition Action rule | Using the PlanDefinition resource to represent an event-condition-action rule in FHIR.
Defining a Referral Request activity | Using the ActivityDefinition resource to define a referral request activity that can be used as part of a knowledge artifact.
Defining a Medication Request activity | Using the ActivityDefinition resource to define a medication request activity that can be used as part of a knowledge artifact.
Defining an Order Set | Using the PlanDefinition resource to represent an order set.
Defining a Protocol | Using the PlanDefinition resource to represent a protocol.
Defining a Questionnaire with dynamic content | Using the Questionnaire resource and expression extensions to add dynamic functionality to a FHIR Questionnaire.
Obtaining guidance from a Decision Support Service | Using the $evaluate operation to request and process guidance from a decision support service.
Defining a Measure | Using the Measure resource to represent a clinical quality measure.
Evaluating a Measure | Using the $evaluate-measure operation to request calculation of a clinical quality measure.
Applying an ActivityDefinition | Using the $apply operation to realize the intent resource defined by an ActivityDefinition.
Applying a PlanDefinition | Using the $apply operation to realize a plan definition for a specific context.
Representing Quality of Evidence/Strength of Recommendation | Using the qualityOfEvidence and strengthOfRecommendation extensions to indicate ratings associated with evidence for a particular artifact or recommendation.

14.0.5 Developmental Roadmap

The resources defined for the Clinical Reasoning module are the result of the combined efforts of multiple communities working on the shared goal of harmonized standards and specifications for clinical decision support and quality measurement artifacts. The current state of the module reflects changes incorporated both from previous ballots on the FHIR-specific material, as well as content derived from several other balloted specifications in the CDS and CQM domains. The content at this point is capable of supporting the two primary use cases of sharing and evaluation in both domains and for a broad variety of artifacts.

In particular, the use of Clinical Quality Language (CQL) as a foundational mechanism for representing clinical quality logic enables decision support and quality measurement artifacts to share common definitions. For example, a Chlamydia Screening measure and related decision support artifacts focused on improving the measure can share a common library that describes the criteria for detecting when Chlamydia Screening is required in a patient. The decision support rule applies these criteria to determine when and how to impact a workflow, while the quality measure uses these same criteria to determine whether the screening goal has been met for a patient or population. In addition, the resources defined in this module use common patterns for describing the structure of artifacts and their associated metadata, enabling a consistent approach to the sharing and distribution of clinical knowledge artifacts.

Over the past year of trial use, the Clinical Reasoning module has been used in both the quality measurement and decision support domains to represent, exchange, and evaluate knowledge artifacts. This usage has generated feedback resulting in the addition of several new profiles, as well as guidance for using the Clincal Reasoning module with CDS Hooks and for quality reporting. Although this feedback resulted in some substantive changes, there were comparatively few, and the goals of the module for the next year are to push towards higher maturity levels and continue to seek implementation feedback.

The goals of the module over the next year are to provide a stable basis for implementation of the sharing use case, as well as unification with the CDS Hooks specification in support of the evaluation use case. The Clinical Quality Framework Initiative will use these resources as the basis for implementation projects, targeting an

FMM level of 3 or 4 for all module resources within a year.

Trial-Use Note: We are actively seeking comments on all areas of the module, with particular focus on supporting the following scenarios:

• FHIR-Based Knowledge Artifact Repository

• FHIR-Based Clinical Decision Support Service

• FHIR-Based Quality Measure Evaluation and Reporting

• FHIR-Based Order Set Catalog/Ordering Services

• The use of Measure and MeasureReport for public health indicator reporting

• Enabling population level query

• The use of FHIR Bulk Data as a mechanism to enable population-level quality reporting

• The use of FHIR interactions in general as a mechanism for enabling knowledge artifact repository and distribution functionality.

Feedback is welcome here .

14.0.6 Background

The FHIR Clinical Reasoning module is sponsored by the Clinical Decision Support (CDS) and Clinical Quality Information (CQI) HL7 Work Groups, with input and coordination from the FHIR Infrastructure and Service Oriented Architecture HL7 Work Groups.

The guidance in this module was prepared as a Universal Realm Specification with support from the Clinical Quality Framework (CQF) initiative , which was a public-private partnership sponsored by the Centers for Medicare & Medicaid Services (CMS) and the U.S. Office of the National Coordinator for Health Information Technology (ONC) to identify, develop, and harmonize standards for clinical decision support and electronic clinical quality measurement. The Clinical Quality Framework effort transitioned to HL7's Clinical Quality Information (CQI) and Clinical Decision Support (CDS) Work Groups in 2016.

The Clinical Quality Framework is focused on harmonizing the historically disjointed specifications used by the Clinical Quality Measurement and Clinical Decision Support communities. Specifically, the initiative has focused on the specifications used to represent knowledge artifacts within the two communities. The strategy employed has been to break the conceptual content of knowledge artifacts into three core components, to define common standards for these core components, and to re-use these common standards for both clinical decision support and clinical quality measurement:

• Metadata - Descriptive information about the artifact and its content

• Clinical Information - Information about a patient or population of concern within a given artifact

• Logic - The clinical reasoning involved in an artifact

The first component has resulted in the Clinical Quality Common Metadata Conceptual Model , an informative document harmonizing metadata requirements between Quality Measurement and Decision Support artifacts.

The second component has resulted in the QUICK Conceptual and Logical Models, a harmonization of the Virtual Medical Record (vMR) used in Decision Support and the Quality Data Model (QDM) used in Quality Measurement, and with its core requirements realized in FHIR as the Quality Improvement Core (QICore) profiles . Ongoing work in this area is focusing on coordination with the Clinical Information Modeling Initiative (CIMI) and a methodology for producing FHIR profiles from CIMI models. Currently, the QICore FHIR profiles (which are in turn derived from the US-Core profiles) can be used to model clinical quality data, and to present a consistent model for use in authoring and evaluating clinical quality artifacts.

Finally, the third component has resulted in the Clinical Quality Language specification , a harmonization of the expressive capabilities of the Clinical Decision Support Knowledge Artifact Specification (CDS KAS) (produced by the Health eDecisions (HeD) Standards and Interoperability (S&I) initiative), and the Health Quality Measures Format (HQMF) .

As part of the ongoing CQF initiative pilot efforts, these developing specifications are being used to support knowledge artifact sharing, as well as evaluation of knowledge artifacts as part of decision support request/response and measure evaluation.

This module continues the harmonization of quality domain specifications by defining an approach to using a FHIR server as a component of a knowledge system in both the Knowledge Repository and Knowledge Evaluation Service roles.

14.0.7 Related Specifications

The approach and representations within this guide are derived from and intended to be consistent with the following specifications:

• HL7 Version 3 Standard: Clinical Decision Support Knowledge Artifact Specification, Release 1.3

• HL7 Version 3 Standard: Representation of the Health Quality Measure Format (eMeasure) Normative, Release 1

• HL7 Version 3 Standard: Decision Support Service, Release 2

• HL7 Version 3 Implementation Guide: Decision Support Service, Release 1

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14.0.9 Acknowledgements

The guidance in this module is the work of a joint project between the HL7 Clinical Quality Information and Clinical Decision Support Work Groups with the co-sponsorship of the FHIR Infrastructure, Implementable Technology Specifications, and Service Oriented Architecture Work Groups.

14.1 Using Expressions

URL: https://hl7.org/fhir/R4/clinicalreasoning-topics-using-expressions.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.1 Using Expressions

This topic discusses the use of expression logic within FHIR resources using expression languages such as FHIRPath and Clinical Quality Language (CQL) . The clinical reasoning module resources use the Expression data type to represent logic that enables dynamic functionality to be utilized directly within the Clinical Reasoning resources, as well as within FHIR resources more generally using extensions.

14.1.1 Expression Elements

In general, the use of expressions involves the following elements:

• Description - A natural language description of the logic.

• Language - The expression language used to represent the logic.

• Expression - The expression itself in the identified language.

This general pattern is used to represent expression logic throughout the resources in the Clinical Reasoning module and allows expression logic to be represented at different levels:

• Unstructured - The logic in this case can be communicated using natural language without the expectation that it is computable. This information is useful for communicating the intended dynamic semantics to environments that do not support computable evaluation.

• Structured - The logic is specified as a computable expression in a formal language. This information is useful for communicating the logic to environments that are expected to support the computation.

• Mixed - The logic in this case is communicated using both a natural language description, as well as a computable expression in a formal language. This information is useful for communicating to environments that may support the computation as well as to environments that do not, or to environments that support the computation but desire human-readable information to more easily understand the logic.

For example, the dynamicValue element of the ActivityDefinition and PlanDefinition resources contains an expression element directly. However, for resources that do not define these elements, the cqf-expression extension can be used to enable expression information to be associated with any FHIR resource.

14.1.1.1 Description

The description element can be used either alone, to communicate unstructured logic, or together with the other elements to provide a natural language narrative for the formal representation. The element is typically optional to enable usage in both contexts. Note that if both an expression and a description are provided, the expression is considered to override the description element, although the description should reflect the semantics of the expression as accurately as possible.

```
<dynamicValue>
  <path value="timing.event"/>
  <expression>
    <description value="Now"/>
  </expression>
```

```
</dynamicValue>
```

14.1.1.2 Expression Language

The language element identifies the expression language used to represent the logic. The ExpressionLanguage code system defines support for FHIRPath, Clinical Quality Language, and FHIR Query (FHIR's RESTful query syntax), but support for other languages can also be provided.

The language element is specified using the Media Type of the language. For inline expressions using FHIRPath and Clinical Quality Language, the media type SHALL be specified as follows:

• FHIRPath - text/fhirpath

• Clinical Quality Language - text/cql

For example, the following fragment illustrates the use of CQL to define a dynamicValue as part of an activity definition:

```
<dynamicValue>
  <path value="timing.event"/>
  <expression>
    <language value="text/cql"/>
    <expression value="Now()"/>
  </expression>
</dynamicValue>
```

Note that when extensions are used, the same Expression structure is used for the extension. For example:

```
<event>
  <extension url="http://hl7.org/fhir/StructureDefinition/cqf-expression">
    <valueExpression>
      <language value="text/cql"/>
      <expression value="Now()"/>
    >/valueExpression>
  </extension>
</event>
```

14.1.1.3 Expression

The expression element specifes the formal representation of the expression logic in the identified expression language. The expression may be inline, as in the above examples, or it may be a reference to a named expression defined in a logic library. For example, the following fragment illustrates the use of a named expression to define a dynamicValue as part of an activity definition:

```
<dynamicValue>
  <path value="reasonCode"/>
  <expression>
    <language value="text/cql"/>
    <expression value="RiskAssessmentScore"/>
  </expression>
</dynamicValue>
```

The example specifies that the reasonCode element should be set to the result of evaluating the RiskAssessmentScore expression. This expression is expected to be present in a Library referenced by the containing resource:

```
<library value="Library/mmi-suiciderisk-orderset-logic"/>
```

If the containing resource has a library element (such as ActivityDefinition and PlanDefinition), and only specifies a single library, the expression is evaluated as though it is in scope in that library. However, if the resource references multiple libraries, the referenced expression must be qualified with the name of the library in order to ensure unambiguous resolution. For example, the following fragment illustrates multiple libraries being referenced by the containing resource:

```
<library value="Library/mmi-suiciderisk-orderset-logic"/>
<library value="Library/another-orderset-logic"/>
```

The following fragment illustrates how an expression reference (reasonCode) would be qualified in the case of multiple library references:

```
<dynamicValue>
  <path value="reasonCode"/>
  <expression>
    <language value="text/cql"/>
    <expression value="&amp;quot;mmi-suiciderisk-orderset-logic&amp;quot;.RiskAssessmentScore"/>
  </expression>
</dynamicValue>
```

Note the use of the " escape sequence on the library scope. This escape sequence is needed for XML attributes delimited with double quotes to avoid premature closure of the attribute, which would result in malformed XML. The following is an example of how to escape double quotes in JSON:

```
"dynamicValue": [
  "path": "reasonCode",
  "expression": {
    "language": "text/cql",
    "expression": "\"mmi-suiciderisk-orderset-logic\".RiskAssessmentScore"
  }
]
```

For resources that do not have a library element, the library extension can be used to reference a library from any resource. The library extension enables a resource to add a reference to a library when it could not otherwise.

For simplicity, resources that use logic libraries SHOULD reference at most one library to avoid the need to qualify expression references. Take, for example, the case of updating a library reference in a resource. If the resource has multiple library references, then every expression qualifier that referred to the previous library must be updated. However, if the libraries were combined into a single reference, then only the reference would need to be updated.

14.1.2 Evaluation Context

In general, when an expression is evaluated, it is evaluated in the context of the containing resource. This means that the resource is available within the expression so that the current values of the resource can be accessed.

Note that this is the general case for establishing the evaluation context for expressions. Many of the clinical reasoning resources, such as the Measure, will define specific behavior for the evaluation context that overrides this general behavior.

For FHIRPath expressions, the expression is evaluated with the containing resource using the %context environment variable. For example, when evaluated against a Patient resource, the following expression will return all given elements of all name elements for the patient:

```
%context.name.given
```

And because FHIRPath expressions can be used within CQL, the same approach works for accessing the context in a CQL expression.

14.1.2.1 Parameters

In addition to the evaluation context, the expression may access parameters defined by referenced libraries. As with expression identifier resolution, the name of a parameter can be used directly so long as there is only one library referenced by the resource. Otherwise, the name of the library must be used to qualify the parameter name to avoid ambiguity.

If the expression uses parameters, a Library SHALL be used to define the required parameters.

Note that the way that parameters are bound to the expression from the evaluation context varies based on where the evaluation is taking place, but in general, the parameters are bound by name to the parameters defined in the evaluation operation. For example, in the $apply operation, parameters defined by libraries can be provided as named parameters in the operation invocation. For more details on how parameter binding occurs within each context, refer to the specific documentation for the $apply, $evaluate, and $evaluate-measure operations.

14.1.2.2 Data Requirements

The data requirements for a given expression describe the minimum data required in order to achieve a successful evaluation of the expression. More data may in general be provided, but not less. For example, an expression may reference laboratory test results for hemoglobin A1c tests over the past two years. If a system provides data for the last three years, the expression can still be successfully evaluated, but if a system provides data for only the last year, the expression may produce incorrect results based on the absence of the expected data.

In the scenario that an evaluation service is not colocated with the clinical information, the service has no general way of knowing whether or not a request has fulfilled the stated data requirements for an expression. As such, this is a critical aspect of implementation. The service assumes the stated data requirements will be provided as part of a request, and requesters shall provide at least the data specified by the data requirements when requesting evaluation for an expression.

Within CQL, data requirements can be inferred based on the retrieve expressions used. This process is described in detail in the Clinical Quality Language Specification, but in general, the set of data requirements is represented by elements of the DataRequirement type. This type is similar to a parameter definition, with the following differences:

• No name element - Because there is no need for a name to inference on the data, there is no need to specify a name for the data required.

• Code and date filters - Filtering by terminology and date range are widely used mechanisms for searching within clinical data, but profiling to achieve this is less dynamic and potentially a significant amount of overhead. The DataRequirement type provides a lightweight mechanism to represent these filters via the codeFilter and dateFilter elements.

• Referenced elements - It is often important to communicate not only the type of data required, but which specific elements within that type will be referenced by a particular knowledge artifact or service module. This is accomplished in the DataRequirement type using the mustSupport element.

For CQL expressions that contain retrieves, a Library SHOULD be used to describe the data requirements. However, the $data-requirements operation can also be used to infer the data requirements based on usage

within the expression.

14.1.3 FHIRPath

FHIRPath is a lightweight path-based navigation language intended to provide simple but flexible access to graph-structured data. It is defined as a general purpose specification available here .

FHIRPath is used throughout the FHIR specification whenever path selection is needed, such as in the definition of search parameters, or when describing invariants as part of the definition of resources and profiles. FHIRPath expressions generally provide a simple and effective means of navigating and accessing the elements of FHIR resources to retrieve data.

14.1.4 Clinical Quality Language

Clinical Quality Language (CQL) defines a high-level clinically-oriented syntax to enable formal representation of clinical logic. It is defined as a general purpose specification available here:

HL7 Standard: Clinical Quality Language Specification, Release 1.3 .

CQL is used to describe the logic used by knowledge artifacts such as clinical decision support rules, quality measure logic, and conditions and actions within order sets and protocols. CQL expressions generally provide a means of computing new information from existing data.

14.2 Definitional Resources

URL: https://hl7.org/fhir/R4/clinicalreasoning-topics-definitional-resources.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.2 Definitional Resources

Part of defining knowledge artifacts such as order sets, protocols, and decision support rules is describing in a patient- or context-independent way the activities to be performed. For example, when defining an order set, the orderable items must be described with enough detail to enable the creation of the items when the order set is applied. These descriptions can be thought of as templates for the creation of patient- or context-specific resources and are often referred to as definitional resources, to distinguish them from intent resources (that signal an intention to take some action for a specific patient), as well as event resources (that signal that some action has actually been taken for a patient).

In the most general case, these definitional resources only need to describe the most basic aspects of the activity to be performed, such as:

• Who - Which person should perform the activity, often specified as a participant role such as a provider specialty.

• What - What type of activity should be performed, often specified simply as a category or a concept.

• When - When the activity should be performed, often specified relative to the time the plan is applied.

• Where - Where the activity should take place, often specified as a facility role.

• Why - Why the activity should be performed

However, this level of conceptual description often does not carry enough information to enable computable description of activities. For example, medication activities will often involve specific drug and dosage information that must be captured as part of the definition. Further, it is often the case that the values for the elements of resources to be created cannot be specified exactly as part of the definition, but must be specified using a formula that allows for the calculation to be based on patient- or context-specific information.

The ActivityDefinition resource supports the description of definitional resources within FHIR:

Who | participantType | Specifies the type of participant that should perform the activity.
What | kind and code | Specifies the type of activity to be performed.
When | timing | Specifies when the activity should be performed.
Where | location | Specifies where the activity should be performed.
Why | reason, documentation | Specifies why the activity should be performed.

For example, the following fragment illustrates a definition to create a referral request:

```
<ActivityDefinition>
  <description value="refer to primary care mental-health integrated care program for evaluation and treatment
   of mental health conditions now"/>
  <kind value="ServiceRequest"/>
  <code>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="306206005"/>
    </coding>
  </code>
  <timingTiming>
    <event>
      <extension url="http://hl7.org/fhir/StructureDefinition/cqf-expression">
        <valueExpression>
          <language value="text/cql"/>
```

```
          <expression value="Now()"/>
        </valueExpression>
      </extension>
    </event>
  </timingTiming>
  <participantType value="practitioner"/>
</ActivityDefinition>
```

Note the use of an expression to represent the value of the timing element as Now().

For medication activities, the ActivityDefinition resource has some basic elements such as the product and quantity, and dosageInstruction but there are cases where elements that need to be set on the resulting MedicationRequest are not present on the ActivityDefinition (such as dispenseRequest). In those cases, the dynamicValue expression elements can be used to describe the values for elements that are present on the target resource, but not in the definitional resource. For example:

```
<ActivityDefinition>
  <id value="citalopramPrescription"/>
  <status value="draft"/>
  <category value="drug"/>
  <productReference>
    <reference value="#citalopramMedication"/>
  </productReference>
  <dynamicValue>
    <path value="dispenseRequest.numberOfRepeatsAllowed"/>
    <expression>
      <language value="text/cql"/>
      <expression value="3"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="dispenseRequest.quantity"/>
    <expression>
      <language value="text/cql"/>
      <expression value="30 '{tbl}'"/>
    </expression>
  </dynamicValue>
</ActivityDefinition>
```

Note to implementers: Although there is currently only one definitional resource defined (ActivityDefinition), the concept is general, and this was a deliberate design decision to avoid the overhead of defining and maintaining a different definitional resource for every category of request. We anticipate that as the use cases of ActivityDefinition require more specialized elements to be added, the resource may be split.

14.3 Representing Knowledge Artifacts

URL: https://hl7.org/fhir/R4/clinicalreasoning-knowledge-artifact-representation.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.3.0 Representing Knowledge Artifacts

For the purposes of the Clinical Reasoning module, a Knowledge Artifact is a structured, computable, and shareable representation of clinical knowledge. A comprehensive definition of clinical knowledge is beyond the scope of this module, but we effectively mean any knowledge relevant to improving patient care. For example, "Patients with diabetes should have regular comprehensive foot examinations to identify risk factors predictive of ulcers and amputations" is a statement of clinical knowledge. A knowledge artifact would then be some representation that can be used to integrate that knowledge into existing healthcare delivery systems.

There are any number of ways to accomplish this integration of clinical knowledge, ranging from direct implementation within an existing EHR system to the use of high-level representation and reasoning systems that can adapt clinical workflow to provide cognitive support to clinicians. Within this module, we aim to provide a representation that is both flexible enough to represent a broad range of knowledge artifacts, but simple and specific enough to enable automated integration of the content.

14.3.0.1 Knowledge Artifact Components

To achieve these goals, we define several components that are present in a broad variety of knowledge artifacts:

Component | Description
Artifact Identity | Information used to uniquely identify an artifact.
Metadata | Information about the knowledge artifact such as lifecycle status, publisher, documentation, and supporting evidence.
Action Definitions | Descriptions of actions to be taken as part of the implementation of knowledge
Trigger Definitions | Information about what events should trigger the use of the artifact
Expression Logic | Expressions used to represent reasoning such as whether or not some criteria is satisfied, or calculation of new values from existing ones.

14.3.0.1.1 Artifact Identity

As with any FHIR resource, the instance identity is specified by the id element. Knowledge artifacts also have a canonical identifier, specified by the url element that serves as the globally unique, "canonical URL" for the artifact that always identifes the resource. The canonical URL is a special kind of logical (or business) identifier, and artifacts may define any number of additional logical identifiers based on the content or behavior they provide, such as the CMS or NQF identifiers for measure content. These logical identifiers can be provided in the identifier element defined on each resource.

In addition to identity, the version element can be used to specify a content version for the artifact. In general, version numbers are required for published, non-experimental artifacts, and optional otherwise.

The following fragment shows the use of the logical identifier and version elements:

```
<url value="http://motivemi.com/artifacts/PlanDefinition/low-suicide-risk-order-set"/>
  <identifier>
  <use value="official"/>
  <system value="http://motivemi.com/artifacts"/>
  <value value="mmi:low-suicide-risk-order-set"/>
</identifier>
<version value="1.0.0"/>
```

When using the identifier element to provide the Entity Identifier for a service module consistent with the Decision Support Service (DSS) specification, the following mappings apply:

DSS Model Element | FHIR Element | Example
businessId | Identifier.value | hemoglobin-control-rule
scopingEntityUri | Identifier.system | com.clinica
version | Artifact.version | 1.0.0

14.3.0.1.2 Metadata

Each of the various knowledge artifact resources defines a set of elements that provide consistent representation of metadata such as title, description, and usage, repository indexing and publication information, as well as additional supporting documentation and evidence:

Element | Description
name | A machine-friendly name for the artifact example
title | A user-friendly title for the artifact example
status | The lifecycle status for the artifact (draft, active, or inactive) example
experimental | Whether the artifact is for testing purposes, rather than production usage example
description | A natural language description of the artifact example
purpose | Describes the purpose of the artifact example
usage | Describes the clinical usage of the artifact example
approvalDate | The date when this artifact was approved by the publisher example
lastReviewDate | The date this version of the artifact was last reviewed example
effectivePeriod | The effective date range for the artifact example
useContext | Describes the context of use for this artifact example
topic | Descriptional topics for the artifact example
contributor | A content contributor example
publisher | The name of the publisher (either an organization or an individual) example
contact | Contact details for the publisher example
copyright | Legal copyright description for the artifact example
relatedArtifact | Related resources for the artifact, including additional documentation, supporting evidence, or knowledge management dependencies example

14.3.0.1.2.1 Documentation

General documentation elements, including description, purpose, and usage, provide immediately available documentation as part of the artifact. Additional documentation can be provided using the relatedArtifact element to link to more detailed information, supporting evidence, and to additional required artifacts as appropriate.

```
<title value="Appropriate Testing for Children with Pharyngitis"/>
<status value="active"/>
<experimental value="true"/>
<description value="Percentage of children 2-18 years of age who were diagnosed with pharyngitis, ordered an antibiotic
    and
  received a group A streptococcus (strep) test for the episode."/>
<purpose value="The Infectious Diseases Society of America (IDSA) "recommends swabbing the throat and testing for GAS
  pharyngitis by rapid antigen detection test (RADT) and/or culture because the clinical features alone do not reliably
    discriminate
  between GAS and viral pharyngitis except when overt viral features like rhinorrhea, cough, oral ulcers, and/or
    hoarseness are present""/>
<topic>
  <coding>
    <system value="http://loinc.org" />
    <code value="57024-2" />
  </coding>
</topic>
```

For information on the useContext and topic elements, refer to the Knowledge Artifact Distribution section.

14.3.0.1.2.2 Publication Information

The status element describes the lifecycle status of the artifact, indicating whether it is a draft, active, or retired artifact. This representation is consistent with the states described by the Quality Metadata Conceptual Model. The following table provides a mapping from the states defined here, to the states as defined in the Clinical Decision Support Knowledge Artifact Specification (CDS KAS), as well as the Decision Support Service (DSS) standard:

Experimental? | Status | CDS KAS Mapping | DSS Mapping
true | draft | Draft, InTest | DRAFT, DEFINED
true | active | InTest | DEFINED
true | retired | Inactive | REJECTED
false | draft | Draft | DRAFT
false | active | Active | APPROVED, PROMOTED
false | retired | Inactive | REJECTED, RETIRED

Note that the experimental element can be used independent of the lifecycle status, allowing the full lifecycle of an artifact to be tested.

14.3.0.1.3 Action Definitions

Actions describe the "output" actions of the knowledge artifact. These actions may be in the form of messages (such as reminders), structured clinical acts (e.g., a laboratory test order) that can be implemented via clinical systems such as a computerized provider entry system or a documentation system, or they may create new events (e.g., declaration of a patient state such as failure of a treatment). Each action definition contains the following basic information:

Description | Descriptive information such as identifier, label, title, and textEquivalent.
Documentation | Supporting documentation, citations, or evidence in support of the action.
Related Action Definitions | Each action definition can specify relationships to other actions, allowing timings and offsets to be expressed.
Trigger Definitions | Triggering events for the action. See the section on trigger definitions below for more information.
Condition | An optional condition specifying whether the action definition applies.
Timing | A description of when the action definition should occur.
Behavior | Various behaviors that communicate the intended semantics of groups and actions (e.g. selection behavior, grouping behavior, etc.)
Type | The type of action to be performed (e.g. create, update, delete, etc.)
Activity Definition | A description of what activity is to be performed. To enable reuse and modularity, ActivityDefinition resources may be reused by multiple action definitions.
Dynamic Values | Expressions that can be used to specify dynamic values for the resulting activity.
Child Actions | Each action may contain any number of child actions, allowing a hierarchy of actions to be described (for example, sections in an order set).

The following fragment shows a simple referral creation action from an example order set for suicide risk assessment and management:

Note: the provided examples can be used formally, but can also be informational. The intent of the structures is to enable communication of the behavior. For systems that support calculation, they would be formal, but for systems that don't support calculation, the resource should still provide enough information to communicate the behavior.

```
<action>
  <textEquivalent value="Refer to outpatient mental health program for evaluation and treatment of mental health
    conditions now"/>
  <definition value="#referralToMentalHealthCare"/>
  <dynamicValue>
    <path value="timing.event"/>
    <expression>
      <language value="text/cql"/>
      <expression value="Now()"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="specialty"/>
    <expression>
      <language value="text/cql"/>
      <expression value="Code '261QM0850X' from SuicideRiskLogic."NUCC Provider Taxonomy" display 'Adult Mental
    Health'"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="occurrenceDateTime"/>
    <expression>
      <language value="text/cql"/>
      <expression value="SuicideRiskLogic.ServiceRequestFulfillmentTime"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="subject"/>
    <expression>
      <language value="text/cql"/>
      <expression value="SuicideRiskLogic.Patient"/>
    </expression>
  </dynamicValue>
```

```
  <dynamicValue>
    <path value="requester.agent"/>
    <expression>
      <language value="text/cql"/>
      <expression value="SuicideRiskLogic.Practitioner"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="reasonCode"/>
    <expression>
      <language value="text/cql"/>
      <expression value="SuicideRiskLogic.RiskAssessmentScore"/>
    </expression>
  </dynamicValue>
  <dynamicValue>
    <path value="reasonReference"/>
    <expression>
      <language value="text/cql"/>
      <expression value="SuicideRiskLogic.RiskAssessment"/>
    </expression>
  </dynamicValue>
</action>
```

The example describes how to create a ServiceRequest resource using information available from the context in which the order set is applied. The definition includes expressions for retrieving contextual information using the dynamicValue element of the ActivityDefinition resource. For more information on using expressions to provide dynamic values as part of a definition, refer to the Using Expressions topic in this module.

14.3.0.1.3.1 Action Type

Each action may specify the type of action to perform:

- Create
- Update
- Remove
- Fire Event

The Create, Update, and Remove actions indicate that particular resources should be created, updated, or removed, and the Fire Event action indicates that a particular named event should be triggered.

Note that this element is optional, and is used to indicate what type of action is to be taken with the associated activity definition. For example, an action that indicates create with an activity definition that defines a referral request means that the referral request should be realized and the resulting resource created (i.e. applied to the patient's record). An action that indicates update with an activity definition would indicate that an existing resource on the patient's record should be updated based on the new resource content.

14.3.0.1.3.2 Trigger Definitions

Each action may have any number of triggers associated to indicate what events should trigger the action to be taken. The following types of triggers may be defined:

- Named Event
- Periodic Event
- Data Event

A named event is an event identified by the implementation environment. This type of event allows Event-Condition-Action (ECA) rules to be triggered by any event generated within the implementation environment. A periodic event occurs on a fixed or periodic schedule. And finally, a data event occurs in response to some data-related event in the integrated environment such as a record being added or updated.

For triggered PlanDefinitions, the event means the PlanDefinition apply process should be performed. How that actually happens within an implementation is not prescribed.

As examples, the following fragments illustrate the use of a named event, a scheduled event, and a data event as part of an ECA rule definition:

```
<trigger>
  <type value="named-event"/>
  <name value="Admission"/>
</trigger>
```

```
<trigger>
  <type value="periodic"/>
  <timingDate>10/12/2015</timingDate>
</trigger>
```

```
<trigger>
  <type value="data-added"/>
  <data>
    <type value="Condition"/>
    <codeFilter>
      <path value="code"/>
      <valueSet value="urn:oid:2.16.840.1.113883.3.464.1003.111.12.1006"/>
```

```
    </codeFilter>
  </data>
</trigger>
```

14.3.0.1.3.3 Expression Logic

Each action may have associated conditions that specify whether the action is applicable, or when it should be started and stopped. The $apply operation is used to apply a plan definition, and the parameters to apply are available as context to expressions within the plan definition, including the applicable condition and the dynamic value expressions.

For example, the following fragment shows the condition element of an action definition used in an ECA rule. The condition references the named expression "NoScreening":

```
<condition>
  <kind value="applicability"/>
  <expression>
    <language value="text/cql"/>
    <expression value="NoScreening"/>
  </expression>
</condition>
```

14.3.0.2 Knowledge Artifact Resources

The Clinical Reasoning module defines several resources, including Library, PlanDefinition, and Measure to represent each of the components described above. By combining these components in different ways, several different types of knowledge artifacts can be represented:

Artifact Type | Description | Example

Library | A general purpose container for clinical knowledge, potentially defined in non-FHIR structures. For example, a sharable package of clinical logic expressed in Clinical Quality Language. | Expressions that define the criteria that must be met to determine whether or not a patient has diabetes.

Event-Condition-Action (ECA) Rule | A decision support rule of the form [on Event] if Condition then Action, defining actions that should be taken whenever some condition is met in response to a particular event. | On a patient checking in to a clinic appointment, if the patient has diabetes and has not had a recent foot examination, place an unsigned order for a comprehensive foot exam.

Order Set | A pre-defined and approved group of orders related to a particular clinical condition (e.g., hypertension treatment and monitoring) or stage of care (e.g., hospital admission to Coronary Care Unit). Order sets are used within electronic health record systems as a checklist for physicians when treating a patient with a specific condition or in a specific context. An order set is a structured collection of orders presented to the physician in a computerized physician order entry system (CPOE). | The specific orders to be placed when a patient presents with diabetic ketoacidosis.

Protocol | A pre-defined procedural method for standardizing a set of activities. As examples, protocols can be used within laboratory settings to standardize handling of specimens for specific tests; standardize the treatment steps for a particular condition; or standardize the steps to be taken in a particular research trial. | The specific tasks that should be performed to provide effective treatment for a new diagnosis of diabetes.

Documentation Template | A structured form for recording information on a patient into a set of predefined data slots. | A form describing the information that should be gathered during a comprehensive diabetic foot exam.

Measure | A quantitative tool to assess the performance of an individual or organization with respect to a specified process or outcome via the measurement of actions, processes, or outcomes of clinical care. Quality measures are often derived from clinical guidelines and are designed to determine whether the appropriate care has been provided given a set of clinical criteria and an evidence base. | Percentage of patients aged 18-75 years of age with diabetes who had a foot exam during the measurement period.

14.3.0.3 Library

The Library resource is a general purpose container for clinical knowledge that may be defined in non-FHIR representations. For example, a Clinical Quality Language expression library can be represented as a Library resource so that it can be referenced as part of other knowledge artifacts.

For example, the following fragment shows a Library that contains the CQL logic for Chlamydia Screening decision support:

```
<Library>
  <id value="example"/>
  <identifier>
    <use value="official"/>
    <value value="ChalmydiaScreening_Common"/>
  </identifier>
  <version value="2.0.0"/>
  <title value="Chlamydia Screening Common Library"/>
  <status value="draft"/>
  <type>
    <coding>
      <code value="logic-library"/>
    </coding>
  </type>
  <date value="2015-07-22"/>
  <description value="Common Logic for adherence to Chlamydia Screening guidelines"/>
  <topic>
    <text value="Chlamydia Screening"/>
  </topic>
  <relatedArtifact>
    <type value="depends-on"/>
    <resource value="Library/library-quick-model-definition"/>
  </relatedArtifact>
```

```
  <dataRequirement>
    <type value="Condition"/>
    <codeFilter>
      <path value="code"/>
      <valueSet value="urn:oid:2.16.840.1.113883.3.464.1003.111.12.1006"/>
    </codeFilter>
  </dataRequirement>
  <content>
    <contentType value="text/cql"/>
    <url value="library-example-content.cql"/>
  </content>
</Library>
```

14.3.0.4 Event-Condition-Action (ECA) Rule

An Event-Condition-Action (ECA) rule is an artifact with the general syntax "on event, if condition is true, then do action." The event triggers the invocation of the rule; the condition is a logical test that, if satisfied or evaluates "true", causes an action; while the action describes a set of activities to be performed. These actions may in turn cause further events to occur, which may in turn cause other ECA rules to fire.

Note that the "activity" to be performed may be to document a patient-specific assessment performed by a clinical decision support service (e.g. patient has a 37% chance of a heart attack in the next 5 years), not just recommendations/recommended actions. In this case, the action would be to "create" a RiskAssessment.

The PlanDefinition resource can be used to represent an ECA rule using the action element. A single, top-level action represents the overall rule, with the trigger element used to specify the triggering event(s), the condition element used to specify the applicable condition for the rule, and the action element describing the action to be performed.

For example, the following fragment illustrates a simple use of PlanDefinition to expose a Chlamydia Screening ECA rule:

```
<PlanDefinition>
  <id value="chlamydia-screening-intervention"/>
  <identifier>
    <use value="official"/>
    <value value="ChlamydiaScreening_CDS_UsingCommon"/>
  </identifier>
  <version value="2.0.0"/>
  <title value="Chalmydia Screening CDS Example Using Common"/>
  <status value="draft"/>
  <date value="2015-07-22"/>
  <description value="Chlamydia Screening CDS Example Using Common"/>
  <topic>
    <text value="Chlamydia Screening"/>
  </topic>
  <library value="Library/example"/>
  <action>
    <title value="Patient has not had chlamydia screening within the recommended timeframe..."/>
    <condition>
      <kind value="applicability"/>
      <expression>
        <language value="text/cql"/>
        <expression value="NoScreening"/>
      </expression>
    </condition>
    <dynamicValue>
      <path value="$this"/>
      <expression>
        <language value="text/cql"/>
        <expression value="ChlamydiaScreeningRequest"/>
      </expression>
    </dynamicValue>
  </action>
</PlanDefinition>
```

Note that the use of $thisin the path element here indicates that the result of the expression is the entire result for the action, rather than providing the value for a specific element of the result. In other words, the ChlamydiaScreeningRequest expression will result in an entire ServiceRequest resource, which is the action to be performed.

The following is an example of a possible result when invoking the $apply operation on the above ECA Rule:

```
<CarePlan>
  <id value="chlamydia-screening-intervention-cp"/>
  <contained>
    <RequestGroup>
      <id value="chlamydia-screening-intervention-rg"/>
      <contained>
        <ServiceRequest>
          <id value="chlamydia-screening-service-request"/>
          <status value="active"/>
          <intent value="order"/>
          <code>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="412761009"/>
              <display value="Urine screen for chlamydia (procedure)"/>
```

```
          </coding>
        </code>
        <subject>
          <reference value="Patient/example"/>
        </subject>
      </ServiceRequest>
    </contained>
    <status value="active"/>
    <intent value="order"/>
    <action>
      <type>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/action-type"/>
          <code value="create"/>
        </coding>
      </type>
      <resource>
        <reference value="#chlamydia-screening-service-request"/>
      </resource>
    </action>
  </RequestGroup>
</contained>
<status value="active"/>
<intent value="order"/>
<subject>
  <reference value="Patient/example"/>
</subject>
<activity>
  <reference>
    <reference value="#chlamydia-screening-intervention-rg"/>
  </reference>
<activity>
</CarePlan>
```

14.3.0.5 Order Set

An order set is a pre-defined and approved group of orders related to a particular clinical context (e.g., hypertension treatment and monitoring) or stage of care (e.g., hospital admission to Coronary Care Unit) or a particular grouping (e.g. pediatrics, adult, males over 50, etc.). An order set is typically used as a checklist for clinicians when managing a patient with a specific context. It is a structured collection of orders relevant to that condition or context and presented to the clinician in a computerized provider order entry system (CPOE).

The PlanDefinition resource can represent an Order Set using action definitions to define the orderable items in the order set. The items in an order set are typically organized hierarchically, as a set of sections, sub-sections, etc., with the actions themselves at the very bottom of the structure. In the PlanDefinition resource, each section, and sub-section, as well as the individual items are all represented as action definitions. Each group and/or subgroup may have behavior indicators associated with it, e.g., the number of actions that can/should/must be selected from the group of actions.

For example, the following fragment illustrates the items in a suicide risk order set:

Note that this example is a fragment of a larger example for the PlanDefinition resource, available .

here

```
<!-- Suicide Risk Assessment and Outpatient Management -->
<action>
  <title value="Suicide Risk Assessment and Outpatient Management"/>
  <action>
    <title value="Consults and Referrals"/>
    <groupingBehavior value="logical-group"/>
    <selectionBehavior value="any"/>
    <action>
      <textEquivalent value="Refer to outpatient mental health program for evaluation and treatment of mental health
       conditions now"/>
      <definition value="#referralToMentalHealthCare"/>
      <dynamicValue>...</dynamicValue>
    </action>
  </action>
  <action>
    <title value="Medications"/>
    <groupingBehavior value="logical-group"/>
    <selectionBehavior value="at-most-one"/>
    <action>
      <title value="First-Line Antidepressants"/>
        <documentation>
          <type value="citation"/>
          <document>
          <extension url="http://hl7.org/fhir/StructureDefinition/cqf-qualityOfEvidence">
            <valueCodeableConcept>
              <coding>
                <system value="http://terminology.hl7.org/CodeSystem/evidence-quality"/>
                <code value="high"/>
              </coding>
              <text value="High Quality"/>
            </valueCodeableConcept>
          </extension>
          <contentType value="text/html"/>
```

```
          <url value="http://psychiatryonline.org/pb/assets/raw/sitewide/practice_guidelines/guidelines/mdd.pdf"/>
          <title value="Practice Guideline for the Treatment of Patients with Major Depressive Disorder"/>
        </document>
      </documentation>
      <groupingBehavior value="logical-group"/>
      <selectionBehavior value="at-most-one"/>
      <action>
        <title value="Selective Serotonin Reuptake Inhibitors (Choose a mazimum of one or document reasons for
    exception)"/>
        <documentation>
          <type value="citation"/>
          <document>
            <contentType value="text/html"/>
            <url value="http://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=6daeb45c-451d-b135-bf8f-2d6dff4b6b01"/>
            <title value="National Library of Medicine. DailyMed website. CITALOPRAM- citalopram hydrobromide tablet,
    film coated."/>
          </document>
        </documentation>
        <groupingBehavior value="logical-group"/>
        <selectionBehavior value="at-most-one"/>
        <action>
          <textEquivalent value="citalopram 20 mg tablet 1 tablet oral 1 time daily now (30 table; 3 refills)"/>
          <definition value="#citalopramPrescription"/>
          <dynamicValue>...</dynamicValue>
        </action>
        <action>
          <textEquivalent value="escitalopram 10 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills)"/>
        </action>
        <action>
          <textEquivalent value="fluoxetine 20 mg capsule 1 capsule oral 1 time daily now (30 tablet; 3 refills)"/>
        </action>
        <action>
          <textEquivalent value="paroxetine 20 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills)"/>
        </action>
        <action>
          <textEquivalent value="sertraline 50 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills)"/>
        </action>
      </action>
      <action>
        <textEquivalent value="Dopamine Norepinephrine Reuptake Inhibitors (Choose a maximum of one or document reasons
    for exception)"/>
      </action>
      <action>
        <textEquivalent value="Serotonin Norepinephrine Reuptake Inhibitors (Choose a maximum of one or doument reasons
    for exception)"/>
      </action>
      <action>
        <textEquivalent value="Norepinephrine-Serotonin Modulators (Choose a maximum of one or document reasons for
    exception)"/>
      </action>
    </action>
  </action>
</action>
```

In this example, the order set structure consists of:

```
<!-- Suicide Risk Assessment and Outpatient Management -->
  <!-- Consults and Referrals -->
      <!-- Refer to outpatient mental health program for evaluation and treatment of mental health conditions now -->
  <!-- Medications -->
      <!-- First-Line Antidepressants -->
        <!-- Selective Serotonin Reuptake Inhibitors (Choose a maximum of one or document reasons for exception) -->
          <!-- citalopram 20 mg tablet 1 tablet oral 1 time daily now (30 table; 3 refills) -->
          <!-- escitalopram 10 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills) -->
          <!-- fluoxetine 20 mg capsule 1 capsule oral 1 time daily now (30 tablet; 3 refills) -->
          <!-- paroxetine 20 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills) -->
          <!-- sertraline 50 mg tablet 1 tablet oral 1 time daily now (30 tablet; 3 refills) -->
        <!-- Dopamine Norepinephrine Reuptake Inhibitors (Choose a maximum of one or document reasons for exception) -->
        <!-- Serotonin Norepinephrine Reuptake Inhibitors (Choose a maximum of one or doument reasons for exception) -->
        <!-- Norepinephrine-Serotonin Modulators (Choose a maximum of one or document reasons for exception) -->
```

Each group defines the selection and grouping behavior for the items in the group, as well as providing links to supporting documentation for the particular items in the order set.

14.3.0.6 Protocol

A protocol is a pre-defined procedural method for standardizing a set of activities. As examples, protocols can be used within laboratory settings to standardize handling of specimens for specific tests; standardize the treatment steps for a particular condition; or standardize the steps to be taken in a particular research trial.

The PlanDefinition resource can be used to represent a Protocol, using the action definitions to define the steps of the protocol and the relationships between them. Triggering events for each action, as well as applicable conditions can all be specified using the appropriate action elements within the PlanDefinition.

As a simple example, the following fragment shows a portion of a BMI Assessment protocol:

```
<action>
  <!--  step title  -->
  <title value="Measure BMI"/>
  <description value="Measure, Weight, Height, Waist, Circumference; Calculate BMI"/>
```

```
  <!--  description of activity  -->
  <textEquivalent value="Weight must be measured so that the BMI can be calculated. Most charts are based on weights
    obtained with the patient wearing undergarments and no shoes. BMI can be manually calculated
    (kg/[height in meters]2), but is more easily obtained from a nomogram. Waist circumference
    is important because evidence suggests that abdominal fat is a particularly strong determinant
    of cardiovascular risk in those with a BMI of 25 to 34.9 kg/m2. Increased waist circumference
    can also be a marker of increased risk even in persons of normal weight. The technique
    for measuring waist circumference is described in the text. A nutrition assessment will
    also help to assess the diet and physical activity habits of overweight patients"/>
  <goalId value="reduce-bmi-ratio"/>
  <condition>
    <kind value="applicability"/>
    <expression>
      <description value="The practitioner must seek to determine whether the patient has ever been overweight.
        While a technical definition is provided, a simple question such as 'Have you ever been
        overweight?' will accomplish the same goal. Questions directed towards weight history,
        dietary habits, physical activities, and medications may provide useful information about
        the origins of obesity in particular patients. For those who have not been overweight,
        a 2 year interval is appropriate for the reassessment of BMI. While this time span is
        not evidence-based, it is believed to be a reasonable compromise between the need to identify
        weight gain at an early stage and the need to limit the time, effort, and cost of repeated
        measurements."/>
      <language value="text/cql"/>
      <expression value="exists ([Condition: Obesity]) or not exists ([Observation: BMI] O where O.effectiveDateTime 2
    years or less before Today())"/>
    </expression>
  </condition>
  <requiredBehavior value="must-unless-documented"/>
  <cardinalityBehavior value="single"/>
  <definition value="#procedure"/>
</action>
```

14.3.0.7 Documentation Template

A documentation template is a structured form for recording information on a patient into a set of pre-defined data slots. These templates are used to guide structured data entry within an EHR or other clinical information system. Some types of clinical documents that can be represented via the documentation template artifacts are encounter summaries, procedure notes, patient-reported outcomes, and flowsheets.

The Questionnaire resource provides a basis for representing the structure of a documentation template. Using the expression extensions, the resource can be extended to provide dynamic behavior such as calculating values, showing/hiding content based on answers to questions, and other interactive functionality.

Documentation Templates often include not only a description of the data to be collected, but the behavior of the template during evaluation. For example, whether or not to display a particular question or group of questions based on answers to previously asked questions, or calculating the value of an answer based on other answers. The Questionnaire resource by itself does not provide this functionality, so this module introduces a CQF-Questionnaire profile of the Questionnaire resource to allow this behavior to be modeled.

As an example, the PHQ-9 Health Questionnaire contains a question that is answered by totaling the weights from the answers of each of the previous questions. The CQF Questionnaire example illustrates the representation of this questionnaire with the required logic for computation.

The following fragment taken from that example illustrates the use of the expression extension to calculate the TotalScore answer for a PHQ-9 Depression Assessment instrument:

```
<item>
  <extension url="http://hl7.org/fhir/StructureDefinition/cqf-expression">
    <valueExpression>
      <language value="text/cql"/>
      <expression value="CalculateTotalScore"/>
    </valueExpression>
  </extension>
  <linkId value="TotalScore"/>
  <code>
    <system value="http://loinc.org"/>
    <code value="44261-6"/>
  </code>
  <text value="Total score"/>
  <type value="integer"/>
  <required value="true"/>
</item>
```

14.3.0.8 Measure

A quantitative tool to assess the performance of an individual or organization with respect to a specified process or outcome via the measurement of actions, processes, or outcomes of clinical care. Note that the Measure itself does not contain any logic; rather a Library resource is referenced that contains the logic required by the measure, and the various expression elements, such as population criteria, reference named expressions within that library (or libraries).

For example, the following fragment defines how to calculate the controlling high blood pressure measure:

```
<Measure>
  <id value="cbp"/>
  <status value="active"/>
```

```
  <experimental value="true"/>
  <description value="Controlling High Blood Pressure Cohort Definition"/>
  <topic>
    <coding>
      <system value="http://loinc.org"/>
      <code value="57024-2"/>
    </coding>
  </topic>
  <library value="Library/cbp-logic"/>
  <group>
    <population>
      <code value="initial-population"/>
      <criteria value="In Demographic"/>
    </population>
  </group>
  <group>
    <population>
      <code value="initial-population"/>
      <criteria value="BP: Systolic"/>
    </population>
  </group>
  <group>
    <population>
      <code value="initial-population"/>
      <criteria value="BP: Diastolic"/>
    </population>
  </group>
</Measure>
```

14.3.1 Supporting Documentation

URL: https://hl7.org/fhir/R4/clinicalreasoning-topics-supporting-documentation.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.3.1 Supporting Documentation

There are many kinds of supporting documentation that can be provided with clinical quality improvement artifacts, from detailed documentation, to references to the guidelines from which artifacts are derived, to grades and scores indicating quality of evidence or the strength of a recommendation.

The relatedArtifact element present on the Clinical Reasoning Module artifacts allows supporting documentation to be provided, either in the form of a reference to a resource, such as a DocumentReference, or as a reference to external content as an Attachment.

This same mechanism can be used to attach supporting documentation to the actionDefinition elements of a PlanDefinition, allowing supporting documentation for specific steps in a rule or protocol, as well as to the action elements of a RequestGroup to provide supporting documentation for individual recommendations as part of the result of a decision support evaluation.

```
<documentation>
  <type value="justification"/>
  <document>
  <extension url="http://hl7.org/fhir/StructureDefinition/cqf-qualityOfEvidence">
    <valueCodeableConcept>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/evidence-quality"/>
      <code value="high"/>
    </coding>
    <text value="High Quality"/>
    </valueCodeableConcept>
  </extension>
  <contentType value="text/html"/>
  <url value="http://psychiatryonline.org/pb/assets/raw/sitewide/practice_guidelines/guidelines/mdd.pdf"/>
  <title value="Practice Guideline for the Treatment of Patients with Major Depressive Disorder"/>
  </document>
</documentation>
```

In all these cases, the qualityOfEvidence and strengthOfRecommendation extensions can be used to grade the information being provided. For example, the qualityOfEvidence extension can be attached to a DocumentReference to indicate a rating for the quality of the evidence represented in the document. Similarly, a strengthOfRecommendation extension can be attached to the action element of a RequestGroup to indicate the strength of the recommendation being made.

These extensions are bound to example valuesets based on the GRADE rating systems for quality of evidence and strength of recommendation. Other rating systems could be used by defining the appropriate valuesets and profiles to make use of them.

14.4 Sharing Knowledge Artifacts

URL: https://hl7.org/fhir/R4/clinicalreasoning-knowledge-artifact-distribution.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.4 Sharing Knowledge Artifacts

One of the primary use cases of the Clinical Reasoning Module is to enable the sharing and distribution of computable clinical knowledge artifacts:

The distribution use case involves enabling knowledge artifacts to be distributed as resources via a FHIR server. The search and read interactions defined by the FHIR infrastructure can be used for this purpose. The Library, ActivityDefinition, PlanDefinition, Questionnaire, and Measure resources can all be used to represent knowledge artifacts, and so define several search parameters specifically to enable searching based on the various attributes of a knowledge artifact. A FHIR service that supports at least searching and retrieval of these resources is then a basic Knowledge Artifact Repository. More advanced knowledge management capabilities such as change management, semantic indexing, and dependency tracking can all be provided on top of this basic infrastructure.

In particular, the useContext and topic elements are intended to provide both semantic and topical indexing functionality for use in knowledge repositories. For example, the following fragment illustrates the use of the useContext element to indicate semantic usage context for an example suicide risk order set:

```
<coverage>
  <patientAgeGroup>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="133936004"/>
      <display value="Adult"/>
    </coding>
  </patientAgeGroup>
  <clinicalFocus>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="87512008"/>
      <display value="Mild major depression"/>
    </coding>
  </clinicalFocus>
  <clinicalFocus>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="394687007"/>
      <display value="Low suicide risk"/>
    </coding>
  </clinicalFocus>
  <targetUser>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="309343006"/>
      <display value="Physician"/>
    </coding>
  </targetUser>
  <clinicalVenue>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="440655000"/>
      <display value="Outpatient environment"/>
    </coding>
  </clinicalVenue>
</coverage>
```

The useContext element can contain any number of characteristics that define the particular context of use for the artifact. Note that the base resource does not communicate the intended semantics when multiple useContext elements are present. To accomplish this, a core extension is defined, usagecontext-group. All use contexts within a group apply, while groups of use contexts indicate that any group applies.

In addition, the useContext element is intended to convey prescriptive semantics about the appropriate context of use for an artifact, while the topic element is intended to convey descriptive semantics suitable for indexing and searching using the topic search parameter.

The FHIR search interaction specifies that search results are returned in a Bundle, and the entries in that bundle allow a score to be specified, consistent with the Decision Support Service (DSS) relevance result. Note that the DSS score range is 1 to 100, while the FHIR score range is 0..1.

The algorithm for determining relevance is not prescribed by this guidance. When calculating relevance, implementations are encouraged to take advantage of the metadata represented within the artifact to help users understand the relevance of the returned artifacts to their search criteria.

To reduce the size of returned results and ensure efficient searching and retrieval, the Bundle returned SHOULD consist only of resources of the requested type. Dependencies of the resources returned SHOULD NOT be included unless specifically referenced by including the resources using the _include and _revinclude parameters.

Support for two different distribution use cases can be provided using the basic GET interaction:

• Summary - Using the _summary parameter, the resources returned represent only the basic information and metadata about each matching artifact.

• Full - Using the _include and _revinclude search parameters to include related resources in the result.

The following table lists the search criteria elements defined by the Decision Support Service (DSS) standard along with their appropriate representation in FHIR:

DSS Search Criteria | FHIR Equivalent
Maximum Results | _count global search parameter
Minimum Score | KnowledgeArtifact.minScore search parameter
Knowledge Artifact Trait | Knowledge search parameters (identifier, topic, title, description, version)
Knowledge Artifact Status | KnowledgeArtifact.status search parameter
Evaluation Result Semantics | Not Implemented
Data Requirements | Not Implemented
Relationships to specific Knowledge Artifacts | Not Implemented

Support for exclusion criteria as described in the DSS is provided by the :not search parameter modifier of FHIR.

Implementing at least the FHIR search and get interactions for the clinical reasoning resources (Library, ActivityDefinition, PlanDefinition, and Measure) is then a basic implementation of a Knowledge Artifact Repository. For example, the following GET retrieves all measures that reference a specific value set:

```
GET [base]/ValueSet/zika-affected-areas?_revinclude=Library:depends-on&_revinclude:recurse=Measure:depends-on
```

More sophisticated management functionality including semantic indexing, change management, and dependency tracking can be provided on top of this basic functionality using the same resource structures.

14.5 Clinical Decision Support Services

URL: https://hl7.org/fhir/R4/clinicalreasoning-cds-on-fhir.html
Work Group Clinical Decision Support | Maturity Level: 2 | Standards Status: Trial Use

14.5 Clinical Decision Support Services

This section of the clinical reasoning module discusses the evaluation use case for clinical decision support and how the various knowledge artifacts can be integrated into clinical workflow. The topic focuses on two main scenarios:

• Using CDS Hooks to evaluate remote clinical decision support

• Using CDS Hooks to surface clinical decision support behavior

Note that this topic is a very high-level approach to using CDS Hooks to support these two use cases. It is not a normative description of any of the CDS Hooks content. Please refer to the CDS Hooks specification itself for more details. At the time of this publication, the CDS Hooks specification has been balloted but is still in the process of being published. Because the links on this page deep-link the CDS Hooks specification, they are still referencing the original CDS Hooks specification. When it is published, the CDS Hooks specification will be located at http://cds-hooks.hl7.org .

14.5.1 Consuming Decision Support Using CDS Hooks

CDS Hooks is an open source specification focused on user-facing remote clinical decision support. CDS Hooks can use FHIR to represent patient information and recommendations, but is architecturally an independent specification. The basic components of CDS Hooks are:

Service | A decision support service that accepts requests containing patient information, and provides responses
Hook | A defined point within the client system's workflow with well-known contextual information provided as part of the request
EHR | An electronic health record, or other clinical information system that consumes decision support in the form of services
Card | Guidance from decision support services is returned in the form of cards, representing discrete recommendations or suggestions that are presented to the user within the EHR

14.5.1.1 Configuration

The first phase in consuming a CDS Service using CDS Hooks is to configure the integration from the EHR. The CDS Service publishes a set of endpoints to advertise available functionality using the discovery endpoint. For each endpoint, the service declares the hook during which it expects to be invoked, and optionally, any prefetch information that could be provided to the service.

Each hook identifies contextual information that is available within the EHR. For example, the medication-prescribe hook specifies the patient in context, as well as the medications being prescribed. When invoking the service from this hook, the EHR is expected to provide this contextual information as part of the request.

In addition, the CDS Service may specify additional information using prefetch templates. Each prefetch template is a FHIR query URL, parameterized by the data available in context, and describing information needed by the CDS Service to perform its processing. By providing this information as part of the request, the EHR alleviates the need for the CDS Service to request the additional information.

The following example illustrates a typical service discovery response:

```
{
  "services": [
    {
      "hook": "medication-prescribe",
```

```
      "prefetch": {
        "medication": "MedicationOrder?patient\u003d{{context.patientId}}\u0026status\u003dactive"
      },
      "title": "Opioid Morphine Milligram Equivalence (MME) Guidance Service",
      "description": "CDS Service that finds the MME of an opioid medication and provides guidance to the prescriber if
    the MME exceeds the recommended range.",
      "id": "cdc-opioid-guidance"
    },
    {
      "hook": "patient-view",
      "prefetch": {
        "patient": "Patient/{{context.patientId}}"
      },
      "title": "Zika Virus Intervention",
      "description": "Identifies possible Zika exposure and offers suggestions for suggested actions for pregnant
    patients",
      "id": "zika-virus-intervention"
    },
}
```

14.5.1.2 Evaluation

The second phase is the actual request/response call to the CDS Service. Once the integration has been configured using the above information, the EHR can make requests to decision support services at the appropriate times based on the hooks it supports. To make a request, the EHR prepares a request object containing the contextual information required for the hook, as well as any additional prefetch information.

For example, the following request illustrates a call to the cdc-opioid-guidance service:

```
{
  "hookInstance": "d1577c69-dfbe-44ad-ba6d-3e05e953b2ea",
  "fhirServer": "https://example.org/fhir",
  "hook": "medication-prescribe",
  "context":
    {
      "medications": [
        {
          "resourceType": "MedicationOrder",
          "id": "medrx001",
          ... <FHIR Resource - snipped for brevity>
        }
      ],
      "patientId": "Patient/Patient-12214",
      "userId": "Practitioner/example"
    },
  "patient": "Patient/Patient-12214",
  "prefetch": {
    "medication": {
      "resource": {
        "resourceType": "Bundle",
        "entry": [
          {
            "fullUrl": "https://example.org/fhir/open/MedicationOrder/medrx002",
            "resource": {
              "resourceType": "MedicationOrder",
              "id": "medrx002",
              ... <FHIR Resource - snipped for brevity>
          }
        ]
      }
    }
  }
}
```

This request identifies:

• hookInstance - A unique identifier for this instance of the hook invocation.

• fhirServer - The base URL for the FHIR server that the CDS Service can use to request any additional information required

• hook - The hook being invoked, medication-prescribe in this case

• user - The identifier of the current user, a Practitioner in this case

• context - The contextual information as defined by the hook, the MedicationOrder being prescribed in this case

• prefetch - The prefetch information as defined by the service, the active MedicationOrders for the patient in this case

The service responds with a set of cards describing any recommendations or suggestions that should be presented to the user:

```
{
   "cards":[
      {
         "summary":"High risk for opioid overdose - taper now",
         "detail":"Total morphine milligram equivalent (MME) is 110 mg/d. Taper to less than 50.",
         "indicator":"warning",
         "source": {
```

```
          "label": "Centers for Comprehensive Disease Control (CDC)",
          "url": "http://cdc.gov"
        },
        "suggestions":[
           {
              "label": "Total morphine milligram equivalent (MME) is 110 mg/d. Taper to less than 50.",
              "actions":[
                 {
             "type": "update",
                    "description":"Total morphine milligram equivalent (MME) is 110 mg/d. Taper to less than 50.",
                    "resource": { ... <Updated FHIR Resource - snipped for brevity> ... }
                 }
              ]
           }
        ],
        "links":[
           {
              "label":"CDC guideline for prescribing opioids for chronic pain",
              "type": "absolute",
              "url":"https://www.cdc.gov/mmwr/volumes/65/rr/rr6501e1.htm"
           },
           {
              "label":"MME Conversion Tables",
              "type": "absolute",
              "url":"https://www.cdc.gov/drugoverdose/pdf/calculating_total_daily_dose-a.pdf"
           }
        ]
     }
  ]
}
```

Each card contains:

• summary - A short description of the result

• detail - A more detailed description of the information for the card

• indicator - The urgency/importance of the card, info, warning or hard-stop

• source - The source of the information

• suggestions - An array of suggestions, allowing the service to suggest a set of changes in the context of the current activity

• selectionBehavior - A selection behavior among returned suggestions for the card.

• links - A set of links allowing the service to provide links to further information about the response

At this point, the EHR processes the response and determines the most appropriate mechanism for displaying the results to the end-user. However, it is often the case that the results of the decision support interaction need to be persisted for future reference. The GuidanceResponse and RequestGroup resources provide a general mechanism that supports this use case.

In general, a CDS Hooks Response can be captured as a single GuidanceResponse that represents the overall response from the CDS Service, and a single RequestGroup containing the cards and suggestions, as illustrated by the following object-level mappings:

CDS Hooks Object | FHIR Resource Mapping | Description
Response | GuidanceResponse and RequestGroup | A CDS Hooks Response is 1 to 1 with a GuidanceResponse and an associated RequestGroup
Card | RequestGroup.action | Each Card in the response is represented as a top level action in the RequestGroup. The selectionBehavior of the action (i.e. among suggestions on the card) is specified by the selectionBehavior element of the card.
Suggestion | RequestGroup.action.action | Each suggestion on a card is represented as a nested action within the action for the card. The selectionBehavior of the action (i.e. among the actions described in the suggestion) is all, because CDS Hooks specifies that when a suggestion is accepted, all the actions on the suggestion are performed.
Action | RequestGroup.action.action.action | Each CDS Hooks Action on a card is represented as a nested action within the RequestGroup action for the suggestion, and the resource in the CDS Hooks Action populates the resource element of the RequestGroup action.

And the following table lists the element-level mappings:

CDS Hooks Element | FHIR Resource Mapping
Request.hookInstance | GuidanceResponse.requestId & RequestGroup.identifier
Request URL | GuidanceResponse.moduleUri & RequestGroup.instantiatesUri
Response status | GuidanceResponse.status
Request Patient | GuidanceResponse.subject & RequestGroup.subject
Request time | GuidanceResponse.occurrenceDateTime & RequestGroup.authoredOn
Request service | GuidanceResponse.performer & RequestGroup.author (as a Device)
Response.card | RequestGroup.action
Response.card.summary | RequestGroup.action.title
Response.card.detail | RequestGroup.action.description
Response.card.indicator | RequestGroup.priority | RequestGroup.action.resource.priority, using the mapping specified here
Response.card.source | RequestGroup.action.relatedArtifact.where(type = 'documentation')
Response.card.selectionBehavior | RequestGroup.action.selectionBehavior
Response.card.suggestion | RequestGroup.action.action
Response.card.suggestion.label | RequestGroup.action.action.title

Response.card.suggestion.uuid | RequestGroup.action.action.id
Response.card.suggestion.action | RequestGroup.action.action.action
Response.card.suggestion.action.type | RequestGroup.action.action.action.type
Response.card.suggestion.action.description | RequestGroup.action.action.action.description
Response.card.suggestion.action.resource | RequestGroup.action.action.action.resource

Note that these examples all assume a FHIR DSTU2 service is being used.

To support these scenarios, this module defines the CDS Hooks GuidanceResponse and CDS Hooks RequestGroup profiles.

14.5.2 Surfacing Clinical Decision Support Guidance Using CDS Hooks

In addition to supporting the user-facing remote decision support use case, CDS Hooks can be used to surface clinical decision support behavior represented by knowledge artifacts in the Clinical Reasoning module. In this use case, a FHIR server functioning as a knowledge provider exposes CDS Hooks Services using the discovery endpoint, and provides guidance using the CDS Service endpoint. To support this, several mappings from Clinical Reasoning functionality to CDS Hooks services are used:

• Hooks - Hooks in CDS Hooks are mapped to the TriggerDefinition structure in FHIR

• Services - Services in CDS Hooks are mapped to the PlanDefinition resource in FHIR to provide evaluation behavior

• Prefetch - Prefetch templates in CDS Hooks are mapped to the DataRequirement structure in FHIR

14.5.2.1 Representing Hooks in FHIR

A hook in CDS Hooks is a pre-defined point in the workflow of a clinical information system such as an EHR. Each hook defines context, which is the information available as part of the current activity in the system. Each hook represents a point in the workflow that can be augmented by decision support from an external system. Within CDS Hooks, each hook defines the set of available context values, along with whether or not that context value can be used as a prefetch token.

For example, the patient-view hook defines patientId and encounterId as context values and indicates that they are both available for use as prefetch tokens (meaning that they can be used to parameterize prefetch templates).

Within FHIR, the concept of a hook can be represented using a combination of TriggerDefinition and ParameterDefinition:

CDS Hooks Element | FHIR Mapping
Hook.name | TriggerDefinition.where(type = 'named-event').name
Hook.context.field | ParameterDefinition.name
Hook.context.priority | ParameterDefinition.min & ParameterDefinition.max
Hook.context.description | ParameterDefinition.documentation & ParameterDefinition.type & ParameterDefinition.profile

Note that using TriggerDefinition to represent hook information requires that hook details be duplicated everywhere they are used. Another approach would be to use the EventDefinition resource to capture the hook information once, and then reuse that by reference wherever it is needed.

14.5.2.2 Representing Services in FHIR

A service in CDS Hooks is a clinical decision support service that can be used to provide guidance to users at pre-defined points in a workflow. The PlanDefinition resource can be used to describe the behavior of decision support functionality, which can then be exposed via a CDS Hooks service. In the simplest case, there is a one-to-one correspondence between a PlanDefinition and a CDS Service:

CDS Hooks Element | FHIR Mapping
Service.id | PlanDefinition.url (without the base)
Service.title | PlanDefinition.title
Service.description | PlanDefinition.description
Service.hook | PlanDefinition.action.trigger
Service.prefetch | PlanDefinition.data-requirement

To support this representation, this module defines a CDS Hooks Service PlanDefinition profile, which also supports specifying the CDS Hooks endpoint on which the PlanDefinition should be exposed.

The PlanDefinition/$apply operation can then be used to provide the behavior of the CDS Hooks service, as described in the Processing CDS Hooks Requests section below.

14.5.2.3 Representing Prefetch in FHIR

In addition to the contextual information defined by the hook, services in CDS Hooks can request that additional information be supplied with each request in the form of prefetch templates. These templates are parameterized FHIR query URLs that can be fulfilled by the EHR as part of the request, reducing the number of round-trips between the CDS Service and the EHR's FHIR server.

The concept of prefetch data is represented within Clinical Reasoning as a DataRequirement, which can be transformed to an instance level read or type level search interaction as follows:

DataRequirement Element | Mapping to FHIR URL

type | [type]{[id] | ?[parameters]}
subject | subject={{patientId}}
codeFilter | [path]{=|:in}[code|valueSet]
dateFilter | [path]{eq|lt|gt|ge|le}[valueDateTime|valuePeriod|valueDuration]
sort | _sort=[sort]
limit | _limit=[limit]

This prefetch data can be automatically determined from the data requirements of the PlanDefinition and provided as part of the service definition to the CDS Hooks discovery response.

14.5.2.4 Processing CDS Hooks Requests

Once the available PlanDefinition resources are advertised through the discovery endpoint, a CDS Hooks endpoint can be used to perform the actual evaluation, as illustrated in the following diagram:

As depicted in the above diagram, an EHR invokes a CDS Hooks request at the appropriate point in the workflow, providing the requested context and data. The CDS Service responds by performing an $apply operation against the specified PlanDefinition, and transforming the resulting RequestGroup into a CDS Hooks response.

Because PlanDefinitions can be used in a broad range of use cases, this module defines the CQL Library and Computable PlanDefinition profiles to describe the special case of a PlanDefinition being used as an event-condition-action rule with conditions and other dynamic behavior specified in a CQL Library. This arrangement provides a common and consistent pattern for describing decision support that can be easily integrated using the CDS Hooks specification.

14.6 Quality Reporting

URL: https://hl7.org/fhir/R4/clinicalreasoning-quality-reporting.html
Work Group Clinical Quality Information | Maturity Level: 2 | Standards Status: Trial Use

14.6 Quality Reporting

14.6.1 Representing Quality Measures

The Measure resource builds on the general approach to representing knowledge artifacts and adds the metadata and structure information that is specific to quality measures:

Quality measures follow a generally hierarchical structure that defines:

Population Groups: Groups of population criteria that define a particular area of measurement. A given measure may include any number of population groups, each with different criteria for the various measure components.

• Populations: Within each population group, a measure defines criteria appropriate to the type of measure being calculated. Each population type identifies a specific component of the calculation such as the numerator, denominator, and initial population.

• Stratifiers: Additional criteria used to calculate the measure along different dimensions within the population such as age or gender. A measure may define any number of stratifiers for each population group.

• Supplemental Data: Additional information that should be included with the calculated results of the measure such as individual age or gender. A measure may define any number of supplemental data elements.

Population Quality Measures are often focused on evaluating from a patient perspective, but this is not always the case. The subject element of the Measure indicates the intended subjects of a measure. If no subject is specified, the measure subject is Patient, but Practitioners, Organizations, Locations, or even Devices can also be the subject of a measure.

14.6.1.1 Measure Metadata

The following table provides a requirements mapping from the content of an eMeasure to the elements defined in the Measure resource:

eMeasure | Cardinality | Element | Notes
Title | 0..1 | Measure.title
Identifier | 0..1 | Measure.identifier | identifier type code as http://hl7.org/fhir/cqi/ecqm/Measure/Identifier/cms
Version Number | 0..1 | Measure.version
NQF Number | 0..1 | Measure.identifier | identifier type code as http://hl7.org/fhir/cqi/ecqm/Measure/Identifier/nqf
GUID | 0..1 | Measure.identifier | identifier type code as http://hl7.org/fhir/cqi/ecqm/Measure/Identifier/guid
Measure Steward | 0..1 | Measure.publisher
Measure Developer | 0..1 | Measure.contributor | type.code of author
Endorser | 0..1 | Measure.contributor | type.code of endorser
Description | 0..1 | Measure.description
Copyright | 0..1 | Measure.copyright
Reference | 0..* | Measure.relatedArtifact | type.code of citation
Disclaimer | 0..1 | disclaimer | String (containing Markdown)
Measure Scoring | 0..1 | scoring | Code, e.g. proportion, CV

Measure Type | 0..1 | type | Code, e.g. process, outcome
Risk Adjustment | 0..1 | riskAdjustment | String
Rate Aggregation | 0..1 | rateAggregation | String
Rationale | 0..1 | rationale | String (containing Markdown)
Clinical Recommendation Statement | 0..1 | clinicalRecommendationStatement | String (containing Markdown)
Improvement Notation | 0..1 | improvementNotation | String, e.g. Higher score indicates better quality
Definition | 0..1 | definition | String (containing Markdown)
Guidance | 0..1 | guidance | String (containing Markdown)

14.6.1.2 Measure Logic

As with other knowledge artifacts, logic is included by referencing a Library resource. Although the base resource allows for the measure to reference any number of libraries, for simplicity of managing sharing, measures should reference only one Library, the primary measure library, and that library should contain all the named expressions required to define the measure structure.

Note that this approach does not preclude sharing of logic between measures, it only requires that that sharing be explicitly done as dependencies within the referenced libraries, rather than allowing a measure to reference multiple libraries directly.

14.6.1.2.1 Specifying Population Criteria

A measure can specify various types of populations, depending on the type of measure scoring being used. The following table shows which population criteria types are required (R), optional (O), or not permitted (NP) for proportion, ratio, and continuous variable measures. This table is adapted from Table 1 from the HQMF Release 1 Normative specification, and Table 2.1 from the QDM-based HQMF IG .

MeasureType | Initial Population | Denominator | Denominator Exclusion | Denominator Exception | Numerator | Numerator Exclusion | Measure Population | Measure Population Exclusion
Proportion | R | R | O | O | R | O | NP | NP
Ratio | R | R | O | NP | R | O | NP | NP
Continuous Variable | R | NP | NP | NP | NP | NP | R | O
Cohort | R | NP | NP | NP | NP | NP | NP | NP

The Measure resource then identifies specific named expressions within the referenced primary measure library that define the criteria for each population. For example, the following fragment illustrates the population criteria definitions for the CMS146 measure example:

```
<group id="CMS146-group-1">
  <population>
    <code>
      <coding>
        <code value="initial-population"/>
      </coding>
    </code>
    <criteria value="CMS146.InInitialPopulation"/>
  </population>
  <population>
    <code>
      <coding>
        <code value="numerator"/>
      </coding>
    </code>
    <criteria value="CMS146.InNumerator"/>
  </population>
  <population>
    <code>
      <coding>
        <code value="denominator"/>
      </coding>
    </code>
    <criteria value="CMS146.InDenominator"/>
  </population>
  <population>
    <code>
      <coding>
        <code value="denominator-exclusion"/>
      </coding>
    </code>
    <criteria value="CMS146.InDenominatorExclusions"/>
  </population>
</group>
```

14.6.1.2.2 Measures with Multiple Populations

Quality measures often specify multiple rates, with different population crtiteria for each rate. This is different than stratifying the scores for the same population. For quality measures that contain multiple rates, the Measure will contain multiple group elements, where the criteria are specified once for each group. The id attribute of the group element is used to uniquely identify the group within the measure, as well as within the quality reporting results.

14.6.1.2.3 Continuous Variable Measures

Continuous variable measures may include a measure observation section. This section defines variables (for example, time from check-in to time of antibiotic administration) used to measure particular aspects of a process or outcome. Note that measure observations are not population criteria in that they do not filter the population in any way. Rather, measure observations are data elements, to be collected from each subject that satisfies the population criteria, which are used to calculate the results for each member of the population.

14.6.1.3 Stratification and Supplemental Data

Stratifiers and supplemental data are specified using the stratifier and supplementalData elements of the Measure resource. Stratification criteria are specified either as a reference to a CQL named expression within a Library (e.g. CMS146.AgesUpToNine), or as FHIR resource paths (e.g. Patient.gender). When the stratification criteria is an expression, the stratification will yield as many result groups as the expression returns. For example, if the expression returns a boolean, then there would be two stratification groups: true and false. When the stratification criteria is a FHIR resource path, there will be as many stratification groups as possible values for the resource path. For example, specifying Patient.gender will yield four stratification groups since FHIR has four gender codes: male, female, other, and unknown.

Supplemental data elements are also specified using FHIR resource paths, however, supplemental data only results in groups in the summary measure report. For individual-level reports, supplemental data elements are reported as Observation resources and included by reference in the evaluateResource for the individual-level report.

The CMS146 example measure illustrates the stratification and supplemental data described above:

• Measure: measure-cms146-example

• MeasureReport: measurereport-cms146-cat3-example

14.6.1.3.1 Data Criteria

The data criteria for the primary library defines the data of interest in the measure as a set of DataRequirement elements. Each data requirement identifies specific types of data along with constraints that the data must meet. For example, one data requirement for CMS 146 identifies FHIR Condition resources that represent confirmed diagnoses of acute pharyngitis. Other data requirements for this measure include Encounters, DiagnosticReports and other FHIR resources representing specific data that is used to calculate the measure.

Specifying the data criteria in this way enables the following use cases:

• Determining the set of data used by a particular eCQM.

• Limited "scoop-and-filter" for creation of population reports.

• Limited backwards compatibility with existing implementations of previous eCQM IGs.

Data criteria can be specified statically, or they can be inferred from the expressions referenced by the measure. The $data-requirements operation can be invoked to retrieve the aggregate data requirements for the measure. This approach has two advantages:

• When the data requirements for the expression contain dynamic criteria (such as date ranges relative to today), the results returned are based on the evaluation request time.

• When the expressions involve multiple libraries, rather than having to retrieve and deal with the requirements for each library, the $data-requirements operation can aggregate the requirements and return them as a single module-definition library.

14.6.2 Invoking Measures

The Health Quality Measure Format (HQMF) defines the electronic representation of an eMeasure but does not define a mechanism for invoking an eMeasure. FHIR defines both the representation of resources and a general mechanism for interacting with them via the OperationDefinition resource. Prior sections of this specification described the Measure representation of an eMeasure, this section describes the $evaluate-measure operation that is used to invoke an eMeasure and obtain the results.

14.6.2.1 Operation Definition

FHIR defines a standard set of common interactions that include read, update, delete and search. In addition, FHIR defines a standard set of extended operations that can be performed on resources, resource types and system wide. The standard operations include profile validation, concept translation and value set expansion. FHIR also supports custom operations via the FHIR OperationDefinition resource. This resource offers a means to create a formal definition of a custom operation that can be performed on a FHIR server. For the purposes of measure evaluation we define a new custom operation with a code of $evaluate-measure.

The $evaluate-measure operation has the following properties:

• Idempotent The operation may be invoked multiple times without side effects. Note that the result of invoking the operation may vary over time if clinical data used in the eMeasure changes between invocations. Note also that the parameters supplied with the operation invocation can affect the results.

• InvocationTarget The operation can be invoked on instances of the MeasureArtifact resource that represent a particular eMeasure or on the type of the resource with a parameter that specifies the eMeasure to calculate.

The effect of invoking the $evaluate-measure operation is to calculate the quality measure according to the supplied parameters and to return a MeasureReport resource through which the results will be made available. Note that because measure calculation might not be instantaneous, the MeasureReport resource provides a mechanism to handle long running calculations.

14.6.2.2 Examples

```
GET [base]/Measure/$evaluate-measure?measure=CMS146&periodStart=2014&periodEnd=2014
```

```
GET [base]/Measure/CMS146/$evaluate-measure?periodStart=2014&periodEnd=2014
```

The above examples show how to obtain the results of evaluating the eMeasure with id "CMS146" for all patients over a measurement period that consists of all of 2014. Some items of note:

• the first variant evaluates the operation on [base]/Measure which is the type of resource and specifies the eMeasure to evaluate using a parameter

• the second variant evaluates the operation on [base]/Measure/CMS146 which is the Measure instance that represents that measure so there's no need to also include a reference to the eMeasure in the operation parameters

• the HTTP GET method is used since the $evaluate-measure operation is idempotent

• [base] is used as a shortcut for the base URI of the FHIR server

• the period start and end values are both specified to a granularity of a year, the description of the parameters above explains why this results in a measure period that spans the entire year

The next example demonstrates how to obtain the results of evaluating the eMeasure with id "CMS146" for the patient with id "124" over a measurement period that consists of the first three months of 2014.

```
GET [base]/Basic/CMS146/$evaluate-measure?subject=124&periodStart=2014-01&periodEnd=2014-03
```

14.6.3 Measure Report

When eCQMs are represented with the Health Quality Measure Format (HQMF), a single HQMF document represents both the measure itself and the request. Meanwhile, the responses are represented as Quality Reporting Document Architecture (QRDA) documents. QRDA documents come in two flavors: Category I for individual patient reports and Category III for population reports.

When eCQMs are represented with FHIR resources, the measure is represented as a Measure resource, and the request is an HTTP GET conforming to the OperationDefinition described above. Meanwhile, the responses are represented as MeasureReport resources. Like QRDA, the MeasureReport allows for Category I (individual), Category II (subject-list), and Category III (population) reports.

14.6.3.1 Reporting Population Data

A MeasureReport will contain one group of data for each group specified in the corresponding Measure, consisting of a set of population elements, one for each criteria defined in each group.

In addition, each group will contain stratifiers with a value stratum for each value defined by the stratifier criteria, for each criteria defined in the measure.

14.6.3.2 Reporting Individual Data

When using a MeasureReport resource to represent the results of an individual calculation, the MeasureReport SHALL have a type-code of "individual" and SHALL have a reference to the subject of the report. In addition, the result SHOULD include a reference to a Bundle containing the subject-specific resources that were used to calculate the result.

See the MeasureReport examples for a detailed illustration of how the data elements involved in the calculation of the measure are communicated through the evaluatedResources element.

14.6.3.3 Subject-List Reports

When using a MeasureReport resource to represent a subject-list, the MeasureReport SHALL have a type-code of "subject-list" and if a subject reference is present, it SHALL be a reference to a Group. In addition, the resource SHALL include for each population a reference to a List resource that references individual level MeasureReport resources for the same measure, one for each subject in the overall population.

For example, the initial population report, in addition to providing the count, provides a reference to a List resource that identifies each of the subjects that make up that population. For each of those subjects, the List will contain a reference to an individual-level report for that subject. Note that for very large populations, implementations MAY decide to limit the size of the result, either by returning an error indicating the request is too costly, or by returning a partial result, so long as there is an indication that the report is only a partial response. In addition, we are actively seeking feedback on how best to approach evaluation of quality measures on large populations, including the use of bulk data formats.

In addition, implementations may return a MeasureReport with a status of pending, indicating that the evaluation is in progress. In this case, clients can request the MeasureReport resource until the status changes to complete.

14.6.4 References

• HL7 Standard: Clinical Quality Language Specification, Release 1 (Standard for Trial Use, Release 2).

• Quality Improvement Core (QICore) FHIR Implementation Guide.

• Representation of the Health Quality Measures Format (eMeasure) Release 1 Normative.

• HL7 Version 3 Implementation Guide: Quality Data Model (QDM)-based Health Quality Measure Format (HQMF), R1 - US Realm, Volume 1 (Draft Standard for Trial Use)

• HL7 Version 3 Implementation Guide: Quality Data Model (QDM)-based Health Quality Measure Format (HQMF), R1 - US Realm, Volume 2 (Draft Standard for Trial Use)

• HL7 Version 3 Implementation Guide: Clinical Quality Language (CQL)-based Health Quality Measure Format (HQMF), Release 1 - US Realm (Standard for Trial Use)

• Quality Data Model, Version 4.2

• HL7 CDA R2 Implementation Guide: Quality Reporting Document Architecture - Category I (QRDA I) DSTU Release 3 (US Realm)

14.7 Resource EffectEvidenceSynthesis - Content

URL: https://hl7.org/fhir/R4/effectevidencesynthesis.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures states in a population where the effect estimate is derived from a combination of research studies.

14.7.1 Scope and Usage

The scope of the EffectEvidenceSynthesis resource is to describe the estimates of an effect of an exposure on an outcome where the effect estimates are derived from the combination of research studies. Effect estimates are a measure of difference between the exposure state and an alternative exposure state (often called the control or comparator state) Expressing effect estimates is done throughout reporting of biomedical research, systematic reviews, and clinical reference across all disciplines.

14.7.2 Resource Content

• Structure

• UML

• XML

• JSON

• Turtle

• R3 Diff

• All

Structure

Name | Flags | Card. | Type | Description & Constraints
EffectEvidenceSynthesis | ITU | | DomainResource | A quantified estimate of effect based on a body of evidence + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this effect evidence synthesis, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the effect evidence synthesis
version | Σ | 0..1 | string | Business version of the effect evidence synthesis
name | ΣI | 0..1 | string | Name for this effect evidence synthesis (computer friendly)
title | Σ | 0..1 | string | Name for this effect evidence synthesis (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the effect evidence synthesis
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for effect evidence synthesis (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the effect evidence synthesis was approved by publisher
lastReviewDate | | 0..1 | date | When the effect evidence synthesis was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the effect evidence synthesis is expected to be used

topic | | 0..* | CodeableConcept | The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
synthesisType | | 0..1 | CodeableConcept | Type of synthesis SynthesisType (Extensible)
studyType | | 0..1 | CodeableConcept | Type of study StudyType (Extensible)
population | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposure | Σ | 1..1 | Reference(EvidenceVariable) | What exposure?
exposureAlternative | Σ | 1..1 | Reference(EvidenceVariable) | What comparison exposure?
outcome | Σ | 1..1 | Reference(EvidenceVariable) | What outcome?
sampleSize | | 0..1 | BackboneElement | What sample size was involved?
description | | 0..1 | string | Description of sample size
numberOfStudies | | 0..1 | integer | How many studies?
numberOfParticipants | | 0..1 | integer | How many participants?
resultsByExposure | | 0..* | BackboneElement | What was the result per exposure?
description | | 0..1 | string | Description of results by exposure
exposureState | | 0..1 | code | exposure | exposure-alternative ExposureState (Required)
variantState | | 0..1 | CodeableConcept | Variant exposure states EvidenceVariantState (Extensible)
riskEvidenceSynthesis | | 1..1 | Reference(RiskEvidenceSynthesis) | Risk evidence synthesis
effectEstimate | Σ | 0..* | BackboneElement | What was the estimated effect
description | | 0..1 | string | Description of effect estimate
type | | 0..1 | CodeableConcept | Type of efffect estimate EffectEstimateType (Extensible)
variantState | | 0..1 | CodeableConcept | Variant exposure states EvidenceVariantState (Extensible)
value | | 0..1 | decimal | Point estimate
unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)
precisionEstimate | | 0..* | BackboneElement | How precise the estimate is
type | | 0..1 | CodeableConcept | Type of precision estimate PrecisionEstimateType (Extensible)
level | | 0..1 | decimal | Level of confidence interval
from | | 0..1 | decimal | Lower bound
to | | 0..1 | decimal | Upper bound
certainty | | 0..* | BackboneElement | How certain is the effect
rating | | 0..* | CodeableConcept | Certainty rating QualityOfEvidenceRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
certaintySubcomponent | | 0..* | BackboneElement | A component that contributes to the overall certainty
type | | 0..1 | CodeableConcept | Type of subcomponent of certainty rating CertaintySubcomponentType (Extensible)
rating | | 0..* | CodeableConcept | Subcomponent certainty rating CertaintySubcomponentRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EffectEvidenceSynthesis xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this effect evidence synthesis, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the effect evidence synthesis --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the effect evidence synthesis -->
 <name value="[string]"/><!--  0..1 Name for this effect evidence synthesis (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this effect evidence synthesis (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the effect evidence synthesis -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for effect evidence synthesis (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the effect evidence synthesis was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the effect evidence synthesis was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the effect evidence synthesis is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment,
     etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <synthesisType><!-- 0..1 CodeableConcept Type of synthesis --></synthesisType>
 <studyType><!-- 0..1 CodeableConcept Type of study --></studyType>
 <population><!-- 1..1 Reference(EvidenceVariable) What population? --></population>
 <exposure><!-- 1..1 Reference(EvidenceVariable) What exposure? --></exposure>
 <exposureAlternative><!-- 1..1 Reference(EvidenceVariable) What comparison exposure? --></exposureAlternative>
 <outcome><!-- 1..1 Reference(EvidenceVariable) What outcome? --></outcome>
```

```
 <sampleSize>  <!-- 0..1 What sample size was involved? -->
  <description value="[string]"/><!-- 0..1 Description of sample size -->
  <numberOfStudies value="[integer]"/><!-- 0..1 How many studies? -->
  <numberOfParticipants value="[integer]"/><!-- 0..1 How many participants? -->
 </sampleSize>
 <resultsByExposure>  <!-- 0..* What was the result per exposure? -->
  <description value="[string]"/><!-- 0..1 Description of results by exposure -->
  <exposureState value="[code]"/><!-- 0..1 exposure | exposure-alternative -->
  <variantState><!-- 0..1 CodeableConcept Variant exposure states --></variantState>
  <riskEvidenceSynthesis><!-- 1..1 Reference(RiskEvidenceSynthesis) Risk evidence synthesis --></riskEvidenceSynthesis>
 </resultsByExposure>
 <effectEstimate>  <!-- 0..* What was the estimated effect -->
  <description value="[string]"/><!-- 0..1 Description of effect estimate -->
  <type><!-- 0..1 CodeableConcept Type of efffect estimate --></type>
  <variantState><!-- 0..1 CodeableConcept Variant exposure states --></variantState>
  <value value="[decimal]"/><!-- 0..1 Point estimate -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <precisionEstimate>  <!-- 0..* How precise the estimate is -->
   <type><!-- 0..1 CodeableConcept Type of precision estimate --></type>
   <level value="[decimal]"/><!-- 0..1 Level of confidence interval -->
   <from value="[decimal]"/><!-- 0..1 Lower bound -->
   <to value="[decimal]"/><!-- 0..1 Upper bound -->
  </precisionEstimate>
 </effectEstimate>
 <certainty>  <!-- 0..* How certain is the effect -->
  <rating><!-- 0..* CodeableConcept Certainty rating --></rating>
  <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  <certaintySubcomponent>  <!-- 0..* A component that contributes to the overall certainty -->
   <type><!-- 0..1 CodeableConcept Type of subcomponent of certainty rating --></type>
   <rating><!-- 0..* CodeableConcept Subcomponent certainty rating --></rating>
   <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  </certaintySubcomponent>
 </certainty>
</EffectEvidenceSynthesis>
```

JSON Template

```
{
  "resourceType" : "EffectEvidenceSynthesis",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this effect evidence synthesis, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the effect evidence synthesis
  "version" : "<string>", // Business version of the effect evidence synthesis
  "name" : "<string>", // C? Name for this effect evidence synthesis (computer friendly)
  "title" : "<string>", // Name for this effect evidence synthesis (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the effect evidence synthesis
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for effect evidence synthesis (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the effect evidence synthesis was approved by publisher
  "lastReviewDate" : "<date>", // When the effect evidence synthesis was last reviewed
  "effectivePeriod" : { Period }, // When the effect evidence synthesis is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EffectEvidenceSynthesis, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "synthesisType" : { CodeableConcept }, // Type of synthesis
  "studyType" : { CodeableConcept }, // Type of study
  "population" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposure" : { Reference(EvidenceVariable) }, // R!  What exposure?
  "exposureAlternative" : { Reference(EvidenceVariable) }, // R!  What comparison exposure?
  "outcome" : { Reference(EvidenceVariable) }, // R!  What outcome?
  "sampleSize" : { // What sample size was involved?
    "description" : "<string>", // Description of sample size
    "numberOfStudies" : <integer>, // How many studies?
    "numberOfParticipants" : <integer> // How many participants?
  },
  "resultsByExposure" : [{ // What was the result per exposure?
    "description" : "<string>", // Description of results by exposure
    "exposureState" : "<code>", // exposure | exposure-alternative
    "variantState" : { CodeableConcept }, // Variant exposure states
    "riskEvidenceSynthesis" : { Reference(RiskEvidenceSynthesis) } // R!  Risk evidence synthesis
  }],
  "effectEstimate" : [{ // What was the estimated effect
    "description" : "<string>", // Description of effect estimate
    "type" : { CodeableConcept }, // Type of efffect estimate
    "variantState" : { CodeableConcept }, // Variant exposure states
    "value" : <decimal>, // Point estimate
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "precisionEstimate" : [{ // How precise the estimate is
      "type" : { CodeableConcept }, // Type of precision estimate
```

```
      "level" : <decimal>, // Level of confidence interval
      "from" : <decimal>, // Lower bound
      "to" : <decimal> // Upper bound
    }]
  }],
  "certainty" : [{ // How certain is the effect
    "rating" : [{ CodeableConcept }], // Certainty rating
    "note" : [{ Annotation }], // Used for footnotes or explanatory notes
    "certaintySubcomponent" : [{ // A component that contributes to the overall certainty
      "type" : { CodeableConcept }, // Type of subcomponent of certainty rating
      "rating" : [{ CodeableConcept }], // Subcomponent certainty rating
      "note" : [{ Annotation }] // Used for footnotes or explanatory notes
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EffectEvidenceSynthesis;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EffectEvidenceSynthesis.url [ uri ]; # 0..1 Canonical identifier for this effect evidence synthesis, represented
    as a URI (globally unique)
  fhir:EffectEvidenceSynthesis.identifier [ Identifier ], ... ; # 0..* Additional identifier for the effect evidence
    synthesis
  fhir:EffectEvidenceSynthesis.version [ string ]; # 0..1 Business version of the effect evidence synthesis
  fhir:EffectEvidenceSynthesis.name [ string ]; # 0..1 Name for this effect evidence synthesis (computer friendly)
  fhir:EffectEvidenceSynthesis.title [ string ]; # 0..1 Name for this effect evidence synthesis (human friendly)
  fhir:EffectEvidenceSynthesis.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EffectEvidenceSynthesis.date [ dateTime ]; # 0..1 Date last changed
  fhir:EffectEvidenceSynthesis.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EffectEvidenceSynthesis.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EffectEvidenceSynthesis.description [ markdown ]; # 0..1 Natural language description of the effect evidence
    synthesis
  fhir:EffectEvidenceSynthesis.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:EffectEvidenceSynthesis.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:EffectEvidenceSynthesis.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for effect evidence
    synthesis (if applicable)
  fhir:EffectEvidenceSynthesis.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EffectEvidenceSynthesis.approvalDate [ date ]; # 0..1 When the effect evidence synthesis was approved by
    publisher
  fhir:EffectEvidenceSynthesis.lastReviewDate [ date ]; # 0..1 When the effect evidence synthesis was last reviewed
  fhir:EffectEvidenceSynthesis.effectivePeriod [ Period ]; # 0..1 When the effect evidence synthesis is expected to be
    used
  fhir:EffectEvidenceSynthesis.topic [ CodeableConcept ], ... ; # 0..* The category of the EffectEvidenceSynthesis, such
     as Education, Treatment, Assessment, etc.
  fhir:EffectEvidenceSynthesis.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EffectEvidenceSynthesis.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EffectEvidenceSynthesis.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EffectEvidenceSynthesis.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EffectEvidenceSynthesis.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations,
    etc.
  fhir:EffectEvidenceSynthesis.synthesisType [ CodeableConcept ]; # 0..1 Type of synthesis
  fhir:EffectEvidenceSynthesis.studyType [ CodeableConcept ]; # 0..1 Type of study
  fhir:EffectEvidenceSynthesis.population [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:EffectEvidenceSynthesis.exposure [ Reference(EvidenceVariable) ]; # 1..1 What exposure?
  fhir:EffectEvidenceSynthesis.exposureAlternative [ Reference(EvidenceVariable) ]; # 1..1 What comparison exposure?
  fhir:EffectEvidenceSynthesis.outcome [ Reference(EvidenceVariable) ]; # 1..1 What outcome?
  fhir:EffectEvidenceSynthesis.sampleSize [ # 0..1 What sample size was involved?
    fhir:EffectEvidenceSynthesis.sampleSize.description [ string ]; # 0..1 Description of sample size
    fhir:EffectEvidenceSynthesis.sampleSize.numberOfStudies [ integer ]; # 0..1 How many studies?
    fhir:EffectEvidenceSynthesis.sampleSize.numberOfParticipants [ integer ]; # 0..1 How many participants?
  ];
  fhir:EffectEvidenceSynthesis.resultsByExposure [ # 0..* What was the result per exposure?
    fhir:EffectEvidenceSynthesis.resultsByExposure.description [ string ]; # 0..1 Description of results by exposure
    fhir:EffectEvidenceSynthesis.resultsByExposure.exposureState [ code ]; # 0..1 exposure | exposure-alternative
    fhir:EffectEvidenceSynthesis.resultsByExposure.variantState [ CodeableConcept ]; # 0..1 Variant exposure states
    fhir:EffectEvidenceSynthesis.resultsByExposure.riskEvidenceSynthesis [ Reference(RiskEvidenceSynthesis) ]; # 1..1
    Risk evidence synthesis
  ], ...;
  fhir:EffectEvidenceSynthesis.effectEstimate [ # 0..* What was the estimated effect
    fhir:EffectEvidenceSynthesis.effectEstimate.description [ string ]; # 0..1 Description of effect estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.type [ CodeableConcept ]; # 0..1 Type of efffect estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.variantState [ CodeableConcept ]; # 0..1 Variant exposure states
    fhir:EffectEvidenceSynthesis.effectEstimate.value [ decimal ]; # 0..1 Point estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome
    described in?
    fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate [ # 0..* How precise the estimate is
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.type [ CodeableConcept ]; # 0..1 Type of precision
    estimate
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.level [ decimal ]; # 0..1 Level of confidence
    interval
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.from [ decimal ]; # 0..1 Lower bound
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.to [ decimal ]; # 0..1 Upper bound
```

```
    ], ...;
  ], ...;
  fhir:EffectEvidenceSynthesis.certainty [ # 0..* How certain is the effect
    fhir:EffectEvidenceSynthesis.certainty.rating [ CodeableConcept ], ... ; # 0..* Certainty rating
    fhir:EffectEvidenceSynthesis.certainty.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
    fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent [ # 0..* A component that contributes to the overall
    certainty
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.type [ CodeableConcept ]; # 0..1 Type of subcomponent
     of certainty rating
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.rating [ CodeableConcept ], ... ; # 0..* Subcomponent
     certainty rating
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.note [ Annotation ], ... ; # 0..* Used for footnotes
    or explanatory notes
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
EffectEvidenceSynthesis | ITU | | DomainResource | A quantified estimate of effect based on a body of evidence + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this effect evidence synthesis, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the effect evidence synthesis
version | Σ | 0..1 | string | Business version of the effect evidence synthesis
name | ΣI | 0..1 | string | Name for this effect evidence synthesis (computer friendly)
title | Σ | 0..1 | string | Name for this effect evidence synthesis (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the effect evidence synthesis
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for effect evidence synthesis (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the effect evidence synthesis was approved by publisher
lastReviewDate | | 0..1 | date | When the effect evidence synthesis was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the effect evidence synthesis is expected to be used
topic | | 0..* | CodeableConcept | The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
synthesisType | | 0..1 | CodeableConcept | Type of synthesis SynthesisType (Extensible)
studyType | | 0..1 | CodeableConcept | Type of study StudyType (Extensible)
population | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposure | Σ | 1..1 | Reference(EvidenceVariable) | What exposure?
exposureAlternative | Σ | 1..1 | Reference(EvidenceVariable) | What comparison exposure?
outcome | Σ | 1..1 | Reference(EvidenceVariable) | What outcome?
sampleSize | | 0..1 | BackboneElement | What sample size was involved?
description | | 0..1 | string | Description of sample size
numberOfStudies | | 0..1 | integer | How many studies?
numberOfParticipants | | 0..1 | integer | How many participants?
resultsByExposure | | 0..* | BackboneElement | What was the result per exposure?
description | | 0..1 | string | Description of results by exposure
exposureState | | 0..1 | code | exposure | exposure-alternative ExposureState (Required)
variantState | | 0..1 | CodeableConcept | Variant exposure states EvidenceVariantState (Extensible)
riskEvidenceSynthesis | | 1..1 | Reference(RiskEvidenceSynthesis) | Risk evidence synthesis
effectEstimate | Σ | 0..* | BackboneElement | What was the estimated effect
description | | 0..1 | string | Description of effect estimate
type | | 0..1 | CodeableConcept | Type of efffect estimate EffectEstimateType (Extensible)
variantState | | 0..1 | CodeableConcept | Variant exposure states EvidenceVariantState (Extensible)
value | | 0..1 | decimal | Point estimate
unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)
precisionEstimate | | 0..* | BackboneElement | How precise the estimate is
type | | 0..1 | CodeableConcept | Type of precision estimate PrecisionEstimateType (Extensible)
level | | 0..1 | decimal | Level of confidence interval
from | | 0..1 | decimal | Lower bound
to | | 0..1 | decimal | Upper bound

certainty | | 0..* | BackboneElement | How certain is the effect
rating | | 0..* | CodeableConcept | Certainty rating QualityOfEvidenceRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
certaintySubcomponent | | 0..* | BackboneElement | A component that contributes to the overall certainty
type | | 0..1 | CodeableConcept | Type of subcomponent of certainty rating CertaintySubcomponentType (Extensible)
rating | | 0..* | CodeableConcept | Subcomponent certainty rating CertaintySubcomponentRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EffectEvidenceSynthesis xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this effect evidence synthesis, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the effect evidence synthesis --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the effect evidence synthesis -->
 <name value="[string]"/><!--  0..1 Name for this effect evidence synthesis (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this effect evidence synthesis (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the effect evidence synthesis -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for effect evidence synthesis (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the effect evidence synthesis was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the effect evidence synthesis was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the effect evidence synthesis is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment,
     etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <synthesisType><!-- 0..1 CodeableConcept Type of synthesis --></synthesisType>
 <studyType><!-- 0..1 CodeableConcept Type of study --></studyType>
 <population><!-- 1..1 Reference(EvidenceVariable) What population? --></population>
 <exposure><!-- 1..1 Reference(EvidenceVariable) What exposure? --></exposure>
 <exposureAlternative><!-- 1..1 Reference(EvidenceVariable) What comparison exposure? --></exposureAlternative>
 <outcome><!-- 1..1 Reference(EvidenceVariable) What outcome? --></outcome>
 <sampleSize>  <!-- 0..1 What sample size was involved? -->
  <description value="[string]"/><!-- 0..1 Description of sample size -->
  <numberOfStudies value="[integer]"/><!-- 0..1 How many studies? -->
  <numberOfParticipants value="[integer]"/><!-- 0..1 How many participants? -->
 </sampleSize>
 <resultsByExposure>  <!-- 0..* What was the result per exposure? -->
  <description value="[string]"/><!-- 0..1 Description of results by exposure -->
  <exposureState value="[code]"/><!-- 0..1 exposure | exposure-alternative -->
  <variantState><!-- 0..1 CodeableConcept Variant exposure states --></variantState>
  <riskEvidenceSynthesis><!-- 1..1 Reference(RiskEvidenceSynthesis) Risk evidence synthesis --></riskEvidenceSynthesis>
 </resultsByExposure>
 <effectEstimate>  <!-- 0..* What was the estimated effect -->
  <description value="[string]"/><!-- 0..1 Description of effect estimate -->
  <type><!-- 0..1 CodeableConcept Type of efffect estimate --></type>
  <variantState><!-- 0..1 CodeableConcept Variant exposure states --></variantState>
  <value value="[decimal]"/><!-- 0..1 Point estimate -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <precisionEstimate>  <!-- 0..* How precise the estimate is -->
   <type><!-- 0..1 CodeableConcept Type of precision estimate --></type>
   <level value="[decimal]"/><!-- 0..1 Level of confidence interval -->
   <from value="[decimal]"/><!-- 0..1 Lower bound -->
   <to value="[decimal]"/><!-- 0..1 Upper bound -->
  </precisionEstimate>
 </effectEstimate>
 <certainty>  <!-- 0..* How certain is the effect -->
  <rating><!-- 0..* CodeableConcept Certainty rating --></rating>
  <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  <certaintySubcomponent>  <!-- 0..* A component that contributes to the overall certainty -->
   <type><!-- 0..1 CodeableConcept Type of subcomponent of certainty rating --></type>
   <rating><!-- 0..* CodeableConcept Subcomponent certainty rating --></rating>
   <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  </certaintySubcomponent>
 </certainty>
</EffectEvidenceSynthesis>
```

JSON Template

```
{
  "resourceType" : "EffectEvidenceSynthesis",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
```

```
  "url" : "<uri>", // Canonical identifier for this effect evidence synthesis, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the effect evidence synthesis
  "version" : "<string>", // Business version of the effect evidence synthesis
  "name" : "<string>", // C? Name for this effect evidence synthesis (computer friendly)
  "title" : "<string>", // Name for this effect evidence synthesis (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the effect evidence synthesis
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for effect evidence synthesis (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the effect evidence synthesis was approved by publisher
  "lastReviewDate" : "<date>", // When the effect evidence synthesis was last reviewed
  "effectivePeriod" : { Period }, // When the effect evidence synthesis is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EffectEvidenceSynthesis, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "synthesisType" : { CodeableConcept }, // Type of synthesis
  "studyType" : { CodeableConcept }, // Type of study
  "population" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposure" : { Reference(EvidenceVariable) }, // R!  What exposure?
  "exposureAlternative" : { Reference(EvidenceVariable) }, // R!  What comparison exposure?
  "outcome" : { Reference(EvidenceVariable) }, // R!  What outcome?
  "sampleSize" : { // What sample size was involved?
    "description" : "<string>", // Description of sample size
    "numberOfStudies" : <integer>, // How many studies?
    "numberOfParticipants" : <integer> // How many participants?
  },
  "resultsByExposure" : [{ // What was the result per exposure?
    "description" : "<string>", // Description of results by exposure
    "exposureState" : "<code>", // exposure | exposure-alternative
    "variantState" : { CodeableConcept }, // Variant exposure states
    "riskEvidenceSynthesis" : { Reference(RiskEvidenceSynthesis) } // R!  Risk evidence synthesis
  }],
  "effectEstimate" : [{ // What was the estimated effect
    "description" : "<string>", // Description of effect estimate
    "type" : { CodeableConcept }, // Type of efffect estimate
    "variantState" : { CodeableConcept }, // Variant exposure states
    "value" : <decimal>, // Point estimate
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "precisionEstimate" : [{ // How precise the estimate is
      "type" : { CodeableConcept }, // Type of precision estimate
      "level" : <decimal>, // Level of confidence interval
      "from" : <decimal>, // Lower bound
      "to" : <decimal> // Upper bound
    }]
  }],
  "certainty" : [{ // How certain is the effect
    "rating" : [{ CodeableConcept }], // Certainty rating
    "note" : [{ Annotation }], // Used for footnotes or explanatory notes
    "certaintySubcomponent" : [{ // A component that contributes to the overall certainty
      "type" : { CodeableConcept }, // Type of subcomponent of certainty rating
      "rating" : [{ CodeableConcept }], // Subcomponent certainty rating
      "note" : [{ Annotation }] // Used for footnotes or explanatory notes
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EffectEvidenceSynthesis;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EffectEvidenceSynthesis.url [ uri ]; # 0..1 Canonical identifier for this effect evidence synthesis, represented
    as a URI (globally unique)
  fhir:EffectEvidenceSynthesis.identifier [ Identifier ], ... ; # 0..* Additional identifier for the effect evidence
    synthesis
  fhir:EffectEvidenceSynthesis.version [ string ]; # 0..1 Business version of the effect evidence synthesis
  fhir:EffectEvidenceSynthesis.name [ string ]; # 0..1 Name for this effect evidence synthesis (computer friendly)
  fhir:EffectEvidenceSynthesis.title [ string ]; # 0..1 Name for this effect evidence synthesis (human friendly)
  fhir:EffectEvidenceSynthesis.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EffectEvidenceSynthesis.date [ dateTime ]; # 0..1 Date last changed
  fhir:EffectEvidenceSynthesis.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EffectEvidenceSynthesis.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EffectEvidenceSynthesis.description [ markdown ]; # 0..1 Natural language description of the effect evidence
    synthesis
  fhir:EffectEvidenceSynthesis.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:EffectEvidenceSynthesis.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
```

```
    support
  fhir:EffectEvidenceSynthesis.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for effect evidence
    synthesis (if applicable)
  fhir:EffectEvidenceSynthesis.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EffectEvidenceSynthesis.approvalDate [ date ]; # 0..1 When the effect evidence synthesis was approved by
    publisher
  fhir:EffectEvidenceSynthesis.lastReviewDate [ date ]; # 0..1 When the effect evidence synthesis was last reviewed
  fhir:EffectEvidenceSynthesis.effectivePeriod [ Period ]; # 0..1 When the effect evidence synthesis is expected to be
    used
  fhir:EffectEvidenceSynthesis.topic [ CodeableConcept ], ... ; # 0..* The category of the EffectEvidenceSynthesis, such
     as Education, Treatment, Assessment, etc.
  fhir:EffectEvidenceSynthesis.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EffectEvidenceSynthesis.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EffectEvidenceSynthesis.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EffectEvidenceSynthesis.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EffectEvidenceSynthesis.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations,
    etc.
  fhir:EffectEvidenceSynthesis.synthesisType [ CodeableConcept ]; # 0..1 Type of synthesis
  fhir:EffectEvidenceSynthesis.studyType [ CodeableConcept ]; # 0..1 Type of study
  fhir:EffectEvidenceSynthesis.population [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:EffectEvidenceSynthesis.exposure [ Reference(EvidenceVariable) ]; # 1..1 What exposure?
  fhir:EffectEvidenceSynthesis.exposureAlternative [ Reference(EvidenceVariable) ]; # 1..1 What comparison exposure?
  fhir:EffectEvidenceSynthesis.outcome [ Reference(EvidenceVariable) ]; # 1..1 What outcome?
  fhir:EffectEvidenceSynthesis.sampleSize [ # 0..1 What sample size was involved?
    fhir:EffectEvidenceSynthesis.sampleSize.description [ string ]; # 0..1 Description of sample size
    fhir:EffectEvidenceSynthesis.sampleSize.numberOfStudies [ integer ]; # 0..1 How many studies?
    fhir:EffectEvidenceSynthesis.sampleSize.numberOfParticipants [ integer ]; # 0..1 How many participants?
  ];
  fhir:EffectEvidenceSynthesis.resultsByExposure [ # 0..* What was the result per exposure?
    fhir:EffectEvidenceSynthesis.resultsByExposure.description [ string ]; # 0..1 Description of results by exposure
    fhir:EffectEvidenceSynthesis.resultsByExposure.exposureState [ code ]; # 0..1 exposure | exposure-alternative
    fhir:EffectEvidenceSynthesis.resultsByExposure.variantState [ CodeableConcept ]; # 0..1 Variant exposure states
    fhir:EffectEvidenceSynthesis.resultsByExposure.riskEvidenceSynthesis [ Reference(RiskEvidenceSynthesis) ]; # 1..1
    Risk evidence synthesis
  ], ...;
  fhir:EffectEvidenceSynthesis.effectEstimate [ # 0..* What was the estimated effect
    fhir:EffectEvidenceSynthesis.effectEstimate.description [ string ]; # 0..1 Description of effect estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.type [ CodeableConcept ]; # 0..1 Type of efffect estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.variantState [ CodeableConcept ]; # 0..1 Variant exposure states
    fhir:EffectEvidenceSynthesis.effectEstimate.value [ decimal ]; # 0..1 Point estimate
    fhir:EffectEvidenceSynthesis.effectEstimate.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome
    described in?
    fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate [ # 0..* How precise the estimate is
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.type [ CodeableConcept ]; # 0..1 Type of precision
    estimate
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.level [ decimal ]; # 0..1 Level of confidence
    interval
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.from [ decimal ]; # 0..1 Lower bound
      fhir:EffectEvidenceSynthesis.effectEstimate.precisionEstimate.to [ decimal ]; # 0..1 Upper bound
    ], ...;
  ], ...;
  fhir:EffectEvidenceSynthesis.certainty [ # 0..* How certain is the effect
    fhir:EffectEvidenceSynthesis.certainty.rating [ CodeableConcept ], ... ; # 0..* Certainty rating
    fhir:EffectEvidenceSynthesis.certainty.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
    fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent [ # 0..* A component that contributes to the overall
    certainty
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.type [ CodeableConcept ]; # 0..1 Type of subcomponent
     of certainty rating
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.rating [ CodeableConcept ], ... ; # 0..* Subcomponent
     certainty rating
      fhir:EffectEvidenceSynthesis.certainty.certaintySubcomponent.note [ Annotation ], ... ; # 0..* Used for footnotes
    or explanatory notes
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.7.2.1 Terminology Bindings

Path | Definition | Type | Reference

EffectEvidenceSynthesis.status | The lifecycle status of an artifact. | Required | PublicationStatus

EffectEvidenceSynthesis.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

EffectEvidenceSynthesis.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic

EffectEvidenceSynthesis.synthesisType | Types of combining results from a body of evidence (eg. summary data meta-analysis). | Extensible | SynthesisType

EffectEvidenceSynthesis.studyType | Types of research studies (types of research methods). | Extensible | StudyType

EffectEvidenceSynthesis.resultsByExposure.exposureState | Whether the results by exposure is describing the results for the primary exposure of interest (exposure) or the alternative state (exposureAlternative). | Required | ExposureState

EffectEvidenceSynthesis.resultsByExposure.variantState EffectEvidenceSynthesis.effectEstimate.variantState | Used for results by exposure in variant states such as low-risk, medium-risk and high-risk states. | Extensible | EvidenceVariantState
EffectEvidenceSynthesis.effectEstimate.type | Whether the effect estimate is an absolute effect estimate (absolute difference) or a relative effect estimate (relative difference), and the specific type of effect estimate (eg relative risk or median difference). | Extensible | EffectEstimateType
EffectEvidenceSynthesis.effectEstimate.unitOfMeasure | Unified Code for Units of Measure (UCUM). | Required | UCUMCodes
EffectEvidenceSynthesis.effectEstimate.precisionEstimate.type | Method of reporting variability of estimates, such as confidence intervals, interquartile range or standard deviation. | Extensible | PrecisionEstimateType
EffectEvidenceSynthesis.certainty.rating | The quality of the evidence described. The code system used specifies the quality scale used to grade this evidence source while the code specifies the actual quality score (represented as a coded value) associated with the evidence. | Extensible | QualityOfEvidenceRating
EffectEvidenceSynthesis.certainty.certaintySubcomponent.type | The subcomponent classification of quality of evidence rating systems. | Extensible | CertaintySubcomponentType
EffectEvidenceSynthesis.certainty.certaintySubcomponent.rating | The quality rating of the subcomponent of a quality of evidence rating. | Extensible | CertaintySubcomponentRating

14.7.2.2 Constraints

id | Level | Location | Description | Expression
ees-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.7.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the effect evidence synthesis | (EffectEvidenceSynthesis.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the effect evidence synthesis | (EffectEvidenceSynthesis.useContext.value as Quantity) | (EffectEvidenceSynthesis.useContext.value as Range)
context-type | token | A type of use context assigned to the effect evidence synthesis | EffectEvidenceSynthesis.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the effect evidence synthesis | On EffectEvidenceSynthesis.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the effect evidence synthesis | On EffectEvidenceSynthesis.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The effect evidence synthesis publication date | EffectEvidenceSynthesis.date
description | string | The description of the effect evidence synthesis | EffectEvidenceSynthesis.description
effective | date | The time during which the effect evidence synthesis is intended to be in use | EffectEvidenceSynthesis.effectivePeriod
identifier | token | External identifier for the effect evidence synthesis | EffectEvidenceSynthesis.identifier
jurisdiction | token | Intended jurisdiction for the effect evidence synthesis | EffectEvidenceSynthesis.jurisdiction
name | string | Computationally friendly name of the effect evidence synthesis | EffectEvidenceSynthesis.name
publisher | string | Name of the publisher of the effect evidence synthesis | EffectEvidenceSynthesis.publisher
status | token | The current status of the effect evidence synthesis | EffectEvidenceSynthesis.status
title | string | The human-friendly name of the effect evidence synthesis | EffectEvidenceSynthesis.title
url | uri | The uri that identifies the effect evidence synthesis | EffectEvidenceSynthesis.url
version | token | The business version of the effect evidence synthesis | EffectEvidenceSynthesis.version

14.7.4 Resource EffectEvidenceSynthesis - Examples

URL: https://hl7.org/fhir/R4/effectevidencesynthesis-examples.html

14.7.4 Resource EffectEvidenceSynthesis - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

TODO

Example Name | id | Format
Example of effectevidencesynthesis | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.7.5 Resource EffectEvidenceSynthesis - Detailed Descriptions

URL: https://hl7.org/fhir/R4/effectevidencesynthesis-definitions.html

14.7.5 Resource EffectEvidenceSynthesis - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the EffectEvidenceSynthesis resource.

EffectEvidenceSynthesis
Element Id | EffectEvidenceSynthesis
Definition | The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures states in a population where the effect estimate is derived from a combination of research studies.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element ees-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
ees-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
EffectEvidenceSynthesis.url
Element Id | EffectEvidenceSynthesis.url
Definition | An absolute URI that is used to identify this effect evidence synthesis when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this effect evidence synthesis is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the effect evidence synthesis is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the effect evidence synthesis to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
EffectEvidenceSynthesis.identifier
Element Id | EffectEvidenceSynthesis.identifier
Definition | A formal identifier that is used to identify this effect evidence synthesis when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this effect evidence synthesis outside of FHIR, where it is not possible to use the logical URI.
EffectEvidenceSynthesis.version
Element Id | EffectEvidenceSynthesis.version
Definition | The identifier that is used to identify this version of the effect evidence synthesis when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the effect evidence synthesis author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different effect evidence synthesis instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the effect evidence synthesis with the format [url]|[version].
EffectEvidenceSynthesis.name
Element Id | EffectEvidenceSynthesis.name
Definition | A natural language name identifying the effect evidence synthesis. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EffectEvidenceSynthesis.title
Element Id | EffectEvidenceSynthesis.title
Definition | A short, descriptive, user-friendly title for the effect evidence synthesis.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.

To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EffectEvidenceSynthesis.status
Element Id | EffectEvidenceSynthesis.status
Definition | The status of this effect evidence synthesis. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of effect evidence synthesiss that are appropriate for use versus not.
EffectEvidenceSynthesis.date
Element Id | EffectEvidenceSynthesis.date
Definition | The date (and optionally time) when the effect evidence synthesis was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the effect evidence synthesis changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the effect evidence synthesis. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
EffectEvidenceSynthesis.publisher
Element Id | EffectEvidenceSynthesis.publisher
Definition | The name of the organization or individual that published the effect evidence synthesis.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the effect evidence synthesis. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the effect evidence synthesis is the organization or individual primarily responsible for the maintenance and upkeep of the effect evidence synthesis. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the effect evidence synthesis. This item SHOULD be populated unless the information is available from context.
EffectEvidenceSynthesis.contact
Element Id | EffectEvidenceSynthesis.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
EffectEvidenceSynthesis.description
Element Id | EffectEvidenceSynthesis.description
Definition | A free text natural language description of the effect evidence synthesis from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the effect evidence synthesis was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the effect evidence synthesis as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the effect evidence synthesis is presumed to be the predominant language in the place the effect evidence synthesis was created).
EffectEvidenceSynthesis.note
Element Id | EffectEvidenceSynthesis.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation
EffectEvidenceSynthesis.useContext
Element Id | EffectEvidenceSynthesis.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate effect evidence synthesis instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EffectEvidenceSynthesis.jurisdiction
Element Id | EffectEvidenceSynthesis.jurisdiction
Definition | A legal or geographic region in which the effect evidence synthesis is intended to be used.

Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the effect evidence synthesis to be used in jurisdictions other than those for which it was originally designed or intended.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EffectEvidenceSynthesis.copyright
Element Id | EffectEvidenceSynthesis.copyright
Definition | A copyright statement relating to the effect evidence synthesis and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the effect evidence synthesis.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the effect evidence synthesis and/or its content.
Alternate Names | License; Restrictions
EffectEvidenceSynthesis.approvalDate
Element Id | EffectEvidenceSynthesis.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
EffectEvidenceSynthesis.lastReviewDate
Element Id | EffectEvidenceSynthesis.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
EffectEvidenceSynthesis.effectivePeriod
Element Id | EffectEvidenceSynthesis.effectivePeriod
Definition | The period during which the effect evidence synthesis content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the effect evidence synthesis are or are expected to be used instead.
Summary | true
Comments | The effective period for a effect evidence synthesis determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
EffectEvidenceSynthesis.topic
Element Id | EffectEvidenceSynthesis.topic
Definition | Descriptive topics related to the content of the EffectEvidenceSynthesis. Topics provide a high-level categorization grouping types of EffectEvidenceSynthesiss that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the EffectEvidenceSynthesis so that it can be found by topical searches.
EffectEvidenceSynthesis.author
Element Id | EffectEvidenceSynthesis.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
EffectEvidenceSynthesis.editor
Element Id | EffectEvidenceSynthesis.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
EffectEvidenceSynthesis.reviewer
Element Id | EffectEvidenceSynthesis.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
EffectEvidenceSynthesis.endorser
Element Id | EffectEvidenceSynthesis.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail

EffectEvidenceSynthesis.relatedArtifact
Element Id | EffectEvidenceSynthesis.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | EffectEvidenceSynthesiss must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
EffectEvidenceSynthesis.synthesisType
Element Id | EffectEvidenceSynthesis.synthesisType
Definition | Type of synthesis eg meta-analysis.
Cardinality | 0..1
Terminology Binding | SynthesisType (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.studyType
Element Id | EffectEvidenceSynthesis.studyType
Definition | Type of study eg randomized trial.
Cardinality | 0..1
Terminology Binding | StudyType (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.population
Element Id | EffectEvidenceSynthesis.population
Definition | A reference to a EvidenceVariable resource that defines the population for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
EffectEvidenceSynthesis.exposure
Element Id | EffectEvidenceSynthesis.exposure
Definition | A reference to a EvidenceVariable resource that defines the exposure for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
EffectEvidenceSynthesis.exposureAlternative
Element Id | EffectEvidenceSynthesis.exposureAlternative
Definition | A reference to a EvidenceVariable resource that defines the comparison exposure for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
EffectEvidenceSynthesis.outcome
Element Id | EffectEvidenceSynthesis.outcome
Definition | A reference to a EvidenceVariable resomece that defines the outcome for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
EffectEvidenceSynthesis.sampleSize
Element Id | EffectEvidenceSynthesis.sampleSize
Definition | A description of the size of the sample involved in the synthesis.
Cardinality | 0..1
EffectEvidenceSynthesis.sampleSize.description
Element Id | EffectEvidenceSynthesis.sampleSize.description
Definition | Human-readable summary of sample size.
Cardinality | 0..1
Type | string
EffectEvidenceSynthesis.sampleSize.numberOfStudies
Element Id | EffectEvidenceSynthesis.sampleSize.numberOfStudies
Definition | Number of studies included in this evidence synthesis.
Cardinality | 0..1
Type | integer
EffectEvidenceSynthesis.sampleSize.numberOfParticipants
Element Id | EffectEvidenceSynthesis.sampleSize.numberOfParticipants
Definition | Number of participants included in this evidence synthesis.
Cardinality | 0..1
Type | integer
EffectEvidenceSynthesis.resultsByExposure
Element Id | EffectEvidenceSynthesis.resultsByExposure
Definition | A description of the results for each exposure considered in the effect estimate.
Cardinality | 0..*
EffectEvidenceSynthesis.resultsByExposure.description
Element Id | EffectEvidenceSynthesis.resultsByExposure.description
Definition | Human-readable summary of results by exposure state.
Cardinality | 0..1

Type | string
EffectEvidenceSynthesis.resultsByExposure.exposureState
Element Id | EffectEvidenceSynthesis.resultsByExposure.exposureState
Definition | Whether these results are for the exposure state or alternative exposure state.
Cardinality | 0..1
Terminology Binding | ExposureState (Required)
Type | code
EffectEvidenceSynthesis.resultsByExposure.variantState
Element Id | EffectEvidenceSynthesis.resultsByExposure.variantState
Definition | Used to define variant exposure states such as low-risk state.
Cardinality | 0..1
Terminology Binding | EvidenceVariantState (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.resultsByExposure.riskEvidenceSynthesis
Element Id | EffectEvidenceSynthesis.resultsByExposure.riskEvidenceSynthesis
Definition | Reference to a RiskEvidenceSynthesis resource.
Cardinality | 1..1
Type | Reference(RiskEvidenceSynthesis)
EffectEvidenceSynthesis.effectEstimate
Element Id | EffectEvidenceSynthesis.effectEstimate
Definition | The estimated effect of the exposure variant.
Cardinality | 0..*
Summary | true
To Do | Note: Cardinality was relaxed to 0..* to get the initial draft building.
EffectEvidenceSynthesis.effectEstimate.description
Element Id | EffectEvidenceSynthesis.effectEstimate.description
Definition | Human-readable summary of effect estimate.
Cardinality | 0..1
Type | string
EffectEvidenceSynthesis.effectEstimate.type
Element Id | EffectEvidenceSynthesis.effectEstimate.type
Definition | Examples include relative risk and mean difference.
Cardinality | 0..1
Terminology Binding | EffectEstimateType (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.effectEstimate.variantState
Element Id | EffectEvidenceSynthesis.effectEstimate.variantState
Definition | Used to define variant exposure states such as low-risk state.
Cardinality | 0..1
Terminology Binding | EvidenceVariantState (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.effectEstimate.value
Element Id | EffectEvidenceSynthesis.effectEstimate.value
Definition | The point estimate of the effect estimate.
Cardinality | 0..1
Type | decimal
EffectEvidenceSynthesis.effectEstimate.unitOfMeasure
Element Id | EffectEvidenceSynthesis.effectEstimate.unitOfMeasure
Definition | Specifies the UCUM unit for the outcome.
Cardinality | 0..1
Terminology Binding | UCUM Codes (Required)
Type | CodeableConcept
EffectEvidenceSynthesis.effectEstimate.precisionEstimate
Element Id | EffectEvidenceSynthesis.effectEstimate.precisionEstimate
Definition | A description of the precision of the estimate for the effect.
Cardinality | 0..*
EffectEvidenceSynthesis.effectEstimate.precisionEstimate.type
Element Id | EffectEvidenceSynthesis.effectEstimate.precisionEstimate.type
Definition | Examples include confidence interval and interquartile range.
Cardinality | 0..1
Terminology Binding | PrecisionEstimateType (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.effectEstimate.precisionEstimate.level
Element Id | EffectEvidenceSynthesis.effectEstimate.precisionEstimate.level
Definition | Use 95 for a 95% confidence interval.
Cardinality | 0..1
Type | decimal
EffectEvidenceSynthesis.effectEstimate.precisionEstimate.from
Element Id | EffectEvidenceSynthesis.effectEstimate.precisionEstimate.from
Definition | Lower bound of confidence interval.
Cardinality | 0..1
Type | decimal

EffectEvidenceSynthesis.effectEstimate.precisionEstimate.to
Element Id | EffectEvidenceSynthesis.effectEstimate.precisionEstimate.to
Definition | Upper bound of confidence interval.
Cardinality | 0..1
Type | decimal
EffectEvidenceSynthesis.certainty
Element Id | EffectEvidenceSynthesis.certainty
Definition | A description of the certainty of the effect estimate.
Cardinality | 0..*
EffectEvidenceSynthesis.certainty.rating
Element Id | EffectEvidenceSynthesis.certainty.rating
Definition | A rating of the certainty of the effect estimate.
Cardinality | 0..*
Terminology Binding | QualityOfEvidenceRating (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.certainty.note
Element Id | EffectEvidenceSynthesis.certainty.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation
EffectEvidenceSynthesis.certainty.certaintySubcomponent
Element Id | EffectEvidenceSynthesis.certainty.certaintySubcomponent
Definition | A description of a component of the overall certainty.
Cardinality | 0..*
EffectEvidenceSynthesis.certainty.certaintySubcomponent.type
Element Id | EffectEvidenceSynthesis.certainty.certaintySubcomponent.type
Definition | Type of subcomponent of certainty rating.
Cardinality | 0..1
Terminology Binding | CertaintySubcomponentType (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.certainty.certaintySubcomponent.rating
Element Id | EffectEvidenceSynthesis.certainty.certaintySubcomponent.rating
Definition | A rating of a subcomponent of rating certainty.
Cardinality | 0..*
Terminology Binding | CertaintySubcomponentRating (Extensible)
Type | CodeableConcept
EffectEvidenceSynthesis.certainty.certaintySubcomponent.note
Element Id | EffectEvidenceSynthesis.certainty.certaintySubcomponent.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation

14.7.6 Resource EffectEvidenceSynthesis - Mappings

URL: https://hl7.org/fhir/R4/effectevidencesynthesis-mappings.html

14.7.6 Resource EffectEvidenceSynthesis - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the effectevidencesynthesis resource (see Mappings to Other Standards for further information & status).

14.7.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

EffectEvidenceSynthesis
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
date | FiveWs.recorded
publisher | FiveWs.witness

14.7.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

EffectEvidenceSynthesis | Definition
url | Definition.url
identifier | Definition.identifier

version | Definition.version
title | Definition.title
status | Definition.status
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.7.6.3 RIM Mapping (http://hl7.org/v3)

EffectEvidenceSynthesis
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.7.7 Resource EffectEvidenceSynthesis - Extensions & Profiles

URL: https://hl7.org/fhir/R4/effectevidencesynthesis-profiles.html

14.7.7 Resource EffectEvidenceSynthesis - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the EffectEvidenceSynthesis resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.7.7.1 Profiles

No Profiles defined for this resource

14.7.7.2 Extensions

replaces | EffectEvidenceSynthesis | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.7.7.3 Search Extensions

No Search Extensions defined for this resource

14.7.8 JSON Schema for EffectEvidenceSynthesis

URL: https://hl7.org/fhir/R4/effectevidencesynthesis.schema.json.html

14.7.8 JSON Schema for EffectEvidenceSynthesis

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for EffectEvidenceSynthesis, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/EffectEvidenceSynthesis",
  "$ref": "#/definitions/EffectEvidenceSynthesis",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
```

```
"definitions": {
  "EffectEvidenceSynthesis": {
    "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
  states in a population where the effect estimate is derived from a combination of research studies.",
    "properties": {
      "resourceType": {
        "description": "This is a EffectEvidenceSynthesis resource",
        "const": "EffectEvidenceSynthesis"
      },
      "id": {
        "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
   never changes.",
        "$ref": "id.schema.json#/definitions/id"
      },
      "meta": {
        "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
  Changes to the content might not always be associated with version changes to the resource.",
        "$ref": "Meta.schema.json#/definitions/Meta"
      },
      "implicitRules": {
        "description": "A reference to a set of rules that were followed when the resource was constructed, and which
  must be understood when processing the content. Often, this is a reference to an implementation guide that defines
  the special rules along with other profiles etc.",
        "$ref": "#/definitions/uri"
      },
      "_implicitRules": {
        "description": "Extensions for implicitRules",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "language": {
        "description": "The base language in which the resource is written.",
        "$ref": "#/definitions/code"
      },
      "_language": {
        "description": "Extensions for language",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "text": {
        "description": "A human-readable narrative that contains a summary of the resource and can be used to
  represent the content of the resource to a human. The narrative need not encode all the structured data, but is
  required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
  Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
        "$ref": "Narrative.schema.json#/definitions/Narrative"
      },
      "contained": {
        "description": "These resources do not have an independent existence apart from the resource that contains
  them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
        "items": {
          "$ref": "ResourceList.schema.json#/definitions/ResourceList"
        },
        "type": "array"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
   definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
  that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of
  the resource and that modifies the understanding of the element that contains it and/or the understanding of the
  containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
  of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
  extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
  met as part of the definition of the extension. Applications processing a resource are required to check for
  modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
  DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "url": {
        "description": "An absolute URI that is used to identify this effect evidence synthesis when it is referenced
  in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally
  unique and SHOULD be a literal address at which at which an authoritative instance of this effect evidence synthesis
   is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the
  effect evidence synthesis is stored on different servers.",
        "$ref": "#/definitions/uri"
      },
      "_url": {
        "description": "Extensions for url",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "identifier": {
```

```
      "description": "A formal identifier that is used to identify this effect evidence synthesis when it is
represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the effect evidence synthesis when it
is referenced in a specification, model, design or instance. This is an arbitrary value managed by the effect
evidence synthesis author and is not expected to be globally unique. For example, it might be a timestamp (e.g.
yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a
lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the effect evidence synthesis. This name should be usable
as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the effect evidence synthesis.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this effect evidence synthesis. Enables tracking the life-cycle of the
content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the effect evidence synthesis was published. The date
must change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the effect evidence synthesis changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the effect evidence synthesis.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the effect evidence synthesis from a
consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
```

```
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate effect evidence synthesis instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the effect evidence synthesis is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "copyright": {
      "description": "A copyright statement relating to the effect evidence synthesis and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the effect evidence synthesis.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the effect evidence synthesis content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the EffectEvidenceSynthesis. Topics provide a high-level categorization grouping types of EffectEvidenceSynthesiss that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
```

```
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "synthesisType": {
      "description": "Type of synthesis eg meta-analysis.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "studyType": {
      "description": "Type of study eg randomized trial.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "population": {
      "description": "A reference to a EvidenceVariable resource that defines the population for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "exposure": {
      "description": "A reference to a EvidenceVariable resource that defines the exposure for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "exposureAlternative": {
      "description": "A reference to a EvidenceVariable resource that defines the comparison exposure for the
research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "outcome": {
      "description": "A reference to a EvidenceVariable resomece that defines the outcome for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "sampleSize": {
      "description": "A description of the size of the sample involved in the synthesis.",
      "$ref": "#/definitions/EffectEvidenceSynthesis_SampleSize"
    },
    "resultsByExposure": {
      "description": "A description of the results for each exposure considered in the effect estimate.",
      "items": {
        "$ref": "#/definitions/EffectEvidenceSynthesis_ResultsByExposure"
      },
      "type": "array"
    },
    "effectEstimate": {
      "description": "The estimated effect of the exposure variant.",
      "items": {
        "$ref": "#/definitions/EffectEvidenceSynthesis_EffectEstimate"
      },
      "type": "array"
    },
    "certainty": {
      "description": "A description of the certainty of the effect estimate.",
      "items": {
        "$ref": "#/definitions/EffectEvidenceSynthesis_Certainty"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "exposureAlternative",
    "exposure",
    "outcome",
    "resourceType",
    "population"
  ]
},
"EffectEvidenceSynthesis_SampleSize": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"

    },
```

```
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable summary of sample size.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfStudies": {
      "description": "Number of studies included in this evidence synthesis.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfStudies": {
      "description": "Extensions for numberOfStudies",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfParticipants": {
      "description": "Number of participants included in this evidence synthesis.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfParticipants": {
      "description": "Extensions for numberOfParticipants",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"EffectEvidenceSynthesis_ResultsByExposure": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable summary of results by exposure state.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "exposureState": {
      "description": "Whether these results are for the exposure state or alternative exposure state.",
      "enum": [
        "exposure",
        "exposure-alternative"
      ]
    },
```

```
    "_exposureState": {
      "description": "Extensions for exposureState",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "variantState": {
      "description": "Used to define variant exposure states such as low-risk state.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "riskEvidenceSynthesis": {
      "description": "Reference to a RiskEvidenceSynthesis resource.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false,
  "required": [
    "riskEvidenceSynthesis"
  ]
},
"EffectEvidenceSynthesis_EffectEstimate": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable summary of effect estimate.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Examples include relative risk and mean difference.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "variantState": {
      "description": "Used to define variant exposure states such as low-risk state.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "value": {
      "description": "The point estimate of the effect estimate.",
      "$ref": "#/definitions/decimal"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "unitOfMeasure": {
      "description": "Specifies the UCUM unit for the outcome.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "precisionEstimate": {
      "description": "A description of the precision of the estimate for the effect.",
      "items": {
        "$ref": "#/definitions/EffectEvidenceSynthesis_PrecisionEstimate"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
```

```
"EffectEvidenceSynthesis_PrecisionEstimate": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Examples include confidence interval and interquartile range.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "level": {
      "description": "Use 95 for a 95% confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_level": {
      "description": "Extensions for level",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "from": {
      "description": "Lower bound of confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_from": {
      "description": "Extensions for from",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "to": {
      "description": "Upper bound of confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_to": {
      "description": "Extensions for to",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"EffectEvidenceSynthesis_Certainty": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
```

```
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "rating": {
      "description": "A rating of the certainty of the effect estimate.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "certaintySubcomponent": {
      "description": "A description of a component of the overall certainty.",
      "items": {
        "$ref": "#/definitions/EffectEvidenceSynthesis_CertaintySubcomponent"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"EffectEvidenceSynthesis_CertaintySubcomponent": {
  "description": "The EffectEvidenceSynthesis resource describes the difference in an outcome between exposures
states in a population where the effect estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Type of subcomponent of certainty rating.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "rating": {
      "description": "A rating of a subcomponent of rating certainty.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
}
  }
}
```

14.7.9 Resource EffectEvidenceSynthesis - Maturity Dependency View

URL: https://hl7.org/fhir/R4/effectevidencesynthesis-dependencies.html

14.7.9 Resource EffectEvidenceSynthesis - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for EffectEvidenceSynthesis FMM level 0

Name | Card. | Type | Dependency Analysis
EffectEvidenceSynthesis | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
note | 0..* | Annotation
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
synthesisType | 0..1 | CodeableConcept
studyType | 0..1 | CodeableConcept
population | 1..1 | Reference(EvidenceVariable)
exposure | 1..1 | Reference(EvidenceVariable)
exposureAlternative | 1..1 | Reference(EvidenceVariable)
outcome | 1..1 | Reference(EvidenceVariable)
sampleSize | 0..1 | BackboneElement
description | 0..1 | string
numberOfStudies | 0..1 | integer
numberOfParticipants | 0..1 | integer
resultsByExposure | 0..* | BackboneElement
description | 0..1 | string
exposureState | 0..1 | code
variantState | 0..1 | CodeableConcept
riskEvidenceSynthesis | 1..1 | Reference(RiskEvidenceSynthesis)
effectEstimate | 0..* | BackboneElement
description | 0..1 | string
type | 0..1 | CodeableConcept
variantState | 0..1 | CodeableConcept
value | 0..1 | decimal
unitOfMeasure | 0..1 | CodeableConcept
precisionEstimate | 0..* | BackboneElement
type | 0..1 | CodeableConcept
level | 0..1 | decimal
from | 0..1 | decimal
to | 0..1 | decimal
certainty | 0..* | BackboneElement
rating | 0..* | CodeableConcept
note | 0..* | Annotation
certaintySubcomponent | 0..* | BackboneElement
type | 0..1 | CodeableConcept
rating | 0..* | CodeableConcept
note | 0..* | Annotation

14.8 Resource EventDefinition - Content

URL: https://hl7.org/fhir/R4/eventdefinition.html

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The EventDefinition resource provides a reusable description of when a particular event can occur.

14.8.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

The EventDefinition provides a reusable description of an event. The resource supports describing different kinds of events, including named events, periodic events, and data-based events. For each of these, the resource also supports a formal description of the event. For example, a 'monitor-emergency-admissions' event can be a named event, but also provide a formal description of the event as monitoring for encounters that occur in emergency department locations.

14.8.2 Boundaries and Relationships

The EventDefinition resource is intended to be used to provide a reusable description of events for use by other resources that require event definition, such as Subscription. EventDefinition is a definition of an event that could occur, whereas Subscription is asking for notification of events when they occur.

14.8.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
EventDefinition | ITU | | DomainResource | A description of when an event can occur + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this event definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the event definition
version | Σ | 0..1 | string | Business version of the event definition
name | ΣI | 0..1 | string | Name for this event definition (computer friendly)
title | Σ | 0..1 | string | Name for this event definition (human friendly)
subtitle | | 0..1 | string | Subordinate title of the event definition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | Type of individual the event definition is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the event definition
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for event definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this event definition is defined
usage | | 0..1 | string | Describes the clinical usage of the event definition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | Σ | 0..1 | date | When the event definition was approved by publisher
lastReviewDate | Σ | 0..1 | date | When the event definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the event definition is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
trigger | Σ | 1..* | TriggerDefinition | "when" the event occurs (multiple = 'or')
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EventDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
```

```
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this event definition, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the event definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the event definition -->
 <name value="[string]"/><!--  0..1 Name for this event definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this event definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the event definition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the event definition is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the event definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for event definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this event definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the event definition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the event definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the event definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the event definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <trigger><!-- 1..* TriggerDefinition "when" the event occurs (multiple = 'or') --></trigger>
</EventDefinition>
```

JSON Template

```
{
  "resourceType" : "EventDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this event definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the event definition
  "version" : "<string>", // Business version of the event definition
  "name" : "<string>", // C? Name for this event definition (computer friendly)
  "title" : "<string>", // Name for this event definition (human friendly)
  "subtitle" : "<string>", // Subordinate title of the event definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: Type of individual the event definition is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the event definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for event definition (if applicable)
  "purpose" : "<markdown>", // Why this event definition is defined
  "usage" : "<string>", // Describes the clinical usage of the event definition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the event definition was approved by publisher
  "lastReviewDate" : "<date>", // When the event definition was last reviewed
  "effectivePeriod" : { Period }, // When the event definition is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "trigger" : [{ TriggerDefinition }] // R!  "when" the event occurs (multiple = 'or')
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EventDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EventDefinition.url [ uri ]; # 0..1 Canonical identifier for this event definition, represented as a URI
    (globally unique)
  fhir:EventDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the event definition
  fhir:EventDefinition.version [ string ]; # 0..1 Business version of the event definition
  fhir:EventDefinition.name [ string ]; # 0..1 Name for this event definition (computer friendly)
  fhir:EventDefinition.title [ string ]; # 0..1 Name for this event definition (human friendly)
  fhir:EventDefinition.subtitle [ string ]; # 0..1 Subordinate title of the event definition
  fhir:EventDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EventDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # EventDefinition.subject[x] : 0..1 Type of individual the event definition is focused on. One of these 2
    fhir:EventDefinition.subjectCodeableConcept [ CodeableConcept ]
```

```
    fhir:EventDefinition.subjectReference [ Reference(Group) ]
  fhir:EventDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:EventDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EventDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EventDefinition.description [ markdown ]; # 0..1 Natural language description of the event definition
  fhir:EventDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:EventDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for event definition (if
    applicable)
  fhir:EventDefinition.purpose [ markdown ]; # 0..1 Why this event definition is defined
  fhir:EventDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the event definition
  fhir:EventDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EventDefinition.approvalDate [ date ]; # 0..1 When the event definition was approved by publisher
  fhir:EventDefinition.lastReviewDate [ date ]; # 0..1 When the event definition was last reviewed
  fhir:EventDefinition.effectivePeriod [ Period ]; # 0..1 When the event definition is expected to be used
  fhir:EventDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:EventDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EventDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EventDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EventDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EventDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:EventDefinition.trigger [ TriggerDefinition ], ... ; # 1..* "when" the event occurs (multiple = 'or')
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

EventDefinition | ITU | | DomainResource | A description of when an event can occur + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this event definition, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the event definition

version | Σ | 0..1 | string | Business version of the event definition

name | ΣI | 0..1 | string | Name for this event definition (computer friendly)

title | Σ | 0..1 | string | Name for this event definition (human friendly)

subtitle | | 0..1 | string | Subordinate title of the event definition

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

subject[x] | | 0..1 | | Type of individual the event definition is focused on Subject type (Extensible)

subjectCodeableConcept | | | CodeableConcept

subjectReference | | | Reference(Group)

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the event definition

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for event definition (if applicable) Jurisdiction (Extensible)

purpose | | 0..1 | markdown | Why this event definition is defined

usage | | 0..1 | string | Describes the clinical usage of the event definition

copyright | | 0..1 | markdown | Use and/or publishing restrictions

approvalDate | Σ | 0..1 | date | When the event definition was approved by publisher

lastReviewDate | Σ | 0..1 | date | When the event definition was last reviewed

effectivePeriod | Σ | 0..1 | Period | When the event definition is expected to be used

topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)

author | | 0..* | ContactDetail | Who authored the content

editor | | 0..* | ContactDetail | Who edited the content

reviewer | | 0..* | ContactDetail | Who reviewed the content

endorser | | 0..* | ContactDetail | Who endorsed the content

relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.

trigger | Σ | 1..* | TriggerDefinition | "when" the event occurs (multiple = 'or')

Documentation for this format

UML Diagram (Legend)

XML Template

```
<EventDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this event definition, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the event definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the event definition -->
 <name value="[string]"/><!--  0..1 Name for this event definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this event definition (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the event definition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the event definition is focused on
```

```
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the event definition -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for event definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this event definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the event definition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the event definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the event definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the event definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <trigger><!-- 1..* TriggerDefinition "when" the event occurs (multiple = 'or') --></trigger>
</EventDefinition>
```

JSON Template

```
{
  "resourceType" : "EventDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this event definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the event definition
  "version" : "<string>", // Business version of the event definition
  "name" : "<string>", // C? Name for this event definition (computer friendly)
  "title" : "<string>", // Name for this event definition (human friendly)
  "subtitle" : "<string>", // Subordinate title of the event definition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: Type of individual the event definition is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the event definition
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for event definition (if applicable)
  "purpose" : "<markdown>", // Why this event definition is defined
  "usage" : "<string>", // Describes the clinical usage of the event definition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the event definition was approved by publisher
  "lastReviewDate" : "<date>", // When the event definition was last reviewed
  "effectivePeriod" : { Period }, // When the event definition is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "trigger" : [{ TriggerDefinition }] // R!  "when" the event occurs (multiple = 'or')
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EventDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EventDefinition.url [ uri ]; # 0..1 Canonical identifier for this event definition, represented as a URI
    (globally unique)
  fhir:EventDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the event definition
  fhir:EventDefinition.version [ string ]; # 0..1 Business version of the event definition
  fhir:EventDefinition.name [ string ]; # 0..1 Name for this event definition (computer friendly)
  fhir:EventDefinition.title [ string ]; # 0..1 Name for this event definition (human friendly)
  fhir:EventDefinition.subtitle [ string ]; # 0..1 Subordinate title of the event definition
  fhir:EventDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EventDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # EventDefinition.subject[x] : 0..1 Type of individual the event definition is focused on. One of these 2
    fhir:EventDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:EventDefinition.subjectReference [ Reference(Group) ]
  fhir:EventDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:EventDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EventDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EventDefinition.description [ markdown ]; # 0..1 Natural language description of the event definition
  fhir:EventDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:EventDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for event definition (if
    applicable)
  fhir:EventDefinition.purpose [ markdown ]; # 0..1 Why this event definition is defined
  fhir:EventDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the event definition
```

```
  fhir:EventDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EventDefinition.approvalDate [ date ]; # 0..1 When the event definition was approved by publisher
  fhir:EventDefinition.lastReviewDate [ date ]; # 0..1 When the event definition was last reviewed
  fhir:EventDefinition.effectivePeriod [ Period ]; # 0..1 When the event definition is expected to be used
  fhir:EventDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:EventDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EventDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EventDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EventDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EventDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:EventDefinition.trigger [ TriggerDefinition ], ... ; # 1..* "when" the event occurs (multiple = 'or')
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.8.3.1 Terminology Bindings

Path | Definition | Type | Reference
EventDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus
EventDefinition.subject[x] | The possible types of subjects for an event (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
EventDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
EventDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic

14.8.3.2 Constraints

id | Level | Location | Description | Expression
evd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.8.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | EventDefinition.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the event definition | (EventDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the event definition | (EventDefinition.useContext.value as Quantity) | (EventDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the event definition | EventDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the event definition | On EventDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the event definition | On EventDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The event definition publication date | EventDefinition.date
depends-on | reference | What resource is being referenced | EventDefinition.relatedArtifact.where(type='depends-on').resource (Any)
derived-from | reference | What resource is being referenced | EventDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the event definition | EventDefinition.description
effective | date | The time during which the event definition is intended to be in use | EventDefinition.effectivePeriod
identifier | token | External identifier for the event definition | EventDefinition.identifier
jurisdiction | token | Intended jurisdiction for the event definition | EventDefinition.jurisdiction
name | string | Computationally friendly name of the event definition | EventDefinition.name
predecessor | reference | What resource is being referenced | EventDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the event definition | EventDefinition.publisher
status | token | The current status of the event definition | EventDefinition.status
successor | reference | What resource is being referenced | EventDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the event definition | EventDefinition.title
topic | token | Topics associated with the module | EventDefinition.topic
url | uri | The uri that identifies the event definition | EventDefinition.url
version | token | The business version of the event definition | EventDefinition.version

14.8.5 Resource EventDefinition - Examples

URL: https://hl7.org/fhir/R4/eventdefinition-examples.html

14.8.5 Resource EventDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of eventdefinition | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.8.6 Resource EventDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/eventdefinition-definitions.html

14.8.6 Resource EventDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the EventDefinition resource.

EventDefinition
Element Id | EventDefinition
Definition | The EventDefinition resource provides a reusable description of when a particular event can occur.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element evd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
evd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
EventDefinition.url
Element Id | EventDefinition.url
Definition | An absolute URI that is used to identify this event definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this event definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the event definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the event definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
EventDefinition.identifier
Element Id | EventDefinition.identifier
Definition | A formal identifier that is used to identify this event definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this event definition outside of FHIR, where it is not possible to use the logical URI.
EventDefinition.version
Element Id | EventDefinition.version
Definition | The identifier that is used to identify this version of the event definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the event definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different event definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the event definition with the format [url]|[version].
EventDefinition.name
Element Id | EventDefinition.name
Definition | A natural language name identifying the event definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
EventDefinition.title
Element Id | EventDefinition.title
Definition | A short, descriptive, user-friendly title for the event definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
EventDefinition.subtitle
Element Id | EventDefinition.subtitle
Definition | An explanatory or alternate title for the event definition giving additional information about its content.
Cardinality | 0..1
Type | string
EventDefinition.status
Element Id | EventDefinition.status
Definition | The status of this event definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of event definitions that are appropriate for use versus not.
EventDefinition.experimental
Element Id | EventDefinition.experimental
Definition | A Boolean value to indicate that this event definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level event definition.
Summary | true
Comments | Allows filtering of event definitions that are appropriate for use versus not.
EventDefinition.subject[x]
Element Id | EventDefinition.subject[x]
Definition | A code or group definition that describes the intended subject of the event definition.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
EventDefinition.date
Element Id | EventDefinition.date
Definition | The date (and optionally time) when the event definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the event definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the event definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
EventDefinition.publisher
Element Id | EventDefinition.publisher
Definition | The name of the organization or individual that published the event definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the event definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the event definition is the organization or individual primarily responsible for the maintenance and upkeep of the event definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the event definition. This item SHOULD be populated unless the information is available from context.
EventDefinition.contact
Element Id | EventDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.

Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
EventDefinition.description
Element Id | EventDefinition.description
Definition | A free text natural language description of the event definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the event definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the event definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the event definition is presumed to be the predominant language in the place the event definition was created).
EventDefinition.useContext
Element Id | EventDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate event definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
EventDefinition.jurisdiction
Element Id | EventDefinition.jurisdiction
Definition | A legal or geographic region in which the event definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the event definition to be used in jurisdictions other than those for which it was originally designed or intended.
EventDefinition.purpose
Element Id | EventDefinition.purpose
Definition | Explanation of why this event definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the event definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this event definition.
EventDefinition.usage
Element Id | EventDefinition.usage
Definition | A detailed description of how the event definition is used from a clinical perspective.
Cardinality | 0..1
Type | string
To Do | Does this apply?
EventDefinition.copyright
Element Id | EventDefinition.copyright
Definition | A copyright statement relating to the event definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the event definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the event definition and/or its content.
Alternate Names | License; Restrictions
EventDefinition.approvalDate
Element Id | EventDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Summary | true
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
EventDefinition.lastReviewDate
Element Id | EventDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Summary | true
Comments | If specified, this date follows the original approval date.

EventDefinition.effectivePeriod
Element Id | EventDefinition.effectivePeriod
Definition | The period during which the event definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the event definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a event definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
EventDefinition.topic
Element Id | EventDefinition.topic
Definition | Descriptive topics related to the module. Topics provide a high-level categorization of the module that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the module so that it can be found by topical searches.
To Do | Does this apply?
EventDefinition.author
Element Id | EventDefinition.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Does this apply?
EventDefinition.editor
Element Id | EventDefinition.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Does this apply?
EventDefinition.reviewer
Element Id | EventDefinition.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Does this apply?
EventDefinition.endorser
Element Id | EventDefinition.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
To Do | Does this apply?
EventDefinition.relatedArtifact
Element Id | EventDefinition.relatedArtifact
Definition | Related resources such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Modules must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related resource is either an attachment, or a reference to another resource, but not both.
To Do | Does this apply?
EventDefinition.trigger
Element Id | EventDefinition.trigger
Definition | The trigger element defines when the event occurs. If more than one trigger condition is specified, the event fires whenever any one of the trigger conditions is met.
Cardinality | 1..*
Type | TriggerDefinition
Summary | true

14.8.7 Resource EventDefinition - Mappings

URL: https://hl7.org/fhir/R4/eventdefinition-mappings.html

14.8.7 Resource EventDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the eventdefinition resource (see Mappings to Other Standards for further information & status).

14.8.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

EventDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

14.8.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

EventDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
subject[x] | Definition.subject
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.8.7.3 RIM Mapping (http://hl7.org/v3)

EventDefinition
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.8.8 Resource EventDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/eventdefinition-profiles.html

14.8.8 Resource EventDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the EventDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.8.8.1 Profiles

No Profiles defined for this resource

14.8.8.2 Extensions

replaces | EventDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions

workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.8.8.3 Search Extensions

No Search Extensions defined for this resource

14.8.9 JSON Schema for EventDefinition

URL: https://hl7.org/fhir/R4/eventdefinition.schema.json.html

14.8.9 JSON Schema for EventDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for EventDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/EventDefinition",
  "$ref": "#/definitions/EventDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "EventDefinition": {
      "description": "The EventDefinition resource provides a reusable description of when a particular event can
    occur.",
      "properties": {
        "resourceType": {
          "description": "This is a EventDefinition resource",
          "const": "EventDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
```

```
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this event definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this event definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the event definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this event definition when it is represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the event definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the event definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the event definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the event definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the event definition giving additional information about its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this event definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this event definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "subjectCodeableConcept": {
      "description": "A code or group definition that describes the intended subject of the event definition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "A code or group definition that describes the intended subject of the event definition.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the event definition was published. The date must change
when the business version changes and it must change if the status code changes. In addition, it should change when
the substantive content of the event definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the event definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the event definition from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate event definition
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the event definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this event definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description of how the event definition is used from a clinical perspective.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the event definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the event definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
```

```
        "_approvalDate": {
          "description": "Extensions for approvalDate",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "lastReviewDate": {
          "description": "The date on which the resource content was last reviewed. Review happens periodically after
    approval but does not change the original approval date.",
          "$ref": "#/definitions/date"
        },
        "_lastReviewDate": {
          "description": "Extensions for lastReviewDate",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "effectivePeriod": {
          "description": "The period during which the event definition content was or is planned to be in active use.",
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "topic": {
          "description": "Descriptive topics related to the module. Topics provide a high-level categorization of the
    module that can be useful for filtering and searching.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "author": {
          "description": "An individiual or organization primarily involved in the creation and maintenance of the
    content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "editor": {
          "description": "An individual or organization primarily responsible for internal coherence of the content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "reviewer": {
          "description": "An individual or organization primarily responsible for review of some aspect of the
    content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "endorser": {
          "description": "An individual or organization responsible for officially endorsing the content for use in some
     setting.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "relatedArtifact": {
          "description": "Related resources such as additional documentation, justification, or bibliographic
    references.",
          "items": {
            "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
          },
          "type": "array"
        },
        "trigger": {
          "description": "The trigger element defines when the event occurs. If more than one trigger condition is
    specified, the event fires whenever any one of the trigger conditions is met.",
          "items": {
            "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "trigger",
        "resourceType"
      ]
    }
  }
}
```


14.8.10 Resource EventDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/eventdefinition-dependencies.html

14.8.10 Resource EventDefinition - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for EventDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
EventDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
trigger | 1..* | TriggerDefinition

14.9 Resource Evidence - Content

URL: https://hl7.org/fhir/R4/evidence.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The Evidence resource describes the conditional state (population and any exposures being compared within the population) and outcome (if specified) that the knowledge (evidence, assertion, recommendation) is about.

14.9.1 Scope and Usage

The scope of the Evidence resource is to describe the conditional state (population and exposures being considered or compared within a population) with or without the outcomes that knowledge (evidence, assertion, recommendation) is about. The Evidence resource is a commonly bundled series of evidence variables from the "PICO" specification of medical knowledge: Population - the population within which exposures and outcomes are being considered Intervention - the conditional state or exposure state being described for which one is expressing the risk of or effect on an outcome Comparison - the alternative conditional state or alternative exposure state when one is comparing states for the effect on an outcome Outcome - the result or effect of an intervention or exposure in a population PICO (and its variants like PECO or PICOT) is a universal acronym used in evidence-based medicine communities to clearly express research questions and evidence findings. The Evidence resource allows expression of the higher-order concept for recommendations in healthcare where multiple outcomes are considered, and the higher-order concept/combination for framing research questions.

14.9.2 Boundaries and Relationships

The Evidence resource references the EvidenceVariable resource two or more times to clearly specify the relationship among evidence variable concepts in framing a research question, evidence assertion, or recommendation.

14.9.3 Resource Content

- Structure
- UML
- XML

- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Evidence | ITU | | DomainResource | A research context or question + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the evidence
version | Σ | 0..1 | string | Business version of the evidence
name | ΣI | 0..1 | string | Name for this evidence (computer friendly)
title | Σ | 0..1 | string | Name for this evidence (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the Evidence
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the evidence
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for evidence (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | | 0..1 | date | When the evidence was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the evidence is expected to be used
topic | | 0..* | CodeableConcept | The category of the Evidence, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | Σ | 0..* | Reference(EvidenceVariable) | What exposure?
outcome | Σ | 0..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Evidence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this evidence, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the evidence --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the evidence -->
 <name value="[string]"/><!--  0..1 Name for this evidence (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this evidence (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the Evidence -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the evidence -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for evidence (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the evidence was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the evidence was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the evidence is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the Evidence, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <exposureBackground><!-- 1..1 Reference(EvidenceVariable) What population? --></exposureBackground>
 <exposureVariant><!-- 0..* Reference(EvidenceVariable) What exposure? --></exposureVariant>
 <outcome><!-- 0..* Reference(EvidenceVariable) What outcome? --></outcome>
</Evidence>
```

JSON Template

```
{
  "resourceType" : "Evidence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this evidence, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the evidence
  "version" : "<string>", // Business version of the evidence
  "name" : "<string>", // C? Name for this evidence (computer friendly)
  "title" : "<string>", // Name for this evidence (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the Evidence
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the evidence
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for evidence (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the evidence was approved by publisher
  "lastReviewDate" : "<date>", // When the evidence was last reviewed
  "effectivePeriod" : { Period }, // When the evidence is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the Evidence, such as Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "exposureBackground" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposureVariant" : [{ Reference(EvidenceVariable) }], // What exposure?
  "outcome" : [{ Reference(EvidenceVariable) }] // What outcome?
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Evidence;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Evidence.url [ uri ]; # 0..1 Canonical identifier for this evidence, represented as a URI (globally unique)
  fhir:Evidence.identifier [ Identifier ], ... ; # 0..* Additional identifier for the evidence
  fhir:Evidence.version [ string ]; # 0..1 Business version of the evidence
  fhir:Evidence.name [ string ]; # 0..1 Name for this evidence (computer friendly)
  fhir:Evidence.title [ string ]; # 0..1 Name for this evidence (human friendly)
  fhir:Evidence.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:Evidence.subtitle [ string ]; # 0..1 Subordinate title of the Evidence
  fhir:Evidence.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Evidence.date [ dateTime ]; # 0..1 Date last changed
  fhir:Evidence.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Evidence.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Evidence.description [ markdown ]; # 0..1 Natural language description of the evidence
  fhir:Evidence.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:Evidence.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Evidence.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for evidence (if applicable)
  fhir:Evidence.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Evidence.approvalDate [ date ]; # 0..1 When the evidence was approved by publisher
  fhir:Evidence.lastReviewDate [ date ]; # 0..1 When the evidence was last reviewed
  fhir:Evidence.effectivePeriod [ Period ]; # 0..1 When the evidence is expected to be used
  fhir:Evidence.topic [ CodeableConcept ], ... ; # 0..* The category of the Evidence, such as Education, Treatment,
    Assessment, etc.
  fhir:Evidence.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Evidence.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Evidence.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Evidence.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Evidence.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Evidence.exposureBackground [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:Evidence.exposureVariant [ Reference(EvidenceVariable) ], ... ; # 0..* What exposure?
  fhir:Evidence.outcome [ Reference(EvidenceVariable) ], ... ; # 0..* What outcome?
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Evidence | ITU | | DomainResource | A research context or question + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the evidence

version | Σ | 0..1 | string | Business version of the evidence

name | ΣI | 0..1 | string | Name for this evidence (computer friendly)
title | Σ | 0..1 | string | Name for this evidence (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the Evidence
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the evidence
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for evidence (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | | 0..1 | date | When the evidence was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the evidence is expected to be used
topic | | 0..* | CodeableConcept | The category of the Evidence, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | Σ | 0..* | Reference(EvidenceVariable) | What exposure?
outcome | Σ | 0..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Evidence xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this evidence, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the evidence --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the evidence -->
 <name value="[string]"/><!--  0..1 Name for this evidence (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this evidence (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the Evidence -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the evidence -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for evidence (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the evidence was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the evidence was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the evidence is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the Evidence, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <exposureBackground><!-- 1..1 Reference(EvidenceVariable) What population? --></exposureBackground>
 <exposureVariant><!-- 0..* Reference(EvidenceVariable) What exposure? --></exposureVariant>
 <outcome><!-- 0..* Reference(EvidenceVariable) What outcome? --></outcome>
</Evidence>
```

JSON Template

```
{
  "resourceType" : "Evidence",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this evidence, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the evidence
  "version" : "<string>", // Business version of the evidence
  "name" : "<string>", // C? Name for this evidence (computer friendly)
  "title" : "<string>", // Name for this evidence (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the Evidence
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the evidence
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
```

```
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for evidence (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the evidence was approved by publisher
  "lastReviewDate" : "<date>", // When the evidence was last reviewed
  "effectivePeriod" : { Period }, // When the evidence is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the Evidence, such as Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "exposureBackground" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposureVariant" : [{ Reference(EvidenceVariable) }], // What exposure?
  "outcome" : [{ Reference(EvidenceVariable) }] // What outcome?
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Evidence;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Evidence.url [ uri ]; # 0..1 Canonical identifier for this evidence, represented as a URI (globally unique)
  fhir:Evidence.identifier [ Identifier ], ... ; # 0..* Additional identifier for the evidence
  fhir:Evidence.version [ string ]; # 0..1 Business version of the evidence
  fhir:Evidence.name [ string ]; # 0..1 Name for this evidence (computer friendly)
  fhir:Evidence.title [ string ]; # 0..1 Name for this evidence (human friendly)
  fhir:Evidence.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:Evidence.subtitle [ string ]; # 0..1 Subordinate title of the Evidence
  fhir:Evidence.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Evidence.date [ dateTime ]; # 0..1 Date last changed
  fhir:Evidence.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Evidence.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Evidence.description [ markdown ]; # 0..1 Natural language description of the evidence
  fhir:Evidence.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:Evidence.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Evidence.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for evidence (if applicable)
  fhir:Evidence.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Evidence.approvalDate [ date ]; # 0..1 When the evidence was approved by publisher
  fhir:Evidence.lastReviewDate [ date ]; # 0..1 When the evidence was last reviewed
  fhir:Evidence.effectivePeriod [ Period ]; # 0..1 When the evidence is expected to be used
  fhir:Evidence.topic [ CodeableConcept ], ... ; # 0..* The category of the Evidence, such as Education, Treatment,
    Assessment, etc.
  fhir:Evidence.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Evidence.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Evidence.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Evidence.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Evidence.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Evidence.exposureBackground [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:Evidence.exposureVariant [ Reference(EvidenceVariable) ], ... ; # 0..* What exposure?
  fhir:Evidence.outcome [ Reference(EvidenceVariable) ], ... ; # 0..* What outcome?
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.9.3.1 Terminology Bindings

Path	Definition	Type	Reference
Evidence.status	The lifecycle status of an artifact.	Required	PublicationStatus
Evidence.jurisdiction	Countries and regions within which this artifact is targeted for use.	Extensible	Jurisdiction ValueSet
Evidence.topic	High-level categorization of the definition, used for searching, sorting, and filtering.	Example	DefinitionTopic

14.9.3.2 Constraints

id	Level	Location	Description	Expression
evi-0	Warning	(base)	Name should be usable as an identifier for the module by machine processing applications such as code generation	name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
composed-of	reference	What resource is being referenced	Evidence.relatedArtifact.where(type='composed-of').resource (Any)	
context	token	A use context assigned to the evidence	(Evidence.useContext.value as CodeableConcept)	

context-quantity | quantity | A quantity- or range-valued use context assigned to the evidence | (Evidence.useContext.value as Quantity) | (Evidence.useContext.value as Range)
context-type | token | A type of use context assigned to the evidence | Evidence.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the evidence | On Evidence.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the evidence | On Evidence.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The evidence publication date | Evidence.date
depends-on | reference | What resource is being referenced | Evidence.relatedArtifact.where(type='depends-on').resource (Any)
derived-from | reference | What resource is being referenced | Evidence.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the evidence | Evidence.description
effective | date | The time during which the evidence is intended to be in use | Evidence.effectivePeriod
identifier | token | External identifier for the evidence | Evidence.identifier
jurisdiction | token | Intended jurisdiction for the evidence | Evidence.jurisdiction
name | string | Computationally friendly name of the evidence | Evidence.name
predecessor | reference | What resource is being referenced | Evidence.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the evidence | Evidence.publisher
status | token | The current status of the evidence | Evidence.status
successor | reference | What resource is being referenced | Evidence.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the evidence | Evidence.title
topic | token | Topics associated with the Evidence | Evidence.topic
url | uri | The uri that identifies the evidence | Evidence.url
version | token | The business version of the evidence | Evidence.version

14.9.5 Resource Evidence - Examples

URL: https://hl7.org/fhir/R4/evidence-examples.html

14.9.5 Resource Evidence - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

TODO

Example Name | id | Format
Example of evidence | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.9.6 Resource Evidence - Detailed Descriptions

URL: https://hl7.org/fhir/R4/evidence-definitions.html

14.9.6 Resource Evidence - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Evidence resource.

Evidence
Element Id | Evidence
Definition | The Evidence resource describes the conditional state (population and any exposures being compared within the population) and outcome (if specified) that the knowledge (evidence, assertion, recommendation) is about.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element evi-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
evi-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Evidence.url
Element Id | Evidence.url
Definition | An absolute URI that is used to identify this evidence when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this evidence is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the evidence is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the evidence to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the

meta.source element to indicate where the current master source of the resource can be found.
Evidence.identifier
Element Id | Evidence.identifier
Definition | A formal identifier that is used to identify this evidence when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this evidence outside of FHIR, where it is not possible to use the logical URI.
Evidence.version
Element Id | Evidence.version
Definition | The identifier that is used to identify this version of the evidence when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the evidence author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different evidence instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the evidence with the format [url]|[version].
Evidence.name
Element Id | Evidence.name
Definition | A natural language name identifying the evidence. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
Evidence.title
Element Id | Evidence.title
Definition | A short, descriptive, user-friendly title for the evidence.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Evidence.shortTitle
Element Id | Evidence.shortTitle
Definition | The short title provides an alternate title for use in informal descriptive contexts where the full, formal title is not necessary.
Cardinality | 0..1
Type | string
Requirements | Need to be able to reference the content by a short description, but still provide a longer, more formal title for the content.
Evidence.subtitle
Element Id | Evidence.subtitle
Definition | An explanatory or alternate title for the Evidence giving additional information about its content.
Cardinality | 0..1
Type | string
Evidence.status
Element Id | Evidence.status
Definition | The status of this evidence. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of evidences that are appropriate for use versus not.

Evidence.date
Element Id | Evidence.date
Definition | The date (and optionally time) when the evidence was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the evidence changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the evidence. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Evidence.publisher
Element Id | Evidence.publisher
Definition | The name of the organization or individual that published the evidence.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the evidence. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the evidence is the organization or individual primarily responsible for the maintenance and upkeep of the evidence. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the evidence. This item SHOULD be populated unless the information is available from context.
Evidence.contact
Element Id | Evidence.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Evidence.description
Element Id | Evidence.description
Definition | A free text natural language description of the evidence from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the evidence was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the evidence as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the evidence is presumed to be the predominant language in the place the evidence was created).
Evidence.note
Element Id | Evidence.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation
Evidence.useContext
Element Id | Evidence.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate evidence instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
Evidence.jurisdiction
Element Id | Evidence.jurisdiction
Definition | A legal or geographic region in which the evidence is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the evidence to be used in jurisdictions other than those for which it was originally designed or intended.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
Evidence.copyright
Element Id | Evidence.copyright
Definition | A copyright statement relating to the evidence and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the evidence.
Cardinality | 0..1

Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the evidence and/or its content.
Alternate Names | License; Restrictions
Evidence.approvalDate
Element Id | Evidence.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
To Do | Consider how this applies to this resource.
Evidence.lastReviewDate
Element Id | Evidence.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
To Do | Consider how this applies to this resource.
Evidence.effectivePeriod
Element Id | Evidence.effectivePeriod
Definition | The period during which the evidence content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the evidence are or are expected to be used instead.
Summary | true
Comments | The effective period for a evidence determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
To Do | Consider how this applies to this resource.
Evidence.topic
Element Id | Evidence.topic
Definition | Descriptive topics related to the content of the Evidence. Topics provide a high-level categorization grouping types of Evidences that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the Evidence so that it can be found by topical searches.
To Do | Consider how this applies to this resource.
Evidence.author
Element Id | Evidence.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
Evidence.editor
Element Id | Evidence.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
Evidence.reviewer
Element Id | Evidence.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
Evidence.endorser
Element Id | Evidence.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
Evidence.relatedArtifact
Element Id | Evidence.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Evidences must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.

Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
To Do | Consider how this applies to this resource.
Evidence.exposureBackground
Element Id | Evidence.exposureBackground
Definition | A reference to a EvidenceVariable resource that defines the population for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
Evidence.exposureVariant
Element Id | Evidence.exposureVariant
Definition | A reference to a EvidenceVariable resource that defines the exposure for the research.
Cardinality | 0..*
Type | Reference(EvidenceVariable)
Summary | true
Evidence.outcome
Element Id | Evidence.outcome
Definition | A reference to a EvidenceVariable resomece that defines the outcome for the research.
Cardinality | 0..*
Type | Reference(EvidenceVariable)
Summary | true

14.9.7 Resource Evidence - Mappings

URL: https://hl7.org/fhir/R4/evidence-mappings.html

14.9.7 Resource Evidence - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the evidence resource (see Mappings to Other Standards for further information & status).

14.9.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Evidence
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
date | FiveWs.recorded
publisher | FiveWs.witness

14.9.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Evidence | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.9.7.3 RIM Mapping (http://hl7.org/v3)

Evidence
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.9.8 Resource Evidence - Extensions & Profiles

URL: https://hl7.org/fhir/R4/evidence-profiles.html

14.9.8 Resource Evidence - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Evidence resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.9.8.1 Profiles

Profile | Description | Context
Synthesis | Explanation of what this profile contains/is for. | Evidence Synthesis Profile

14.9.8.2 Extensions

replaces | Evidence | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.9.8.3 Search Extensions

No Search Extensions defined for this resource

14.9.9 Evidence Synthesis Profile

URL: https://hl7.org/fhir/R4/evidence-synthesis.html

14.9.9 Evidence Synthesis Profile

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.9.9.1 Content

Profiles:
Synthesis | Explanation of what this profile contains/is for.

14.9.9.2 StructureDefinition: Synthesis

URL: https://hl7.org/fhir/R4/synthesis.html

14.9.9.2 StructureDefinition: Synthesis

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/synthesis`

Explanation of what this profile contains/is for.

This profile was published on Wed, Oct 31, 2018 00:00+1100 as a draft by Health Level Seven, Inc. - Clinical Decision Support Workgroup.

14.9.9.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Evidence.

Summary

Mandatory: 4 elements

This structure is derived from Evidence.

Name | Flags | Card. | Type | Description & Constraints
Evidence | | 0..*
url | Σ | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the evidence
version | Σ | 0..1 | string | Business version of the evidence
title | Σ | 0..1 | string | Name for this evidence (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the Evidence
status | ?!Σ | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the evidence
note | | 0..* | Annotation | Used for footnotes or explanatory notes
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | | 0..1 | date | When the evidence was last reviewed
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | | 1..2 | Reference(EvidenceVariable) | What exposure?
outcome | | 1..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Evidence | I | 0..* | | A research context or question
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the evidence
version | ΣI | 0..1 | string | Business version of the evidence
name | ΣI | 0..1 | string | Name for this evidence (computer friendly)
title | ΣI | 0..1 | string | Name for this evidence (human friendly)
shortTitle | I | 0..1 | string | Title for use in informal contexts
subtitle | I | 0..1 | string | Subordinate title of the Evidence
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the evidence
note | I | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for evidence (if applicable) Binding: Jurisdiction (extensible)
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | I | 0..1 | date | When the evidence was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the evidence is expected to be used
topic | I | 0..* | CodeableConcept | The category of the Evidence, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | ΣI | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | ΣI | 1..2 | Reference(EvidenceVariable) | What exposure?
outcome | ΣI | 1..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

```
<!-- Evidence Synthesis Profile -->

<Evidence xmlns="http://hl7.org/fhir"
```

```
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1 Canonical identifier for this evidence, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the evidence --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the evidence -->
 <name value="[string]"/><!--  0..1 Name for this evidence (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this evidence (human friendly) -->
 <shortTitle value="[string]"/><!--  0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the Evidence -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the evidence -->
 <note><!--  0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for evidence (if applicable)   --></jurisdiction>
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the evidence was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the evidence was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the evidence is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the Evidence, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <exposureBackground><!--  1..1 Reference(EvidenceVariable) What population? --></exposureBackground>
 <exposureVariant><!--  1..2 Reference(EvidenceVariable) What exposure? --></exposureVariant>
 <outcome><!--  1..* Reference(EvidenceVariable) What outcome? --></outcome>
</Evidence>
```

```
{ // Evidence Synthesis Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this evidence, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the evidence
    "version" : "<string>", // C?Business version of the evidence
    "name" : "<string>", // C?Name for this evidence (computer friendly)
    "title" : "<string>", // C?Name for this evidence (human friendly)
    "shortTitle" : "<string>", // C?Title for use in informal contexts
    "subtitle" : "<string>", // C?Subordinate title of the Evidence
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the evidence
    "note" : [{ Annotation }], // C?Used for footnotes or explanatory notes
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for evidence (if applicable)
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the evidence was approved by publisher
    "lastReviewDate" : "<date>", // C?When the evidence was last reviewed
    "effectivePeriod" : { Period }, // C?When the evidence is expected to be used
    "topic" : [{ CodeableConcept }], // C?The category of the Evidence, such as Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "exposureBackground" : { Reference(EvidenceVariable) }, // C? R! What population?
    "exposureVariant" : { Reference(EvidenceVariable) }, // C? R! What exposure?
    "outcome" : [{ Reference(EvidenceVariable) }] // C? R! What outcome?
  }
```

This structure is derived from Evidence.

Summary

Mandatory: 4 elements

Differential View

This structure is derived from Evidence.

Name | Flags | Card. | Type | Description & Constraints
Evidence | | 0..*
url | Σ | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the evidence
version | Σ | 0..1 | string | Business version of the evidence
title | Σ | 0..1 | string | Name for this evidence (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the Evidence
status | ?!Σ | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the evidence
note | | 0..* | Annotation | Used for footnotes or explanatory notes
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | | 0..1 | date | When the evidence was last reviewed
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | | 1..2 | Reference(EvidenceVariable) | What exposure?
outcome | | 1..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Evidence | I | 0..* | | A research context or question
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this evidence, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the evidence
version | ΣI | 0..1 | string | Business version of the evidence
name | ΣI | 0..1 | string | Name for this evidence (computer friendly)
title | ΣI | 0..1 | string | Name for this evidence (human friendly)
shortTitle | I | 0..1 | string | Title for use in informal contexts
subtitle | I | 0..1 | string | Subordinate title of the Evidence
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the evidence
note | I | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for evidence (if applicable) Binding: Jurisdiction (extensible)
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the evidence was approved by publisher
lastReviewDate | I | 0..1 | date | When the evidence was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the evidence is expected to be used
topic | I | 0..* | CodeableConcept | The category of the Evidence, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
exposureBackground | ΣI | 1..1 | Reference(EvidenceVariable) | What population?
exposureVariant | ΣI | 1..2 | Reference(EvidenceVariable) | What exposure?
outcome | ΣI | 1..* | Reference(EvidenceVariable) | What outcome?
Documentation for this format

XML Template

```
<!-- Evidence Synthesis Profile -->

<Evidence xmlns="http://hl7.org/fhir"
```

```
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1 Canonical identifier for this evidence, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the evidence --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the evidence -->
 <name value="[string]"/><!--  0..1 Name for this evidence (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this evidence (human friendly) -->
 <shortTitle value="[string]"/><!--  0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the Evidence -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the evidence -->
 <note><!--  0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for evidence (if applicable)   --></jurisdiction>
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the evidence was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the evidence was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the evidence is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the Evidence, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <exposureBackground><!--  1..1 Reference(EvidenceVariable) What population? --></exposureBackground>
 <exposureVariant><!--  1..2 Reference(EvidenceVariable) What exposure? --></exposureVariant>
 <outcome><!--  1..* Reference(EvidenceVariable) What outcome? --></outcome>
</Evidence>
```

JSON Template

```
{ // Evidence Synthesis Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this evidence, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the evidence
    "version" : "<string>", // C?Business version of the evidence
    "name" : "<string>", // C?Name for this evidence (computer friendly)
    "title" : "<string>", // C?Name for this evidence (human friendly)
    "shortTitle" : "<string>", // C?Title for use in informal contexts
    "subtitle" : "<string>", // C?Subordinate title of the Evidence
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the evidence
    "note" : [{ Annotation }], // C?Used for footnotes or explanatory notes
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for evidence (if applicable)
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the evidence was approved by publisher
    "lastReviewDate" : "<date>", // C?When the evidence was last reviewed
    "effectivePeriod" : { Period }, // C?When the evidence is expected to be used
    "topic" : [{ CodeableConcept }], // C?The category of the Evidence, such as Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "exposureBackground" : { Reference(EvidenceVariable) }, // C? R! What population?
    "exposureVariant" : { Reference(EvidenceVariable) }, // C? R! What exposure?
    "outcome" : [{ Reference(EvidenceVariable) }] // C? R! What outcome?
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.9.9.2.2 Terminology Bindings

Path | Name | Conformance | ValueSet

Evidence.language | Common Languages | preferred | Common Languages
Evidence.status | PublicationStatus | required | PublicationStatus
Evidence.jurisdiction | Jurisdiction | extensible | Jurisdiction
Evidence.topic | DefinitionTopic | example | DefinitionTopic

14.9.10 JSON Schema for Evidence

URL: https://hl7.org/fhir/R4/evidence.schema.json.html

14.9.10 JSON Schema for Evidence

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Evidence, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Evidence",
  "$ref": "#/definitions/Evidence",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Evidence": {
      "description": "The Evidence resource describes the conditional state (population and any exposures being compared
     within the population) and outcome (if specified) that the knowledge (evidence, assertion, recommendation) is
    about.",
      "properties": {
        "resourceType": {
          "description": "This is a Evidence resource",
          "const": "Evidence"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
```

containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this evidence when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this evidence is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the evidence is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this evidence when it is represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the evidence when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the evidence author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the evidence. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the evidence.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "shortTitle": {
      "description": "The short title provides an alternate title for use in informal descriptive contexts where the full, formal title is not necessary.",
      "$ref": "#/definitions/string"
    },
    "_shortTitle": {
      "description": "Extensions for shortTitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the Evidence giving additional information about its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this evidence. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },

```
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the evidence was published. The date must change when the
 business version changes and it must change if the status code changes. In addition, it should change when the
substantive content of the evidence changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the evidence.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the evidence from a consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate evidence instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the evidence is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "copyright": {
      "description": "A copyright statement relating to the evidence and/or its contents. Copyright statements are
generally legal restrictions on the use and publishing of the evidence.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the evidence content was or is planned to be in active use.",
```

```
          "$ref": "Period.schema.json#/definitions/Period"
        },
        "topic": {
          "description": "Descriptive topics related to the content of the Evidence. Topics provide a high-level
    categorization grouping types of Evidences that can be useful for filtering and searching.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "author": {
          "description": "An individiual or organization primarily involved in the creation and maintenance of the
    content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "editor": {
          "description": "An individual or organization primarily responsible for internal coherence of the content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "reviewer": {
          "description": "An individual or organization primarily responsible for review of some aspect of the
    content.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "endorser": {
          "description": "An individual or organization responsible for officially endorsing the content for use in some
     setting.",
          "items": {
            "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
          },
          "type": "array"
        },
        "relatedArtifact": {
          "description": "Related artifacts such as additional documentation, justification, or bibliographic
    references.",
          "items": {
            "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
          },
          "type": "array"
        },
        "exposureBackground": {
          "description": "A reference to a EvidenceVariable resource that defines the population for the research.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "exposureVariant": {
          "description": "A reference to a EvidenceVariable resource that defines the exposure for the research.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        },
        "outcome": {
          "description": "A reference to a EvidenceVariable resomece that defines the outcome for the research.",
          "items": {
            "$ref": "Reference.schema.json#/definitions/Reference"
          },
          "type": "array"
        }
      },
      "additionalProperties": false,
      "required": [
        "exposureBackground",
        "resourceType"
      ]
    }
  }
}
```


14.9.11 Resource Evidence - Maturity Dependency View

URL: https://hl7.org/fhir/R4/evidence-dependencies.html

14.9.11 Resource Evidence - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Evidence FMM level 0

Name | Card. | Type | Dependency Analysis

Evidence | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
shortTitle | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
note | 0..* | Annotation
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
exposureBackground | 1..1 | Reference(EvidenceVariable)
exposureVariant | 0..* | Reference(EvidenceVariable)
outcome | 0..* | Reference(EvidenceVariable)

14.10 Resource EvidenceVariable - Content

URL: https://hl7.org/fhir/R4/evidencevariable.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The EvidenceVariable resource describes a "PICO" element that knowledge (evidence, assertion, recommendation) is about.

14.10.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

The scope of the EvidenceVariable resource is to describe a "PICO" element that knowledge (evidence, assertion, recommendation) is about. PICO stands for: Population - the population within which exposures and outcomes are being considered Intervention - the conditional state or exposure state being described for which one is expressing the risk of or effect on an outcome Comparison - the alternative conditional state or alternative exposure state when one is comparing states for the effect on an outcome Outcome - the result or effect of an intervention or exposure in a population PICO (and its variants like PECO or PICOT) is a universal acronym used in evidence-based medicine communities to clearly express research questions and evidence findings. The EvidenceVariable resource allows expression of the components of a PICO question in codeable and reusable formats.

14.10.2 Boundaries and Relationships

Multiple resources describing evidence findings (Evidence, EffectEvidenceSynthesis, EffectEvidenceBit, RiskEvidenceSynthesis, RiskEvidenceBit) will reference EvidenceVariable to describe what the evidence findings are about.

This resource is referenced by EffectEvidenceSynthesis, Evidence and RiskEvidenceSynthesis

14.10.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
EvidenceVariable | ITU | | DomainResource | A population, intervention, or exposure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this evidence variable, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the evidence variable
version | Σ | 0..1 | string | Business version of the evidence variable
name | ΣI | 0..1 | string | Name for this evidence variable (computer friendly)
title | Σ | 0..1 | string | Name for this evidence variable (human friendly)
shortTitle | Σ | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the EvidenceVariable
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the evidence variable
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for evidence variable (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the evidence variable was approved by publisher
lastReviewDate | | 0..1 | date | When the evidence variable was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the evidence variable is expected to be used
topic | | 0..* | CodeableConcept | The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
type | Σ | 0..1 | code | dichotomous | continuous | descriptive EvidenceVariableType (Required)
characteristic | Σ | 1..* | BackboneElement | What defines the members of the evidence element
description | | 0..1 | string | Natural language description of the characteristic
definition[x] | Σ | 1..1 | | What code or expression defines members?
definitionReference | | | Reference(Group)
definitionCanonical | | | canonical(ActivityDefinition)
definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | | 0..* | UsageContext | What code/value pairs define members?
exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | | 0..1 | Duration | Observation time from study start
groupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-median GroupMeasure (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EvidenceVariable xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this evidence variable, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the evidence variable --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the evidence variable -->
 <name value="[string]"/><!--  0..1 Name for this evidence variable (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this evidence variable (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the EvidenceVariable -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the evidence variable -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for evidence variable (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the evidence variable was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the evidence variable was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the evidence variable is expected to be used --></effectivePeriod>
```

```
 <topic><!-- 0..* CodeableConcept The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <type value="[code]"/><!-- 0..1 dichotomous | continuous | descriptive -->
 <characteristic>  <!-- 1..* What defines the members of the evidence element -->
  <description value="[string]"/><!-- 0..1 Natural language description of the characteristic -->
  <definition[x]><!-- 1..1 Reference(Group)|canonical(ActivityDefinition)|
    CodeableConcept|Expression|DataRequirement|TriggerDefinition What code or expression defines members?
    --></definition[x]>
  <usageContext><!-- 0..* UsageContext What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!-- 0..1 Whether the characteristic includes or excludes members -->
  <participantEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period do participants cover
    --></participantEffective[x]>
  <timeFromStart><!-- 0..1 Duration Observation time from study start --></timeFromStart>
  <groupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-
    median -->
 </characteristic>
</EvidenceVariable>
```

JSON Template

```
{
  "resourceType" : "EvidenceVariable",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this evidence variable, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the evidence variable
  "version" : "<string>", // Business version of the evidence variable
  "name" : "<string>", // C? Name for this evidence variable (computer friendly)
  "title" : "<string>", // Name for this evidence variable (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the EvidenceVariable
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the evidence variable
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for evidence variable (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the evidence variable was approved by publisher
  "lastReviewDate" : "<date>", // When the evidence variable was last reviewed
  "effectivePeriod" : { Period }, // When the evidence variable is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EvidenceVariable, such as Education, Treatment, Assessment,
    etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "type" : "<code>", // dichotomous | continuous | descriptive
  "characteristic" : [{ // R!  What defines the members of the evidence element
    "description" : "<string>", // Natural language description of the characteristic
    // definition[x]: What code or expression defines members?. One of these 6:
    "definitionReference" : { Reference(Group) },
    "definitionCanonical" : { canonical(ActivityDefinition) },
    "definitionCodeableConcept" : { CodeableConcept },
    "definitionExpression" : { Expression },
    "definitionDataRequirement" : { DataRequirement },
    "definitionTriggerDefinition" : { TriggerDefinition },
    "usageContext" : [{ UsageContext }], // What code/value pairs define members?
    "exclude" : <boolean>, // Whether the characteristic includes or excludes members
    // participantEffective[x]: What time period do participants cover. One of these 4:
    "participantEffectiveDateTime" : "<dateTime>",
    "participantEffectivePeriod" : { Period },
    "participantEffectiveDuration" : { Duration },
    "participantEffectiveTiming" : { Timing },
    "timeFromStart" : { Duration }, // Observation time from study start
    "groupMeasure" : "<code>" // mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EvidenceVariable;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EvidenceVariable.url [ uri ]; # 0..1 Canonical identifier for this evidence variable, represented as a URI
    (globally unique)
  fhir:EvidenceVariable.identifier [ Identifier ], ... ; # 0..* Additional identifier for the evidence variable
  fhir:EvidenceVariable.version [ string ]; # 0..1 Business version of the evidence variable
```

```
  fhir:EvidenceVariable.name [ string ]; # 0..1 Name for this evidence variable (computer friendly)
  fhir:EvidenceVariable.title [ string ]; # 0..1 Name for this evidence variable (human friendly)
  fhir:EvidenceVariable.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:EvidenceVariable.subtitle [ string ]; # 0..1 Subordinate title of the EvidenceVariable
  fhir:EvidenceVariable.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EvidenceVariable.date [ dateTime ]; # 0..1 Date last changed
  fhir:EvidenceVariable.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EvidenceVariable.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EvidenceVariable.description [ markdown ]; # 0..1 Natural language description of the evidence variable
  fhir:EvidenceVariable.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:EvidenceVariable.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:EvidenceVariable.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for evidence variable (if
    applicable)
  fhir:EvidenceVariable.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EvidenceVariable.approvalDate [ date ]; # 0..1 When the evidence variable was approved by publisher
  fhir:EvidenceVariable.lastReviewDate [ date ]; # 0..1 When the evidence variable was last reviewed
  fhir:EvidenceVariable.effectivePeriod [ Period ]; # 0..1 When the evidence variable is expected to be used
  fhir:EvidenceVariable.topic [ CodeableConcept ], ... ; # 0..* The category of the EvidenceVariable, such as Education,
     Treatment, Assessment, etc.
  fhir:EvidenceVariable.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EvidenceVariable.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EvidenceVariable.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EvidenceVariable.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EvidenceVariable.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:EvidenceVariable.type [ code ]; # 0..1 dichotomous | continuous | descriptive
  fhir:EvidenceVariable.characteristic [ # 1..* What defines the members of the evidence element
    fhir:EvidenceVariable.characteristic.description [ string ]; # 0..1 Natural language description of the
    characteristic
    # EvidenceVariable.characteristic.definition[x] : 1..1 What code or expression defines members?. One of these 6
      fhir:EvidenceVariable.characteristic.definitionReference [ Reference(Group) ]
      fhir:EvidenceVariable.characteristic.definitionCanonical [ canonical(ActivityDefinition) ]
      fhir:EvidenceVariable.characteristic.definitionCodeableConcept [ CodeableConcept ]
      fhir:EvidenceVariable.characteristic.definitionExpression [ Expression ]
      fhir:EvidenceVariable.characteristic.definitionDataRequirement [ DataRequirement ]
      fhir:EvidenceVariable.characteristic.definitionTriggerDefinition [ TriggerDefinition ]
    fhir:EvidenceVariable.characteristic.usageContext [ UsageContext ], ... ; # 0..* What code/value pairs define
    members?
    fhir:EvidenceVariable.characteristic.exclude [ boolean ]; # 0..1 Whether the characteristic includes or excludes
    members
    # EvidenceVariable.characteristic.participantEffective[x] : 0..1 What time period do participants cover. One of
    these 4
      fhir:EvidenceVariable.characteristic.participantEffectiveDateTime [ dateTime ]
      fhir:EvidenceVariable.characteristic.participantEffectivePeriod [ Period ]
      fhir:EvidenceVariable.characteristic.participantEffectiveDuration [ Duration ]
      fhir:EvidenceVariable.characteristic.participantEffectiveTiming [ Timing ]
    fhir:EvidenceVariable.characteristic.timeFromStart [ Duration ]; # 0..1 Observation time from study start
    fhir:EvidenceVariable.characteristic.groupMeasure [ code ]; # 0..1 mean | median | mean-of-mean | mean-of-median |
    median-of-mean | median-of-median
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

EvidenceVariable | ITU | | DomainResource | A population, intervention, or exposure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this evidence variable, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the evidence variable

version | Σ | 0..1 | string | Business version of the evidence variable

name | ΣI | 0..1 | string | Name for this evidence variable (computer friendly)

title | Σ | 0..1 | string | Name for this evidence variable (human friendly)

shortTitle | Σ | 0..1 | string | Title for use in informal contexts

subtitle | | 0..1 | string | Subordinate title of the EvidenceVariable

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | Σ | 0..1 | markdown | Natural language description of the evidence variable

note | | 0..* | Annotation | Used for footnotes or explanatory notes

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for evidence variable (if applicable) Jurisdiction (Extensible)

copyright | | 0..1 | markdown | Use and/or publishing restrictions

approvalDate | | 0..1 | date | When the evidence variable was approved by publisher

lastReviewDate | | 0..1 | date | When the evidence variable was last reviewed

effectivePeriod | Σ | 0..1 | Period | When the evidence variable is expected to be used

topic | | 0..* | CodeableConcept | The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)

author | | 0..* | ContactDetail | Who authored the content

editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
type | Σ | 0..1 | code | dichotomous | continuous | descriptive EvidenceVariableType (Required)
characteristic | Σ | 1..* | BackboneElement | What defines the members of the evidence element
description | | 0..1 | string | Natural language description of the characteristic
definition[x] | Σ | 1..1 | | What code or expression defines members?
definitionReference | | | Reference(Group)
definitionCanonical | | | canonical(ActivityDefinition)
definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | | 0..* | UsageContext | What code/value pairs define members?
exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | | 0..1 | Duration | Observation time from study start
groupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median
GroupMeasure (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EvidenceVariable xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this evidence variable, represented as a URI (globally unique)
    -->
 <identifier><!-- 0..* Identifier Additional identifier for the evidence variable --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the evidence variable -->
 <name value="[string]"/><!--  0..1 Name for this evidence variable (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this evidence variable (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the EvidenceVariable -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the evidence variable -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for evidence variable (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the evidence variable was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the evidence variable was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the evidence variable is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <type value="[code]"/><!-- 0..1 dichotomous | continuous | descriptive -->
 <characteristic>  <!-- 1..* What defines the members of the evidence element -->
  <description value="[string]"/><!-- 0..1 Natural language description of the characteristic -->
  <definition[x]><!-- 1..1 Reference(Group)|canonical(ActivityDefinition)|
    CodeableConcept|Expression|DataRequirement|TriggerDefinition What code or expression defines members?
    --></definition[x]>
  <usageContext><!-- 0..* UsageContext What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!-- 0..1 Whether the characteristic includes or excludes members -->
  <participantEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period do participants cover
    --></participantEffective[x]>
  <timeFromStart><!-- 0..1 Duration Observation time from study start --></timeFromStart>
  <groupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-
    median -->
 </characteristic>
</EvidenceVariable>
```

JSON Template

```
{
  "resourceType" : "EvidenceVariable",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this evidence variable, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the evidence variable
  "version" : "<string>", // Business version of the evidence variable
```

```
  "name" : "<string>", // C? Name for this evidence variable (computer friendly)
  "title" : "<string>", // Name for this evidence variable (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the EvidenceVariable
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the evidence variable
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for evidence variable (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the evidence variable was approved by publisher
  "lastReviewDate" : "<date>", // When the evidence variable was last reviewed
  "effectivePeriod" : { Period }, // When the evidence variable is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EvidenceVariable, such as Education, Treatment, Assessment,
    etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "type" : "<code>", // dichotomous | continuous | descriptive
  "characteristic" : [{ // R!  What defines the members of the evidence element
    "description" : "<string>", // Natural language description of the characteristic
    // definition[x]: What code or expression defines members?. One of these 6:
    "definitionReference" : { Reference(Group) },
    "definitionCanonical" : { canonical(ActivityDefinition) },
    "definitionCodeableConcept" : { CodeableConcept },
    "definitionExpression" : { Expression },
    "definitionDataRequirement" : { DataRequirement },
    "definitionTriggerDefinition" : { TriggerDefinition },
    "usageContext" : [{ UsageContext }], // What code/value pairs define members?
    "exclude" : <boolean>, // Whether the characteristic includes or excludes members
    // participantEffective[x]: What time period do participants cover. One of these 4:
    "participantEffectiveDateTime" : "<dateTime>",
    "participantEffectivePeriod" : { Period },
    "participantEffectiveDuration" : { Duration },
    "participantEffectiveTiming" : { Timing },
    "timeFromStart" : { Duration }, // Observation time from study start
    "groupMeasure" : "<code>" // mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EvidenceVariable;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EvidenceVariable.url [ uri ]; # 0..1 Canonical identifier for this evidence variable, represented as a URI
    (globally unique)
  fhir:EvidenceVariable.identifier [ Identifier ], ... ; # 0..* Additional identifier for the evidence variable
  fhir:EvidenceVariable.version [ string ]; # 0..1 Business version of the evidence variable
  fhir:EvidenceVariable.name [ string ]; # 0..1 Name for this evidence variable (computer friendly)
  fhir:EvidenceVariable.title [ string ]; # 0..1 Name for this evidence variable (human friendly)
  fhir:EvidenceVariable.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:EvidenceVariable.subtitle [ string ]; # 0..1 Subordinate title of the EvidenceVariable
  fhir:EvidenceVariable.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:EvidenceVariable.date [ dateTime ]; # 0..1 Date last changed
  fhir:EvidenceVariable.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:EvidenceVariable.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:EvidenceVariable.description [ markdown ]; # 0..1 Natural language description of the evidence variable
  fhir:EvidenceVariable.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:EvidenceVariable.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:EvidenceVariable.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for evidence variable (if
    applicable)
  fhir:EvidenceVariable.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:EvidenceVariable.approvalDate [ date ]; # 0..1 When the evidence variable was approved by publisher
  fhir:EvidenceVariable.lastReviewDate [ date ]; # 0..1 When the evidence variable was last reviewed
  fhir:EvidenceVariable.effectivePeriod [ Period ]; # 0..1 When the evidence variable is expected to be used
  fhir:EvidenceVariable.topic [ CodeableConcept ], ... ; # 0..* The category of the EvidenceVariable, such as Education,
     Treatment, Assessment, etc.
  fhir:EvidenceVariable.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:EvidenceVariable.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:EvidenceVariable.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:EvidenceVariable.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:EvidenceVariable.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:EvidenceVariable.type [ code ]; # 0..1 dichotomous | continuous | descriptive
  fhir:EvidenceVariable.characteristic [ # 1..* What defines the members of the evidence element
    fhir:EvidenceVariable.characteristic.description [ string ]; # 0..1 Natural language description of the
    characteristic
    # EvidenceVariable.characteristic.definition[x] : 1..1 What code or expression defines members?. One of these 6
      fhir:EvidenceVariable.characteristic.definitionReference [ Reference(Group) ]
      fhir:EvidenceVariable.characteristic.definitionCanonical [ canonical(ActivityDefinition) ]
```

```
      fhir:EvidenceVariable.characteristic.definitionCodeableConcept [ CodeableConcept ]
      fhir:EvidenceVariable.characteristic.definitionExpression [ Expression ]
      fhir:EvidenceVariable.characteristic.definitionDataRequirement [ DataRequirement ]
      fhir:EvidenceVariable.characteristic.definitionTriggerDefinition [ TriggerDefinition ]
    fhir:EvidenceVariable.characteristic.usageContext [ UsageContext ], ... ; # 0..* What code/value pairs define
    members?
    fhir:EvidenceVariable.characteristic.exclude [ boolean ]; # 0..1 Whether the characteristic includes or excludes
    members
    # EvidenceVariable.characteristic.participantEffective[x] : 0..1 What time period do participants cover. One of
    these 4
      fhir:EvidenceVariable.characteristic.participantEffectiveDateTime [ dateTime ]
      fhir:EvidenceVariable.characteristic.participantEffectivePeriod [ Period ]
      fhir:EvidenceVariable.characteristic.participantEffectiveDuration [ Duration ]
      fhir:EvidenceVariable.characteristic.participantEffectiveTiming [ Timing ]
    fhir:EvidenceVariable.characteristic.timeFromStart [ Duration ]; # 0..1 Observation time from study start
    fhir:EvidenceVariable.characteristic.groupMeasure [ code ]; # 0..1 mean | median | mean-of-mean | mean-of-median |
    median-of-mean | median-of-median
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.10.3.1 Terminology Bindings

Path | Definition | Type | Reference
EvidenceVariable.status | The lifecycle status of an artifact. | Required | PublicationStatus
EvidenceVariable.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
EvidenceVariable.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic
EvidenceVariable.type | The possible types of variables for exposures or outcomes (E.g. Dichotomous, Continuous, Descriptive). | Required | EvidenceVariableType
EvidenceVariable.characteristic.groupMeasure | Possible group measure aggregates (E.g. Mean, Median). | Required | GroupMeasure

14.10.3.2 Constraints

id | Level | Location | Description | Expression
evv-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.10.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the evidence variable | (EvidenceVariable.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the evidence variable | (EvidenceVariable.useContext.value as Quantity) | (EvidenceVariable.useContext.value as Range)
context-type | token | A type of use context assigned to the evidence variable | EvidenceVariable.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the evidence variable | On EvidenceVariable.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the evidence variable | On EvidenceVariable.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The evidence variable publication date | EvidenceVariable.date
depends-on | reference | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='depends-on').resource (Any)
derived-from | reference | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the evidence variable | EvidenceVariable.description
effective | date | The time during which the evidence variable is intended to be in use | EvidenceVariable.effectivePeriod
identifier | token | External identifier for the evidence variable | EvidenceVariable.identifier
jurisdiction | token | Intended jurisdiction for the evidence variable | EvidenceVariable.jurisdiction
name | string | Computationally friendly name of the evidence variable | EvidenceVariable.name
predecessor | reference | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the evidence variable | EvidenceVariable.publisher
status | token | The current status of the evidence variable | EvidenceVariable.status
successor | reference | What resource is being referenced | EvidenceVariable.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the evidence variable | EvidenceVariable.title
topic | token | Topics associated with the EvidenceVariable | EvidenceVariable.topic

url | uri | The uri that identifies the evidence variable | EvidenceVariable.url
version | token | The business version of the evidence variable | EvidenceVariable.version

14.10.5 Resource EvidenceVariable - Examples

URL: https://hl7.org/fhir/R4/evidencevariable-examples.html

14.10.5 Resource EvidenceVariable - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

TODO

Example Name | id | Format
Example of evidencevariable | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.10.6 Resource EvidenceVariable - Detailed Descriptions

URL: https://hl7.org/fhir/R4/evidencevariable-definitions.html

14.10.6 Resource EvidenceVariable - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the EvidenceVariable resource.

EvidenceVariable
Element Id | EvidenceVariable
Definition | The EvidenceVariable resource describes a "PICO" element that knowledge (evidence, assertion, recommendation) is about.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to be able to define and reuse the definition of individual elements of a research question.
Comments | PICO stands for Population (the population within which exposures are being compared), Intervention (the conditional state or exposure state being described for its effect on outcomes), Comparison (the alternative conditional state or alternative exposure state being compared against), and Outcome (the result or effect of the intervention in the population).
Invariants | Defined on this element evv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
evv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
EvidenceVariable.url
Element Id | EvidenceVariable.url
Definition | An absolute URI that is used to identify this evidence variable when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this evidence variable is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the evidence variable is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the evidence variable to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
EvidenceVariable.identifier
Element Id | EvidenceVariable.identifier
Definition | A formal identifier that is used to identify this evidence variable when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this evidence variable outside of FHIR, where it is not possible to use the logical URI.
EvidenceVariable.version
Element Id | EvidenceVariable.version
Definition | The identifier that is used to identify this version of the evidence variable when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the evidence variable author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service

specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different evidence variable instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the evidence variable with the format [url]|[version].
EvidenceVariable.name
Element Id | EvidenceVariable.name
Definition | A natural language name identifying the evidence variable. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EvidenceVariable.title
Element Id | EvidenceVariable.title
Definition | A short, descriptive, user-friendly title for the evidence variable.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
EvidenceVariable.shortTitle
Element Id | EvidenceVariable.shortTitle
Definition | The short title provides an alternate title for use in informal descriptive contexts where the full, formal title is not necessary.
Cardinality | 0..1
Type | string
Requirements | Need to be able to reference the content by a short description, but still provide a longer, more formal title for the content.
Summary | true
EvidenceVariable.subtitle
Element Id | EvidenceVariable.subtitle
Definition | An explanatory or alternate title for the EvidenceVariable giving additional information about its content.
Cardinality | 0..1
Type | string
EvidenceVariable.status
Element Id | EvidenceVariable.status
Definition | The status of this evidence variable. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of evidence variables that are appropriate for use versus not.
EvidenceVariable.date
Element Id | EvidenceVariable.date
Definition | The date (and optionally time) when the evidence variable was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the evidence variable changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the evidence variable. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
EvidenceVariable.publisher
Element Id | EvidenceVariable.publisher
Definition | The name of the organization or individual that published the evidence variable.
Cardinality | 0..1
Type | string

Requirements | Helps establish the "authority/credibility" of the evidence variable. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the evidence variable is the organization or individual primarily responsible for the maintenance and upkeep of the evidence variable. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the evidence variable. This item SHOULD be populated unless the information is available from context.
EvidenceVariable.contact
Element Id | EvidenceVariable.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
EvidenceVariable.description
Element Id | EvidenceVariable.description
Definition | A free text natural language description of the evidence variable from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the evidence variable was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the evidence variable as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the evidence variable is presumed to be the predominant language in the place the evidence variable was created).
EvidenceVariable.note
Element Id | EvidenceVariable.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation
EvidenceVariable.useContext
Element Id | EvidenceVariable.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate evidence variable instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EvidenceVariable.jurisdiction
Element Id | EvidenceVariable.jurisdiction
Definition | A legal or geographic region in which the evidence variable is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the evidence variable to be used in jurisdictions other than those for which it was originally designed or intended.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
EvidenceVariable.copyright
Element Id | EvidenceVariable.copyright
Definition | A copyright statement relating to the evidence variable and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the evidence variable.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the evidence variable and/or its content.
Alternate Names | License; Restrictions
EvidenceVariable.approvalDate
Element Id | EvidenceVariable.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
To Do | Consider how this applies to this resource.
EvidenceVariable.lastReviewDate
Element Id | EvidenceVariable.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1

Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
To Do | Consider how this applies to this resource.
EvidenceVariable.effectivePeriod
Element Id | EvidenceVariable.effectivePeriod
Definition | The period during which the evidence variable content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the evidence variable are or are expected to be used instead.
Summary | true
Comments | The effective period for a evidence variable determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
To Do | Consider how this applies to this resource.
EvidenceVariable.topic
Element Id | EvidenceVariable.topic
Definition | Descriptive topics related to the content of the EvidenceVariable. Topics provide a high-level categorization grouping types of EvidenceVariables that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the EvidenceVariable so that it can be found by topical searches.
To Do | Consider how this applies to this resource.
EvidenceVariable.author
Element Id | EvidenceVariable.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
EvidenceVariable.editor
Element Id | EvidenceVariable.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
EvidenceVariable.reviewer
Element Id | EvidenceVariable.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
EvidenceVariable.endorser
Element Id | EvidenceVariable.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
EvidenceVariable.relatedArtifact
Element Id | EvidenceVariable.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | EvidenceVariables must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
To Do | Consider how this applies to this resource.
EvidenceVariable.type
Element Id | EvidenceVariable.type
Definition | The type of evidence element, a population, an exposure, or an outcome.
Cardinality | 0..1
Terminology Binding | EvidenceVariableType (Required)
Type | code
Summary | true
EvidenceVariable.characteristic
Element Id | EvidenceVariable.characteristic
Definition | A characteristic that defines the members of the evidence element. Multiple characteristics are applied with "and" semantics.
Cardinality | 1..*
Summary | true
Comments | Characteristics can be defined flexibly to accommodate different use cases for membership criteria, ranging from simple codes, all the way to using an expression language to express the criteria.
EvidenceVariable.characteristic.description
Element Id | EvidenceVariable.characteristic.description

Definition | A short, natural language description of the characteristic that could be used to communicate the criteria to an end-user.
Cardinality | 0..1
Type | string
Requirements | Need to be able to describe characteristics in natural language so that end users can understand the criteria.
EvidenceVariable.characteristic.definition[x]
Element Id | EvidenceVariable.characteristic.definition[x]
Definition | Define members of the evidence element using Codes (such as condition, medication, or observation), Expressions (using an expression language such as FHIRPath or CQL) or DataRequirements (such as Diabetes diagnosis onset in the last year).
Cardinality | 1..1
Type | Reference(Group)|canonical(ActivityDefinition)|CodeableConcept|Expression|DataRequirement|TriggerDefinition
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Need to be able to define members in simple codes when the membership aligns well with terminology, with common criteria such as observations in a value set or lab tests within a year, or with expression language to support criteria that do not fit in the above.
Summary | true
EvidenceVariable.characteristic.usageContext
Element Id | EvidenceVariable.characteristic.usageContext
Definition | Use UsageContext to define the members of the population, such as Age Ranges, Genders, Settings.
Cardinality | 0..*
Type | UsageContext
Requirements | Need to be able to define members more structurally when more information can be communicated such as age range.
To Do | Define semantics when multiple definition types are present, or disallow mixtures.
EvidenceVariable.characteristic.exclude
Element Id | EvidenceVariable.characteristic.exclude
Definition | When true, members with this characteristic are excluded from the element.
Cardinality | 0..1
Type | boolean
Meaning if Missing | False
EvidenceVariable.characteristic.participantEffective[x]
Element Id | EvidenceVariable.characteristic.participantEffective[x]
Definition | Indicates what effective period the study covers.
Cardinality | 0..1
Type | dateTime|Period|Duration|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
EvidenceVariable.characteristic.timeFromStart
Element Id | EvidenceVariable.characteristic.timeFromStart
Definition | Indicates duration from the participant's study entry.
Cardinality | 0..1
Type | Duration
EvidenceVariable.characteristic.groupMeasure
Element Id | EvidenceVariable.characteristic.groupMeasure
Definition | Indicates how elements are aggregated within the study effective period.
Cardinality | 0..1
Terminology Binding | GroupMeasure (Required)
Type | code

14.10.7 Resource EvidenceVariable - Mappings

URL: https://hl7.org/fhir/R4/evidencevariable-mappings.html

14.10.7 Resource EvidenceVariable - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the evidencevariable resource (see Mappings to Other Standards for further information & status).

14.10.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

EvidenceVariable
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
date | FiveWs.recorded
publisher | FiveWs.witness

14.10.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

EvidenceVariable | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.10.7.3 RIM Mapping (http://hl7.org/v3)

EvidenceVariable
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.10.8 Resource EvidenceVariable - Extensions & Profiles

URL: https://hl7.org/fhir/R4/evidencevariable-profiles.html

14.10.8 Resource EvidenceVariable - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the EvidenceVariable resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.10.8.1 Profiles

Profile | Description | Context
PICOElement | Explanation of what this profile contains/is for. | PICO Element Profile

14.10.8.2 Extensions

replaces | EvidenceVariable | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.10.8.3 Search Extensions

No Search Extensions defined for this resource

14.10.9 PICO Element Profile

URL: https://hl7.org/fhir/R4/evidencevaraible-picoelement.html

14.10.9 PICO Element Profile

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.10.9.1 Content

Profiles:

PICOElement | Explanation of what this profile contains/is for.

14.10.9.2 StructureDefinition: PICOElement

URL: https://hl7.org/fhir/R4/picoelement.html

14.10.9.2 StructureDefinition: PICOElement

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/picoelement`

Explanation of what this profile contains/is for.

This profile was published on Wed, Oct 31, 2018 00:00+1100 as a draft by Health Level Seven, Inc. - Clinical Decision Support Workgroup.

14.10.9.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from EvidenceVariable.

Summary

Mandatory: 2 elements

This structure is derived from EvidenceVariable.

Name | Flags | Card. | Type | Description & Constraints
EvidenceVariable | | 0..*
identifier | Σ | 0..* | Identifier | Additional identifier for the evidence variable
title | Σ | 0..1 | string | Name for this evidence variable (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
date | Σ | 0..1 | dateTime | Date last changed
description | | 0..1 | markdown | Natural language description of the evidence variable
note | | 0..* | Annotation | Used for footnotes or explanatory notes
type | | 0..1 | code | dichotomous | continuous | descriptive
characteristic | | 1..* | BackboneElement | What defines the members of the evidence element
description | | 0..1 | string | Natural language description of the characteristic
definition[x] | | 1..1 | | What code or expression defines members?
definitionReference | | | Reference(group.html)
definitionCanonical | | | canonical(activitydefinition.html)
definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | | 0..* | UsageContext | What code/value pairs define members?
exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | | 0..1 | Duration | Observation time from study start
groupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
EvidenceVariable | I | 0..* | | A population, intervention, or exposure definition
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this evidence variable, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the evidence variable

version | ΣI | 0..1 | string | Business version of the evidence variable
name | ΣI | 0..1 | string | Name for this evidence variable (computer friendly)
title | ΣI | 0..1 | string | Name for this evidence variable (human friendly)
shortTitle | ΣI | 0..1 | string | Title for use in informal contexts
subtitle | I | 0..1 | string | Subordinate title of the EvidenceVariable
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the evidence variable
note | I | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for evidence variable (if applicable) Binding: Jurisdiction (extensible)
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the evidence variable was approved by publisher
lastReviewDate | I | 0..1 | date | When the evidence variable was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the evidence variable is expected to be used
topic | I | 0..* | CodeableConcept | The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
type | ΣI | 0..1 | code | dichotomous | continuous | descriptive Binding: EvidenceVariableType (required)
characteristic | ΣI | 1..* | BackboneElement | What defines the members of the evidence element
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
description | I | 0..1 | string | Natural language description of the characteristic
definition[x] | ΣI | 1..1 | | | What code or expression defines members?
definitionReference | | | Reference(group.html)
definitionCanonical | | | canonical(activitydefinition.html)
definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | I | 0..* | UsageContext | What code/value pairs define members?
exclude | I | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | I | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | I | 0..1 | Duration | Observation time from study start
groupMeasure | I | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-median Binding: GroupMeasure (required)
Documentation for this format

```
<!-- PICO Element Profile -->

<EvidenceVariable xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this evidence variable, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the evidence variable --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the evidence variable -->
 <name value="[string]"/><!--  0..1 Name for this evidence variable (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this evidence variable (human friendly) -->
 <shortTitle value="[string]"/><!--  0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the EvidenceVariable -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the evidence variable -->
 <note><!--  0..* Annotation
```

```
    Used for footnotes or explanatory notes --></note>
 <useContext><!--  0..* UsageContext
    The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for evidence variable (if applicable)
   --></jurisdiction>
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the evidence variable was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the evidence variable was last reviewed -->
 <effectivePeriod><!--  0..1 Period
    When the evidence variable is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc.
      --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail
    Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail
    Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
    Additional documentation, citations, etc. --></relatedArtifact>
 <type value="[code]"/><!--  0..1 dichotomous | continuous | descriptive   -->
 <characteristic>  1..* BackboneElement  <!--  1..* What defines the members of the evidence element -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <description value="[string]"/><!--  0..1 Natural language description of the characteristic -->
  <definition[x]><!--  1..1 Reference(Group)|canonical|
    CodeableConcept|Expression|DataRequirement|TriggerDefinition
      What code or expression defines members? --></definition[x]>
  <usageContext><!--  0..* UsageContext
      What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!--  0..1 Whether the characteristic includes or excludes members -->
  <participantEffective[x]><!--  0..1 dateTime|
    Period|Duration|Timing
      What time period do participants cover --></participantEffective[x]>
  <timeFromStart><!--  0..1 Duration
      Observation time from study start --></timeFromStart>
  <groupMeasure value="[code]"/><!--  0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-
    median   -->
 </characteristic>
</EvidenceVariable>
```

```
{ // PICO Element Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this evidence variable, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the evidence variable
    "version" : "<string>", // C?Business version of the evidence variable
    "name" : "<string>", // C?Name for this evidence variable (computer friendly)
    "title" : "<string>", // C?Name for this evidence variable (human friendly)
    "shortTitle" : "<string>", // C?Title for use in informal contexts
    "subtitle" : "<string>", // C?Subordinate title of the EvidenceVariable
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the evidence variable
    "note" : [{ Annotation }], // C?Used for footnotes or explanatory notes
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for evidence variable (if applicable)
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the evidence variable was approved by publisher
    "lastReviewDate" : "<date>", // C?When the evidence variable was last reviewed
    "effectivePeriod" : { Period }, // C?When the evidence variable is expected to be used
    "topic" : [{ CodeableConcept }], // C?The category of the EvidenceVariable, such as Education, Treatment,
    Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "type" : "<code>", // C?dichotomous | continuous | descriptive
    "characteristic" : [{ BackboneElement }] // C? R! What defines the members of the evidence element
  }
```

This structure is derived from EvidenceVariable.

Summary

Mandatory: 2 elements

Differential View

This structure is derived from EvidenceVariable.

Name	Flags	Card.	Type	Description & Constraints
EvidenceVariable		0..*		

identifier | Σ | 0..* | Identifier | Additional identifier for the evidence variable
title | Σ | 0..1 | string | Name for this evidence variable (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
date | Σ | 0..1 | dateTime | Date last changed
description | | 0..1 | markdown | Natural language description of the evidence variable
note | | 0..* | Annotation | Used for footnotes or explanatory notes
type | | 0..1 | code | dichotomous | continuous | descriptive
characteristic | | 1..* | BackboneElement | What defines the members of the evidence element
description | | 0..1 | string | Natural language description of the characteristic
definition[x] | | 1..1 | | What code or expression defines members?
definitionReference | | | Reference(group.html)
definitionCanonical | | | canonical(activitydefinition.html)
definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | | 0..* | UsageContext | What code/value pairs define members?
exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | | 0..1 | Duration | Observation time from study start
groupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
EvidenceVariable | I | 0..* | | A population, intervention, or exposure definition
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this evidence variable, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the evidence variable
version | ΣI | 0..1 | string | Business version of the evidence variable
name | ΣI | 0..1 | string | Name for this evidence variable (computer friendly)
title | ΣI | 0..1 | string | Name for this evidence variable (human friendly)
shortTitle | ΣI | 0..1 | string | Title for use in informal contexts
subtitle | I | 0..1 | string | Subordinate title of the EvidenceVariable
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the evidence variable
note | I | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for evidence variable (if applicable) Binding: Jurisdiction (extensible)
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the evidence variable was approved by publisher
lastReviewDate | I | 0..1 | date | When the evidence variable was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the evidence variable is expected to be used
topic | I | 0..* | CodeableConcept | The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
type | ΣI | 0..1 | code | dichotomous | continuous | descriptive Binding: EvidenceVariableType (required)
characteristic | ΣI | 1..* | BackboneElement | What defines the members of the evidence element
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
description | I | 0..1 | string | Natural language description of the characteristic
definition[x] | ΣI | 1..1 | | What code or expression defines members?
definitionReference | | | Reference(group.html)
definitionCanonical | | | canonical(activitydefinition.html)

definitionCodeableConcept | | | CodeableConcept
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
definitionTriggerDefinition | | | TriggerDefinition
usageContext | I | 0..* | UsageContext | What code/value pairs define members?
exclude | I | 0..1 | boolean | Whether the characteristic includes or excludes members
participantEffective[x] | I | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
timeFromStart | I | 0..1 | Duration | Observation time from study start
groupMeasure | I | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median Binding: GroupMeasure (required)
Documentation for this format

XML Template

```
<!-- PICO Element Profile -->

<EvidenceVariable xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1
     Canonical identifier for this evidence variable, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier
     Additional identifier for the evidence variable --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the evidence variable -->
 <name value="[string]"/><!--  0..1 Name for this evidence variable (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this evidence variable (human friendly) -->
 <shortTitle value="[string]"/><!--  0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the EvidenceVariable -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail
     Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the evidence variable -->
 <note><!--  0..* Annotation
     Used for footnotes or explanatory notes --></note>
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for evidence variable (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the evidence variable was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the evidence variable was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the evidence variable is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the EvidenceVariable, such as Education, Treatment, Assessment, etc.
       --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail
     Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail
     Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <type value="[code]"/><!--  0..1 dichotomous | continuous | descriptive   -->
 <characteristic>  1..* BackboneElement  <!--  1..* What defines the members of the evidence element -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <description value="[string]"/><!--  0..1 Natural language description of the characteristic -->
  <definition[x]><!--  1..1 Reference(Group)|canonical|
    CodeableConcept|Expression|DataRequirement|TriggerDefinition
      What code or expression defines members? --></definition[x]>
  <usageContext><!--  0..* UsageContext
      What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!--  0..1 Whether the characteristic includes or excludes members -->
  <participantEffective[x]><!--  0..1 dateTime|
    Period|Duration|Timing
      What time period do participants cover --></participantEffective[x]>
  <timeFromStart><!--  0..1 Duration
      Observation time from study start --></timeFromStart>
  <groupMeasure value="[code]"/><!--  0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-
    median   -->
 </characteristic>
```

```
</EvidenceVariable>
```

JSON Template

```
{ // PICO Element Profile
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this evidence variable, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the evidence variable
    "version" : "<string>", // C?Business version of the evidence variable
    "name" : "<string>", // C?Name for this evidence variable (computer friendly)
    "title" : "<string>", // C?Name for this evidence variable (human friendly)
    "shortTitle" : "<string>", // C?Title for use in informal contexts
    "subtitle" : "<string>", // C?Subordinate title of the EvidenceVariable
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the evidence variable
    "note" : [{ Annotation }], // C?Used for footnotes or explanatory notes
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for evidence variable (if applicable)
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the evidence variable was approved by publisher
    "lastReviewDate" : "<date>", // C?When the evidence variable was last reviewed
    "effectivePeriod" : { Period }, // C?When the evidence variable is expected to be used
    "topic" : [{ CodeableConcept }], // C?The category of the EvidenceVariable, such as Education, Treatment,
    Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "type" : "<code>", // C?dichotomous | continuous | descriptive
    "characteristic" : [{ BackboneElement }] // C? R! What defines the members of the evidence element
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.10.9.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
EvidenceVariable.language	Common Languages	preferred	Common Languages
EvidenceVariable.status	PublicationStatus	required	PublicationStatus
EvidenceVariable.jurisdiction	Jurisdiction	extensible	Jurisdiction
EvidenceVariable.topic	DefinitionTopic	example	DefinitionTopic
EvidenceVariable.type	EvidenceVariableType	required	EvidenceVariableType
EvidenceVariable.characteristic.groupMeasure	GroupMeasure	required	GroupMeasure

14.10.10 JSON Schema for EvidenceVariable

URL: https://hl7.org/fhir/R4/evidencevariable.schema.json.html

14.10.10 JSON Schema for EvidenceVariable

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for EvidenceVariable, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/EvidenceVariable",
  "$ref": "#/definitions/EvidenceVariable",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "EvidenceVariable": {
      "description": "The EvidenceVariable resource describes a \"PICO\" element that knowledge (evidence, assertion,
    recommendation) is about.",
      "properties": {
        "resourceType": {
          "description": "This is a EvidenceVariable resource",
          "const": "EvidenceVariable"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
```

```
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this evidence variable when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this evidence variable is (or will
be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the evidence
variable is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this evidence variable when it is represented in
other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the evidence variable when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the evidence
variable author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a
 managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence. To provide a version consistent with the Decision Support Service specification, use the format
Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision
Support Service specification. Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
```

```
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the evidence variable. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the evidence variable.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "shortTitle": {
      "description": "The short title provides an alternate title for use in informal descriptive contexts where the
 full, formal title is not necessary.",
      "$ref": "#/definitions/string"
    },
    "_shortTitle": {
      "description": "Extensions for shortTitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the EvidenceVariable giving additional information about
 its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this evidence variable. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the evidence variable was published. The date must change
 when the business version changes and it must change if the status code changes. In addition, it should change when
 the substantive content of the evidence variable changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the evidence variable.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the evidence variable from a consumer\u0027s
perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
```

```
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate evidence variable
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the evidence variable is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "copyright": {
      "description": "A copyright statement relating to the evidence variable and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the evidence variable.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the evidence variable content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the EvidenceVariable. Topics provide a high-level
 categorization grouping types of EvidenceVariables that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
```

```
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of evidence element, a population, an exposure, or an outcome.",
      "enum": [
        "dichotomous",
        "continuous",
        "descriptive"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "characteristic": {
      "description": "A characteristic that defines the members of the evidence element. Multiple characteristics
are applied with \"and\" semantics.",
      "items": {
        "$ref": "#/definitions/EvidenceVariable_Characteristic"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "characteristic",
    "resourceType"
  ]
},
"EvidenceVariable_Characteristic": {
  "description": "The EvidenceVariable resource describes a \"PICO\" element that knowledge (evidence, assertion,
recommendation) is about.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "A short, natural language description of the characteristic that could be used to communicate
the criteria to an end-user.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definitionReference": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "definitionCanonical": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "pattern": "^\\S*$",
```

```
      "type": "string"
    },
    "_definitionCanonical": {
      "description": "Extensions for definitionCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definitionCodeableConcept": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "definitionExpression": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "Expression.schema.json#/definitions/Expression"
    },
    "definitionDataRequirement": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
    },
    "definitionTriggerDefinition": {
      "description": "Define members of the evidence element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "TriggerDefinition.schema.json#/definitions/TriggerDefinition"
    },
    "usageContext": {
      "description": "Use UsageContext to define the members of the population, such as Age Ranges, Genders,
Settings.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "exclude": {
      "description": "When true, members with this characteristic are excluded from the element.",
      "$ref": "#/definitions/boolean"
    },
    "_exclude": {
      "description": "Extensions for exclude",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participantEffectiveDateTime": {
      "description": "Indicates what effective period the study covers.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_participantEffectiveDateTime": {
      "description": "Extensions for participantEffectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participantEffectivePeriod": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "participantEffectiveDuration": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "participantEffectiveTiming": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "timeFromStart": {
      "description": "Indicates duration from the participant\u0027s study entry.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "groupMeasure": {
      "description": "Indicates how elements are aggregated within the study effective period.",
      "enum": [
        "mean",
        "median",
        "mean-of-mean",
        "mean-of-median",
        "median-of-mean",
        "median-of-median"
      ]
    },
    "_groupMeasure": {
      "description": "Extensions for groupMeasure",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
```

```
    }
  }
}
```

14.10.11 Resource EvidenceVariable - Maturity Dependency View

URL: https://hl7.org/fhir/R4/evidencevariable-dependencies.html

14.10.11 Resource EvidenceVariable - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for EvidenceVariable FMM level 0

Name | Card. | Type | Dependency Analysis
EvidenceVariable | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
shortTitle | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
note | 0..* | Annotation
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
type | 0..1 | code
characteristic | 1..* | BackboneElement
description | 0..1 | string
definition[x] | 1..1
definitionReference | | Reference(Group
definitionCanonical | | canonical
definitionCodeableConcept | | CodeableConcept
definitionExpression | | Expression
definitionDataRequirement | | DataRequirement
definitionTriggerDefinition | | TriggerDefinition
usageContext | 0..* | UsageContext
exclude | 0..1 | boolean
participantEffective[x] | 0..1
participantEffectiveDateTime | | dateTime
participantEffectivePeriod | | Period
participantEffectiveDuration | | Duration
participantEffectiveTiming | | Timing
timeFromStart | 0..1 | Duration
groupMeasure | 0..1 | code

14.11 Resource Library - Content

URL: https://hl7.org/fhir/R4/library.html
Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The Library resource is a general-purpose container for knowledge asset definitions. It can be used to describe and expose existing knowledge assets such as logic libraries and information model descriptions, as well as to describe a collection of knowledge assets.

14.11.1 Scope and Usage

The Library resource is a general purpose container for clinical knowledge assets. These assets may be defined using a non-FHIR representation, such as a shareable library of clinical logic, written in Clinical Quality Language (CQL), or the XML Schema for an information model, or they may be defined using FHIR resources such as the description of a protocol represented by a PlanDefinition resource. In addition to representing the metadata of the library, the resource has elements for tracking dependencies, as well as for representing the parameters and data requirements for any expression functionality provided by the library.

For non-FHIR assets, the actual content of the library is represented using the Attachment data type, and may either be referenced with a URL, or the content may be embedded as a base-64 encoded string. Either way, the contentType element of the attachment is used to indicate the representation of the library content.

Note that because the library content may be embedded as well as be retrievable from an external repository via the attachment URL, the possibility exists for the embedded content to be different from the content on the repository. With proper versioning and governance, this should never occur, but to minimize the potential impact of this possibility, implementers SHALL give precedence to the embedded content of a library when it is present.

14.11.2 Boundaries and Relationships

The Library resource is definitional in nature, meaning it is intended to represent shareable knowledge independent of any particular patient. This is in contrast to the DocumentReference resource, which captures non-FHIR content related to a specific patient, and the Media and Binary resources which capture multimedia content and raw binary content, respectively.

This resource is referenced by ActivityDefinition, Measure, PlanDefinition, ResearchDefinition and ResearchElementDefinition

14.11.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
Library | ITU | | DomainResource | Represents a library of quality improvement components + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the library
version | Σ | 0..1 | string | Business version of the library
name | ΣI | 0..1 | string | Name for this library (computer friendly)
title | Σ | 0..1 | string | Name for this library (human friendly)
subtitle | | 0..1 | string | Subordinate title of the library
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
type | Σ | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition LibraryType (Extensible)
subject[x] | | 0..1 | | Type of individual the library content is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the library
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this library is defined
usage | | 0..1 | string | Describes the clinical usage of the library
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the library was approved by publisher
lastReviewDate | | 0..1 | date | When the library was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the library is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.

parameter | | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | | 0..* | DataRequirement | What data is referenced by this library
content | Σ | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Library xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the library -->
 <name value="[string]"/><!--  0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this library (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the library -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <type><!-- 1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition --></type>
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the library -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for library (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this library is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the library was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the library is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!-- 0..* ParameterDefinition Parameters defined by the library --></parameter>
 <dataRequirement><!-- 0..* DataRequirement What data is referenced by this library --></dataRequirement>
 <content><!-- 0..* Attachment Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{
  "resourceType" : "Library",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this library, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the library
  "version" : "<string>", // Business version of the library
  "name" : "<string>", // C? Name for this library (computer friendly)
  "title" : "<string>", // Name for this library (human friendly)
  "subtitle" : "<string>", // Subordinate title of the library
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "type" : { CodeableConcept }, // R!  logic-library | model-definition | asset-collection | module-definition
  // subject[x]: Type of individual the library content is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the library
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for library (if applicable)
  "purpose" : "<markdown>", // Why this library is defined
  "usage" : "<string>", // Describes the clinical usage of the library
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the library was approved by publisher
  "lastReviewDate" : "<date>", // When the library was last reviewed
  "effectivePeriod" : { Period }, // When the library is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "parameter" : [{ ParameterDefinition }], // Parameters defined by the library
  "dataRequirement" : [{ DataRequirement }], // What data is referenced by this library
  "content" : [{ Attachment }] // Contents of the library, either embedded or referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:Library;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Library.url [ uri ]; # 0..1 Canonical identifier for this library, represented as a URI (globally unique)
  fhir:Library.identifier [ Identifier ], ... ; # 0..* Additional identifier for the library
  fhir:Library.version [ string ]; # 0..1 Business version of the library
  fhir:Library.name [ string ]; # 0..1 Name for this library (computer friendly)
  fhir:Library.title [ string ]; # 0..1 Name for this library (human friendly)
  fhir:Library.subtitle [ string ]; # 0..1 Subordinate title of the library
  fhir:Library.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Library.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Library.type [ CodeableConcept ]; # 1..1 logic-library | model-definition | asset-collection | module-definition
  # Library.subject[x] : 0..1 Type of individual the library content is focused on. One of these 2
    fhir:Library.subjectCodeableConcept [ CodeableConcept ]
    fhir:Library.subjectReference [ Reference(Group) ]
  fhir:Library.date [ dateTime ]; # 0..1 Date last changed
  fhir:Library.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Library.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Library.description [ markdown ]; # 0..1 Natural language description of the library
  fhir:Library.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Library.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for library (if applicable)
  fhir:Library.purpose [ markdown ]; # 0..1 Why this library is defined
  fhir:Library.usage [ string ]; # 0..1 Describes the clinical usage of the library
  fhir:Library.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Library.approvalDate [ date ]; # 0..1 When the library was approved by publisher
  fhir:Library.lastReviewDate [ date ]; # 0..1 When the library was last reviewed
  fhir:Library.effectivePeriod [ Period ]; # 0..1 When the library is expected to be used
  fhir:Library.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:Library.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Library.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Library.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Library.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Library.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Library.parameter [ ParameterDefinition ], ... ; # 0..* Parameters defined by the library
  fhir:Library.dataRequirement [ DataRequirement ], ... ; # 0..* What data is referenced by this library
  fhir:Library.content [ Attachment ], ... ; # 0..* Contents of the library, either embedded or referenced
]
```

Changes since R3

Library
Library.subtitle | Added Element
Library.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
Library.experimental | No longer marked as Modifier
Library.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/library-type" to "http://terminology.hl7.org/CodeSystem/library-type"
Library.subject[x] | Added Element
Library.author | Added Element
Library.editor | Added Element
Library.reviewer | Added Element
Library.endorser | Added Element
Library.contributor | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 11 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Library | ITU | | DomainResource | Represents a library of quality improvement components + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the library
version | Σ | 0..1 | string | Business version of the library
name | ΣI | 0..1 | string | Name for this library (computer friendly)
title | Σ | 0..1 | string | Name for this library (human friendly)
subtitle | | 0..1 | string | Subordinate title of the library
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
type | Σ | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition LibraryType (Extensible)
subject[x] | | 0..1 | | Type of individual the library content is focused on Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the library

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this library is defined
usage | | 0..1 | string | Describes the clinical usage of the library
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the library was approved by publisher
lastReviewDate | | 0..1 | date | When the library was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the library is expected to be used
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | | 0..* | DataRequirement | What data is referenced by this library
content | Σ | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Library xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the library -->
 <name value="[string]"/><!--  0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this library (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the library -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <type><!-- 1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition --></type>
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the library -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for library (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this library is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the library was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the library is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!-- 0..* ParameterDefinition Parameters defined by the library --></parameter>
 <dataRequirement><!-- 0..* DataRequirement What data is referenced by this library --></dataRequirement>
 <content><!-- 0..* Attachment Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{
  "resourceType" : "Library",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this library, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the library
  "version" : "<string>", // Business version of the library
  "name" : "<string>", // C? Name for this library (computer friendly)
  "title" : "<string>", // Name for this library (human friendly)
  "subtitle" : "<string>", // Subordinate title of the library
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "type" : { CodeableConcept }, // R!  logic-library | model-definition | asset-collection | module-definition
  // subject[x]: Type of individual the library content is focused on. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the library
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for library (if applicable)
  "purpose" : "<markdown>", // Why this library is defined
  "usage" : "<string>", // Describes the clinical usage of the library
  "copyright" : "<markdown>", // Use and/or publishing restrictions
```

```
  "approvalDate" : "<date>", // When the library was approved by publisher
  "lastReviewDate" : "<date>", // When the library was last reviewed
  "effectivePeriod" : { Period }, // When the library is expected to be used
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "parameter" : [{ ParameterDefinition }], // Parameters defined by the library
  "dataRequirement" : [{ DataRequirement }], // What data is referenced by this library
  "content" : [{ Attachment }] // Contents of the library, either embedded or referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Library;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Library.url [ uri ]; # 0..1 Canonical identifier for this library, represented as a URI (globally unique)
  fhir:Library.identifier [ Identifier ], ... ; # 0..* Additional identifier for the library
  fhir:Library.version [ string ]; # 0..1 Business version of the library
  fhir:Library.name [ string ]; # 0..1 Name for this library (computer friendly)
  fhir:Library.title [ string ]; # 0..1 Name for this library (human friendly)
  fhir:Library.subtitle [ string ]; # 0..1 Subordinate title of the library
  fhir:Library.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Library.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Library.type [ CodeableConcept ]; # 1..1 logic-library | model-definition | asset-collection | module-definition
  # Library.subject[x] : 0..1 Type of individual the library content is focused on. One of these 2
    fhir:Library.subjectCodeableConcept [ CodeableConcept ]
    fhir:Library.subjectReference [ Reference(Group) ]
  fhir:Library.date [ dateTime ]; # 0..1 Date last changed
  fhir:Library.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Library.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Library.description [ markdown ]; # 0..1 Natural language description of the library
  fhir:Library.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Library.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for library (if applicable)
  fhir:Library.purpose [ markdown ]; # 0..1 Why this library is defined
  fhir:Library.usage [ string ]; # 0..1 Describes the clinical usage of the library
  fhir:Library.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Library.approvalDate [ date ]; # 0..1 When the library was approved by publisher
  fhir:Library.lastReviewDate [ date ]; # 0..1 When the library was last reviewed
  fhir:Library.effectivePeriod [ Period ]; # 0..1 When the library is expected to be used
  fhir:Library.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc.
  fhir:Library.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Library.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Library.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Library.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Library.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Library.parameter [ ParameterDefinition ], ... ; # 0..* Parameters defined by the library
  fhir:Library.dataRequirement [ DataRequirement ], ... ; # 0..* What data is referenced by this library
  fhir:Library.content [ Attachment ], ... ; # 0..* Contents of the library, either embedded or referenced
]
```

Changes since Release 3

Library
Library.subtitle | Added Element
Library.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1
Library.experimental | No longer marked as Modifier
Library.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/library-type" to "http://terminology.hl7.org/CodeSystem/library-type"
Library.subject[x] | Added Element
Library.author | Added Element
Library.editor | Added Element
Library.reviewer | Added Element
Library.endorser | Added Element
Library.contributor | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 11 tests that all execute ok. All tests pass round-trip testing and 11 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.11.3.1 Terminology Bindings

Path | Definition | Type | Reference
Library.status | The lifecycle status of an artifact. | Required | PublicationStatus

Library.type | The type of knowledge asset this library contains. | Extensible | LibraryType
Library.subject[x] | The possible types of subjects for a library (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
Library.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
Library.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic

14.11.3.2 Constraints

id | Level | Location | Description | Expression
lib-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.11.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | Library.relatedArtifact.where(type='composed-of').resource (Any)
content-type | token | The type of content in the library (e.g. text/cql) | Library.content.contentType
context | token | A use context assigned to the library | (Library.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the library | (Library.useContext.value as Quantity) | (Library.useContext.value as Range)
context-type | token | A type of use context assigned to the library | Library.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the library | On Library.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the library | On Library.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The library publication date | Library.date
depends-on | reference | What resource is being referenced | Library.relatedArtifact.where(type='depends-on').resource (Any)
derived-from | reference | What resource is being referenced | Library.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the library | Library.description
effective | date | The time during which the library is intended to be in use | Library.effectivePeriod
identifier | token | External identifier for the library | Library.identifier
jurisdiction | token | Intended jurisdiction for the library | Library.jurisdiction
name | string | Computationally friendly name of the library | Library.name
predecessor | reference | What resource is being referenced | Library.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the library | Library.publisher
status | token | The current status of the library | Library.status
successor | reference | What resource is being referenced | Library.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the library | Library.title
topic | token | Topics associated with the module | Library.topic
type | token | The type of the library (e.g. logic-library, model-definition, asset-collection, module-definition) | Library.type
url | uri | The uri that identifies the library | Library.url
version | token | The business version of the library | Library.version

14.11.5 Resource Library - Examples

URL: https://hl7.org/fhir/R4/library-examples.html

14.11.5 Resource Library - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

These examples illustrate the use of the Library resource to share quality measurement and improvement logic as a FHIR resource. Note that the actual content of each library (the Clinical Quality Language (CQL) source, or the XML for the information model definition, etc.) should be contained in the content element as an Attachment. For illustration purposes, the content of each library is included in its raw form, and linked here.

Example | Description | Content
Chlamydia Screening Common Library | Common Chlamydia Screening library developed as the running example in the Author's Guide of the Clinical Quality Language specification. | CQL Source
Exclusive Breastfeeding Intervention Logic | Clinical Decision Support rules derived from CMS9v4 - Exclusive Breastfeeding measure | CQL Source
Exclusive Breastfeeding Measure Logic | Quality Measurement logic used to express the CMS9v4 measure. | CQL Source
CMS 146 Measure Logic | Measure logic for CMS 146 used as the running example for Quality Reporting guidance in the Clinical Reasoning module. | CQL Source
QUICK Model Definition | The QUICK model information file used by the CQL-to-ELM translator to validate CQL written against the QUICK data model. | XML Information Model
FHIR Model Definition | The FHIR model information file used by the CQL-to-ELM translator to validate CQL written against the FHIR data model. | XML Information Model

Example Name | id | Format
Chlamydia Screening Common Library | example | XML | JSON | Turtle

Exclusive Breastfeeding Intervention Logic | library-exclusive-breastfeeding-cds-logic | XML | JSON | Turtle
Exclusive Breastfeeding Measurement Logic | library-exclusive-breastfeeding-cqm-logic | XML | JSON | Turtle
CMS146 Measure Logic | library-cms146-example | XML | JSON | Turtle
QUICK Model Definition | library-quick-model-definition | XML | JSON | Turtle
FHIR Model Definition | library-fhir-model-definition | XML | JSON | Turtle
Suicide Risk Order Set Logic | suiciderisk-orderset-logic | XML | JSON | Turtle
FHIR Helpers | library-fhir-helpers | XML | JSON | Turtle
Zika Virus Management Intervention Logic | zika-virus-intervention-logic | XML | JSON | Turtle
Zika Virus Management Library | composition-example | XML | JSON | Turtle
Predecessor library example | library-fhir-helpers-predecessor | XML | JSON | Turtle
OMTK Logic | omtk-logic | XML | JSON | Turtle
OMTK Model | omtk-modelinfo | XML | JSON | Turtle
Opioid CDS Common Library | opioidcds-common | XML | JSON | Turtle
Opioid CDS Recommendation 4 Logic | opioidcds-recommendation-04 | XML | JSON | Turtle
Opioid CDS Recommendation 5 Logic | opioidcds-recommendation-05 | XML | JSON | Turtle
Opioid CDS Recommendation 7 Logic | opioidcds-recommendation-07 | XML | JSON | Turtle
Opioid CDS Recommendation 8 Logic | opioidcds-recommendation-08 | XML | JSON | Turtle
Opioid CDS Recommendation 10 Logic | opioidcds-recommendation-10 | XML | JSON | Turtle
Opioid CDS Recommendation 11 Logic | opioidcds-recommendation-11 | XML | JSON | Turtle
HIV Indicators Reporting Example | hiv-indicators | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.11.6 Resource Library - Detailed Descriptions

URL: https://hl7.org/fhir/R4/library-definitions.html

14.11.6 Resource Library - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Library resource.

Library
Element Id | Library
Definition | The Library resource is a general-purpose container for knowledge asset definitions. It can be used to describe and expose existing knowledge assets such as logic libraries and information model descriptions, as well as to describe a collection of knowledge assets.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element lib-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
lib-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Library.url
Element Id | Library.url
Definition | An absolute URI that is used to identify this library when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this library is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the library is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the library to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
Library.identifier
Element Id | Library.identifier
Definition | A formal identifier that is used to identify this library when it is represented in other formats, or referenced in a specification, model, design or an instance. e.g. CMS or NQF identifiers for a measure artifact. Note that at least one identifier is required for non-experimental active artifacts.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this library outside of FHIR, where it is not possible to use the logical URI.
Library.version
Element Id | Library.version

Definition | The identifier that is used to identify this version of the library when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the library author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different library instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the library with the format [url]|[version].
Library.name
Element Id | Library.name
Definition | A natural language name identifying the library. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Library.title
Element Id | Library.title
Definition | A short, descriptive, user-friendly title for the library.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Library.subtitle
Element Id | Library.subtitle
Definition | An explanatory or alternate title for the library giving additional information about its content.
Cardinality | 0..1
Type | string
Library.status
Element Id | Library.status
Definition | The status of this library. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of libraries that are appropriate for use vs. not.
Library.experimental
Element Id | Library.experimental
Definition | A Boolean value to indicate that this library is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level library.
Summary | true
Comments | Allows filtering of librarys that are appropriate for use versus not.
Library.type
Element Id | Library.type
Definition | Identifies the type of library such as a Logic Library, Model Definition, Asset Collection, or Module Definition.
Cardinality | 1..1
Terminology Binding | LibraryType (Extensible)
Type | CodeableConcept
Summary | true
Library.subject[x]
Element Id | Library.subject[x]
Definition | A code or group definition that describes the intended subject of the contents of the library.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]

Meaning if Missing | Patient
Library.date
Element Id | Library.date
Definition | The date (and optionally time) when the library was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the library changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the library. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Library.publisher
Element Id | Library.publisher
Definition | The name of the organization or individual that published the library.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the library. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the library is the organization or individual primarily responsible for the maintenance and upkeep of the library. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the library. This item SHOULD be populated unless the information is available from context.
Library.contact
Element Id | Library.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Library.description
Element Id | Library.description
Definition | A free text natural language description of the library from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the library was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the library as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the library is presumed to be the predominant language in the place the library was created).
Library.useContext
Element Id | Library.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate library instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
Library.jurisdiction
Element Id | Library.jurisdiction
Definition | A legal or geographic region in which the library is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the library to be used in jurisdictions other than those for which it was originally designed or intended.
Library.purpose
Element Id | Library.purpose
Definition | Explanation of why this library is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the library. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this library.
Library.usage
Element Id | Library.usage
Definition | A detailed description of how the library is used from a clinical perspective.
Cardinality | 0..1
Type | string
Library.copyright

Element Id | Library.copyright
Definition | A copyright statement relating to the library and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the library.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the library and/or its content.
Alternate Names | License; Restrictions
Library.approvalDate
Element Id | Library.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
Library.lastReviewDate
Element Id | Library.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
Library.effectivePeriod
Element Id | Library.effectivePeriod
Definition | The period during which the library content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the library are or are expected to be used instead.
Summary | true
Comments | The effective period for a library determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
Library.topic
Element Id | Library.topic
Definition | Descriptive topics related to the content of the library. Topics provide a high-level categorization of the library that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the library so that it can be found by topical searches.
Library.author
Element Id | Library.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
Library.editor
Element Id | Library.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
Library.reviewer
Element Id | Library.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
Library.endorser
Element Id | Library.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
Library.relatedArtifact
Element Id | Library.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Libraries must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
Library.parameter
Element Id | Library.parameter
Definition | The parameter element defines parameters used by the library.
Cardinality | 0..*

Type | ParameterDefinition
Library.dataRequirement
Element Id | Library.dataRequirement
Definition | Describes a set of data that must be provided in order to be able to successfully perform the computations defined by the library.
Cardinality | 0..*
Type | DataRequirement
Library.content
Element Id | Library.content
Definition | The content of the library as an Attachment. The content may be a reference to a url, or may be directly embedded as a base-64 string. Either way, the contentType of the attachment determines how to interpret the content.
Cardinality | 0..*
Type | Attachment
Summary | true

14.11.7 Resource Library - Mappings

URL: https://hl7.org/fhir/R4/library-mappings.html

14.11.7 Resource Library - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the library resource (see Mappings to Other Standards for further information & status).

14.11.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Library
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
type | FiveWs.what[x]
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

14.11.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Library
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.11.7.3 RIM Mapping (http://hl7.org/v3)

Library | Act[classCode=GROUPER;moodCode=EVN]
url | .identifier[scope=BUSN;reliability=ISS]
identifier | .identifier
version | N/A (to add?)
name | N/A
title | .title
status | .status
experimental | N/A (to add?)

type | .code
date | .participation[typeCode=AUT].time
publisher | .participation[typeCode=AUT].role
contact | .participation[typeCode=CALLBCK].role
description | .text
useContext | N/A (to add?)
jurisdiction | N/A (to add?)
purpose | .reasonCode.text
usage | N/A (to add?)
copyright | N/A (to add?)
approvalDate | .outboundRelationship[typeCode="SUBJ"].act[classCode=CACT;moodCode=EVN;code="approval"].effectiveTime
lastReviewDate | .outboundRelationship[typeCode="SUBJ";
subsetCode="RECENT"].act[classCode=CACT;moodCode=EVN;code="review"].effectiveTime
effectivePeriod | N/A (to add?)
topic | N/A (to add?)
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
relatedArtifact | .outboundRelationship[typeCode=DOC,RSON,PREV, DRIV, USE, COMP] {successor would be PREV w/
inversionInd=true; No support for citation}
parameter | N/A (to add?)
dataRequirement | N/A (to add?)
content | .text

14.11.8 Resource Library - Extensions & Profiles

URL: https://hl7.org/fhir/R4/library-profiles.html

14.11.8 Resource Library - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Library resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.11.8.1 Profiles

Profile | Description | Context
CQLLibrary | Represents a CQL logic library | CQL Library
ShareableLibrary | Enforces the minimum information set for the library metadata required by HL7 and other organizations that share and publish libraries | Shareable Library

14.11.8.2 Extensions

replaces | Library | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

cqf-library | library | for Clinical Reasoning Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.11.8.3 Search Extensions

No Search Extensions defined for this resource

14.11.9 Resource Library - Operations

URL: https://hl7.org/fhir/R4/library-operations.html

14.11.9 Resource Library - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 1 operation operations associated with it:

$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

14.11.9.1 Operation $data-requirements on Library

URL: https://hl7.org/fhir/R4/library-operation-data-requirements.html

14.11.9.1 Operation $data-requirements on Library

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The data-requirements operation aggregates and returns the parameters and data requirements for a resource and all its dependencies as a single module definition.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Library-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/$data-requirements

URL: [base]/Library/[id]/$data-requirements

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
target	0..1	string			The target of the data requirements operation

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
return	1..1	Library			The result of the requirements gathering Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for a given target resource. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the target resource and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

14.11.9.1.1 Examples

Request: Request data requirements for the CMS146 library

```
GET [base]/Library/library-cms146-example/$data-requirements
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK
```

```
{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "resource": "Library/library-quick-model-definition"
    }
  ],
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.111.12.1006"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "code": [
            { "code": "diagnosis" }
          ]
        },
        {
          "path": "clinicalStatus",
```

```
      "code": [
        { "code": "confirmed" }
      ]
    },
    {
      "path": "code",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.102.12.1011"
    }
  ]
},
{
  "type": "Condition",
  "codeFilter": [
    {
      "path": "category",
      "code": [
        { "code": "diagnosis" }
      ]
    },
    {
      "path": "clinicalStatus",
      "code": [
        { "code": "confirmed" }
      ]
    },
    {
      "path": "code",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.102.12.1012"
    }
  ]
},
{
  "type": "Encounter",
  "codeFilter": [
    {
      "path": "status",
      "code": [
        { "code": "finished" }
      ]
    },
    {
      "path": "class",
      "code": [
        { "code": "ambulatory" }
      ]
    },
    {
      "path": "type",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.101.12.1061"
    }
  ]
},
{
  "type": "DiagnosticReport",
  "codeFilter": [
    {
      "path": "diagnosis",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.198.12.1012"
    }
  ]
},
{
  "type": "Medication",
  "codeFilter": [
    {
      "path": "code",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
  "type": "MedicationRequest",
  "codeFilter": [
    {
      "path": "status",
      "code": [
        { "code": "active" }
      ]
    },
    {
      "path": "medication.code",
      "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
  "type": "MedicationStatement",
  "codeFilter": [
    {
```

```
          "path": "status",
          "code": [
            { "code": "completed" }
          ]
        },
        {
          "path": "medication.code",
          "valueSet": "urn:oid:2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

14.11.10 Resource Library - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/library-version-maps.html

No validation errors - all conversions are clean

14.11.10 Resource Library - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Library.

Functional status for this map: 11 tests that all execute ok. All tests pass round-trip testing and 11 r3 resources are invalid (0 errors). (see documentation)

14.11.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Library3to4" = "R3 to R4 Conversions for Library"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Library" alias LibraryR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Library" alias Library as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Library(source src : LibraryR3, target tgt : Library) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.type -> tgt.type;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.contributor as vs where type = 'author' -> tgt.author as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'editor' -> tgt.editor as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'reviewer' -> tgt.reviewer as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'endorser' -> tgt.endorser as vt then Contributor(vs, vt);
  src.relatedArtifact -> tgt.relatedArtifact;
  src.parameter -> tgt.parameter;
  src.dataRequirement -> tgt.dataRequirement;
  src.content -> tgt.content;
}
```

14.11.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Library4to3" = "R4 to R3 Conversion for Library"

uses "http://hl7.org/fhir/StructureDefinition/Library" alias Library as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Library" alias LibraryR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Library(source src : LibraryR3, target tgt : Library) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
```

```
  src.type -> tgt.type;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.author as vs ->  tgt.contributor as vt,  vt.type = 'author' then Contributor(vs, vt);
  src.editor as vs ->  tgt.contributor as vt,  vt.type = 'editor',  vt.contact = vs;
  src.reviewer as vs ->  tgt.contributor as vt,  vt.type = 'reviewer',  vt.contact = vs;
  src.endorser as vs ->  tgt.contributor as vt,  vt.type = 'endorser',  vt.contact = vs;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.parameter -> tgt.parameter;
  src.dataRequirement -> tgt.dataRequirement;
  src.content -> tgt.content;
}
```

14.11.10.3 R4 Validation Errors

14.11.11 Shareable Library

URL: https://hl7.org/fhir/R4/library-shareable.html

Enforces the minimum information set for the library metadata required by HL7 and other organizations that share and publish libraries

14.11.11 Shareable Library

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.11.11.1 Content

Profiles:

ShareableLibrary | Enforces the minimum information set for the library metadata required by HL7 and other organizations that share and publish libraries

14.11.11.2 StructureDefinition: ShareableLibrary

URL: https://hl7.org/fhir/R4/shareablelibrary.html

14.11.11.2 StructureDefinition: ShareableLibrary

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareablelibrary`

Enforces the minimum information set for the library metadata required by HL7 and other organizations that share and publish libraries

This profile was published on [no date] as a draft by HL7.

14.11.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table
• Snapshot Table
• XML Template
• JSON Template
• All

This structure is derived from Library.

Summary

Mandatory: 6 elements Must-Support: 6 elements

This structure is derived from Library.

Name | Flags | Card. | Type | Description & Constraints
Library | | 1..1
url | | 1..1 | uri
identifier | S | 0..* | Identifier
version | | 1..1 | string
name | | 1..1 | string

title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Library | I | 1..1 | | Represents a library of quality improvement components
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the library
version | ΣI | 1..1 | string | Business version of the library
name | ΣI | 1..1 | string | Name for this library (computer friendly)
title | SΣI | 0..1 | string | Name for this library (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the library
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
type | ΣI | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition Binding: LibraryType (extensible)
subject[x] | I | 0..1 | | Type of individual the library content is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the library
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this library is defined
usage | I | 0..1 | string | Describes the clinical usage of the library
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the library was approved by publisher
lastReviewDate | I | 0..1 | date | When the library was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the library is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | I | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | I | 0..* | DataRequirement | What data is referenced by this library
content | ΣI | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

```
<!-- Shareable Library -->

<Library xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the library -->
 <name value="[string]"/><!--  1..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this library (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the library -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <type><!--  1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition   --></type>
```

```
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the library -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for library (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this library is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the library was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the library is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!--  0..* ParameterDefinition
     Parameters defined by the library --></parameter>
 <dataRequirement><!--  0..* DataRequirement
     What data is referenced by this library --></dataRequirement>
 <content><!--  0..* Attachment
     Contents of the library, either embedded or referenced --></content>
</Library>
```

```
{ // Shareable Library
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this library, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the library
    "version" : "<string>", // C? R! Business version of the library
    "name" : "<string>", // C? R! Name for this library (computer friendly)
    "title" : "<string>", // C?Name for this library (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the library
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "type" : { CodeableConcept }, // C? R! logic-library | model-definition | asset-collection | module-definition
// value[x]: Type of individual the library content is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the library content is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the library content is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the library
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for library (if applicable)
    "purpose" : "<markdown>", // C?Why this library is defined
    "usage" : "<string>", // C?Describes the clinical usage of the library
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the library was approved by publisher
    "lastReviewDate" : "<date>", // C?When the library was last reviewed
    "effectivePeriod" : { Period }, // C?When the library is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "parameter" : [{ ParameterDefinition }], // C?Parameters defined by the library
    "dataRequirement" : [{ DataRequirement }] // C?What data is referenced by this library
    "content" : [{ Attachment }], // C?Contents of the library, either embedded or referenced
  }
```

This structure is derived from Library.

Summary

Mandatory: 6 elements Must-Support: 6 elements

Differential View

This structure is derived from Library.

Name	Flags	Card.	Type	Description & Constraints
Library		1..1		
url		1..1	uri	
identifier	S	0..*	Identifier	
version		1..1	string	
name		1..1	string	

title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Library | I | 1..1 | | Represents a library of quality improvement components
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the library
version | ΣI | 1..1 | string | Business version of the library
name | ΣI | 1..1 | string | Name for this library (computer friendly)
title | SΣI | 0..1 | string | Name for this library (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the library
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
type | ΣI | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition Binding: LibraryType (extensible)
subject[x] | I | 0..1 | | Type of individual the library content is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the library
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this library is defined
usage | I | 0..1 | string | Describes the clinical usage of the library
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the library was approved by publisher
lastReviewDate | I | 0..1 | date | When the library was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the library is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | I | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | I | 0..* | DataRequirement | What data is referenced by this library
content | ΣI | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

XML Template

```
<!-- Shareable Library -->

<Library xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the library -->
 <name value="[string]"/><!--  1..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this library (human friendly) -->
```

```
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the library -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <type><!--  1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition   --></type>
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the library -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for library (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this library is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the library was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the library is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!--  0..* ParameterDefinition
     Parameters defined by the library --></parameter>
 <dataRequirement><!--  0..* DataRequirement
     What data is referenced by this library --></dataRequirement>
 <content><!--  0..* Attachment
     Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{ // Shareable Library
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this library, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the library
    "version" : "<string>", // C? R! Business version of the library
    "name" : "<string>", // C? R! Name for this library (computer friendly)
    "title" : "<string>", // C?Name for this library (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the library
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
    "type" : { CodeableConcept }, // C? R! logic-library | model-definition | asset-collection | module-definition
// value[x]: Type of individual the library content is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the library content is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the library content is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the library
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for library (if applicable)
    "purpose" : "<markdown>", // C?Why this library is defined
    "usage" : "<string>", // C?Describes the clinical usage of the library
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the library was approved by publisher
    "lastReviewDate" : "<date>", // C?When the library was last reviewed
    "effectivePeriod" : { Period }, // C?When the library is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "parameter" : [{ ParameterDefinition }], // C?Parameters defined by the library
    "dataRequirement" : [{ DataRequirement }] // C?What data is referenced by this library
    "content" : [{ Attachment }], // C?Contents of the library, either embedded or referenced
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.11.11.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Library.language	Common Languages	preferred	Common Languages
Library.status	PublicationStatus	required	PublicationStatus
Library.type	LibraryType	extensible	LibraryType
Library.subject[x]	Subject type	extensible	Subject type

Library.jurisdiction | Jurisdiction | extensible | Jurisdiction
Library.topic | DefinitionTopic | example | DefinitionTopic

14.11.12 CQL Library

URL: https://hl7.org/fhir/R4/library-cql.html

Represents a CQL logic library

14.11.12 CQL Library

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.11.12.1 Content

Profiles:
CQLLibrary | Represents a CQL logic library

14.11.12.2 StructureDefinition: CQLLibrary

URL: https://hl7.org/fhir/R4/cqllibrary.html

14.11.12.2 StructureDefinition: CQLLibrary

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cqllibrary`

Represents a CQL logic library

This profile was published on [no date] as a draft by HL7.

14.11.12.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Library.

Summary

Mandatory: 1 element Must-Support: 2 elements Fixed Value: 1 element

This structure is derived from Library.

Name | Flags | Card. | Type | Description & Constraints
Library | | 1..1
type | | 1..1 | CodeableConcept | Fixed Value: As shown
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/library-type
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: logic-library
display | | 1..1 | string | Representation defined by the system Fixed Value: Logic Library
parameter | S | 0..* | ParameterDefinition
dataRequirement | S | 0..* | DataRequirement
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Library | I | 1..1 | | Represents a library of quality improvement components
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 0..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the library
version | ΣI | 0..1 | string | Business version of the library
name | ΣI | 0..1 | string | Name for this library (computer friendly)
title | ΣI | 0..1 | string | Name for this library (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the library
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)

experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
type | ΣI | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition Binding: LibraryType (extensible) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/library-type
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: logic-library
display | | 1..1 | string | Representation defined by the system Fixed Value: Logic Library
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject[x] | I | 0..1 | | Type of individual the library content is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the library
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this library is defined
usage | I | 0..1 | string | Describes the clinical usage of the library
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the library was approved by publisher
lastReviewDate | I | 0..1 | date | When the library was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the library is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | SI | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | SI | 0..* | DataRequirement | What data is referenced by this library
content | ΣI | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

```
<!-- CQL Library -->

<Library xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the library -->
 <name value="[string]"/><!--  0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this library (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the library -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <type><!--  1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition   --></type>
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the library -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for library (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this library is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the library was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the library is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
```

```
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!--  0..* ParameterDefinition
     Parameters defined by the library --></parameter>
 <dataRequirement><!--  0..* DataRequirement
     What data is referenced by this library --></dataRequirement>
 <content><!--  0..* Attachment
     Contents of the library, either embedded or referenced --></content>
</Library>
```

```
{ // CQL Library
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this library, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the library
    "version" : "<string>", // C?Business version of the library
    "name" : "<string>", // C?Name for this library (computer friendly)
    "title" : "<string>", // C?Name for this library (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the library
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
    "type" : { CodeableConcept }, // C? R!
// value[x]: Type of individual the library content is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the library content is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the library content is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the library
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for library (if applicable)
    "purpose" : "<markdown>", // C?Why this library is defined
    "usage" : "<string>", // C?Describes the clinical usage of the library
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the library was approved by publisher
    "lastReviewDate" : "<date>", // C?When the library was last reviewed
    "effectivePeriod" : { Period }, // C?When the library is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "parameter" : [{ ParameterDefinition }], // C?Parameters defined by the library
    "dataRequirement" : [{ DataRequirement }] // C?What data is referenced by this library
    "content" : [{ Attachment }], // C?Contents of the library, either embedded or referenced
  }
```

This structure is derived from Library.

Summary

Mandatory: 1 element Must-Support: 2 elements Fixed Value: 1 element

Differential View

This structure is derived from Library.

Name	Flags	Card.	Type	Description & Constraints
Library		1..1		
type		1..1	CodeableConcept	Fixed Value: As shown
coding		1..1	Coding	Code defined by a terminology system Fixed Value: (complex)
system		1..1	uri	Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/library-type
code		1..1	code	Symbol in syntax defined by the system Fixed Value: logic-library
display		1..1	string	Representation defined by the system Fixed Value: Logic Library
parameter	S	0..*	ParameterDefinition	
dataRequirement	S	0..*	DataRequirement	
Documentation for this format				

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
Library	I	1..1		Represents a library of quality improvement components
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Additional content defined by implementations
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored

url | ΣI | 0..1 | uri | Canonical identifier for this library, represented as a URI (globally unique)
identifier | ΣI | 0..* | Identifier | Additional identifier for the library
version | ΣI | 0..1 | string | Business version of the library
name | ΣI | 0..1 | string | Name for this library (computer friendly)
title | ΣI | 0..1 | string | Name for this library (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the library
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 0..1 | boolean | For testing purposes, not real usage
type | ΣI | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition Binding: LibraryType (extensible) Fixed Value: As shown
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
coding | | 1..1 | Coding | Code defined by a terminology system Fixed Value: (complex)
id | | 0..0 | string | Unique id for inter-element referencing
extension | | 0..0 | Extension | Additional content defined by implementations
system | | 1..1 | uri | Identity of the terminology system Fixed Value: http://terminology.hl7.org/CodeSystem/library-type
version | | 0..0 | string | Version of the system - if relevant
code | | 1..1 | code | Symbol in syntax defined by the system Fixed Value: logic-library
display | | 1..1 | string | Representation defined by the system Fixed Value: Logic Library
userSelected | | 0..0 | boolean | If this coding was chosen directly by the user
text | | 0..0 | string | Plain text representation of the concept
subject[x] | I | 0..1 | | Type of individual the library content is focused on Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | ΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 0..1 | string | Name of the publisher (organization or individual)
contact | ΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 0..1 | markdown | Natural language description of the library
useContext | ΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | ΣI | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this library is defined
usage | I | 0..1 | string | Describes the clinical usage of the library
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the library was approved by publisher
lastReviewDate | I | 0..1 | date | When the library was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the library is expected to be used
topic | I | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | SI | 0..* | ParameterDefinition | Parameters defined by the library
dataRequirement | SI | 0..* | DataRequirement | What data is referenced by this library
content | ΣI | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

XML Template

```
<!-- CQL Library -->

<Library xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  0..1 Canonical identifier for this library, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!--  0..1 Business version of the library -->
 <name value="[string]"/><!--  0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this library (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the library -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  0..1 For testing purposes, not real usage -->
 <type><!--  1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition   --></type>
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) Type of individual the library content is focused on
    --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  0..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the library -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
```

```
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for library (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this library is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the library -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the library was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the library is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept E.g. Education, Treatment, Assessment, etc.   --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!--  0..* ParameterDefinition
     Parameters defined by the library --></parameter>
 <dataRequirement><!--  0..* DataRequirement
     What data is referenced by this library --></dataRequirement>
 <content><!--  0..* Attachment
     Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{ // CQL Library
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C?Canonical identifier for this library, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the library
    "version" : "<string>", // C?Business version of the library
    "name" : "<string>", // C?Name for this library (computer friendly)
    "title" : "<string>", // C?Name for this library (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the library
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C?For testing purposes, not real usage
    "type" : { CodeableConcept }, // C? R!
// value[x]: Type of individual the library content is focused on. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?Type of individual the library content is focused on
    "subjectReference" : { Reference(Group) }, // C?Type of individual the library content is focused on
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C?Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C?Natural language description of the library
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for library (if applicable)
    "purpose" : "<markdown>", // C?Why this library is defined
    "usage" : "<string>", // C?Describes the clinical usage of the library
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the library was approved by publisher
    "lastReviewDate" : "<date>", // C?When the library was last reviewed
    "effectivePeriod" : { Period }, // C?When the library is expected to be used
    "topic" : [{ CodeableConcept }], // C?E.g. Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "parameter" : [{ ParameterDefinition }], // C?Parameters defined by the library
    "dataRequirement" : [{ DataRequirement }] // C?What data is referenced by this library
    "content" : [{ Attachment }], // C?Contents of the library, either embedded or referenced
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.11.12.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Library.language	Common Languages	preferred	Common Languages
Library.status	PublicationStatus	required	PublicationStatus
Library.type	LibraryType	extensible	LibraryType
Library.subject[x]	Subject type	extensible	Subject type
Library.jurisdiction	Jurisdiction	extensible	Jurisdiction
Library.topic	DefinitionTopic	example	DefinitionTopic

14.11.13 JSON Schema for Library

URL: https://hl7.org/fhir/R4/library.schema.json.html

14.11.13 JSON Schema for Library

This is a representation of the json schema for Library, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Library",
  "$ref": "#/definitions/Library",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Library": {
      "description": "The Library resource is a general-purpose container for knowledge asset definitions. It can be
    used to describe and expose existing knowledge assets such as logic libraries and information model descriptions, as
     well as to describe a collection of knowledge assets.",
      "properties": {
        "resourceType": {
          "description": "This is a Library resource",
          "const": "Library"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "url": {
          "description": "An absolute URI that is used to identify this library when it is referenced in a
    specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
```

```
and SHOULD be a literal address at which at which an authoritative instance of this library is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the library is stored
on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this library when it is represented in other
formats, or referenced in a specification, model, design or an instance. e.g. CMS or NQF identifiers for a measure
artifact. Note that at least one identifier is required for non-experimental active artifacts.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the library when it is referenced in a
 specification, model, design or instance. This is an arbitrary value managed by the library author and is not
expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not
available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a
version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g.
1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification.
Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the library. This name should be usable as an identifier
for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the library.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the library giving additional information about its
content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this library. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this library is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Identifies the type of library such as a Logic Library, Model Definition, Asset Collection, or
 Module Definition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectCodeableConcept": {
      "description": "A code or group definition that describes the intended subject of the contents of the
library.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
```

```
    },
    "subjectReference": {
      "description": "A code or group definition that describes the intended subject of the contents of the
library.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the library was published. The date must change when the
business version changes and it must change if the status code changes. In addition, it should change when the
substantive content of the library changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the library.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the library from a consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate library instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the library is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this library is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description of how the library is used from a clinical perspective.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the library and/or its contents. Copyright statements are
generally legal restrictions on the use and publishing of the library.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

```
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the library content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the library. Topics provide a high-level
categorization of the library that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "parameter": {
      "description": "The parameter element defines parameters used by the library.",
      "items": {
        "$ref": "ParameterDefinition.schema.json#/definitions/ParameterDefinition"
      },
      "type": "array"
    },
    "dataRequirement": {
      "description": "Describes a set of data that must be provided in order to be able to successfully perform the
computations defined by the library.",
      "items": {
        "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
      },
      "type": "array"
    },
    "content": {
      "description": "The content of the library as an Attachment. The content may be a reference to a url, or may
be directly embedded as a base-64 string. Either way, the contentType of the attachment determines how to interpret
the content.",
      "items": {
        "$ref": "Attachment.schema.json#/definitions/Attachment"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "type",
    "resourceType"
```

```
      ]
    }
  }
}
```

14.11.14 Resource Library - Maturity Dependency View

URL: https://hl7.org/fhir/R4/library-dependencies.html

14.11.14 Resource Library - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Library FMM level 2

Name | Card. | Type | Dependency Analysis
Library | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
type | 1..1 | CodeableConcept
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
parameter | 0..* | ParameterDefinition
dataRequirement | 0..* | DataRequirement
content | 0..* | Attachment

14.12 Resource GuidanceResponse - Content

URL: https://hl7.org/fhir/R4/guidanceresponse.html
Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Not linked to any defined compartments

A guidance response is the formal response to a guidance request, including any output parameters returned by the evaluation, as well as the description of any proposed actions to be taken.

14.12.1 Scope and Usage

The GuidanceResponse resource is used to represent the result of invoking a decision support service. It provides a container for the status of the response, any warnings or messages returned by the service, as well as any output data from the service and any suggested actions to be performed.

Use cases for the GuidanceResponse resource include:

• Keeping a record of past service requests, including the ability to reference the context of the request such as the disease state

• Representing additional data requirements for subsequent service requests

• Representing Public Health case reporting and recording

• Sharing the results of decision support guidance and determination of reportability with clinicians and public health agencies

In addition, the GuidanceResponse resource is aligned with the CDS Hooks response structure and so provides a mechanism to record the results of CDS Hooks interactions. For a detailed discussion of how the GuidanceResponse resource can be used for this purpose, refer to the Evaluation topic in the Clinical Reasoning module.

14.12.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
GuidanceResponse | TU | | DomainResource | The formal response to a guidance request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
requestIdentifier | Σ | 0..1 | Identifier | The identifier of the request associated with this response, if any
identifier | Σ | 0..* | Identifier | Business identifier
module[x] | Σ | 1..1 | | What guidance was requested
moduleUri | | | uri
moduleCanonical | | | canonical()
moduleCodeableConcept | | | CodeableConcept
status | ?!Σ | 1..1 | code | success | data-requested | data-required | in-progress | failure | entered-in-error
GuidanceResponseStatus (Required)
subject | | 0..1 | Reference(Patient | Group) | Patient the request was performed for
encounter | | 0..1 | Reference(Encounter) | Encounter during which the response was returned
occurrenceDateTime | | 0..1 | dateTime | When the guidance response was processed
performer | | 0..1 | Reference(Device) | Device returning the guidance
reasonCode | | 0..* | CodeableConcept | Why guidance is needed
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why guidance is needed
note | | 0..* | Annotation | Additional notes about the response
evaluationMessage | | 0..* | Reference(OperationOutcome) | Messages resulting from the evaluation of the artifact or artifacts
outputParameters | | 0..1 | Reference(Parameters) | The output parameters of the evaluation, if any
result | | 0..1 | Reference(CarePlan | RequestGroup) | Proposed actions, if any
dataRequirement | | 0..* | DataRequirement | Additional required data
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GuidanceResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <requestIdentifier><!-- 0..1 Identifier The identifier of the request associated with this response, if any
    --></requestIdentifier>
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <module[x]><!-- 1..1 uri|canonical|CodeableConcept What guidance was requested --></module[x]>
 <status value="[code]"/><!-- 1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
    -->
 <subject><!-- 0..1 Reference(Patient|Group) Patient the request was performed for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter during which the response was returned --></encounter>
 <occurrenceDateTime value="[dateTime]"/><!-- 0..1 When the guidance response was processed -->
 <performer><!-- 0..1 Reference(Device) Device returning the guidance --></performer>
 <reasonCode><!-- 0..* CodeableConcept Why guidance is needed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why guidance is needed --></reasonReference>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <evaluationMessage><!-- 0..* Reference(OperationOutcome) Messages resulting from the evaluation of the artifact or
    artifacts --></evaluationMessage>
 <outputParameters><!-- 0..1 Reference(Parameters) The output parameters of the evaluation, if any
    --></outputParameters>
 <result><!-- 0..1 Reference(CarePlan|RequestGroup) Proposed actions, if any --></result>
 <dataRequirement><!-- 0..* DataRequirement Additional required data --></dataRequirement>
</GuidanceResponse>
```

JSON Template

```
{
  "resourceType" : "GuidanceResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "requestIdentifier" : { Identifier }, // The identifier of the request associated with this response, if any
  "identifier" : [{ Identifier }], // Business identifier
  // module[x]: What guidance was requested. One of these 3:
  "moduleUri" : "<uri>",
  "moduleCanonical" : "<canonical>",
```

```
  "moduleCodeableConcept" : { CodeableConcept },
  "status" : "<code>", // R!  success | data-requested | data-required | in-progress | failure | entered-in-error
  "subject" : { Reference(Patient|Group) }, // Patient the request was performed for
  "encounter" : { Reference(Encounter) }, // Encounter during which the response was returned
  "occurrenceDateTime" : "<dateTime>", // When the guidance response was processed
  "performer" : { Reference(Device) }, // Device returning the guidance
  "reasonCode" : [{ CodeableConcept }], // Why guidance is needed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why guidance is needed
  "note" : [{ Annotation }], // Additional notes about the response
  "evaluationMessage" : [{ Reference(OperationOutcome) }], // Messages resulting from the evaluation of the artifact or
    artifacts
  "outputParameters" : { Reference(Parameters) }, // The output parameters of the evaluation, if any
  "result" : { Reference(CarePlan|RequestGroup) }, // Proposed actions, if any
  "dataRequirement" : [{ DataRequirement }] // Additional required data
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GuidanceResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GuidanceResponse.requestIdentifier [ Identifier ]; # 0..1 The identifier of the request associated with this
    response, if any
  fhir:GuidanceResponse.identifier [ Identifier ], ... ; # 0..* Business identifier
  # GuidanceResponse.module[x] : 1..1 What guidance was requested. One of these 3
    fhir:GuidanceResponse.moduleUri [ uri ]
    fhir:GuidanceResponse.moduleCanonical [ canonical ]
    fhir:GuidanceResponse.moduleCodeableConcept [ CodeableConcept ]
  fhir:GuidanceResponse.status [ code ]; # 1..1 success | data-requested | data-required | in-progress | failure |
    entered-in-error
  fhir:GuidanceResponse.subject [ Reference(Patient|Group) ]; # 0..1 Patient the request was performed for
  fhir:GuidanceResponse.encounter [ Reference(Encounter) ]; # 0..1 Encounter during which the response was returned
  fhir:GuidanceResponse.occurrenceDateTime [ dateTime ]; # 0..1 When the guidance response was processed
  fhir:GuidanceResponse.performer [ Reference(Device) ]; # 0..1 Device returning the guidance
  fhir:GuidanceResponse.reasonCode [ CodeableConcept ], ... ; # 0..* Why guidance is needed
  fhir:GuidanceResponse.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
     0..* Why guidance is needed
  fhir:GuidanceResponse.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:GuidanceResponse.evaluationMessage [ Reference(OperationOutcome) ], ... ; # 0..* Messages resulting from the
    evaluation of the artifact or artifacts
  fhir:GuidanceResponse.outputParameters [ Reference(Parameters) ]; # 0..1 The output parameters of the evaluation, if
    any
  fhir:GuidanceResponse.result [ Reference(CarePlan|RequestGroup) ]; # 0..1 Proposed actions, if any
  fhir:GuidanceResponse.dataRequirement [ DataRequirement ], ... ; # 0..* Additional required data
]
```

Changes since R3

GuidanceResponse

GuidanceResponse.requestIdentifier | Renamed from requestId to requestIdentifier Type changed from id to Identifier

GuidanceResponse.identifier | Max Cardinality changed from 1 to *

GuidanceResponse.module[x] | Renamed from module to module[x] Add Types uri, canonical, CodeableConcept Remove Type Reference(ServiceDefinition)

GuidanceResponse.status | Change value set from http://hl7.org/fhir/ValueSet/guidance-response-status to http://hl7.org/fhir/ValueSet/guidance-response-status|4.0.1

GuidanceResponse.encounter | Added Element

GuidanceResponse.reasonCode | Added Element

GuidanceResponse.reasonReference | Added Element

GuidanceResponse.context | deleted

GuidanceResponse.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

Structure

Name | Flags | Card. | Type | Description & Constraints

GuidanceResponse | TU | | DomainResource | The formal response to a guidance request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

requestIdentifier | Σ | 0..1 | Identifier | The identifier of the request associated with this response, if any

identifier | Σ | 0..* | Identifier | Business identifier

module[x] | Σ | 1..1 | | What guidance was requested

moduleUri | | | uri

moduleCanonical | | | canonical()

moduleCodeableConcept | | | CodeableConcept

status | ?!Σ | 1..1 | code | success | data-requested | data-required | in-progress | failure | entered-in-error

GuidanceResponseStatus (Required)

subject | | 0..1 | Reference(Patient | Group) | Patient the request was performed for
encounter | | 0..1 | Reference(Encounter) | Encounter during which the response was returned
occurrenceDateTime | | 0..1 | dateTime | When the guidance response was processed
performer | | 0..1 | Reference(Device) | Device returning the guidance
reasonCode | | 0..* | CodeableConcept | Why guidance is needed
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why guidance is needed
note | | 0..* | Annotation | Additional notes about the response
evaluationMessage | | 0..* | Reference(OperationOutcome) | Messages resulting from the evaluation of the artifact or artifacts
outputParameters | | 0..1 | Reference(Parameters) | The output parameters of the evaluation, if any
result | | 0..1 | Reference(CarePlan | RequestGroup) | Proposed actions, if any
dataRequirement | | 0..* | DataRequirement | Additional required data
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GuidanceResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <requestIdentifier><!-- 0..1 Identifier The identifier of the request associated with this response, if any
    --></requestIdentifier>
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <module[x]><!-- 1..1 uri|canonical|CodeableConcept What guidance was requested --></module[x]>
 <status value="[code]"/><!-- 1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
    -->
 <subject><!-- 0..1 Reference(Patient|Group) Patient the request was performed for --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Encounter during which the response was returned --></encounter>
 <occurrenceDateTime value="[dateTime]"/><!-- 0..1 When the guidance response was processed -->
 <performer><!-- 0..1 Reference(Device) Device returning the guidance --></performer>
 <reasonCode><!-- 0..* CodeableConcept Why guidance is needed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why guidance is needed --></reasonReference>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <evaluationMessage><!-- 0..* Reference(OperationOutcome) Messages resulting from the evaluation of the artifact or
    artifacts --></evaluationMessage>
 <outputParameters><!-- 0..1 Reference(Parameters) The output parameters of the evaluation, if any
    --></outputParameters>
 <result><!-- 0..1 Reference(CarePlan|RequestGroup) Proposed actions, if any --></result>
 <dataRequirement><!-- 0..* DataRequirement Additional required data --></dataRequirement>
</GuidanceResponse>
```

JSON Template

```
{
  "resourceType" : "GuidanceResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "requestIdentifier" : { Identifier }, // The identifier of the request associated with this response, if any
  "identifier" : [{ Identifier }], // Business identifier
  // module[x]: What guidance was requested. One of these 3:
  "moduleUri" : "<uri>",
  "moduleCanonical" : "<canonical>",
  "moduleCodeableConcept" : { CodeableConcept },
  "status" : "<code>", // R!  success | data-requested | data-required | in-progress | failure | entered-in-error
  "subject" : { Reference(Patient|Group) }, // Patient the request was performed for
  "encounter" : { Reference(Encounter) }, // Encounter during which the response was returned
  "occurrenceDateTime" : "<dateTime>", // When the guidance response was processed
  "performer" : { Reference(Device) }, // Device returning the guidance
  "reasonCode" : [{ CodeableConcept }], // Why guidance is needed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why guidance is needed
  "note" : [{ Annotation }], // Additional notes about the response
  "evaluationMessage" : [{ Reference(OperationOutcome) }], // Messages resulting from the evaluation of the artifact or
    artifacts
  "outputParameters" : { Reference(Parameters) }, // The output parameters of the evaluation, if any
  "result" : { Reference(CarePlan|RequestGroup) }, // Proposed actions, if any
  "dataRequirement" : [{ DataRequirement }] // Additional required data
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GuidanceResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GuidanceResponse.requestIdentifier [ Identifier ]; # 0..1 The identifier of the request associated with this
    response, if any
  fhir:GuidanceResponse.identifier [ Identifier ], ... ; # 0..* Business identifier
  # GuidanceResponse.module[x] : 1..1 What guidance was requested. One of these 3
    fhir:GuidanceResponse.moduleUri [ uri ]
    fhir:GuidanceResponse.moduleCanonical [ canonical ]
    fhir:GuidanceResponse.moduleCodeableConcept [ CodeableConcept ]
  fhir:GuidanceResponse.status [ code ]; # 1..1 success | data-requested | data-required | in-progress | failure |
    entered-in-error
```

```
  fhir:GuidanceResponse.subject [ Reference(Patient|Group) ]; # 0..1 Patient the request was performed for
  fhir:GuidanceResponse.encounter [ Reference(Encounter) ]; # 0..1 Encounter during which the response was returned
  fhir:GuidanceResponse.occurrenceDateTime [ dateTime ]; # 0..1 When the guidance response was processed
  fhir:GuidanceResponse.performer [ Reference(Device) ]; # 0..1 Device returning the guidance
  fhir:GuidanceResponse.reasonCode [ CodeableConcept ], ... ; # 0..* Why guidance is needed
  fhir:GuidanceResponse.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
     0..* Why guidance is needed
  fhir:GuidanceResponse.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:GuidanceResponse.evaluationMessage [ Reference(OperationOutcome) ], ... ; # 0..* Messages resulting from the
    evaluation of the artifact or artifacts
  fhir:GuidanceResponse.outputParameters [ Reference(Parameters) ]; # 0..1 The output parameters of the evaluation, if
    any
  fhir:GuidanceResponse.result [ Reference(CarePlan|RequestGroup) ]; # 0..1 Proposed actions, if any
  fhir:GuidanceResponse.dataRequirement [ DataRequirement ], ... ; # 0..* Additional required data
]
```

Changes since Release 3

GuidanceResponse

GuidanceResponse.requestIdentifier | Renamed from requestId to requestIdentifier Type changed from id to Identifier

GuidanceResponse.identifier | Max Cardinality changed from 1 to *

GuidanceResponse.module[x] | Renamed from module to module[x] Add Types uri, canonical, CodeableConcept Remove Type Reference(ServiceDefinition)

GuidanceResponse.status | Change value set from http://hl7.org/fhir/ValueSet/guidance-response-status to http://hl7.org/fhir/ValueSet/guidance-response-status|4.0.1

GuidanceResponse.encounter | Added Element

GuidanceResponse.reasonCode | Added Element

GuidanceResponse.reasonReference | Added Element

GuidanceResponse.context | deleted

GuidanceResponse.reason[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.12.2.1 Terminology Bindings

Path | Definition | Type | Reference

GuidanceResponse.status | The status of a guidance response. | Required | GuidanceResponseStatus

14.12.3 GuidanceResponse Search

Although the GuidanceResponse resource does define search parameters, it is up to the individual decision support service whether or not historical records of guidance requests are preserved. A service might not support searching on GuidanceResponses at all, or it may support searching only for a pre-defined expiration period. The search parameters are defined to provide consumers with a consistent interface to searching if it is available for a specific service implementation. In any case, services should detail the support they do provide using a CapabilityStatement.

14.12.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

identifier | token | The identifier of the guidance response | GuidanceResponse.identifier

patient | reference | The identity of a patient to search for guidance response results | GuidanceResponse.subject.where(resolve() is Patient) (Patient)

request | token | The identifier of the request associated with the response | GuidanceResponse.requestIdentifier

subject | reference | The subject that the guidance response is about | GuidanceResponse.subject (Group, Patient)

14.12.5 Resource GuidanceResponse - Examples

URL: https://hl7.org/fhir/R4/guidanceresponse-examples.html

This example illustrates the use of the GuidanceResponse resource to represent the response returned by an $evaluate operation against the Guideline Appropriate Ordering decision support service module.

14.12.5 Resource GuidanceResponse - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

Example Name | id | Format

Example of guidanceresponse | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.12.6 Resource GuidanceResponse - Detailed Descriptions

URL: https://hl7.org/fhir/R4/guidanceresponse-definitions.html

14.12.6 Resource GuidanceResponse - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Patient | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the GuidanceResponse resource.

GuidanceResponse
Element Id | GuidanceResponse
Definition | A guidance response is the formal response to a guidance request, including any output parameters returned by the evaluation, as well as the description of any proposed actions to be taken.
Cardinality | 0..*
Type | DomainResource
Requirements | The GuidanceResponse resource supports recording the results of decision support interactions, reportability determination for public health, as well as the communication of additional data requirements for subsequent interactions.
GuidanceResponse.requestIdentifier
Element Id | GuidanceResponse.requestIdentifier
Definition | The identifier of the request associated with this response. If an identifier was given as part of the request, it will be reproduced here to enable the requester to more easily identify the response in a multi-request scenario.
Cardinality | 0..1
Type | Identifier
Summary | true
GuidanceResponse.identifier
Element Id | GuidanceResponse.identifier
Definition | Allows a service to provide unique, business identifiers for the response.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Summary | true
GuidanceResponse.module[x]
Element Id | GuidanceResponse.module[x]
Definition | An identifier, CodeableConcept or canonical reference to the guidance that was requested.
Cardinality | 1..1
Type | uri|canonical|CodeableConcept
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
GuidanceResponse.status
Element Id | GuidanceResponse.status
Definition | The status of the response. If the evaluation is completed successfully, the status will indicate success. However, in order to complete the evaluation, the engine may require more information. In this case, the status will be data-required, and the response will contain a description of the additional required information. If the evaluation completed successfully, but the engine determines that a potentially more accurate response could be provided if more data was available, the status will be data-requested, and the response will contain a description of the additional requested information.
Cardinality | 1..1
Terminology Binding | GuidanceResponseStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
GuidanceResponse.subject
Element Id | GuidanceResponse.subject
Definition | The patient for which the request was processed.
Cardinality | 0..1
Type | Reference(Patient | Group)
GuidanceResponse.encounter
Element Id | GuidanceResponse.encounter
Definition | The encounter during which this response was created or to which the creation of this record is tightly associated.
Cardinality | 0..1
Type | Reference(Encounter)
Comments | This will typically be the encounter the event occurred within, but some activities may be initiated prior to or after the official copmletion of an encounter but still be tied to the context of the encounter.
GuidanceResponse.occurrenceDateTime
Element Id | GuidanceResponse.occurrenceDateTime
Definition | Indicates when the guidance response was processed.
Cardinality | 0..1
Type | dateTime
GuidanceResponse.performer

Element Id | GuidanceResponse.performer
Definition | Provides a reference to the device that performed the guidance.
Cardinality | 0..1
Type | Reference(Device)
GuidanceResponse.reasonCode
Element Id | GuidanceResponse.reasonCode
Definition | Describes the reason for the guidance response in coded or textual form.
Cardinality | 0..*
Type | CodeableConcept
GuidanceResponse.reasonReference
Element Id | GuidanceResponse.reasonReference
Definition | Indicates the reason the request was initiated. This is typically provided as a parameter to the evaluation and echoed by the service, although for some use cases, such as subscription- or event-based scenarios, it may provide an indication of the cause for the response.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
GuidanceResponse.note
Element Id | GuidanceResponse.note
Definition | Provides a mechanism to communicate additional information about the response.
Cardinality | 0..*
Type | Annotation
GuidanceResponse.evaluationMessage
Element Id | GuidanceResponse.evaluationMessage
Definition | Messages resulting from the evaluation of the artifact or artifacts. As part of evaluating the request, the engine may produce informational or warning messages. These messages will be provided by this element.
Cardinality | 0..*
Type | Reference(OperationOutcome)
GuidanceResponse.outputParameters
Element Id | GuidanceResponse.outputParameters
Definition | The output parameters of the evaluation, if any. Many modules will result in the return of specific resources such as procedure or communication requests that are returned as part of the operation result. However, modules may define specific outputs that would be returned as the result of the evaluation, and these would be returned in this element.
Cardinality | 0..1
Type | Reference(Parameters)
GuidanceResponse.result
Element Id | GuidanceResponse.result
Definition | The actions, if any, produced by the evaluation of the artifact.
Cardinality | 0..1
Type | Reference(CarePlan | RequestGroup)
GuidanceResponse.dataRequirement
Element Id | GuidanceResponse.dataRequirement
Definition | If the evaluation could not be completed due to lack of information, or additional information would potentially result in a more accurate response, this element will a description of the data required in order to proceed with the evaluation. A subsequent request to the service should include this data.
Cardinality | 0..*
Type | DataRequirement

14.12.7 Resource GuidanceResponse - Mappings

URL: https://hl7.org/fhir/R4/guidanceresponse-mappings.html

14.12.7 Resource GuidanceResponse - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

Mappings for the guidanceresponse resource (see Mappings to Other Standards for further information & status).

14.12.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

GuidanceResponse
identifier | FiveWs.identifier
status | FiveWs.status
subject | FiveWs.subject[x]
encounter | FiveWs.context
occurrenceDateTime | FiveWs.done[x]
performer | FiveWs.actor
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]

14.12.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

GuidanceResponse | Event
requestIdentifier | Event.basedOn
identifier | Event.identifier
module[x] | Event.instantiatesUri | Event.instantiatesCanonical
status | Event.status
subject | Event.subject
encounter | Event.encounter
occurrenceDateTime | Event.occurrence[x]
performer | Event.performer.actor
reasonCode | Event.reasonCode
reasonReference | Event.reasonReference
note | Event.note

14.12.8 Resource GuidanceResponse - Extensions & Profiles

URL: https://hl7.org/fhir/R4/guidanceresponse-profiles.html

14.12.8 Resource GuidanceResponse - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the GuidanceResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.12.8.1 Profiles

Profile | Description | Context
CDSHooksGuidanceResponse | Defines a GuidanceResponse that represents the response container for a CDS Hooks response | CDS Hooks GuidanceResponse

14.12.8.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.12.8.3 Search Extensions

No Search Extensions defined for this resource

14.12.9 Resource GuidanceResponse - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/guidanceresponse-version-maps.html

No validation errors - all conversions are clean

14.12.9 Resource GuidanceResponse - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Patient | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for GuidanceResponse.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

14.12.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/GuidanceResponse3to4" = "R3 to R4 Conversions for GuidanceResponse"

uses "http://hl7.org/fhir/3.0/StructureDefinition/GuidanceResponse" alias GuidanceResponseR3 as source
uses "http://hl7.org/fhir/StructureDefinition/GuidanceResponse" alias GuidanceResponse as target

imports "http://hl7.org/fhir/StructureMap/*3to4"
```

```
group GuidanceResponse(source src : GuidanceResponseR3, target tgt : GuidanceResponse) extends DomainResource <<type+>>
    {
  src.requestId as v ->  tgt.requestIdentifier as ii,  ii.value = v;
  src.identifier -> tgt.identifier;
  src.module as vs -> tgt.module = create('canonical') as vt then Reference2Canonical(vs, vt);
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.occurrenceDateTime -> tgt.occurrenceDateTime;
  src.performer -> tgt.performer;
  src.reason : CodeableConcept as vs -> tgt.reasonCode as vt then CodeableConcept(vs, vt);
  src.reason : Reference as vs -> tgt.reasonReference as vt then Reference(vs, vt);
  src.note -> tgt.note;
  src.evaluationMessage -> tgt.evaluationMessage;
  src.outputParameters -> tgt.outputParameters;
  src.result -> tgt.result;
  src.dataRequirement -> tgt.dataRequirement;
}
```

14.12.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/GuidanceResponse4to3" = "R4 to R3 Conversion for GuidanceResponse"

uses "http://hl7.org/fhir/StructureDefinition/GuidanceResponse" alias GuidanceResponse as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/GuidanceResponse" alias GuidanceResponseR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group GuidanceResponse(source src : GuidanceResponseR3, target tgt : GuidanceResponse) extends DomainResource <<type+>>
    {
  src.requestIdentifier as ii then {
    ii.value -> tgt.requestId;
  };
  src.identifier -> tgt.identifier;
  src.module : canonical as vs -> tgt.module as vt then canonical2Reference(vs, vt);
  src.status -> tgt.status;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.occurrenceDateTime -> tgt.occurrenceDateTime;
  src.performer -> tgt.performer;
  src.reasonCode as vs -> tgt.reason = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.reasonReference as vs -> tgt.reason = create('Reference') as vt then Reference(vs, vt);
  src.note -> tgt.note;
  src.evaluationMessage -> tgt.evaluationMessage;
  src.outputParameters -> tgt.outputParameters;
  src.result -> tgt.result;
  src.dataRequirement -> tgt.dataRequirement;
}
```

14.12.9.3 R4 Validation Errors

14.12.10 CDS Hooks GuidanceResponse

URL: https://hl7.org/fhir/R4/guidanceresponse-cdshooks.html

Defines a GuidanceResponse that represents the response container for a CDS Hooks response

14.12.10 CDS Hooks GuidanceResponse

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.12.10.1 Content

Profiles:
CDSHooksGuidanceResponse | Defines a GuidanceResponse that represents the response container for a CDS Hooks response

14.12.10.2 StructureDefinition: CDSHooksGuidanceResponse

URL: https://hl7.org/fhir/R4/cdshooksguidanceresponse.html

14.12.10.2 StructureDefinition: CDSHooksGuidanceResponse

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cdshooksguidanceresponse`

Defines a GuidanceResponse that represents the response container for a CDS Hooks response

This profile was published on [no date] as a draft by HL7.

14.12.10.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

• Text Summary
• Differential Table

- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from GuidanceResponse.

Summary

Mandatory: 4 elements Must-Support: 4 elements

Extensions

This structure refers to these extensions:

- http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint

This structure is derived from GuidanceResponse.

Name	Flags	Card.	Type	Description & Constraints
GuidanceResponse		1..1		
cqf-cdsHooksEndpoint		1..1	uri	Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
requestIdentifier		1..1	Identifier	
identifier		1..1	Identifier	
moduleUri		1..1	uri	
subject	S	0..1	Reference(Patient | Group)	
occurrenceDateTime	S	0..1	dateTime	
performer	S	0..1	Reference(Device)	
result	S	0..1	Reference(CarePlan | RequestGroup)	

Documentation for this format

Name	Flags	Card.	Type	Description & Constraints
GuidanceResponse	I	1..1		The formal response to a guidance request
id	Σ	0..1	string	Logical id of this artifact
meta	ΣI	0..1	Meta	Metadata about the resource
implicitRules	?!ΣI	0..1	uri	A set of rules under which this content was created
language	I	0..1	code	Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text	I	0..1	Narrative	Text summary of the resource, for human interpretation
contained		0..*	Resource	Contained, inline Resources
extension	I	0..*	Extension	Extension Slice: Unordered, Open by value:url
cqf-cdsHooksEndpoint	I	1..1	uri	Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
modifierExtension	?!I	0..*	Extension	Extensions that cannot be ignored
requestIdentifier	ΣI	1..1	Identifier	The identifier of the request associated with this response, if any
identifier	ΣI	1..1	Identifier	Business identifier
module[x]	ΣI		(Slice Definition)	What guidance was requested Slice: Unordered, Closed by type:$this
module[x]:moduleUri	ΣI	1..1	uri	What guidance was requested
status	?!ΣI	1..1	code	success | data-requested | data-required | in-progress | failure | entered-in-error Binding: GuidanceResponseStatus (required)
subject	SI	0..1	Reference(Patient | Group)	Patient the request was performed for
encounter	I	0..1	Reference(Encounter)	Encounter during which the response was returned
occurrenceDateTime	SI	0..1	dateTime	When the guidance response was processed
performer	SI	0..1	Reference(Device)	Device returning the guidance
reasonCode	I	0..*	CodeableConcept	Why guidance is needed
reasonReference	I	0..*	Reference(Condition | Observation | DiagnosticReport | DocumentReference)	Why guidance is needed
note	I	0..*	Annotation	Additional notes about the response
evaluationMessage	I	0..*	Reference(OperationOutcome)	Messages resulting from the evaluation of the artifact or artifacts
outputParameters	I	0..1	Reference(Parameters)	The output parameters of the evaluation, if any
result	SI	0..1	Reference(CarePlan | RequestGroup)	Proposed actions, if any
dataRequirement	I	0..*	DataRequirement	Additional required data

Documentation for this format

```
<!-- CDS Hooks GuidanceResponse -->

<GuidanceResponse xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <requestIdentifier><!--  1..1 Identifier
     The identifier of the request associated with this response, if any --></requestIdentifier>
 <identifier><!--  1..1 Identifier Business identifier --></identifier>
 <-- moduleUri sliced by type:$this  in the specified orderClosed-->
 <moduleUri value="[uri]"/><!--  1..1 What guidance was requested -->
 <status value="[code]"/><!--  1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
```

```
      -->
 <subject><!--  0..1 Reference(Patient)
     Patient the request was performed for --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Encounter during which the response was returned --></encounter>
 <occurrenceDateTime value="[dateTime]"/><!--  0..1 When the guidance response was processed -->
 <performer><!--  0..1 Reference(Device)
     Device returning the guidance --></performer>
 <reasonCode><!--  0..* CodeableConcept
     Why guidance is needed --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why guidance is needed --></reasonReference>
 <note><!--  0..* Annotation
     Additional notes about the response --></note>
 <evaluationMessage><!--  0..* Reference(OperationOutcome)
     Messages resulting from the evaluation of the artifact or artifacts --></evaluationMessage>
 <outputParameters><!--  0..1 Reference(Parameters)
     The output parameters of the evaluation, if any --></outputParameters>
 <result><!--  0..1 Reference(CarePlan) Proposed actions, if any --></result>
 <dataRequirement><!--  0..* DataRequirement
     Additional required data --></dataRequirement>
</GuidanceResponse>
```

```
{ // CDS Hooks GuidanceResponse
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Service endpoint // C? R!
        // from Element: extension
      }
    ],
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "requestIdentifier" : { Identifier }, // C? R! The identifier of the request associated with this response, if any
    "identifier" : { Identifier }, // C? R! Business identifier
    "moduleUri" : [ //  sliced by type:$this  in the specified order, Closed
      { // What guidance was requested // C? R!
        // from Element: extension
      }
    ],
    "status" : "<code>", // C? R! success | data-requested | data-required | in-progress | failure | entered-in-error
    "subject" : { Reference(Patient) }, // C?Patient the request was performed for
    "encounter" : { Reference(Encounter) }, // C?Encounter during which the response was returned
    "occurrenceDateTime" : "<dateTime>", // C?When the guidance response was processed
    "performer" : { Reference(Device) }, // C?Device returning the guidance
    "reasonCode" : [{ CodeableConcept }], // C?Why guidance is needed
    "reasonReference" : [{ Reference(Condition) }], // C?Why guidance is needed
    "note" : [{ Annotation }], // C?Additional notes about the response
    "evaluationMessage" : [{ Reference(OperationOutcome) }], // C?Messages resulting from the evaluation of the artifact
     or artifacts
    "outputParameters" : { Reference(Parameters) }, // C?The output parameters of the evaluation, if any
    "result" : { Reference(CarePlan) }, // C?Proposed actions, if any
    "dataRequirement" : [{ DataRequirement }], // C?Additional required data
  }
```

This structure is derived from GuidanceResponse.

Summary

Mandatory: 4 elements Must-Support: 4 elements

Extensions

This structure refers to these extensions:

• http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint

Differential View

This structure is derived from GuidanceResponse.

Name	Flags	Card.	Type	Description & Constraints
GuidanceResponse		1..1		
cqf-cdsHooksEndpoint		1..1	uri	Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
requestIdentifier		1..1	Identifier	
identifier		1..1	Identifier	
moduleUri		1..1	uri	
subject	S	0..1	Reference(Patient | Group)	
occurrenceDateTime	S	0..1	dateTime	
performer	S	0..1	Reference(Device)	
result	S	0..1	Reference(CarePlan | RequestGroup)	

Documentation for this format

Snapshot View

Name	Flags	Card.	Type	Description & Constraints
GuidanceResponse	I	1..1		The formal response to a guidance request

id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Extension Slice: Unordered, Open by value:url
cqf-cdsHooksEndpoint | I | 1..1 | uri | Service endpoint URL: http://hl7.org/fhir/StructureDefinition/cqf-cdsHooksEndpoint
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
requestIdentifier | ΣI | 1..1 | Identifier | The identifier of the request associated with this response, if any
identifier | ΣI | 1..1 | Identifier | Business identifier
module[x] | ΣI | | (Slice Definition) | What guidance was requested Slice: Unordered, Closed by type:$this
module[x]:moduleUri | ΣI | 1..1 | uri | What guidance was requested
status | ?!ΣI | 1..1 | code | success | data-requested | data-required | in-progress | failure | entered-in-error Binding: GuidanceResponseStatus (required)
subject | SI | 0..1 | Reference(Patient | Group) | Patient the request was performed for
encounter | I | 0..1 | Reference(Encounter) | Encounter during which the response was returned
occurrenceDateTime | SI | 0..1 | dateTime | When the guidance response was processed
performer | SI | 0..1 | Reference(Device) | Device returning the guidance
reasonCode | I | 0..* | CodeableConcept | Why guidance is needed
reasonReference | I | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why guidance is needed
note | I | 0..* | Annotation | Additional notes about the response
evaluationMessage | I | 0..* | Reference(OperationOutcome) | Messages resulting from the evaluation of the artifact or artifacts
outputParameters | I | 0..1 | Reference(Parameters) | The output parameters of the evaluation, if any
result | SI | 0..1 | Reference(CarePlan | RequestGroup) | Proposed actions, if any
dataRequirement | I | 0..* | DataRequirement | Additional required data
Documentation for this format

XML Template

```
<!-- CDS Hooks GuidanceResponse -->

<GuidanceResponse xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource
     Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <requestIdentifier><!--  1..1 Identifier
     The identifier of the request associated with this response, if any --></requestIdentifier>
 <identifier><!--  1..1 Identifier Business identifier --></identifier>
 <-- moduleUri sliced by type:$this  in the specified orderClosed-->
 <moduleUri value="[uri]"/><!--  1..1 What guidance was requested -->
 <status value="[code]"/><!--  1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
       -->
 <subject><!--  0..1 Reference(Patient)
     Patient the request was performed for --></subject>
 <encounter><!--  0..1 Reference(Encounter)
     Encounter during which the response was returned --></encounter>
 <occurrenceDateTime value="[dateTime]"/><!--  0..1 When the guidance response was processed -->
 <performer><!--  0..1 Reference(Device)
     Device returning the guidance --></performer>
 <reasonCode><!--  0..* CodeableConcept
     Why guidance is needed --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why guidance is needed --></reasonReference>
 <note><!--  0..* Annotation
     Additional notes about the response --></note>
 <evaluationMessage><!--  0..* Reference(OperationOutcome)
     Messages resulting from the evaluation of the artifact or artifacts --></evaluationMessage>
 <outputParameters><!--  0..1 Reference(Parameters)
     The output parameters of the evaluation, if any --></outputParameters>
 <result><!--  0..1 Reference(CarePlan) Proposed actions, if any --></result>
 <dataRequirement><!--  0..* DataRequirement
     Additional required data --></dataRequirement>
</GuidanceResponse>
```

JSON Template

```
{ // CDS Hooks GuidanceResponse
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "extension" : [ //  sliced by value:url  in the specified order, Open
      { // Service endpoint // C? R!
        // from Element: extension
```

```
    }
  ],
  "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
  "requestIdentifier" : { Identifier }, // C? R! The identifier of the request associated with this response, if any
  "identifier" : { Identifier }, // C? R! Business identifier
  "moduleUri" : [ //  sliced by type:$this  in the specified order, Closed
    { // What guidance was requested // C? R!
      // from Element: extension
    }
  ],
  "status" : "<code>", // C? R! success | data-requested | data-required | in-progress | failure | entered-in-error
  "subject" : { Reference(Patient) }, // C?Patient the request was performed for
  "encounter" : { Reference(Encounter) }, // C?Encounter during which the response was returned
  "occurrenceDateTime" : "<dateTime>", // C?When the guidance response was processed
  "performer" : { Reference(Device) }, // C?Device returning the guidance
  "reasonCode" : [{ CodeableConcept }], // C?Why guidance is needed
  "reasonReference" : [{ Reference(Condition) }], // C?Why guidance is needed
  "note" : [{ Annotation }], // C?Additional notes about the response
  "evaluationMessage" : [{ Reference(OperationOutcome) }], // C?Messages resulting from the evaluation of the artifact
   or artifacts
  "outputParameters" : { Reference(Parameters) }, // C?The output parameters of the evaluation, if any
  "result" : { Reference(CarePlan) }, // C?Proposed actions, if any
  "dataRequirement" : [{ DataRequirement }], // C?Additional required data
}
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.12.10.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
GuidanceResponse.language	Common Languages	preferred	Common Languages
GuidanceResponse.status	GuidanceResponseStatus	required	GuidanceResponseStatus

14.12.11 JSON Schema for GuidanceResponse

URL: https://hl7.org/fhir/R4/guidanceresponse.schema.json.html

14.12.11 JSON Schema for GuidanceResponse

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for GuidanceResponse, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/GuidanceResponse",
  "$ref": "#/definitions/GuidanceResponse",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "GuidanceResponse": {
      "description": "A guidance response is the formal response to a guidance request, including any output parameters
    returned by the evaluation, as well as the description of any proposed actions to be taken.",
      "properties": {
        "resourceType": {
          "description": "This is a GuidanceResponse resource",
          "const": "GuidanceResponse"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
```

```
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "requestIdentifier": {
      "description": "The identifier of the request associated with this response. If an identifier was given as part of the request, it will be reproduced here to enable the requester to more easily identify the response in a multi-request scenario.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "identifier": {
      "description": "Allows a service to provide  unique, business identifiers for the response.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "moduleUri": {
      "description": "An identifier, CodeableConcept or canonical reference to the guidance that was requested.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_moduleUri": {
      "description": "Extensions for moduleUri",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "moduleCanonical": {
      "description": "An identifier, CodeableConcept or canonical reference to the guidance that was requested.",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_moduleCanonical": {
      "description": "Extensions for moduleCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "moduleCodeableConcept": {
      "description": "An identifier, CodeableConcept or canonical reference to the guidance that was requested.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "status": {
      "description": "The status of the response. If the evaluation is completed successfully, the status will indicate success. However, in order to complete the evaluation, the engine may require more information. In this case, the status will be data-required, and the response will contain a description of the additional required information. If the evaluation completed successfully, but the engine determines that a potentially more accurate response could be provided if more data was available, the status will be data-requested, and the response will contain a description of the additional requested information.",
      "enum": [
        "success",
        "data-requested",
        "data-required",
        "in-progress",
        "failure",
        "entered-in-error"
      ]
    },
```

```
      "_status": {
        "description": "Extensions for status",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "subject": {
        "description": "The patient for which the request was processed.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "encounter": {
        "description": "The encounter during which this response was created or to which the creation of this record
  is tightly associated.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "occurrenceDateTime": {
        "description": "Indicates when the guidance response was processed.",
        "$ref": "#/definitions/dateTime"
      },
      "_occurrenceDateTime": {
        "description": "Extensions for occurrenceDateTime",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "performer": {
        "description": "Provides a reference to the device that performed the guidance.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "reasonCode": {
        "description": "Describes the reason for the guidance response in coded or textual form.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "reasonReference": {
        "description": "Indicates the reason the request was initiated. This is typically provided as a parameter to
  the evaluation and echoed by the service, although for some use cases, such as subscription- or event-based
  scenarios, it may provide an indication of the cause for the response.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "note": {
        "description": "Provides a mechanism to communicate additional information about the response.",
        "items": {
          "$ref": "Annotation.schema.json#/definitions/Annotation"
        },
        "type": "array"
      },
      "evaluationMessage": {
        "description": "Messages resulting from the evaluation of the artifact or artifacts. As part of evaluating the
   request, the engine may produce informational or warning messages. These messages will be provided by this
  element.",
        "items": {
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "type": "array"
      },
      "outputParameters": {
        "description": "The output parameters of the evaluation, if any. Many modules will result in the return of
  specific resources such as procedure or communication requests that are returned as part of the operation result.
  However, modules may define specific outputs that would be returned as the result of the evaluation, and these would
   be returned in this element.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "result": {
        "description": "The actions, if any, produced by the evaluation of the artifact.",
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "dataRequirement": {
        "description": "If the evaluation could not be completed due to lack of information, or additional information
   would potentially result in a more accurate response, this element will a description of the data required in order
   to proceed with the evaluation. A subsequent request to the service should include this data.",
        "items": {
          "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
        },
        "type": "array"
      }
    },
    "additionalProperties": false,
    "required": [
      "resourceType"
    ]
  }
 }
}
```

14.12.12 Resource GuidanceResponse - Maturity Dependency View

URL: https://hl7.org/fhir/R4/guidanceresponse-dependencies.html

14.12.12 Resource GuidanceResponse - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for GuidanceResponse FMM level 2

Name | Card. | Type | Dependency Analysis
GuidanceResponse | | DomainResource
requestIdentifier | 0..1 | Identifier
identifier | 0..* | Identifier
module[x] | 1..1
moduleUri | | uri
moduleCanonical | | canonical
moduleCodeableConcept | | CodeableConcept
status | 1..1 | code
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
occurrenceDateTime | 0..1 | dateTime
performer | 0..1 | Reference(Device)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
note | 0..* | Annotation
evaluationMessage | 0..* | Reference(OperationOutcome)
outputParameters | 0..1 | Reference(Parameters)
result | 0..1 | Reference(CarePlan | RequestGroup)
dataRequirement | 0..* | DataRequirement

14.12.13 Value Set Sc-valueset-guidance-response-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-guidance-response-status.html

14.12.13 Value Set Sc-valueset-guidance-response-status - Status Code Mapping

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is the narrative for the resource. See also the XML, JSON or Turtle format.

14.12.14 GuidanceResponseStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-guidance-response-status)

Mapping from http://hl7.org/fhir/ValueSet/guidance-response-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of a guidance response."

Source Code | Equivalence | Destination Code
entered-in-error | equivalent | error
data-requested | equivalent | draft
data-required | equivalent | declined
in-progress | equivalent | active
failure | equivalent | failed
success | equivalent | complete

14.13 Resource RequestGroup - Content

URL: https://hl7.org/fhir/R4/requestgroup.html
Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A group of related requests that can be used to capture intended activities that have inter-dependencies such as "give this medication after that one".

14.13.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow, specifically Request.

The RequestGroup resource is used to represent a group of optional activities that may be performed for a specific patient or context. This resource is often, but not always, the result of applying a specific PlanDefinition to a particular patient. Other than differences that tie the RequestGroup to a particular subject and setting, the actionDefinition element of PlanDefinition is identical to the action element of the RequestGroup, allowing the same features and functionality to be used in both places to describe optionality of and relationships between activities in a workflow.

RequestGroups can contain hierarchical groups of actions, where each specific action references the action to be performed (in terms of a Request resource), and each group describes additional behavior, relationships, and applicable conditions between the actions in the overall group.

This resource is referenced by CarePlan and GuidanceResponse

14.13.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | TU | | DomainResource | A group of related requests Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Business identifier
instantiatesCanonical | Σ | 0..* | canonical() | Instantiates FHIR protocol or definition
instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition
basedOn | | 0..* | Reference(Any) | Fulfills plan, proposal, or order
replaces | | 0..* | Reference(Any) | Request(s) replaced by this request
groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of
status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)
priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | Σ | 0..1 | CodeableConcept | What's being requested/ordered
subject | | 0..1 | Reference(Patient | Group) | Who the request group is about
encounter | | 0..1 | Reference(Encounter) | Created as part of
authoredOn | | 0..1 | dateTime | When the request group was authored
author | | 0..1 | Reference(Device | Practitioner | PractitionerRole) | Device or practitioner that authored the request group
reasonCode | | 0..* | CodeableConcept | Why the request group is needed
reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the request group is needed
note | | 0..* | Annotation | Additional notes about the response
action | I | 0..* | BackboneElement | Proposed actions, if any + Rule: Must have resource or action but not both
prefix | | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | Σ | 0..1 | string | Short description of the action
textEquivalent | Σ | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
expression | | 0..1 | Expression | Boolean-valued expression
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action this is related to
relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing

participant | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device) | Who should perform the action
type | | 0..1 | CodeableConcept | create | update | remove | fire-event ActionType (Extensible)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
resource | I | 0..1 | Reference(Any) | The target of the action
action | I | 0..* | see action | Sub action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RequestGroup xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <instantiatesCanonical value="[canonical]"/><!-- 0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) Fulfills plan, proposal, or order --></basedOn>
 <replaces><!-- 0..* Reference(Any) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept What's being requested/ordered --></code>
 <subject><!-- 0..1 Reference(Patient|Group) Who the request group is about --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Created as part of --></encounter>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request group was authored -->
 <author><!-- 0..1 Reference(Device|Practitioner|PractitionerRole) Device or practitioner that authored the request
    group --></author>
 <reasonCode><!-- 0..* CodeableConcept Why the request group is needed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why the request group is needed --></reasonReference>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <action>  <!-- 0..* Proposed actions, if any -->
  <prefix value="[string]"/><!-- 0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <expression><!-- 0..1 Expression Boolean-valued expression --></expression>
  </condition>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action this is related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Age|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Device) Who should perform the action --></participant>
  <type><!-- 0..1 CodeableConcept create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <resource><!--  0..1 Reference(Any) The target of the action --></resource>
  <action><!--  0..* Content as for RequestGroup.action Sub action --></action>
 </action>
</RequestGroup>
```

JSON Template

```
{
  "resourceType" : "RequestGroup",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "instantiatesCanonical" : ["<canonical>"], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // Fulfills plan, proposal, or order
  "replaces" : [{ Reference(Any) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
```

```
  "code" : { CodeableConcept }, // What's being requested/ordered
  "subject" : { Reference(Patient|Group) }, // Who the request group is about
  "encounter" : { Reference(Encounter) }, // Created as part of
  "authoredOn" : "<dateTime>", // When the request group was authored
  "author" : { Reference(Device|Practitioner|PractitionerRole) }, // Device or practitioner that authored the request
    group
  "reasonCode" : [{ CodeableConcept }], // Why the request group is needed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why the request group is needed
  "note" : [{ Annotation }], // Additional notes about the response
  "action" : [{ // Proposed actions, if any
    "prefix" : "<string>", // User-visible prefix for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "priority" : "<code>", // routine | urgent | asap | stat
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "expression" : { Expression } // Boolean-valued expression
    }],
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action this is related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 6:
    "timingDateTime" : "<dateTime>",
    "timingAge" : { Age },
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Device) }], // Who should perform the action
    "type" : { CodeableConcept }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "resource" : { Reference(Any) }, // C? The target of the action
    "action" : [{ Content as for RequestGroup.action }] // C? Sub action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RequestGroup;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RequestGroup.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:RequestGroup.instantiatesCanonical [ canonical ], ... ; # 0..* Instantiates FHIR protocol or definition
  fhir:RequestGroup.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:RequestGroup.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan, proposal, or order
  fhir:RequestGroup.replaces [ Reference(Any) ], ... ; # 0..* Request(s) replaced by this request
  fhir:RequestGroup.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:RequestGroup.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:RequestGroup.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:RequestGroup.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:RequestGroup.code [ CodeableConcept ]; # 0..1 What's being requested/ordered
  fhir:RequestGroup.subject [ Reference(Patient|Group) ]; # 0..1 Who the request group is about
  fhir:RequestGroup.encounter [ Reference(Encounter) ]; # 0..1 Created as part of
  fhir:RequestGroup.authoredOn [ dateTime ]; # 0..1 When the request group was authored
  fhir:RequestGroup.author [ Reference(Device|Practitioner|PractitionerRole) ]; # 0..1 Device or practitioner that
    authored the request group
  fhir:RequestGroup.reasonCode [ CodeableConcept ], ... ; # 0..* Why the request group is needed
  fhir:RequestGroup.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why the request group is needed
  fhir:RequestGroup.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:RequestGroup.action [ # 0..* Proposed actions, if any
    fhir:RequestGroup.action.prefix [ string ]; # 0..1 User-visible prefix for the action (e.g. 1. or A.)
    fhir:RequestGroup.action.title [ string ]; # 0..1 User-visible title
    fhir:RequestGroup.action.description [ string ]; # 0..1 Short description of the action
    fhir:RequestGroup.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the dynamic
     aspects cannot be interpreted by the receiving system
    fhir:RequestGroup.action.priority [ code ]; # 0..1 routine | urgent | asap | stat
    fhir:RequestGroup.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or sub-
    actions
```

```
    fhir:RequestGroup.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
    performer of the action
    fhir:RequestGroup.action.condition [ # 0..* Whether or not the action is applicable
      fhir:RequestGroup.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:RequestGroup.action.condition.expression [ Expression ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:RequestGroup.action.relatedAction [ # 0..* Relationship to another action
      fhir:RequestGroup.action.relatedAction.actionId [ id ]; # 1..1 What action this is related to
      fhir:RequestGroup.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # RequestGroup.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:RequestGroup.action.relatedAction.offsetDuration [ Duration ]
        fhir:RequestGroup.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # RequestGroup.action.timing[x] : 0..1 When the action should take place. One of these 6
      fhir:RequestGroup.action.timingDateTime [ dateTime ]
      fhir:RequestGroup.action.timingAge [ Age ]
      fhir:RequestGroup.action.timingPeriod [ Period ]
      fhir:RequestGroup.action.timingDuration [ Duration ]
      fhir:RequestGroup.action.timingRange [ Range ]
      fhir:RequestGroup.action.timingTiming [ Timing ]
    fhir:RequestGroup.action.participant [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device) ], ...
    ; # 0..* Who should perform the action
    fhir:RequestGroup.action.type [ CodeableConcept ]; # 0..1 create | update | remove | fire-event
    fhir:RequestGroup.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:RequestGroup.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:RequestGroup.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:RequestGroup.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:RequestGroup.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:RequestGroup.action.resource [ Reference(Any) ]; # 0..1 The target of the action
    fhir:RequestGroup.action.action [ See RequestGroup.action ], ... ; # 0..* Sub action
  ], ...;
]
```

Changes since R3

RequestGroup

RequestGroup.instantiatesCanonical | Added Element

RequestGroup.instantiatesUri | Added Element

RequestGroup.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1

RequestGroup.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/request-intent|4.0.1

RequestGroup.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1

RequestGroup.code | Added Element

RequestGroup.encounter | Added Element

RequestGroup.author | Type Reference: Added Target Type PractitionerRole

RequestGroup.reasonCode | Added Element

RequestGroup.reasonReference | Added Element

RequestGroup.action.prefix | Added Element

RequestGroup.action.priority | Added Element

RequestGroup.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1

RequestGroup.action.condition.expression | Type changed from string to Expression

RequestGroup.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1

RequestGroup.action.timing[x] | Add Type Age

RequestGroup.action.participant | Type Reference: Added Target Types PractitionerRole, Device Type Reference: Removed Target Type Person

RequestGroup.action.type | Type changed from Coding to CodeableConcept Change code system for extensibly bound codes from "http://hl7.org/fhir/action-type" to "http://terminology.hl7.org/CodeSystem/action-type"

RequestGroup.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1

RequestGroup.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1

RequestGroup.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1

RequestGroup.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1

RequestGroup.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1

RequestGroup.definition | deleted

RequestGroup.context | deleted

RequestGroup.reason[x] | deleted

RequestGroup.action.label | deleted

RequestGroup.action.condition.description | deleted

RequestGroup.action.condition.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints

RequestGroup | TU | | DomainResource | A group of related requests Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | Σ | 0..* | Identifier | Business identifier

instantiatesCanonical | Σ | 0..* | canonical() | Instantiates FHIR protocol or definition

instantiatesUri | Σ | 0..* | uri | Instantiates external protocol or definition

basedOn | | 0..* | Reference(Any) | Fulfills plan, proposal, or order

replaces | | 0..* | Reference(Any) | Request(s) replaced by this request

groupIdentifier | Σ | 0..1 | Identifier | Composite request this is part of

status | ?!Σ | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown RequestStatus (Required)

intent | ?!Σ | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option RequestIntent (Required)

priority | Σ | 0..1 | code | routine | urgent | asap | stat Request priority (Required)

code | Σ | 0..1 | CodeableConcept | What's being requested/ordered

subject | | 0..1 | Reference(Patient | Group) | Who the request group is about

encounter | | 0..1 | Reference(Encounter) | Created as part of

authoredOn | | 0..1 | dateTime | When the request group was authored

author | | 0..1 | Reference(Device | Practitioner | PractitionerRole) | Device or practitioner that authored the request group

reasonCode | | 0..* | CodeableConcept | Why the request group is needed

reasonReference | | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the request group is needed

note | | 0..* | Annotation | Additional notes about the response

action | I | 0..* | BackboneElement | Proposed actions, if any + Rule: Must have resource or action but not both

prefix | | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)

title | | 0..1 | string | User-visible title

description | Σ | 0..1 | string | Short description of the action

textEquivalent | Σ | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system

priority | | 0..1 | code | routine | urgent | asap | stat Request priority (Required)

code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions

documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action

condition | | 0..* | BackboneElement | Whether or not the action is applicable

kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)

expression | | 0..1 | Expression | Boolean-valued expression

relatedAction | | 0..* | BackboneElement | Relationship to another action

actionId | | 1..1 | id | What action this is related to

relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)

offset[x] | | 0..1 | | Time offset for the relationship

offsetDuration | | | Duration

offsetRange | | | Range

timing[x] | | 0..1 | | When the action should take place

timingDateTime | | | dateTime

timingAge | | | Age

timingPeriod | | | Period

timingDuration | | | Duration

timingRange | | | Range

timingTiming | | | Timing

participant | | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device) | Who should perform the action

type | | 0..1 | CodeableConcept | create | update | remove | fire-event ActionType (Extensible)

groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)

selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)

requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)

precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)

cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)

resource | I | 0..1 | Reference(Any) | The target of the action

action | I | 0..* | see action | Sub action

Documentation for this format

UML Diagram (Legend)

XML Template

```
<RequestGroup xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <instantiatesCanonical value="[canonical]"/><!-- 0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!-- 0..* Instantiates external protocol or definition -->
 <basedOn><!-- 0..* Reference(Any) Fulfills plan, proposal, or order --></basedOn>
```

```
 <replaces><!-- 0..* Reference(Any) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <code><!-- 0..1 CodeableConcept What's being requested/ordered --></code>
 <subject><!-- 0..1 Reference(Patient|Group) Who the request group is about --></subject>
 <encounter><!-- 0..1 Reference(Encounter) Created as part of --></encounter>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request group was authored -->
 <author><!-- 0..1 Reference(Device|Practitioner|PractitionerRole) Device or practitioner that authored the request
    group --></author>
 <reasonCode><!-- 0..* CodeableConcept Why the request group is needed --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) Why the request group is needed --></reasonReference>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <action>  <!-- 0..* Proposed actions, if any -->
  <prefix value="[string]"/><!-- 0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <expression><!-- 0..1 Expression Boolean-valued expression --></expression>
  </condition>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action this is related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Age|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant><!-- 0..* Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Device) Who should perform the action --></participant>
  <type><!-- 0..1 CodeableConcept create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <resource><!--  0..1 Reference(Any) The target of the action --></resource>
  <action><!--  0..* Content as for RequestGroup.action Sub action --></action>
 </action>
</RequestGroup>
```

JSON Template

```
{
  "resourceType" : "RequestGroup",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "instantiatesCanonical" : ["<canonical>"], // Instantiates FHIR protocol or definition
  "instantiatesUri" : ["<uri>"], // Instantiates external protocol or definition
  "basedOn" : [{ Reference(Any) }], // Fulfills plan, proposal, or order
  "replaces" : [{ Reference(Any) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
  "priority" : "<code>", // routine | urgent | asap | stat
  "code" : { CodeableConcept }, // What's being requested/ordered
  "subject" : { Reference(Patient|Group) }, // Who the request group is about
  "encounter" : { Reference(Encounter) }, // Created as part of
  "authoredOn" : "<dateTime>", // When the request group was authored
  "author" : { Reference(Device|Practitioner|PractitionerRole) }, // Device or practitioner that authored the request
    group
  "reasonCode" : [{ CodeableConcept }], // Why the request group is needed
  "reasonReference" : [{ Reference(Condition|Observation|DiagnosticReport|
   DocumentReference) }], // Why the request group is needed
  "note" : [{ Annotation }], // Additional notes about the response
  "action" : [{ // Proposed actions, if any
    "prefix" : "<string>", // User-visible prefix for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "priority" : "<code>", // routine | urgent | asap | stat
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "expression" : { Expression } // Boolean-valued expression
    }],
    "relatedAction" : [{ // Relationship to another action
```

```
      "actionId" : "<id>", // R!  What action this is related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 6:
    "timingDateTime" : "<dateTime>",
    "timingAge" : { Age },
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ Reference(Patient|Practitioner|PractitionerRole|
    RelatedPerson|Device) }], // Who should perform the action
    "type" : { CodeableConcept }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "resource" : { Reference(Any) }, // C? The target of the action
    "action" : [{ Content as for RequestGroup.action }] // C? Sub action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RequestGroup;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RequestGroup.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:RequestGroup.instantiatesCanonical [ canonical ], ... ; # 0..* Instantiates FHIR protocol or definition
  fhir:RequestGroup.instantiatesUri [ uri ], ... ; # 0..* Instantiates external protocol or definition
  fhir:RequestGroup.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan, proposal, or order
  fhir:RequestGroup.replaces [ Reference(Any) ], ... ; # 0..* Request(s) replaced by this request
  fhir:RequestGroup.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:RequestGroup.status [ code ]; # 1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown
  fhir:RequestGroup.intent [ code ]; # 1..1 proposal | plan | directive | order | original-order | reflex-order |
    filler-order | instance-order | option
  fhir:RequestGroup.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:RequestGroup.code [ CodeableConcept ]; # 0..1 What's being requested/ordered
  fhir:RequestGroup.subject [ Reference(Patient|Group) ]; # 0..1 Who the request group is about
  fhir:RequestGroup.encounter [ Reference(Encounter) ]; # 0..1 Created as part of
  fhir:RequestGroup.authoredOn [ dateTime ]; # 0..1 When the request group was authored
  fhir:RequestGroup.author [ Reference(Device|Practitioner|PractitionerRole) ]; # 0..1 Device or practitioner that
    authored the request group
  fhir:RequestGroup.reasonCode [ CodeableConcept ], ... ; # 0..* Why the request group is needed
  fhir:RequestGroup.reasonReference [ Reference(Condition|Observation|DiagnosticReport|DocumentReference) ], ... ; #
    0..* Why the request group is needed
  fhir:RequestGroup.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:RequestGroup.action [ # 0..* Proposed actions, if any
    fhir:RequestGroup.action.prefix [ string ]; # 0..1 User-visible prefix for the action (e.g. 1. or A.)
    fhir:RequestGroup.action.title [ string ]; # 0..1 User-visible title
    fhir:RequestGroup.action.description [ string ]; # 0..1 Short description of the action
    fhir:RequestGroup.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the dynamic
     aspects cannot be interpreted by the receiving system
    fhir:RequestGroup.action.priority [ code ]; # 0..1 routine | urgent | asap | stat
    fhir:RequestGroup.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or sub-
    actions
    fhir:RequestGroup.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
    performer of the action
    fhir:RequestGroup.action.condition [ # 0..* Whether or not the action is applicable
      fhir:RequestGroup.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:RequestGroup.action.condition.expression [ Expression ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:RequestGroup.action.relatedAction [ # 0..* Relationship to another action
      fhir:RequestGroup.action.relatedAction.actionId [ id ]; # 1..1 What action this is related to
      fhir:RequestGroup.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # RequestGroup.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:RequestGroup.action.relatedAction.offsetDuration [ Duration ]
        fhir:RequestGroup.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # RequestGroup.action.timing[x] : 0..1 When the action should take place. One of these 6
      fhir:RequestGroup.action.timingDateTime [ dateTime ]
      fhir:RequestGroup.action.timingAge [ Age ]
      fhir:RequestGroup.action.timingPeriod [ Period ]
      fhir:RequestGroup.action.timingDuration [ Duration ]
      fhir:RequestGroup.action.timingRange [ Range ]
      fhir:RequestGroup.action.timingTiming [ Timing ]
    fhir:RequestGroup.action.participant [ Reference(Patient|Practitioner|PractitionerRole|RelatedPerson|Device) ], ...
    ; # 0..* Who should perform the action
    fhir:RequestGroup.action.type [ CodeableConcept ]; # 0..1 create | update | remove | fire-event
```

```
    fhir:RequestGroup.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:RequestGroup.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:RequestGroup.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:RequestGroup.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:RequestGroup.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:RequestGroup.action.resource [ Reference(Any) ]; # 0..1 The target of the action
    fhir:RequestGroup.action.action [ See RequestGroup.action ], ... ; # 0..* Sub action
  ], ...;
]
```

Changes since Release 3

RequestGroup
RequestGroup.instantiatesCanonical | Added Element
RequestGroup.instantiatesUri | Added Element
RequestGroup.status | Change value set from http://hl7.org/fhir/ValueSet/request-status to http://hl7.org/fhir/ValueSet/request-status|4.0.1
RequestGroup.intent | Change value set from http://hl7.org/fhir/ValueSet/request-intent to http://hl7.org/fhir/ValueSet/request-intent|4.0.1
RequestGroup.priority | Change value set from http://hl7.org/fhir/ValueSet/request-priority to http://hl7.org/fhir/ValueSet/request-priority|4.0.1
RequestGroup.code | Added Element
RequestGroup.encounter | Added Element
RequestGroup.author | Type Reference: Added Target Type PractitionerRole
RequestGroup.reasonCode | Added Element
RequestGroup.reasonReference | Added Element
RequestGroup.action.prefix | Added Element
RequestGroup.action.priority | Added Element
RequestGroup.action.condition.kind | Change value set from http://hl7.org/fhir/ValueSet/action-condition-kind to http://hl7.org/fhir/ValueSet/action-condition-kind|4.0.1
RequestGroup.action.condition.expression | Type changed from string to Expression
RequestGroup.action.relatedAction.relationship | Change value set from http://hl7.org/fhir/ValueSet/action-relationship-type to http://hl7.org/fhir/ValueSet/action-relationship-type|4.0.1
RequestGroup.action.timing[x] | Add Type Age
RequestGroup.action.participant | Type Reference: Added Target Types PractitionerRole, Device Type Reference: Removed Target Type Person
RequestGroup.action.type | Type changed from Coding to CodeableConcept Change code system for extensibly bound codes from "http://hl7.org/fhir/action-type" to "http://terminology.hl7.org/CodeSystem/action-type"
RequestGroup.action.groupingBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-grouping-behavior to http://hl7.org/fhir/ValueSet/action-grouping-behavior|4.0.1
RequestGroup.action.selectionBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-selection-behavior to http://hl7.org/fhir/ValueSet/action-selection-behavior|4.0.1
RequestGroup.action.requiredBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-required-behavior to http://hl7.org/fhir/ValueSet/action-required-behavior|4.0.1
RequestGroup.action.precheckBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-precheck-behavior to http://hl7.org/fhir/ValueSet/action-precheck-behavior|4.0.1
RequestGroup.action.cardinalityBehavior | Change value set from http://hl7.org/fhir/ValueSet/action-cardinality-behavior to http://hl7.org/fhir/ValueSet/action-cardinality-behavior|4.0.1
RequestGroup.definition | deleted
RequestGroup.context | deleted
RequestGroup.reason[x] | deleted
RequestGroup.action.label | deleted
RequestGroup.action.condition.description | deleted
RequestGroup.action.condition.language | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.13.2.1 Terminology Bindings

Path | Definition | Type | Reference
RequestGroup.status | Codes identifying the lifecycle stage of a request. | Required | RequestStatus
RequestGroup.intent | Codes indicating the degree of authority/intentionality associated with a request. | Required | RequestIntent
RequestGroup.priority RequestGroup.action.priority | Identifies the level of importance to be assigned to actioning the request. | Required | RequestPriority
RequestGroup.action.condition.kind | The kind of condition for the action. | Required | ActionConditionKind
RequestGroup.action.relatedAction.relationship | Defines the types of relationships between actions. | Required | ActionRelationshipType
RequestGroup.action.type | The type of action to be performed. | Extensible | ActionType
RequestGroup.action.groupingBehavior | Defines organization behavior of a group. | Required | ActionGroupingBehavior
RequestGroup.action.selectionBehavior | Defines selection behavior of a group. | Required | ActionSelectionBehavior

RequestGroup.action.requiredBehavior | Defines expectations around whether an action or action group is required. | Required | ActionRequiredBehavior
RequestGroup.action.precheckBehavior | Defines selection frequency behavior for an action or group. | Required | ActionPrecheckBehavior
RequestGroup.action.cardinalityBehavior | Defines behavior for an action or a group for how many times that item may be repeated. | Required | ActionCardinalityBehavior

14.13.2.2 Constraints

id | Level | Location | Description | Expression
rqg-1 | Rule | RequestGroup.action | Must have resource or action but not both | resource.exists() != action.exists()

14.13.3 Usage

The RequestGroup resource is used when there are temporal, co-occurrence or other dependencies between one or more steps of an overall workflow. For example, "do procedure A or procedure B, but not both" or "do procedure A after procedure B" or "Act on this ServiceRequest, then use the value of that observation in the calculation of the dose of this subsequent MedicationRequest". RequestGroups that define actions (i.e. that are more than just narrative representations) will always reference other Request resources with an intent of "option".

Each "option" request can only be interpreted in the context of a RequestGroup that references it. This is because the RequestGroup defines the context in which the option request may/should/must occur, including any triggers, timing constraints, choices, sequencing requirements, etc. Typically such "option" requests will be contained resources due to this dependency. However, in some cases "option" requests may be stand-alone if they are immutable or tightly tied to a ActivityDefinition such that the option resources can safely be referenced without a risk of their content/intent changing

Elements in the "option" requests may include extensions for timing or other elements that allow calculation based on information found in the RequestGroup or other referenced "option" resources, as well as to expose elements within the "option" resource for referencing in other "option" resources. These extensions are:

• TODO

The RequestGroup and all of its referenced "option" Requests are treated as a single integrated Request whose status is the status of the RequestGroup. If there is a need to manage statuses of the different parts, separately, refer to the guidance here.

14.13.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The author of the request group | RequestGroup.author (Practitioner, Device, PractitionerRole)
authored | date | The date the request group was authored | RequestGroup.authoredOn
code | token | The code of the request group | RequestGroup.code
encounter | reference | The encounter the request group applies to | RequestGroup.encounter (Encounter)
group-identifier | token | The group identifier for the request group | RequestGroup.groupIdentifier
identifier | token | External identifiers for the request group | RequestGroup.identifier
instantiates-canonical | reference | The FHIR-based definition from which the request group is realized | RequestGroup.instantiatesCanonical
instantiates-uri | uri | The external definition from which the request group is realized | RequestGroup.instantiatesUri
intent | token | The intent of the request group | RequestGroup.intent
participant | reference | The participant in the requests in the group | RequestGroup.action.participant (Practitioner, Device, Patient, PractitionerRole, RelatedPerson)
patient | reference | The identity of a patient to search for request groups | RequestGroup.subject.where(resolve() is Patient) (Patient)
priority | token | The priority of the request group | RequestGroup.priority
status | token | The status of the request group | RequestGroup.status
subject | reference | The subject that the request group is about | RequestGroup.subject (Group, Patient)

14.13.5 Resource RequestGroup - Examples

URL: https://hl7.org/fhir/R4/requestgroup-examples.html

14.13.5 Resource RequestGroup - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example Name | id | Format
Simple example request group illustrating related actions with offsets | example | XML | JSON | Turtle
Example realization of the KDN5 PlanDefinition example | kdn5-example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.13.6 Resource RequestGroup - Detailed Descriptions

URL: https://hl7.org/fhir/R4/requestgroup-definitions.html

14.13.6 Resource RequestGroup - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the RequestGroup resource.

RequestGroup
Element Id | RequestGroup
Definition | A group of related requests that can be used to capture intended activities that have inter-dependencies such as "give this medication after that one".
Cardinality | 0..*
Type | DomainResource
RequestGroup.identifier
Element Id | RequestGroup.identifier
Definition | Allows a service to provide a unique, business identifier for the request.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows identification of the request as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
RequestGroup.instantiatesCanonical
Element Id | RequestGroup.instantiatesCanonical
Definition | A canonical URL referencing a FHIR-defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this request.
Cardinality | 0..*
Type | canonical
Summary | true
RequestGroup.instantiatesUri
Element Id | RequestGroup.instantiatesUri
Definition | A URL referencing an externally defined protocol, guideline, orderset or other definition that is adhered to in whole or in part by this request.
Cardinality | 0..*
Type | uri
Summary | true
RequestGroup.basedOn
Element Id | RequestGroup.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this request.
Cardinality | 0..*
Type | Reference(Any)
Requirements | Allows tracing of authorization for the request and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
RequestGroup.replaces
Element Id | RequestGroup.replaces
Definition | Completed or terminated request(s) whose function is taken by this new request.
Cardinality | 0..*
Type | Reference(Any)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple requests.
Alternate Names | supersedes; prior; renewed order
Comments | The replacement could be because the initial request was immediately rejected (due to an issue) or because the previous request was completed, but the need for the action described by the request remains ongoing.
RequestGroup.groupIdentifier
Element Id | RequestGroup.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition, prescription or similar form.
Cardinality | 0..1
Type | Identifier
Requirements | Some business processes need to know if multiple items were ordered as part of the same "prescription" or "requisition" for billing or other purposes.
Alternate Names | grouperid; requisition
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
RequestGroup.status
Element Id | RequestGroup.status
Definition | The current state of the request. For request groups, the status reflects the status of all the requests in the group.
Cardinality | 1..1
Terminology Binding | RequestStatus (Required)
Type | code

Is Modifier | true (Reason: This element is labeled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
RequestGroup.intent
Element Id | RequestGroup.intent
Definition | Indicates the level of authority/intentionality associated with the request and where the request fits into the workflow chain.
Cardinality | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true (Reason: This element changes the interpretation of all descriptive attributes. For example "the time the request is recommended to occur" vs. "the time the request is authorized to occur" or "who is recommended to perform the request" vs. "who is authorized to perform the request)
Requirements | Proposals/recommendations, plans and orders all use the same structure and can exist in the same fulfillment chain.
Alternate Names | category
Summary | true
RequestGroup.priority
Element Id | RequestGroup.priority
Definition | Indicates how quickly the request should be addressed with respect to other requests.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
Meaning if Missing | If missing, this request should be performed with normal priority
Summary | true
RequestGroup.code
Element Id | RequestGroup.code
Definition | A code that identifies what the overall request group is.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
Comments | This element can be used to provide a code that captures the meaning of the request group as a whole, as opposed to the code of the action element, which captures the meaning of the individual actions within the request group.
RequestGroup.subject
Element Id | RequestGroup.subject
Definition | The subject for which the request group was created.
Cardinality | 0..1
Type | Reference(Patient | Group)
RequestGroup.encounter
Element Id | RequestGroup.encounter
Definition | Describes the context of the request group, if any.
Cardinality | 0..1
Type | Reference(Encounter)
RequestGroup.authoredOn
Element Id | RequestGroup.authoredOn
Definition | Indicates when the request group was created.
Cardinality | 0..1
Type | dateTime
RequestGroup.author
Element Id | RequestGroup.author
Definition | Provides a reference to the author of the request group.
Cardinality | 0..1
Type | Reference(Device | Practitioner | PractitionerRole)
RequestGroup.reasonCode
Element Id | RequestGroup.reasonCode
Definition | Describes the reason for the request group in coded or textual form.
Cardinality | 0..*
Type | CodeableConcept
RequestGroup.reasonReference
Element Id | RequestGroup.reasonReference
Definition | Indicates another resource whose existence justifies this request group.
Cardinality | 0..*
Type | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
RequestGroup.note
Element Id | RequestGroup.note
Definition | Provides a mechanism to communicate additional information about the response.
Cardinality | 0..*
Type | Annotation
RequestGroup.action
Element Id | RequestGroup.action
Definition | The actions, if any, produced by the evaluation of the artifact.
Cardinality | 0..*
Invariants | Defined on this element rqg-1 Rule Must have resource or action but not both resource.exists() != action.exists()

Defined on this element
rqg-1 | Rule | Must have resource or action but not both | resource.exists() != action.exists()
RequestGroup.action.prefix
Element Id | RequestGroup.action.prefix
Definition | A user-visible prefix for the action.
Cardinality | 0..1
Type | string
RequestGroup.action.title
Element Id | RequestGroup.action.title
Definition | The title of the action displayed to a user.
Cardinality | 0..1
Type | string
RequestGroup.action.description
Element Id | RequestGroup.action.description
Definition | A short description of the action used to provide a summary to display to the user.
Cardinality | 0..1
Type | string
Summary | true
RequestGroup.action.textEquivalent
Element Id | RequestGroup.action.textEquivalent
Definition | A text equivalent of the action to be performed. This provides a human-interpretable description of the action when the definition is consumed by a system that might not be capable of interpreting it dynamically.
Cardinality | 0..1
Type | string
Summary | true
RequestGroup.action.priority
Element Id | RequestGroup.action.priority
Definition | Indicates how quickly the action should be addressed with respect to other actions.
Cardinality | 0..1
Terminology Binding | Request priority (Required)
Type | code
RequestGroup.action.code
Element Id | RequestGroup.action.code
Definition | A code that provides meaning for the action or action group. For example, a section may have a LOINC code for a section of a documentation template.
Cardinality | 0..*
Type | CodeableConcept
RequestGroup.action.documentation
Element Id | RequestGroup.action.documentation
Definition | Didactic or other informational resources associated with the action that can be provided to the CDS recipient. Information resources can include inline text commentary and links to web resources.
Cardinality | 0..*
Type | RelatedArtifact
RequestGroup.action.condition
Element Id | RequestGroup.action.condition
Definition | An expression that describes applicability criteria, or start/stop conditions for the action.
Cardinality | 0..*
Comments | When multiple conditions of the same kind are present, the effects are combined using AND semantics, so the overall condition is true only if all of the conditions are true.
RequestGroup.action.condition.kind
Element Id | RequestGroup.action.condition.kind
Definition | The kind of condition.
Cardinality | 1..1
Terminology Binding | ActionConditionKind (Required)
Type | code
Comments | Applicability criteria are used to determine immediate applicability when a plan definition is applied to a given context. Start and stop criteria are carried through application and used to describe enter/exit criteria for an action.
RequestGroup.action.condition.expression
Element Id | RequestGroup.action.condition.expression
Definition | An expression that returns true or false, indicating whether or not the condition is satisfied.
Cardinality | 0..1
Type | Expression
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.
RequestGroup.action.relatedAction
Element Id | RequestGroup.action.relatedAction
Definition | A relationship to another action such as "before" or "30-60 minutes after start of".
Cardinality | 0..*
RequestGroup.action.relatedAction.actionId
Element Id | RequestGroup.action.relatedAction.actionId
Definition | The element id of the action this is related to.
Cardinality | 1..1
Type | id

RequestGroup.action.relatedAction.relationship
Element Id | RequestGroup.action.relatedAction.relationship
Definition | The relationship of this action to the related action.
Cardinality | 1..1
Terminology Binding | ActionRelationshipType (Required)
Type | code
RequestGroup.action.relatedAction.offset[x]
Element Id | RequestGroup.action.relatedAction.offset[x]
Definition | A duration or range of durations to apply to the relationship. For example, 30-60 minutes before.
Cardinality | 0..1
Type | Duration|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
RequestGroup.action.timing[x]
Element Id | RequestGroup.action.timing[x]
Definition | An optional value describing when the action should be performed.
Cardinality | 0..1
Type | dateTime|Age|Period|Duration|Range|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
RequestGroup.action.participant
Element Id | RequestGroup.action.participant
Definition | The participant that should perform or be responsible for this action.
Cardinality | 0..*
Type | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device)
RequestGroup.action.type
Element Id | RequestGroup.action.type
Definition | The type of action to perform (create, update, remove).
Cardinality | 0..1
Terminology Binding | ActionType (Extensible)
Type | CodeableConcept
RequestGroup.action.groupingBehavior
Element Id | RequestGroup.action.groupingBehavior
Definition | Defines the grouping behavior for the action and its children.
Cardinality | 0..1
Terminology Binding | ActionGroupingBehavior (Required)
Type | code
RequestGroup.action.selectionBehavior
Element Id | RequestGroup.action.selectionBehavior
Definition | Defines the selection behavior for the action and its children.
Cardinality | 0..1
Terminology Binding | ActionSelectionBehavior (Required)
Type | code
RequestGroup.action.requiredBehavior
Element Id | RequestGroup.action.requiredBehavior
Definition | Defines expectations around whether an action is required.
Cardinality | 0..1
Terminology Binding | ActionRequiredBehavior (Required)
Type | code
RequestGroup.action.precheckBehavior
Element Id | RequestGroup.action.precheckBehavior
Definition | Defines whether the action should usually be preselected.
Cardinality | 0..1
Terminology Binding | ActionPrecheckBehavior (Required)
Type | code
RequestGroup.action.cardinalityBehavior
Element Id | RequestGroup.action.cardinalityBehavior
Definition | Defines whether the action can be selected multiple times.
Cardinality | 0..1
Terminology Binding | ActionCardinalityBehavior (Required)
Type | code
RequestGroup.action.resource
Element Id | RequestGroup.action.resource
Definition | The resource that is the target of the action (e.g. CommunicationRequest).
Cardinality | 0..1
Type | Reference(Any)
Comments | The target resource SHALL be a Request resource with a Request.intent set to "option".
Invariants | Affect this element rqg-1 Rule Must have resource or action but not both resource.exists() != action.exists()
Affect this element
rqg-1 | Rule | Must have resource or action but not both | resource.exists() != action.exists()
RequestGroup.action.action
Element Id | RequestGroup.action.action
Definition | Sub actions.
Cardinality | 0..*

Type | See RequestGroup.action
Invariants | Affect this element rqg-1 Rule Must have resource or action but not both resource.exists() != action.exists()
Affect this element
rqg-1 | Rule | Must have resource or action but not both | resource.exists() != action.exists()

14.13.7 Resource RequestGroup - Mappings

URL: https://hl7.org/fhir/R4/requestgroup-mappings.html

14.13.7 Resource RequestGroup - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

Mappings for the requestgroup resource (see Mappings to Other Standards for further information & status).

14.13.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

RequestGroup
identifier | FiveWs.identifier
status | FiveWs.status
intent | FiveWs.class
priority | FiveWs.grade
code | FiveWs.what[x]
subject | FiveWs.subject[x]
encounter | FiveWs.context
authoredOn | FiveWs.recorded
author | FiveWs.author
reasonCode | FiveWs.why[x]
reasonReference | FiveWs.why[x]
participant | FiveWs.actor

14.13.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

RequestGroup | Request
identifier | Request.identifier
instantiatesCanonical | Request.instantiatesCanonical
instantiatesUri | Request.instantiatesUri
basedOn | Request.basedOn
replaces | Request.replaces
groupIdentifier | Request.groupIdentifier
status | Request.status
intent | Request.intent
priority | Request.priority
code | Request.code
subject | Request.subject
encounter | Request.encounter
authoredOn | Request.authoredOn
author | Request.requester
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
note | Request.note
priority | Request.priority
code | Request.code
documentation | Request.supportingInfo
timing[x] | Request.occurrence[x]
participant | Request.performer

14.13.8 Resource RequestGroup - Extensions & Profiles

URL: https://hl7.org/fhir/R4/requestgroup-profiles.html

14.13.8 Resource RequestGroup - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This table lists profiles and extensions for the RequestGroup resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.13.8.1 Profiles

Profile | Description | Context
CDSHooksRequestGroup | Defines a RequestGroup that can represent a CDS Hooks response | CDS Hooks RequestGroup

14.13.8.2 Extensions

timing-daysOfCycle | RequestGroup.action | daysOfCycle | for Common extensions for Timing data type
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.13.8.3 Search Extensions

No Search Extensions defined for this resource

14.13.9 Resource RequestGroup - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/requestgroup-version-maps.html

No validation errors - all conversions are clean

14.13.9 Resource RequestGroup - R3/R4 Conversions

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R3 : R4 Conversion maps for RequestGroup.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (0 errors). (see documentation)

14.13.9.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/RequestGroup3to4" = "R3 to R4 Conversions for RequestGroup"

uses "http://hl7.org/fhir/3.0/StructureDefinition/RequestGroup" alias RequestGroupR3 as source
uses "http://hl7.org/fhir/StructureDefinition/RequestGroup" alias Requestgroup as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group RequestGroup(source src : RequestGroupR3, target tgt : RequestGroup) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.definition -> tgt.instantiatesCanonical;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.priority -> tgt.priority;
  src.subject -> tgt.subject;
  src.context -> tgt.encounter;
  src.authoredOn -> tgt.authoredOn;
  src.author -> tgt.author;
  src.reason : CodeableConcept as v -> tgt.reasonCode = v;
  src.reason : Reference as v -> tgt.reasonReference = v;
  src.note -> tgt.note;
  src.action as vs -> tgt.action as vt then action(vs, vt);
}

group action(source src, target tgt) extends BackboneElement {
  src.label -> tgt.prefix;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.textEquivalent -> tgt.textEquivalent;
  src.code -> tgt.code;
  src.documentation -> tgt.documentation;
  src.condition as s -> tgt.condition as t then condition(s, t);
  src.relatedAction as s -> tgt.relatedAction as t then relatedAction(s, t);
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : Duration as vs -> tgt.timing = create('Duration') as vt then Duration(vs, vt);
  src.timing : Range as vs -> tgt.timing = create('Range') as vt then Range(vs, vt);
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
```

```
  src.participant -> tgt.participant;
  src.type -> tgt.type;
  src.groupingBehavior -> tgt.groupingBehavior;
  src.selectionBehavior -> tgt.selectionBehavior;
  src.requiredBehavior -> tgt.requiredBehavior;
  src.precheckBehavior -> tgt.precheckBehavior;
  src.cardinalityBehavior -> tgt.cardinalityBehavior;
  src.resource -> tgt.resource;
  src.action as vs -> tgt.action as vt then action(vs, vt);
}

group condition(source src, target tgt) extends BackboneElement {
  src.kind -> tgt.kind;
  src.description as v ->  tgt.expression as e,  e.description = v;
  src.language as v ->  tgt.expression as e,  e.language = v;
  src.expression as v ->  tgt.expression as e,  e.expression = v;
}

group relatedAction(source src, target tgt) extends BackboneElement {
  src.actionId -> tgt.actionId;
  src.relationship -> tgt.relationship;
  src.offset : Duration as vs -> tgt.offset = create('Duration') as vt then Duration(vs, vt);
  src.offset : Range as vs -> tgt.offset = create('Range') as vt then Range(vs, vt);
}
```

14.13.9.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/RequestGroup4to3" = "R4 to R3 Conversion for RequestGroup"

uses "http://hl7.org/fhir/StructureDefinition/RequestGroup" alias RequestGroup as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/RequestGroup" alias RequestGroupR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group RequestGroup(source src : RequestGroupR3, target tgt : RequestGroup) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.instantiatesCanonical -> tgt.definition;
  src.basedOn -> tgt.basedOn;
  src.replaces -> tgt.replaces;
  src.groupIdentifier -> tgt.groupIdentifier;
  src.status -> tgt.status;
  src.intent -> tgt.intent;
  src.priority -> tgt.priority;
  src.subject -> tgt.subject;
  src.encounter -> tgt.context;
  src.authoredOn -> tgt.authoredOn;
  src.author -> tgt.author;
  src.reasonCode as vs -> tgt.reason = create('CodeableConcept') as vt then CodeableConcept(vs, vt);
  src.reasonReference as vs -> tgt.reason = create('Reference') as vt then Reference(vs, vt);
  src.note -> tgt.note;
  src.action as vs -> tgt.action as vt then action(vs, vt);
}

group action(source src, target tgt) extends BackboneElement {
  src.prefix -> tgt.label;
  src.title -> tgt.title;
  src.description -> tgt.description;
  src.textEquivalent -> tgt.textEquivalent;
  src.code -> tgt.code;
  src.documentation -> tgt.documentation;
  src.condition as s -> tgt.condition as t then condition(s, t);
  src.relatedAction as s -> tgt.relatedAction as t then relatedAction(s, t);
  src.timing : dateTime as vs -> tgt.timing = create('dateTime') as vt then dateTime(vs, vt);
  src.timing : Period as vs -> tgt.timing = create('Period') as vt then Period(vs, vt);
  src.timing : Duration as vs -> tgt.timing = create('Duration') as vt then Duration(vs, vt);
  src.timing : Range as vs -> tgt.timing = create('Range') as vt then Range(vs, vt);
  src.timing : Timing as vs -> tgt.timing = create('Timing') as vt then Timing(vs, vt);
  src.participant -> tgt.participant;
  src.type -> tgt.type;
  src.groupingBehavior -> tgt.groupingBehavior;
  src.selectionBehavior -> tgt.selectionBehavior;
  src.requiredBehavior -> tgt.requiredBehavior;
  src.precheckBehavior -> tgt.precheckBehavior;
  src.cardinalityBehavior -> tgt.cardinalityBehavior;
  src.resource -> tgt.resource;
  src.action as vs -> tgt.action as vt then action(vs, vt);
}

group condition(source src, target tgt) extends BackboneElement {
  src.kind -> tgt.kind;
  src.expression as e then {
    e.description -> tgt.description;
    e.language -> tgt.language;
    e.expression -> tgt.expression;
  };
}

group relatedAction(source src, target tgt) extends BackboneElement {
  src.actionId -> tgt.actionId;
  src.relationship -> tgt.relationship;
```

```
  src.offset : Duration as vs -> tgt.offset = create('Duration') as vt then Duration(vs, vt);
  src.offset : Range as vs -> tgt.offset = create('Range') as vt then Range(vs, vt);
}
```

14.13.9.3 R4 Validation Errors

14.13.10 CDS Hooks RequestGroup

URL: https://hl7.org/fhir/R4/requestgroup-cdshooks.html

Defines a RequestGroup that can represent a CDS Hooks response

14.13.10 CDS Hooks RequestGroup

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative

14.13.10.1 Content

Profiles:
CDSHooksRequestGroup | Defines a RequestGroup that can represent a CDS Hooks response

14.13.10.2 StructureDefinition: CDSHooksRequestGroup

URL: https://hl7.org/fhir/R4/cdshooksrequestgroup.html

14.13.10.2 StructureDefinition: CDSHooksRequestGroup

Clinical Decision Support Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/cdshooksrequestgroup`

Defines a RequestGroup that can represent a CDS Hooks response

This profile was published on [no date] as a draft by HL7.

14.13.10.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from RequestGroup.

Summary

Mandatory: 2 elements Must-Support: 14 elements

This structure is derived from RequestGroup.

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | | 1..1
identifier | | 1..1 | Identifier
instantiatesUri | | 1..1 | uri
priority | S | 0..1 | code
subject | S | 0..1 | Reference(Patient | Group)
authoredOn | S | 0..1 | dateTime
author | S | 0..1 | Reference(Device | Practitioner | PractitionerRole)
action | | 0..* | BackboneElement
title | S | 0..1 | string
description | S | 0..1 | string
priority | S | 0..1 | code
documentation | S | 0..* | RelatedArtifact
condition | S | 0..* | BackboneElement
timing[x] | S | 0..1 | dateTime, Age, Period, Duration, Range, Timing
participant | S | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device)
type | S | 0..1 | CodeableConcept
selectionBehavior | S | 0..1 | code
resource | S | 0..1 | Reference(Resource)
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | I | 1..1 | | A group of related requests
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created

language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 1..1 | Identifier | Business identifier
instantiatesCanonical | ΣI | 0..* | canonical() | Instantiates FHIR protocol or definition
instantiatesUri | ΣI | 1..1 | uri | Instantiates external protocol or definition
basedOn | I | 0..* | Reference(Resource) | Fulfills plan, proposal, or order
replaces | I | 0..* | Reference(Resource) | Request(s) replaced by this request
groupIdentifier | ΣI | 0..1 | Identifier | Composite request this is part of
status | ?!ΣI | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown Binding: RequestStatus (required)
intent | ?!ΣI | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
priority | SΣI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | ΣI | 0..1 | CodeableConcept | What's being requested/ordered
subject | SI | 0..1 | Reference(Patient | Group) | Who the request group is about
encounter | I | 0..1 | Reference(Encounter) | Created as part of
authoredOn | SI | 0..1 | dateTime | When the request group was authored
author | SI | 0..1 | Reference(Device | Practitioner | PractitionerRole) | Device or practitioner that authored the request group
reasonCode | I | 0..* | CodeableConcept | Why the request group is needed
reasonReference | I | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the request group is needed
note | I | 0..* | Annotation | Additional notes about the response
action | I | 0..* | BackboneElement | Proposed actions, if any
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | SI | 0..1 | string | User-visible title
description | SΣI | 0..1 | string | Short description of the action
textEquivalent | ΣI | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | SI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
documentation | SI | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
condition | SI | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action this is related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | SI | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | SI | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device) | Who should perform the action
type | SI | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | SI | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
resource | SI | 0..1 | Reference(Resource) | The target of the action
action | I | 0..* | | Unknown reference to #RequestGroup.action Sub action
Documentation for this format

```
<!-- CDS Hooks RequestGroup -->
```

```
<RequestGroup xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  1..1 Identifier Business identifier --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  1..1 Instantiates external protocol or definition -->
 <basedOn><!--  0..* Reference(Resource) Fulfills plan, proposal, or order --></basedOn>
 <replaces><!--  0..* Reference(Resource)
     Request(s) replaced by this request --></replaces>
 <groupIdentifier><!--  0..1 Identifier
     Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!--  1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown   -->
 <intent value="[code]"/><!--  1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order |
     instance-order | option   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..1 CodeableConcept What's being requested/ordered --></code>
 <subject><!--  0..1 Reference(Patient) Who the request group is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Created as part of --></encounter>
 <authoredOn value="[dateTime]"/><!--  0..1 When the request group was authored -->
 <author><!--  0..1 Reference(Device)
     Device or practitioner that authored the request group --></author>
 <reasonCode><!--  0..* CodeableConcept
     Why the request group is needed --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why the request group is needed --></reasonReference>
 <note><!--  0..* Annotation Additional notes about the response --></note>
 <action>  0..* BackboneElement  <!--  0..* Proposed actions, if any -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!--  0..1 User-visible title -->
  <description value="[string]"/><!--  0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!--  0..1
      Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
    -->
  <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
  <code><!--  0..* CodeableConcept
      Code representing the meaning of the action or sub-actions --></code>
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the intended performer of the action --></documentation>
  <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
   <expression><!--  0..1 Expression
       Boolean-valued expression --></expression>
  </condition>
  <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <actionId value="[id]"/><!--  1..1 What action this is related to -->
   <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end   -->
   <offset[x]><!--  0..1 Duration|Range
       Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
    Timing When the action should take place --></timing[x]>
  <participant><!--  0..* Reference(Patient)
      Who should perform the action --></participant>
  <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
  <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
  <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
  <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
  <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <resource><!--  0..1 Reference(Resource)
      The target of the action --></resource>
  <action><!-- See #RequestGroup.action  Sub action --></action>
 </action>
</RequestGroup>
```

```
{ // CDS Hooks RequestGroup
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : { Identifier }, // C? R! Business identifier
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
```

```
  "instantiatesUri" : "<uri>", // C? R! Instantiates external protocol or definition
  "basedOn" : [{ Reference(Resource) }], // C?Fulfills plan, proposal, or order
  "replaces" : [{ Reference(Resource) }], // C?Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // C?Composite request this is part of
  "status" : "<code>", // C? R! draft | active | on-hold | revoked | completed | entered-in-error | unknown
  "intent" : "<code>", // C? R! proposal | plan | directive | order | original-order | reflex-order | filler-order |
  instance-order | option
  "priority" : "<code>", // C?routine | urgent | asap | stat
  "code" : { CodeableConcept }, // C?What's being requested/ordered
  "subject" : { Reference(Patient) }, // C?Who the request group is about
  "encounter" : { Reference(Encounter) }, // C?Created as part of
  "authoredOn" : "<dateTime>", // C?When the request group was authored
  "author" : { Reference(Device) }, // C?Device or practitioner that authored the request group
  "reasonCode" : [{ CodeableConcept }], // C?Why the request group is needed
  "reasonReference" : [{ Reference(Condition) }], // C?Why the request group is needed
  "note" : [{ Annotation }], // C?Additional notes about the response
  "action" : [{ BackboneElement }] // C?Proposed actions, if any
}
```

This structure is derived from RequestGroup.

Summary

Mandatory: 2 elements Must-Support: 14 elements

Differential View

This structure is derived from RequestGroup.

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | | 1..1
identifier | | 1..1 | Identifier
instantiatesUri | | 1..1 | uri
priority | S | 0..1 | code
subject | S | 0..1 | Reference(Patient | Group)
authoredOn | S | 0..1 | dateTime
author | S | 0..1 | Reference(Device | Practitioner | PractitionerRole)
action | | 0..* | BackboneElement
title | S | 0..1 | string
description | S | 0..1 | string
priority | S | 0..1 | code
documentation | S | 0..* | RelatedArtifact
condition | S | 0..* | BackboneElement
timing[x] | S | 0..1 | dateTime, Age, Period, Duration, Range, Timing
participant | S | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device)
type | S | 0..1 | CodeableConcept
selectionBehavior | S | 0..1 | code
resource | S | 0..1 | Reference(Resource)
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | I | 1..1 | | A group of related requests
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
identifier | ΣI | 1..1 | Identifier | Business identifier
instantiatesCanonical | ΣI | 0..* | canonical() | Instantiates FHIR protocol or definition
instantiatesUri | ΣI | 1..1 | uri | Instantiates external protocol or definition
basedOn | I | 0..* | Reference(Resource) | Fulfills plan, proposal, or order
replaces | I | 0..* | Reference(Resource) | Request(s) replaced by this request
groupIdentifier | ΣI | 0..1 | Identifier | Composite request this is part of
status | ?!ΣI | 1..1 | code | draft | active | on-hold | revoked | completed | entered-in-error | unknown Binding: RequestStatus (required)
intent | ?!ΣI | 1..1 | code | proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option Binding: RequestIntent (required)
priority | SΣI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | ΣI | 0..1 | CodeableConcept | What's being requested/ordered
subject | SI | 0..1 | Reference(Patient | Group) | Who the request group is about
encounter | I | 0..1 | Reference(Encounter) | Created as part of
authoredOn | SI | 0..1 | dateTime | When the request group was authored
author | SI | 0..1 | Reference(Device | Practitioner | PractitionerRole) | Device or practitioner that authored the request group
reasonCode | I | 0..* | CodeableConcept | Why the request group is needed
reasonReference | I | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference) | Why the request group is needed

note | I | 0..* | Annotation | Additional notes about the response
action | I | 0..* | BackboneElement | Proposed actions, if any
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
prefix | I | 0..1 | string | User-visible prefix for the action (e.g. 1. or A.)
title | SI | 0..1 | string | User-visible title
description | SΣI | 0..1 | string | Short description of the action
textEquivalent | ΣI | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
priority | SI | 0..1 | code | routine | urgent | asap | stat Binding: Request priority (required)
code | I | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
documentation | SI | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
condition | SI | 0..* | BackboneElement | Whether or not the action is applicable
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
kind | I | 1..1 | code | applicability | start | stop Binding: ActionConditionKind (required)
expression | I | 0..1 | Expression | Boolean-valued expression
relatedAction | I | 0..* | BackboneElement | Relationship to another action
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
actionId | I | 1..1 | id | What action this is related to
relationship | I | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end Binding: ActionRelationshipType (required)
offset[x] | I | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | SI | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingAge | | | Age
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | SI | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device) | Who should perform the action
type | SI | 0..1 | CodeableConcept | create | update | remove | fire-event Binding: ActionType (extensible)
groupingBehavior | I | 0..1 | code | visual-group | logical-group | sentence-group Binding: ActionGroupingBehavior (required)
selectionBehavior | SI | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more Binding: ActionSelectionBehavior (required)
requiredBehavior | I | 0..1 | code | must | could | must-unless-documented Binding: ActionRequiredBehavior (required)
precheckBehavior | I | 0..1 | code | yes | no Binding: ActionPrecheckBehavior (required)
cardinalityBehavior | I | 0..1 | code | single | multiple Binding: ActionCardinalityBehavior (required)
resource | SI | 0..1 | Reference(Resource) | The target of the action
action | I | 0..* | | Unknown reference to #RequestGroup.action Sub action
Documentation for this format

XML Template

```
<!-- CDS Hooks RequestGroup -->

<RequestGroup xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <identifier><!--  1..1 Identifier Business identifier --></identifier>
 <instantiatesCanonical value="[canonical]"/><!--  0..* Instantiates FHIR protocol or definition -->
 <instantiatesUri value="[uri]"/><!--  1..1 Instantiates external protocol or definition -->
 <basedOn><!--  0..* Reference(Resource) Fulfills plan, proposal, or order --></basedOn>
 <replaces><!--  0..* Reference(Resource)
     Request(s) replaced by this request --></replaces>
 <groupIdentifier><!--  0..1 Identifier
     Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!--  1..1 draft | active | on-hold | revoked | completed | entered-in-error | unknown   -->
 <intent value="[code]"/><!--  1..1 proposal | plan | directive | order | original-order | reflex-order | filler-order | instance-order | option   -->
 <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
 <code><!--  0..1 CodeableConcept What's being requested/ordered --></code>
 <subject><!--  0..1 Reference(Patient) Who the request group is about --></subject>
 <encounter><!--  0..1 Reference(Encounter) Created as part of --></encounter>
 <authoredOn value="[dateTime]"/><!--  0..1 When the request group was authored -->
```

```
 <author><!--  0..1 Reference(Device)
     Device or practitioner that authored the request group --></author>
 <reasonCode><!--  0..* CodeableConcept
     Why the request group is needed --></reasonCode>
 <reasonReference><!--  0..* Reference(Condition)
     Why the request group is needed --></reasonReference>
 <note><!--  0..* Annotation Additional notes about the response --></note>
 <action>  0..* BackboneElement  <!--  0..* Proposed actions, if any -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <prefix value="[string]"/><!--  0..1 User-visible prefix for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!--  0..1 User-visible title -->
  <description value="[string]"/><!--  0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!--  0..1
      Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
    -->
  <priority value="[code]"/><!--  0..1 routine | urgent | asap | stat   -->
  <code><!--  0..* CodeableConcept
      Code representing the meaning of the action or sub-actions --></code>
  <documentation><!--  0..* RelatedArtifact
      Supporting documentation for the intended performer of the action --></documentation>
  <condition>  0..* BackboneElement  <!--  0..* Whether or not the action is applicable -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <kind value="[code]"/><!--  1..1 applicability | start | stop   -->
   <expression><!--  0..1 Expression
       Boolean-valued expression --></expression>
  </condition>
  <relatedAction>  0..* BackboneElement  <!--  0..* Relationship to another action -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <actionId value="[id]"/><!--  1..1 What action this is related to -->
   <relationship value="[code]"/><!--  1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end   -->
   <offset[x]><!--  0..1 Duration|Range
       Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!--  0..1 dateTime|Age|Period|Duration|Range|
    Timing When the action should take place --></timing[x]>
  <participant><!--  0..* Reference(Patient)
      Who should perform the action --></participant>
  <type><!--  0..1 CodeableConcept create | update | remove | fire-event   --></type>
  <groupingBehavior value="[code]"/><!--  0..1 visual-group | logical-group | sentence-group   -->
  <selectionBehavior value="[code]"/><!--  0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more   -->
  <requiredBehavior value="[code]"/><!--  0..1 must | could | must-unless-documented   -->
  <precheckBehavior value="[code]"/><!--  0..1 yes | no   -->
  <cardinalityBehavior value="[code]"/><!--  0..1 single | multiple   -->
  <resource><!--  0..1 Reference(Resource)
      The target of the action --></resource>
  <action><!-- See #RequestGroup.action  Sub action --></action>
 </action>
</RequestGroup>
```

JSON Template

```
{ // CDS Hooks RequestGroup
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "identifier" : { Identifier }, // C? R! Business identifier
    "instantiatesCanonical" : ["<canonical>"], // C?Instantiates FHIR protocol or definition
    "instantiatesUri" : "<uri>", // C? R! Instantiates external protocol or definition
    "basedOn" : [{ Reference(Resource) }], // C?Fulfills plan, proposal, or order
    "replaces" : [{ Reference(Resource) }], // C?Request(s) replaced by this request
    "groupIdentifier" : { Identifier }, // C?Composite request this is part of
    "status" : "<code>", // C? R! draft | active | on-hold | revoked | completed | entered-in-error | unknown
    "intent" : "<code>", // C? R! proposal | plan | directive | order | original-order | reflex-order | filler-order |
    instance-order | option
    "priority" : "<code>", // C?routine | urgent | asap | stat
    "code" : { CodeableConcept }, // C?What's being requested/ordered
    "subject" : { Reference(Patient) }, // C?Who the request group is about
    "encounter" : { Reference(Encounter) }, // C?Created as part of
    "authoredOn" : "<dateTime>", // C?When the request group was authored
    "author" : { Reference(Device) }, // C?Device or practitioner that authored the request group
    "reasonCode" : [{ CodeableConcept }], // C?Why the request group is needed
    "reasonReference" : [{ Reference(Condition) }], // C?Why the request group is needed
    "note" : [{ Annotation }], // C?Additional notes about the response
    "action" : [{ BackboneElement }] // C?Proposed actions, if any
  }
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.13.10.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
RequestGroup.language	Common Languages	preferred	Common Languages

RequestGroup.status | RequestStatus | required | RequestStatus
RequestGroup.intent | RequestIntent | required | RequestIntent
RequestGroup.priority | Request priority | required | Request priority
RequestGroup.action.priority | Request priority | required | Request priority
RequestGroup.action.condition.kind | ActionConditionKind | required | ActionConditionKind
RequestGroup.action.relatedAction.relationship | ActionRelationshipType | required | ActionRelationshipType
RequestGroup.action.type | ActionType | extensible | ActionType
RequestGroup.action.groupingBehavior | ActionGroupingBehavior | required | ActionGroupingBehavior
RequestGroup.action.selectionBehavior | ActionSelectionBehavior | required | ActionSelectionBehavior
RequestGroup.action.requiredBehavior | ActionRequiredBehavior | required | ActionRequiredBehavior
RequestGroup.action.precheckBehavior | ActionPrecheckBehavior | required | ActionPrecheckBehavior
RequestGroup.action.cardinalityBehavior | ActionCardinalityBehavior | required | ActionCardinalityBehavior

14.13.11 JSON Schema for RequestGroup

URL: https://hl7.org/fhir/R4/requestgroup.schema.json.html

14.13.11 JSON Schema for RequestGroup

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

This is a representation of the json schema for RequestGroup, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/RequestGroup",
  "$ref": "#/definitions/RequestGroup",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "RequestGroup": {
      "description": "A group of related requests that can be used to capture intended activities that have inter-
    dependencies such as \"give this medication after that one\".",
      "properties": {
        "resourceType": {
          "description": "This is a RequestGroup resource",
          "const": "RequestGroup"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
```

```
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "Allows a service to provide a unique, business identifier for the request.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "instantiatesCanonical": {
      "description": "A canonical URL referencing a FHIR-defined protocol, guideline, orderset or other definition
that is adhered to in whole or in part by this request.",
      "items": {
        "$ref": "#/definitions/canonical"
      },
      "type": "array"
    },
    "_instantiatesCanonical": {
      "description": "Extensions for instantiatesCanonical",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "instantiatesUri": {
      "description": "A URL referencing an externally defined protocol, guideline, orderset or other definition that
 is adhered to in whole or in part by this request.",
      "items": {
        "$ref": "#/definitions/uri"
      },
      "type": "array"
    },
    "_instantiatesUri": {
      "description": "Extensions for instantiatesUri",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "basedOn": {
      "description": "A plan, proposal or order that is fulfilled in whole or in part by this request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "replaces": {
      "description": "Completed or terminated request(s) whose function is taken by this new request.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "groupIdentifier": {
      "description": "A shared identifier common to all requests that were authorized more or less simultaneously by
 a single author, representing the identifier of the requisition, prescription or similar form.",
      "$ref": "Identifier.schema.json#/definitions/Identifier"
    },
    "status": {
      "description": "The current state of the request. For request groups, the status reflects the status of all
the requests in the group.",
      "$ref": "#/definitions/code"
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "intent": {
      "description": "Indicates the level of authority/intentionality associated with the request and where the
request fits into the workflow chain.",
      "$ref": "#/definitions/code"
```

```
    },
    "_intent": {
      "description": "Extensions for intent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the request should be addressed with respect to other requests.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A code that identifies what the overall request group is.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subject": {
      "description": "The subject for which the request group was created.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "encounter": {
      "description": "Describes the context of the request group, if any.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "authoredOn": {
      "description": "Indicates when the request group was created.",
      "$ref": "#/definitions/dateTime"
    },
    "_authoredOn": {
      "description": "Extensions for authoredOn",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "author": {
      "description": "Provides a reference to the author of the request group.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "reasonCode": {
      "description": "Describes the reason for the request group in coded or textual form.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "reasonReference": {
      "description": "Indicates another resource whose existence justifies this request group.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "note": {
      "description": "Provides a mechanism to communicate additional information about the response.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "action": {
      "description": "The actions, if any, produced by the evaluation of the artifact.",
      "items": {
        "$ref": "#/definitions/RequestGroup_Action"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"RequestGroup_Action": {
  "description": "A group of related requests that can be used to capture intended activities that have inter-
dependencies such as \"give this medication after that one\".",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "prefix": {
      "description": "A user-visible prefix for the action.",
      "$ref": "#/definitions/string"
    },
    "_prefix": {
      "description": "Extensions for prefix",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "The title of the action displayed to a user.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "description": {
      "description": "A short description of the action used to provide a summary to display to the user.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "textEquivalent": {
      "description": "A text equivalent of the action to be performed. This provides a human-interpretable
description of the action when the definition is consumed by a system that might not be capable of interpreting it
dynamically.",
      "$ref": "#/definitions/string"
    },
    "_textEquivalent": {
      "description": "Extensions for textEquivalent",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "priority": {
      "description": "Indicates how quickly the action should be addressed with respect to other actions.",
      "$ref": "#/definitions/code"
    },
    "_priority": {
      "description": "Extensions for priority",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "code": {
      "description": "A code that provides meaning for the action or action group. For example, a section may have a
 LOINC code for a section of a documentation template.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "documentation": {
      "description": "Didactic or other informational resources associated with the action that can be provided to
the CDS recipient. Information resources can include inline text commentary and links to web resources.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "condition": {
      "description": "An expression that describes applicability criteria, or start/stop conditions for the
action.",
      "items": {
        "$ref": "#/definitions/RequestGroup_Condition"
      },
      "type": "array"
    },
    "relatedAction": {
      "description": "A relationship to another action such as \"before\" or \"30-60 minutes after start of\".",
      "items": {
        "$ref": "#/definitions/RequestGroup_RelatedAction"
      },
      "type": "array"
    },
    "timingDateTime": {
```

```
      "description": "An optional value describing when the action should be performed.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_timingDateTime": {
      "description": "Extensions for timingDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "timingAge": {
      "description": "An optional value describing when the action should be performed.",
      "$ref": "Age.schema.json#/definitions/Age"
    },
    "timingPeriod": {
      "description": "An optional value describing when the action should be performed.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "timingDuration": {
      "description": "An optional value describing when the action should be performed.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "timingRange": {
      "description": "An optional value describing when the action should be performed.",
      "$ref": "Range.schema.json#/definitions/Range"
    },
    "timingTiming": {
      "description": "An optional value describing when the action should be performed.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "participant": {
      "description": "The participant that should perform or be responsible for this action.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of action to perform (create, update, remove).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "groupingBehavior": {
      "description": "Defines the grouping behavior for the action and its children.",
      "$ref": "#/definitions/code"
    },
    "_groupingBehavior": {
      "description": "Extensions for groupingBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "selectionBehavior": {
      "description": "Defines the selection behavior for the action and its children.",
      "$ref": "#/definitions/code"
    },
    "_selectionBehavior": {
      "description": "Extensions for selectionBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "requiredBehavior": {
      "description": "Defines expectations around whether an action is required.",
      "$ref": "#/definitions/code"
    },
    "_requiredBehavior": {
      "description": "Extensions for requiredBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "precheckBehavior": {
      "description": "Defines whether the action should usually be preselected.",
      "$ref": "#/definitions/code"
    },
    "_precheckBehavior": {
      "description": "Extensions for precheckBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "cardinalityBehavior": {
      "description": "Defines whether the action can be selected multiple times.",
      "$ref": "#/definitions/code"
    },
    "_cardinalityBehavior": {
      "description": "Extensions for cardinalityBehavior",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "resource": {
      "description": "The resource that is the target of the action (e.g. CommunicationRequest).",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "action": {
      "description": "Sub actions.",
      "items": {
        "$ref": "#/definitions/RequestGroup_Action"
      },
```

```
      "type": "array"
    }
  },
  "additionalProperties": false
},
"RequestGroup_Condition": {
  "description": "A group of related requests that can be used to capture intended activities that have inter-dependencies such as \"give this medication after that one\".",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "kind": {
      "description": "The kind of condition.",
      "$ref": "#/definitions/code"
    },
    "_kind": {
      "description": "Extensions for kind",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "expression": {
      "description": "An expression that returns true or false, indicating whether or not the condition is satisfied.",
      "$ref": "Expression.schema.json#/definitions/Expression"
    }
  },
  "additionalProperties": false
},
"RequestGroup_RelatedAction": {
  "description": "A group of related requests that can be used to capture intended activities that have inter-dependencies such as \"give this medication after that one\".",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "actionId": {
      "description": "The element id of the action this is related to.",
```

```
          "$ref": "#/definitions/id"
        },
        "_actionId": {
          "description": "Extensions for actionId",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "relationship": {
          "description": "The relationship of this action to the related action.",
          "$ref": "#/definitions/code"
        },
        "_relationship": {
          "description": "Extensions for relationship",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "offsetDuration": {
          "description": "A duration or range of durations to apply to the relationship. For example, 30-60 minutes
    before.",
          "$ref": "Duration.schema.json#/definitions/Duration"
        },
        "offsetRange": {
          "description": "A duration or range of durations to apply to the relationship. For example, 30-60 minutes
    before.",
          "$ref": "Range.schema.json#/definitions/Range"
        }
      },
      "additionalProperties": false
    }
  }
}
```


14.13.12 Resource RequestGroup - Maturity Dependency View

URL: https://hl7.org/fhir/R4/requestgroup-dependencies.html

14.13.12 Resource RequestGroup - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Dependency Graph for RequestGroup FMM level 2

Name | Card. | Type | Dependency Analysis
RequestGroup | | DomainResource
identifier | 0..* | Identifier
instantiatesCanonical | 0..* | canonical()
instantiatesUri | 0..* | uri
basedOn | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
replaces | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
groupIdentifier | 0..1 | Identifier
status | 1..1 | code
intent | 1..1 | code
priority | 0..1 | code
code | 0..1 | CodeableConcept
subject | 0..1 | Reference(Patient | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
encounter | 0..1 | Reference(Encounter)
authoredOn | 0..1 | dateTime
author | 0..1 | Reference(Device | Practitioner | PractitionerRole)
reasonCode | 0..* | CodeableConcept
reasonReference | 0..* | Reference(Condition | Observation | DiagnosticReport | DocumentReference)
note | 0..* | Annotation
action | 0..* | BackboneElement
prefix | 0..1 | string
title | 0..1 | string
description | 0..1 | string
textEquivalent | 0..1 | string
priority | 0..1 | code
code | 0..* | CodeableConcept
documentation | 0..* | RelatedArtifact
condition | 0..* | BackboneElement
kind | 1..1 | code
expression | 0..1 | Expression
relatedAction | 0..* | BackboneElement
actionId | 1..1 | id
relationship | 1..1 | code
offset[x] | 0..1
offsetDuration | | Duration
offsetRange | | Range
timing[x] | 0..1
timingDateTime | | dateTime
timingAge | | Age

timingPeriod | | Period
timingDuration | | Duration
timingRange | | Range
timingTiming | | Timing
participant | 0..* | Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Device)
type | 0..1 | CodeableConcept
groupingBehavior | 0..1 | code
selectionBehavior | 0..1 | code
requiredBehavior | 0..1 | code
precheckBehavior | 0..1 | code
cardinalityBehavior | 0..1 | code
resource | 0..1 | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)
action | 0..* | see action

14.14 Resource Measure - Content

URL: https://hl7.org/fhir/R4/measure.html
Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

The Measure resource provides the definition of a quality measure.

14.14.1 Scope and Usage

The Measure resource represents a structured, computable definition of a health-related measure such as a clinical quality measure, public health indicator, or population analytics measure. A quality measure is a quantitative tool to assess the performance of an individual or organization with respect to a specified process or outcome via the measurement of actions, processes, or outcomes of clinical care. Quality measures are often derived from clinical guidelines and are designed to determine whether the appropriate care has been provided given a set of clinical criteria and an evidence base.

Note that the Measure itself does not typically contain any logic; rather a Library resource is referenced that contains the logic required by the measure, and the various expression elements, such as population criteria, reference named expressions within that library (or libraries). In addition, if the Measure references multiple libraries, then any expression references within the resource must be qualified with the name of the library that contains the referenced expression.

For a detailed discussion of how to use the Measure and MeasureReport resources, refer to the Quality Reporting topic.

14.14.2 Boundaries and Relationships

The Measure resource describes a specific quality measure, or population analytic, providing the structure of the measure in terms of the calculation elements (the populations involved). The Group resource is also capable of describing a population, however, the complexity involved in specifying the criteria in the general case requires the use of a high-level query language such as Clinical Quality Language (CQL). As such, the Measure resource defines only the top-level populations and references expressions for the actual criteria. These expressions are typically provided using a Library resource containing CQL or ELM expressions. In addition, the individual members of a population may be cases such as encounters or procedures and in these cases, the Group resource would be unable to represent the population characteristics accurately.

A Measure is also similar to an Observation resource, with the exception that the Measure is purely definitional, it contains no actual measurements, only a description of how to calculate a particular measurement or set of measurements.

A Measure is also similar to a clinical document, but as with the relationship to Observation, a Document is specific to a particular subject.

This resource is referenced by CarePlan, Communication, FamilyMemberHistory, MeasureReport and Procedure

14.14.3 Resource Content

• Structure
• UML
• XML
• JSON
• Turtle
• R3 Diff
• All

Structure

Name | Flags | Card. | Type | Description & Constraints

Measure | ITU | | DomainResource | A quality measure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Stratifier SHALL be either a single criteria or a set of criteria components Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this measure, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the measure
version | Σ | 0..1 | string | Business version of the measure
name | ΣI | 0..1 | string | Name for this measure (computer friendly)
title | Σ | 0..1 | string | Name for this measure (human friendly)
subtitle | | 0..1 | string | Subordinate title of the measure
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the measure
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this measure is defined
usage | | 0..1 | string | Describes the clinical usage of the measure
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the measure was approved by publisher
lastReviewDate | | 0..1 | date | When the measure was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the measure is expected to be used
topic | | 0..* | CodeableConcept | The category of the measure, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the measure
disclaimer | Σ | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | Σ | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort MeasureScoring (Extensible)
compositeScoring | Σ | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted CompositeMeasureScoring (Extensible)
type | Σ | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite MeasureType (Extensible)
riskAdjustment | Σ | 0..1 | string | How risk adjustment is applied for this measure
rateAggregation | Σ | 0..1 | string | How is rate aggregation performed for this measure
rationale | Σ | 0..1 | markdown | Detailed description of why the measure exists
clinicalRecommendationStatement | Σ | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | Σ | 0..1 | CodeableConcept | increase | decrease MeasureImprovementNotation (Required)
definition | Σ | 0..* | markdown | Defined terms used in the measure documentation
guidance | Σ | 0..1 | markdown | Additional guidance for implementers
group | | 0..* | BackboneElement | Population criteria group
code | | 0..1 | CodeableConcept | Meaning of the group
description | | 0..1 | string | Summary description
population | | 0..* | BackboneElement | Population criteria
code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
description | | 0..1 | string | The human readable description of this population criteria
criteria | | 1..1 | Expression | The criteria that defines this population
stratifier | | 0..* | BackboneElement | Stratifier criteria for the measure
code | | 0..1 | CodeableConcept | Meaning of the stratifier
description | | 0..1 | string | The human readable description of this stratifier
criteria | | 0..1 | Expression | How the measure should be stratified
component | | 0..* | BackboneElement | Stratifier criteria component for the measure
code | | 0..1 | CodeableConcept | Meaning of the stratifier component
description | | 0..1 | string | The human readable description of this stratifier component
criteria | | 1..1 | Expression | Component of how the measure should be stratified
supplementalData | | 0..* | BackboneElement | What other data should be reported with the measure
code | | 0..1 | CodeableConcept | Meaning of the supplemental data
usage | | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor MeasureDataUsage (Extensible)
description | | 0..1 | string | The human readable description of this supplemental data
criteria | | 1..1 | Expression | Expression describing additional data to be reported
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Measure xmlns="http://hl7.org/fhir">
```

```
<!-- from Resource: id, meta, implicitRules, and language -->
<!-- from DomainResource: text, contained, extension, and modifierExtension -->
<url value="[uri]"/><!-- 0..1 Canonical identifier for this measure, represented as a URI (globally unique) -->
<identifier><!-- 0..* Identifier Additional identifier for the measure --></identifier>
<version value="[string]"/><!-- 0..1 Business version of the measure -->
<name value="[string]"/><!--  0..1 Name for this measure (computer friendly) -->
<title value="[string]"/><!-- 0..1 Name for this measure (human friendly) -->
<subtitle value="[string]"/><!-- 0..1 Subordinate title of the measure -->
<status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
<experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
<subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
   Location, Device --></subject[x]>
<date value="[dateTime]"/><!-- 0..1 Date last changed -->
<publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
<contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
<description value="[markdown]"/><!-- 0..1 Natural language description of the measure -->
<useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
<jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for measure (if applicable) --></jurisdiction>
<purpose value="[markdown]"/><!-- 0..1 Why this measure is defined -->
<usage value="[string]"/><!-- 0..1 Describes the clinical usage of the measure -->
<copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
<approvalDate value="[date]"/><!-- 0..1 When the measure was approved by publisher -->
<lastReviewDate value="[date]"/><!-- 0..1 When the measure was last reviewed -->
<effectivePeriod><!-- 0..1 Period When the measure is expected to be used --></effectivePeriod>
<topic><!-- 0..* CodeableConcept The category of the measure, such as Education, Treatment, Assessment, etc.
   --></topic>
<author><!-- 0..* ContactDetail Who authored the content --></author>
<editor><!-- 0..* ContactDetail Who edited the content --></editor>
<reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
<endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
<relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
<library><!-- 0..* canonical(Library) Logic used by the measure --></library>
<disclaimer value="[markdown]"/><!-- 0..1 Disclaimer for use of the measure or its referenced content -->
<scoring><!-- 0..1 CodeableConcept proportion | ratio | continuous-variable | cohort --></scoring>
<compositeScoring><!-- 0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted --></compositeScoring>
<type><!-- 0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite --></type>
<riskAdjustment value="[string]"/><!-- 0..1 How risk adjustment is applied for this measure -->
<rateAggregation value="[string]"/><!-- 0..1 How is rate aggregation performed for this measure -->
<rationale value="[markdown]"/><!-- 0..1 Detailed description of why the measure exists -->
<clinicalRecommendationStatement value="[markdown]"/><!-- 0..1 Summary of clinical guidelines -->
<improvementNotation><!-- 0..1 CodeableConcept increase | decrease --></improvementNotation>
<definition value="[markdown]"/><!-- 0..* Defined terms used in the measure documentation -->
<guidance value="[markdown]"/><!-- 0..1 Additional guidance for implementers -->
<group>  <!-- 0..* Population criteria group -->
 <code><!-- 0..1 CodeableConcept Meaning of the group --></code>
 <description value="[string]"/><!-- 0..1 Summary description -->
 <population>  <!-- 0..* Population criteria -->
  <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
   exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
   --></code>
  <description value="[string]"/><!-- 0..1 The human readable description of this population criteria -->
  <criteria><!-- 1..1 Expression The criteria that defines this population --></criteria>
 </population>
 <stratifier>  <!-- 0..* Stratifier criteria for the measure -->
  <code><!-- 0..1 CodeableConcept Meaning of the stratifier --></code>
  <description value="[string]"/><!-- 0..1 The human readable description of this stratifier -->
  <criteria><!-- 0..1 Expression How the measure should be stratified --></criteria>
  <component>  <!-- 0..* Stratifier criteria component for the measure -->
   <code><!-- 0..1 CodeableConcept Meaning of the stratifier component --></code>
   <description value="[string]"/><!-- 0..1 The human readable description of this stratifier component -->
   <criteria><!-- 1..1 Expression Component of how the measure should be stratified --></criteria>
  </component>
 </stratifier>
</group>
<supplementalData>  <!-- 0..* What other data should be reported with the measure -->
 <code><!-- 0..1 CodeableConcept Meaning of the supplemental data --></code>
 <usage><!-- 0..* CodeableConcept supplemental-data | risk-adjustment-factor --></usage>
 <description value="[string]"/><!-- 0..1 The human readable description of this supplemental data -->
 <criteria><!-- 1..1 Expression Expression describing additional data to be reported --></criteria>
</supplementalData>
</Measure>
```

JSON Template

```
{
  "resourceType" : "Measure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this measure, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the measure
  "version" : "<string>", // Business version of the measure
  "name" : "<string>", // C? Name for this measure (computer friendly)
  "title" : "<string>", // Name for this measure (human friendly)
  "subtitle" : "<string>", // Subordinate title of the measure
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
```

```
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the measure
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for measure (if applicable)
  "purpose" : "<markdown>", // Why this measure is defined
  "usage" : "<string>", // Describes the clinical usage of the measure
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the measure was approved by publisher
  "lastReviewDate" : "<date>", // When the measure was last reviewed
  "effectivePeriod" : { Period }, // When the measure is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the measure, such as Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the measure
  "disclaimer" : "<markdown>", // Disclaimer for use of the measure or its referenced content
  "scoring" : { CodeableConcept }, // proportion | ratio | continuous-variable | cohort
  "compositeScoring" : { CodeableConcept }, // opportunity | all-or-nothing | linear | weighted
  "type" : [{ CodeableConcept }], // process | outcome | structure | patient-reported-outcome | composite
  "riskAdjustment" : "<string>", // How risk adjustment is applied for this measure
  "rateAggregation" : "<string>", // How is rate aggregation performed for this measure
  "rationale" : "<markdown>", // Detailed description of why the measure exists
  "clinicalRecommendationStatement" : "<markdown>", // Summary of clinical guidelines
  "improvementNotation" : { CodeableConcept }, // increase | decrease
  "definition" : ["<markdown>"], // Defined terms used in the measure documentation
  "guidance" : "<markdown>", // Additional guidance for implementers
  "group" : [{ // Population criteria group
    "code" : { CodeableConcept }, // Meaning of the group
    "description" : "<string>", // Summary description
    "population" : [{ // Population criteria
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "description" : "<string>", // The human readable description of this population criteria
      "criteria" : { Expression } // R!  The criteria that defines this population
    }],
    "stratifier" : [{ // Stratifier criteria for the measure
      "code" : { CodeableConcept }, // Meaning of the stratifier
      "description" : "<string>", // The human readable description of this stratifier
      "criteria" : { Expression }, // How the measure should be stratified
      "component" : [{ // Stratifier criteria component for the measure
        "code" : { CodeableConcept }, // Meaning of the stratifier component
        "description" : "<string>", // The human readable description of this stratifier component
        "criteria" : { Expression } // R!  Component of how the measure should be stratified
      }]
    }]
  }],
  "supplementalData" : [{ // What other data should be reported with the measure
    "code" : { CodeableConcept }, // Meaning of the supplemental data
    "usage" : [{ CodeableConcept }], // supplemental-data | risk-adjustment-factor
    "description" : "<string>", // The human readable description of this supplemental data
    "criteria" : { Expression } // R!  Expression describing additional data to be reported
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Measure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Measure.url [ uri ]; # 0..1 Canonical identifier for this measure, represented as a URI (globally unique)
  fhir:Measure.identifier [ Identifier ], ... ; # 0..* Additional identifier for the measure
  fhir:Measure.version [ string ]; # 0..1 Business version of the measure
  fhir:Measure.name [ string ]; # 0..1 Name for this measure (computer friendly)
  fhir:Measure.title [ string ]; # 0..1 Name for this measure (human friendly)
  fhir:Measure.subtitle [ string ]; # 0..1 Subordinate title of the measure
  fhir:Measure.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Measure.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # Measure.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2
    fhir:Measure.subjectCodeableConcept [ CodeableConcept ]
    fhir:Measure.subjectReference [ Reference(Group) ]
  fhir:Measure.date [ dateTime ]; # 0..1 Date last changed
  fhir:Measure.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Measure.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Measure.description [ markdown ]; # 0..1 Natural language description of the measure
  fhir:Measure.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Measure.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for measure (if applicable)
  fhir:Measure.purpose [ markdown ]; # 0..1 Why this measure is defined
  fhir:Measure.usage [ string ]; # 0..1 Describes the clinical usage of the measure
  fhir:Measure.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Measure.approvalDate [ date ]; # 0..1 When the measure was approved by publisher
  fhir:Measure.lastReviewDate [ date ]; # 0..1 When the measure was last reviewed
  fhir:Measure.effectivePeriod [ Period ]; # 0..1 When the measure is expected to be used
```

```
  fhir:Measure.topic [ CodeableConcept ], ... ; # 0..* The category of the measure, such as Education, Treatment,
    Assessment, etc.
  fhir:Measure.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Measure.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Measure.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Measure.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Measure.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Measure.library [ canonical(Library) ], ... ; # 0..* Logic used by the measure
  fhir:Measure.disclaimer [ markdown ]; # 0..1 Disclaimer for use of the measure or its referenced content
  fhir:Measure.scoring [ CodeableConcept ]; # 0..1 proportion | ratio | continuous-variable | cohort
  fhir:Measure.compositeScoring [ CodeableConcept ]; # 0..1 opportunity | all-or-nothing | linear | weighted
  fhir:Measure.type [ CodeableConcept ], ... ; # 0..* process | outcome | structure | patient-reported-outcome |
    composite
  fhir:Measure.riskAdjustment [ string ]; # 0..1 How risk adjustment is applied for this measure
  fhir:Measure.rateAggregation [ string ]; # 0..1 How is rate aggregation performed for this measure
  fhir:Measure.rationale [ markdown ]; # 0..1 Detailed description of why the measure exists
  fhir:Measure.clinicalRecommendationStatement [ markdown ]; # 0..1 Summary of clinical guidelines
  fhir:Measure.improvementNotation [ CodeableConcept ]; # 0..1 increase | decrease
  fhir:Measure.definition [ markdown ], ... ; # 0..* Defined terms used in the measure documentation
  fhir:Measure.guidance [ markdown ]; # 0..1 Additional guidance for implementers
  fhir:Measure.group [ # 0..* Population criteria group
    fhir:Measure.group.code [ CodeableConcept ]; # 0..1 Meaning of the group
    fhir:Measure.group.description [ string ]; # 0..1 Summary description
    fhir:Measure.group.population [ # 0..* Population criteria
      fhir:Measure.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-observation
      fhir:Measure.group.population.description [ string ]; # 0..1 The human readable description of this population
    criteria
      fhir:Measure.group.population.criteria [ Expression ]; # 1..1 The criteria that defines this population
    ], ...;
    fhir:Measure.group.stratifier [ # 0..* Stratifier criteria for the measure
      fhir:Measure.group.stratifier.code [ CodeableConcept ]; # 0..1 Meaning of the stratifier
      fhir:Measure.group.stratifier.description [ string ]; # 0..1 The human readable description of this stratifier
      fhir:Measure.group.stratifier.criteria [ Expression ]; # 0..1 How the measure should be stratified
      fhir:Measure.group.stratifier.component [ # 0..* Stratifier criteria component for the measure
        fhir:Measure.group.stratifier.component.code [ CodeableConcept ]; # 0..1 Meaning of the stratifier component
        fhir:Measure.group.stratifier.component.description [ string ]; # 0..1 The human readable description of this
    stratifier component
        fhir:Measure.group.stratifier.component.criteria [ Expression ]; # 1..1 Component of how the measure should be
    stratified
      ], ...;
    ], ...;
  ], ...;
  fhir:Measure.supplementalData [ # 0..* What other data should be reported with the measure
    fhir:Measure.supplementalData.code [ CodeableConcept ]; # 0..1 Meaning of the supplemental data
    fhir:Measure.supplementalData.usage [ CodeableConcept ], ... ; # 0..* supplemental-data | risk-adjustment-factor
    fhir:Measure.supplementalData.description [ string ]; # 0..1 The human readable description of this supplemental
    data
    fhir:Measure.supplementalData.criteria [ Expression ]; # 1..1 Expression describing additional data to be reported
  ], ...;
]
```

Changes since R3

Measure

Measure.subtitle | Added Element

Measure.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

Measure.experimental | No longer marked as Modifier

Measure.subject[x] | Added Element

Measure.author | Added Element

Measure.editor | Added Element

Measure.reviewer | Added Element

Measure.endorser | Added Element

Measure.library | Type changed from Reference(Library) to canonical(Library)

Measure.scoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-scoring" to "http://terminology.hl7.org/CodeSystem/measure-scoring"

Measure.compositeScoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/composite-measure-scoring" to "http://terminology.hl7.org/CodeSystem/composite-measure-scoring"

Measure.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-type" to "http://terminology.hl7.org/CodeSystem/measure-type"

Measure.improvementNotation | Type changed from string to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/measure-improvement-notation|4.0.1 (required)

Measure.group.code | Added Element

Measure.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"

Measure.group.population.criteria | Type changed from string to Expression

Measure.group.stratifier.code | Added Element

Measure.group.stratifier.description | Added Element

Measure.group.stratifier.criteria | Type changed from string to Expression

Measure.group.stratifier.component | Added Element

Measure.group.stratifier.component.code | Added Element

Measure.group.stratifier.component.description | Added Element

Measure.group.stratifier.component.criteria | Added Mandatory Element
Measure.supplementalData.code | Added Element
Measure.supplementalData.usage | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-data-usage" to "http://terminology.hl7.org/CodeSystem/measure-data-usage"
Measure.supplementalData.description | Added Element
Measure.supplementalData.criteria | Min Cardinality changed from 0 to 1 Type changed from string to Expression
Measure.contributor | deleted
Measure.set | deleted
Measure.group.identifier | deleted
Measure.group.name | deleted
Measure.group.population.identifier | deleted
Measure.group.population.name | deleted
Measure.group.stratifier.identifier | deleted
Measure.group.stratifier.path | deleted
Measure.supplementalData.identifier | deleted
Measure.supplementalData.path | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
Measure | ITU | | DomainResource | A quality measure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation + Rule: Stratifier SHALL be either a single criteria or a set of criteria components Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this measure, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the measure
version | Σ | 0..1 | string | Business version of the measure
name | ΣI | 0..1 | string | Name for this measure (computer friendly)
title | Σ | 0..1 | string | Name for this measure (human friendly)
subtitle | | 0..1 | string | Subordinate title of the measure
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the measure
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this measure is defined
usage | | 0..1 | string | Describes the clinical usage of the measure
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the measure was approved by publisher
lastReviewDate | | 0..1 | date | When the measure was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the measure is expected to be used
topic | | 0..* | CodeableConcept | The category of the measure, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the measure
disclaimer | Σ | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | Σ | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort MeasureScoring (Extensible)
compositeScoring | Σ | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted CompositeMeasureScoring (Extensible)
type | Σ | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite MeasureType (Extensible)
riskAdjustment | Σ | 0..1 | string | How risk adjustment is applied for this measure
rateAggregation | Σ | 0..1 | string | How is rate aggregation performed for this measure
rationale | Σ | 0..1 | markdown | Detailed description of why the measure exists
clinicalRecommendationStatement | Σ | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | Σ | 0..1 | CodeableConcept | increase | decrease MeasureImprovementNotation (Required)
definition | Σ | 0..* | markdown | Defined terms used in the measure documentation
guidance | Σ | 0..1 | markdown | Additional guidance for implementers
group | | 0..* | BackboneElement | Population criteria group
code | | 0..1 | CodeableConcept | Meaning of the group
description | | 0..1 | string | Summary description

population | | 0..* | BackboneElement | Population criteria
code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
description | | 0..1 | string | The human readable description of this population criteria
criteria | | 1..1 | Expression | The criteria that defines this population
stratifier | | 0..* | BackboneElement | Stratifier criteria for the measure
code | | 0..1 | CodeableConcept | Meaning of the stratifier
description | | 0..1 | string | The human readable description of this stratifier
criteria | | 0..1 | Expression | How the measure should be stratified
component | | 0..* | BackboneElement | Stratifier criteria component for the measure
code | | 0..1 | CodeableConcept | Meaning of the stratifier component
description | | 0..1 | string | The human readable description of this stratifier component
criteria | | 1..1 | Expression | Component of how the measure should be stratified
supplementalData | | 0..* | BackboneElement | What other data should be reported with the measure
code | | 0..1 | CodeableConcept | Meaning of the supplemental data
usage | | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor MeasureDataUsage (Extensible)
description | | 0..1 | string | The human readable description of this supplemental data
criteria | | 1..1 | Expression | Expression describing additional data to be reported
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Measure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this measure, represented as a URI (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the measure --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the measure -->
 <name value="[string]"/><!--  0..1 Name for this measure (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this measure (human friendly) -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the measure -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the measure -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for measure (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this measure is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the measure -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the measure was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the measure was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the measure is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the measure, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the measure --></library>
 <disclaimer value="[markdown]"/><!-- 0..1 Disclaimer for use of the measure or its referenced content -->
 <scoring><!-- 0..1 CodeableConcept proportion | ratio | continuous-variable | cohort --></scoring>
 <compositeScoring><!-- 0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted --></compositeScoring>
 <type><!-- 0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite --></type>
 <riskAdjustment value="[string]"/><!-- 0..1 How risk adjustment is applied for this measure -->
 <rateAggregation value="[string]"/><!-- 0..1 How is rate aggregation performed for this measure -->
 <rationale value="[markdown]"/><!-- 0..1 Detailed description of why the measure exists -->
 <clinicalRecommendationStatement value="[markdown]"/><!-- 0..1 Summary of clinical guidelines -->
 <improvementNotation><!-- 0..1 CodeableConcept increase | decrease --></improvementNotation>
 <definition value="[markdown]"/><!-- 0..* Defined terms used in the measure documentation -->
 <guidance value="[markdown]"/><!-- 0..1 Additional guidance for implementers -->
 <group>  <!-- 0..* Population criteria group -->
  <code><!-- 0..1 CodeableConcept Meaning of the group --></code>
  <description value="[string]"/><!-- 0..1 Summary description -->
  <population>  <!-- 0..* Population criteria -->
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <description value="[string]"/><!-- 0..1 The human readable description of this population criteria -->
   <criteria><!-- 1..1 Expression The criteria that defines this population --></criteria>
  </population>
  <stratifier>  <!-- 0..* Stratifier criteria for the measure -->
   <code><!-- 0..1 CodeableConcept Meaning of the stratifier --></code>
   <description value="[string]"/><!-- 0..1 The human readable description of this stratifier -->
   <criteria><!-- 0..1 Expression How the measure should be stratified --></criteria>
   <component>  <!-- 0..* Stratifier criteria component for the measure -->
    <code><!-- 0..1 CodeableConcept Meaning of the stratifier component --></code>
    <description value="[string]"/><!-- 0..1 The human readable description of this stratifier component -->
```

```
    <criteria><!-- 1..1 Expression Component of how the measure should be stratified --></criteria>
   </component>
  </stratifier>
 </group>
 <supplementalData>  <!-- 0..* What other data should be reported with the measure -->
  <code><!-- 0..1 CodeableConcept Meaning of the supplemental data --></code>
  <usage><!-- 0..* CodeableConcept supplemental-data | risk-adjustment-factor --></usage>
  <description value="[string]"/><!-- 0..1 The human readable description of this supplemental data -->
  <criteria><!-- 1..1 Expression Expression describing additional data to be reported --></criteria>
 </supplementalData>
</Measure>
```

JSON Template

```
{
  "resourceType" : "Measure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this measure, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the measure
  "version" : "<string>", // Business version of the measure
  "name" : "<string>", // C? Name for this measure (computer friendly)
  "title" : "<string>", // Name for this measure (human friendly)
  "subtitle" : "<string>", // Subordinate title of the measure
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the measure
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for measure (if applicable)
  "purpose" : "<markdown>", // Why this measure is defined
  "usage" : "<string>", // Describes the clinical usage of the measure
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the measure was approved by publisher
  "lastReviewDate" : "<date>", // When the measure was last reviewed
  "effectivePeriod" : { Period }, // When the measure is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the measure, such as Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the measure
  "disclaimer" : "<markdown>", // Disclaimer for use of the measure or its referenced content
  "scoring" : { CodeableConcept }, // proportion | ratio | continuous-variable | cohort
  "compositeScoring" : { CodeableConcept }, // opportunity | all-or-nothing | linear | weighted
  "type" : [{ CodeableConcept }], // process | outcome | structure | patient-reported-outcome | composite
  "riskAdjustment" : "<string>", // How risk adjustment is applied for this measure
  "rateAggregation" : "<string>", // How is rate aggregation performed for this measure
  "rationale" : "<markdown>", // Detailed description of why the measure exists
  "clinicalRecommendationStatement" : "<markdown>", // Summary of clinical guidelines
  "improvementNotation" : { CodeableConcept }, // increase | decrease
  "definition" : ["<markdown>"], // Defined terms used in the measure documentation
  "guidance" : "<markdown>", // Additional guidance for implementers
  "group" : [{ // Population criteria group
    "code" : { CodeableConcept }, // Meaning of the group
    "description" : "<string>", // Summary description
    "population" : [{ // Population criteria
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "description" : "<string>", // The human readable description of this population criteria
      "criteria" : { Expression } // R!  The criteria that defines this population
    }],
    "stratifier" : [{ // Stratifier criteria for the measure
      "code" : { CodeableConcept }, // Meaning of the stratifier
      "description" : "<string>", // The human readable description of this stratifier
      "criteria" : { Expression }, // How the measure should be stratified
      "component" : [{ // Stratifier criteria component for the measure
        "code" : { CodeableConcept }, // Meaning of the stratifier component
        "description" : "<string>", // The human readable description of this stratifier component
        "criteria" : { Expression } // R!  Component of how the measure should be stratified
      }]
    }]
  }],
  "supplementalData" : [{ // What other data should be reported with the measure
    "code" : { CodeableConcept }, // Meaning of the supplemental data
    "usage" : [{ CodeableConcept }], // supplemental-data | risk-adjustment-factor
    "description" : "<string>", // The human readable description of this supplemental data
    "criteria" : { Expression } // R!  Expression describing additional data to be reported
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Measure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Measure.url [ uri ]; # 0..1 Canonical identifier for this measure, represented as a URI (globally unique)
  fhir:Measure.identifier [ Identifier ], ... ; # 0..* Additional identifier for the measure
  fhir:Measure.version [ string ]; # 0..1 Business version of the measure
  fhir:Measure.name [ string ]; # 0..1 Name for this measure (computer friendly)
  fhir:Measure.title [ string ]; # 0..1 Name for this measure (human friendly)
  fhir:Measure.subtitle [ string ]; # 0..1 Subordinate title of the measure
  fhir:Measure.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Measure.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # Measure.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2
    fhir:Measure.subjectCodeableConcept [ CodeableConcept ]
    fhir:Measure.subjectReference [ Reference(Group) ]
  fhir:Measure.date [ dateTime ]; # 0..1 Date last changed
  fhir:Measure.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Measure.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Measure.description [ markdown ]; # 0..1 Natural language description of the measure
  fhir:Measure.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:Measure.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for measure (if applicable)
  fhir:Measure.purpose [ markdown ]; # 0..1 Why this measure is defined
  fhir:Measure.usage [ string ]; # 0..1 Describes the clinical usage of the measure
  fhir:Measure.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Measure.approvalDate [ date ]; # 0..1 When the measure was approved by publisher
  fhir:Measure.lastReviewDate [ date ]; # 0..1 When the measure was last reviewed
  fhir:Measure.effectivePeriod [ Period ]; # 0..1 When the measure is expected to be used
  fhir:Measure.topic [ CodeableConcept ], ... ; # 0..* The category of the measure, such as Education, Treatment,
    Assessment, etc.
  fhir:Measure.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:Measure.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:Measure.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:Measure.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:Measure.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Measure.library [ canonical(Library) ], ... ; # 0..* Logic used by the measure
  fhir:Measure.disclaimer [ markdown ]; # 0..1 Disclaimer for use of the measure or its referenced content
  fhir:Measure.scoring [ CodeableConcept ]; # 0..1 proportion | ratio | continuous-variable | cohort
  fhir:Measure.compositeScoring [ CodeableConcept ]; # 0..1 opportunity | all-or-nothing | linear | weighted
  fhir:Measure.type [ CodeableConcept ], ... ; # 0..* process | outcome | structure | patient-reported-outcome |
    composite
  fhir:Measure.riskAdjustment [ string ]; # 0..1 How risk adjustment is applied for this measure
  fhir:Measure.rateAggregation [ string ]; # 0..1 How is rate aggregation performed for this measure
  fhir:Measure.rationale [ markdown ]; # 0..1 Detailed description of why the measure exists
  fhir:Measure.clinicalRecommendationStatement [ markdown ]; # 0..1 Summary of clinical guidelines
  fhir:Measure.improvementNotation [ CodeableConcept ]; # 0..1 increase | decrease
  fhir:Measure.definition [ markdown ], ... ; # 0..* Defined terms used in the measure documentation
  fhir:Measure.guidance [ markdown ]; # 0..1 Additional guidance for implementers
  fhir:Measure.group [ # 0..* Population criteria group
    fhir:Measure.group.code [ CodeableConcept ]; # 0..1 Meaning of the group
    fhir:Measure.group.description [ string ]; # 0..1 Summary description
    fhir:Measure.group.population [ # 0..* Population criteria
      fhir:Measure.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-observation
      fhir:Measure.group.population.description [ string ]; # 0..1 The human readable description of this population
    criteria
      fhir:Measure.group.population.criteria [ Expression ]; # 1..1 The criteria that defines this population
    ], ...;
    fhir:Measure.group.stratifier [ # 0..* Stratifier criteria for the measure
      fhir:Measure.group.stratifier.code [ CodeableConcept ]; # 0..1 Meaning of the stratifier
      fhir:Measure.group.stratifier.description [ string ]; # 0..1 The human readable description of this stratifier
      fhir:Measure.group.stratifier.criteria [ Expression ]; # 0..1 How the measure should be stratified
      fhir:Measure.group.stratifier.component [ # 0..* Stratifier criteria component for the measure
        fhir:Measure.group.stratifier.component.code [ CodeableConcept ]; # 0..1 Meaning of the stratifier component
        fhir:Measure.group.stratifier.component.description [ string ]; # 0..1 The human readable description of this
    stratifier component
        fhir:Measure.group.stratifier.component.criteria [ Expression ]; # 1..1 Component of how the measure should be
    stratified
      ], ...;
    ], ...;
  ], ...;
  fhir:Measure.supplementalData [ # 0..* What other data should be reported with the measure
    fhir:Measure.supplementalData.code [ CodeableConcept ]; # 0..1 Meaning of the supplemental data
    fhir:Measure.supplementalData.usage [ CodeableConcept ], ... ; # 0..* supplemental-data | risk-adjustment-factor
    fhir:Measure.supplementalData.description [ string ]; # 0..1 The human readable description of this supplemental
    data
    fhir:Measure.supplementalData.criteria [ Expression ]; # 1..1 Expression describing additional data to be reported
  ], ...;
]
```

Changes since Release 3

Measure

Measure.subtitle | Added Element

Measure.status | Change value set from http://hl7.org/fhir/ValueSet/publication-status to http://hl7.org/fhir/ValueSet/publication-status|4.0.1

Measure.experimental | No longer marked as Modifier

Measure.subject[x] | Added Element

Measure.author | Added Element
Measure.editor | Added Element
Measure.reviewer | Added Element
Measure.endorser | Added Element
Measure.library | Type changed from Reference(Library) to canonical(Library)
Measure.scoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-scoring" to "http://terminology.hl7.org/CodeSystem/measure-scoring"
Measure.compositeScoring | Change code system for extensibly bound codes from "http://hl7.org/fhir/composite-measure-scoring" to "http://terminology.hl7.org/CodeSystem/composite-measure-scoring"
Measure.type | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-type" to "http://terminology.hl7.org/CodeSystem/measure-type"
Measure.improvementNotation | Type changed from string to CodeableConcept Add Binding http://hl7.org/fhir/ValueSet/measure-improvement-notation|4.0.1 (required)
Measure.group.code | Added Element
Measure.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
Measure.group.population.criteria | Type changed from string to Expression
Measure.group.stratifier.code | Added Element
Measure.group.stratifier.description | Added Element
Measure.group.stratifier.criteria | Type changed from string to Expression
Measure.group.stratifier.component | Added Element
Measure.group.stratifier.component.code | Added Element
Measure.group.stratifier.component.description | Added Element
Measure.group.stratifier.component.criteria | Added Mandatory Element
Measure.supplementalData.code | Added Element
Measure.supplementalData.usage | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-data-usage" to "http://terminology.hl7.org/CodeSystem/measure-data-usage"
Measure.supplementalData.description | Added Element
Measure.supplementalData.criteria | Min Cardinality changed from 0 to 1 Type changed from string to Expression
Measure.contributor | deleted
Measure.set | deleted
Measure.group.identifier | deleted
Measure.group.name | deleted
Measure.group.population.identifier | deleted
Measure.group.population.name | deleted
Measure.group.stratifier.identifier | deleted
Measure.group.stratifier.path | deleted
Measure.supplementalData.identifier | deleted
Measure.supplementalData.path | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.14.3.1 Terminology Bindings

Path | Definition | Type | Reference
Measure.status | The lifecycle status of an artifact. | Required | PublicationStatus
Measure.subject[x] | The possible types of subjects for a measure (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType
Measure.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
Measure.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic
Measure.scoring | The scoring type of the measure. | Extensible | MeasureScoring
Measure.compositeScoring | The composite scoring method of the measure. | Extensible | CompositeMeasureScoring
Measure.type | The type of measure (includes codes from 2.16.840.1.113883.1.11.20368). | Extensible | MeasureType
Measure.improvementNotation | Observation values that indicate what change in a measurement value or score is indicative of an improvement in the measured item or scored issue. | Required | MeasureImprovementNotation
Measure.group.population.code | The type of population. | Extensible | MeasurePopulationType
Measure.supplementalData.usage | The intended usage for supplemental data elements in the measure. | Extensible | MeasureDataUsage

14.14.3.2 Constraints

id | Level | Location | Description | Expression
mea-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
mea-1 | Rule | (base) | Stratifier SHALL be either a single criteria or a set of criteria components | group.stratifier.all((code | description | criteria).exists() xor component.exists())

14.14.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | Measure.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the measure | (Measure.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the measure | (Measure.useContext.value as Quantity) | (Measure.useContext.value as Range)
context-type | token | A type of use context assigned to the measure | Measure.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the measure | On Measure.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the measure | On Measure.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The measure publication date | Measure.date
depends-on | reference | What resource is being referenced | Measure.relatedArtifact.where(type='depends-on').resource | Measure.library (Library, Any)
derived-from | reference | What resource is being referenced | Measure.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the measure | Measure.description
effective | date | The time during which the measure is intended to be in use | Measure.effectivePeriod
identifier | token | External identifier for the measure | Measure.identifier
jurisdiction | token | Intended jurisdiction for the measure | Measure.jurisdiction
name | string | Computationally friendly name of the measure | Measure.name
predecessor | reference | What resource is being referenced | Measure.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the measure | Measure.publisher
status | token | The current status of the measure | Measure.status
successor | reference | What resource is being referenced | Measure.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the measure | Measure.title
topic | token | Topics associated with the measure | Measure.topic
url | uri | The uri that identifies the measure | Measure.url
version | token | The business version of the measure | Measure.version

14.14.5 Resource Measure - Examples

URL: https://hl7.org/fhir/R4/measure-examples.html

This example illustrates the use of the Measure resource to describe the CMS9v4 Exclusive Breastfeeding measure. The logic for the measure is specified in the reference library, Exclusive Breatfeeding Measurement Logic.

14.14.5 Resource Measure - Examples

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Example Name | id | Format
Exclusive Breastfeeding Measure | measure-exclusive-breastfeeding | XML | JSON | Turtle
Appropriate Testing for Children with Pharyngitis | measure-cms146-example | XML | JSON | Turtle
Screening for Alcohol Misuse - Component A of a Composite Measure | component-a-example | XML | JSON | Turtle
Screening for Depression - Component B of a Composite Measure Example | component-b-example | XML | JSON | Turtle
Behavioral Health Assessment - Composite Measure Example | composite-example | XML | JSON | Turtle
Predecessor example - Exclusive Breastfeeding | measure-predecessor-example | XML | JSON | Turtle
Example of Measure to define public health reporting data elements | hiv-indicators | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.14.6 Resource Measure - Detailed Descriptions

URL: https://hl7.org/fhir/R4/measure-definitions.html

14.14.6 Resource Measure - Detailed Descriptions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Measure resource.

Measure
Element Id | Measure
Definition | The Measure resource provides the definition of a quality measure.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element mea-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}') mea-1 Rule Stratifier SHALL be either a single criteria or a set of criteria components group.stratifier.all((code | description | criteria).exists() xor component.exists())
Defined on this element

mea-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
mea-1 | Rule | Stratifier SHALL be either a single criteria or a set of criteria components | group.stratifier.all((code | description | criteria).exists() xor component.exists())
Measure.url
Element Id | Measure.url
Definition | An absolute URI that is used to identify this measure when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this measure is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the measure is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the measure to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
Measure.identifier
Element Id | Measure.identifier
Definition | A formal identifier that is used to identify this measure when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this measure outside of FHIR, where it is not possible to use the logical URI.
Measure.version
Element Id | Measure.version
Definition | The identifier that is used to identify this version of the measure when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the measure author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different measure instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the measure with the format [url]|[version].
Measure.name
Element Id | Measure.name
Definition | A natural language name identifying the measure. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Measure.title
Element Id | Measure.title
Definition | A short, descriptive, user-friendly title for the measure.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Measure.subtitle
Element Id | Measure.subtitle
Definition | An explanatory or alternate title for the measure giving additional information about its content.
Cardinality | 0..1
Type | string
Measure.status
Element Id | Measure.status

Definition | The status of this measure. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of measures that are appropriate for use versus not.
Measure.experimental
Element Id | Measure.experimental
Definition | A Boolean value to indicate that this measure is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level measure.
Summary | true
Comments | Allows filtering of measures that are appropriate for use versus not.
Measure.subject[x]
Element Id | Measure.subject[x]
Definition | The intended subjects for the measure. If this element is not provided, a Patient subject is assumed, but the subject of the measure can be anything.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
Comments | The subject of the measure is critical in interpreting the criteria definitions, as the logic in the measures is evaluated with respect to a particular subject. This corresponds roughly to the notion of a Compartment in that it limits what content is available based on its relationship to the subject. In CQL, this corresponds to the context declaration.
Measure.date
Element Id | Measure.date
Definition | The date (and optionally time) when the measure was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the measure changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the measure. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Measure.publisher
Element Id | Measure.publisher
Definition | The name of the organization or individual that published the measure.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the measure. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the measure is the organization or individual primarily responsible for the maintenance and upkeep of the measure. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the measure. This item SHOULD be populated unless the information is available from context.
Measure.contact
Element Id | Measure.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Measure.description
Element Id | Measure.description
Definition | A free text natural language description of the measure from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the measure was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the measure as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the measure is presumed to be the predominant language in the place the measure was created).
Measure.useContext
Element Id | Measure.useContext

Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate measure instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
Measure.jurisdiction
Element Id | Measure.jurisdiction
Definition | A legal or geographic region in which the measure is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the measure to be used in jurisdictions other than those for which it was originally designed or intended.
Measure.purpose
Element Id | Measure.purpose
Definition | Explanation of why this measure is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the measure. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this measure.
Measure.usage
Element Id | Measure.usage
Definition | A detailed description, from a clinical perspective, of how the measure is used.
Cardinality | 0..1
Type | string
Measure.copyright
Element Id | Measure.copyright
Definition | A copyright statement relating to the measure and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the measure.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the measure and/or its content.
Alternate Names | License; Restrictions
Measure.approvalDate
Element Id | Measure.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
Measure.lastReviewDate
Element Id | Measure.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
Measure.effectivePeriod
Element Id | Measure.effectivePeriod
Definition | The period during which the measure content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the measure are or are expected to be used instead.
Summary | true
Comments | The effective period for a measure determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
Measure.topic
Element Id | Measure.topic
Definition | Descriptive topics related to the content of the measure. Topics provide a high-level categorization grouping types of measures that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the measure so that it can be found by topical searches.
Measure.author

Element Id | Measure.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
Measure.editor
Element Id | Measure.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
Measure.reviewer
Element Id | Measure.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
Measure.endorser
Element Id | Measure.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
Measure.relatedArtifact
Element Id | Measure.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | Measures must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
Measure.library
Element Id | Measure.library
Definition | A reference to a Library resource containing the formal logic used by the measure.
Cardinality | 0..*
Type | canonical(Library)
Measure.disclaimer
Element Id | Measure.disclaimer
Definition | Notices and disclaimers regarding the use of the measure or related to intellectual property (such as code systems) referenced by the measure.
Cardinality | 0..1
Type | markdown
Requirements | Some organizations require that notices and disclosures be included when publishing measures that reference their intellecutal property.
Summary | true
Measure.scoring
Element Id | Measure.scoring
Definition | Indicates how the calculation is performed for the measure, including proportion, ratio, continuous-variable, and cohort. The value set is extensible, allowing additional measure scoring types to be represented.
Cardinality | 0..1
Terminology Binding | MeasureScoring (Extensible)
Type | CodeableConcept
Summary | true
Measure.compositeScoring
Element Id | Measure.compositeScoring
Definition | If this is a composite measure, the scoring method used to combine the component measures to determine the composite score.
Cardinality | 0..1
Terminology Binding | CompositeMeasureScoring (Extensible)
Type | CodeableConcept
Summary | true
Measure.type
Element Id | Measure.type
Definition | Indicates whether the measure is used to examine a process, an outcome over time, a patient-reported outcome, or a structure measure such as utilization.
Cardinality | 0..*
Terminology Binding | MeasureType (Extensible)
Type | CodeableConcept
Summary | true
Measure.riskAdjustment
Element Id | Measure.riskAdjustment
Definition | A description of the risk adjustment factors that may impact the resulting score for the measure and how they may be accounted for when computing and reporting measure results.
Cardinality | 0..1
Type | string
Summary | true

Comments | Describes the method of adjusting for clinical severity and conditions present at the start of care that can influence patient outcomes for making valid comparisons of outcome measures across providers. Indicates whether a measure is subject to the statistical process for reducing, removing, or clarifying the influences of confounding factors to allow for more useful comparisons.
Measure.rateAggregation
Element Id | Measure.rateAggregation
Definition | Describes how to combine the information calculated, based on logic in each of several populations, into one summarized result.
Cardinality | 0..1
Type | string
Summary | true
Comments | The measure rate for an organization or clinician is based upon the entities' aggregate data and summarizes the performance of the entity over a given time period (e.g., monthly, quarterly, yearly). The aggregated data are derived from the results of a specific measure algorithm and, if appropriate, the application of specific risk adjustment models. Can also be used to describe how to risk adjust the data based on supplemental data elements described in the eMeasure (e.g., pneumonia hospital measures antibiotic selection in the ICU versus non-ICU and then the roll-up of the two). This could be applied to aggregated cohort measure definitions (e.g., CDC's aggregate reporting for TB at the state level).
Measure.rationale
Element Id | Measure.rationale
Definition | Provides a succinct statement of the need for the measure. Usually includes statements pertaining to importance criterion: impact, gap in care, and evidence.
Cardinality | 0..1
Type | markdown
Requirements | Measure developers need to be able to provide a detailed description of the purpose of the measure. This element provides detailed rationale, where the purpose element provides a summary of the rationale.
Summary | true
Measure.clinicalRecommendationStatement
Element Id | Measure.clinicalRecommendationStatement
Definition | Provides a summary of relevant clinical guidelines or other clinical recommendations supporting the measure.
Cardinality | 0..1
Type | markdown
Summary | true
Measure.improvementNotation
Element Id | Measure.improvementNotation
Definition | Information on whether an increase or decrease in score is the preferred result (e.g., a higher score indicates better quality OR a lower score indicates better quality OR quality is within a range).
Cardinality | 0..1
Terminology Binding | MeasureImprovementNotation (Required)
Type | CodeableConcept
Requirements | Measure consumers and implementers must be able to determine how to interpret a measure score.
Summary | true
Measure.definition
Element Id | Measure.definition
Definition | Provides a description of an individual term used within the measure.
Cardinality | 0..*
Type | markdown
Requirements | Measure developers must be able to provide precise definitions of terms used within measure descriptions and guidance to help communicate the intent of the measure.
Summary | true
Measure.guidance
Element Id | Measure.guidance
Definition | Additional guidance for the measure including how it can be used in a clinical context, and the intent of the measure.
Cardinality | 0..1
Type | markdown
Requirements | Measure developers must be able to provide additional guidance for implementers to understand the intent with greater specificity than that provided in the population criteria of the measure. This element provides detailed guidance, where the usage element provides a summary of the guidance.
Summary | true
Measure.group
Element Id | Measure.group
Definition | A group of population criteria for the measure.
Cardinality | 0..*
Measure.group.code
Element Id | Measure.group.code
Definition | Indicates a meaning for the group. This can be as simple as a unique identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing groups to be correlated across measures.
Cardinality | 0..1
Type | CodeableConcept
Measure.group.description
Element Id | Measure.group.description
Definition | The human readable description of this population group.
Cardinality | 0..1
Type | string
Measure.group.population

Element Id | Measure.group.population
Definition | A population criteria for the measure.
Cardinality | 0..*
Measure.group.population.code
Element Id | Measure.group.population.code
Definition | The type of population criteria.
Cardinality | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
Measure.group.population.description
Element Id | Measure.group.population.description
Definition | The human readable description of this population criteria.
Cardinality | 0..1
Type | string
Measure.group.population.criteria
Element Id | Measure.group.population.criteria
Definition | An expression that specifies the criteria for the population, typically the name of an expression in a library.
Cardinality | 1..1
Type | Expression
Comments | In the case of a continuous-variable or ratio measure, this may be the name of a function that calculates the value of the individual observation for each patient or event in the population. For these types of measures, individual observations are reported as observation resources included in the evaluatedResources bundle for each patient. See the MeasureReport resource or the Quality Reporting topic for more information.
Measure.group.stratifier
Element Id | Measure.group.stratifier
Definition | The stratifier criteria for the measure report, specified as either the name of a valid CQL expression defined within a referenced library or a valid FHIR Resource Path.
Cardinality | 0..*
Measure.group.stratifier.code
Element Id | Measure.group.stratifier.code
Definition | Indicates a meaning for the stratifier. This can be as simple as a unique identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing stratifiers to be correlated across measures.
Cardinality | 0..1
Type | CodeableConcept
Measure.group.stratifier.description
Element Id | Measure.group.stratifier.description
Definition | The human readable description of this stratifier criteria.
Cardinality | 0..1
Type | string
Measure.group.stratifier.criteria
Element Id | Measure.group.stratifier.criteria
Definition | An expression that specifies the criteria for the stratifier. This is typically the name of an expression defined within a referenced library, but it may also be a path to a stratifier element.
Cardinality | 0..1
Type | Expression
Measure.group.stratifier.component
Element Id | Measure.group.stratifier.component
Definition | A component of the stratifier criteria for the measure report, specified as either the name of a valid CQL expression defined within a referenced library or a valid FHIR Resource Path.
Cardinality | 0..*
Comments | Stratifiers are defined either as a single criteria, or as a set of component criteria.
Measure.group.stratifier.component.code
Element Id | Measure.group.stratifier.component.code
Definition | Indicates a meaning for the stratifier component. This can be as simple as a unique identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing stratifiers to be correlated across measures.
Cardinality | 0..1
Type | CodeableConcept
Measure.group.stratifier.component.description
Element Id | Measure.group.stratifier.component.description
Definition | The human readable description of this stratifier criteria component.
Cardinality | 0..1
Type | string
Measure.group.stratifier.component.criteria
Element Id | Measure.group.stratifier.component.criteria
Definition | An expression that specifies the criteria for this component of the stratifier. This is typically the name of an expression defined within a referenced library, but it may also be a path to a stratifier element.
Cardinality | 1..1
Type | Expression
Measure.supplementalData
Element Id | Measure.supplementalData
Definition | The supplemental data criteria for the measure report, specified as either the name of a valid CQL expression within a referenced library, or a valid FHIR Resource Path.
Cardinality | 0..*

Comments | Note that supplemental data are reported as observations for each patient and included in the evaluatedResources bundle. See the MeasureReport resource or the Quality Reporting topic for more information.
Measure.supplementalData.code
Element Id | Measure.supplementalData.code
Definition | Indicates a meaning for the supplemental data. This can be as simple as a unique identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing supplemental data to be correlated across measures.
Cardinality | 0..1
Type | CodeableConcept
Measure.supplementalData.usage
Element Id | Measure.supplementalData.usage
Definition | An indicator of the intended usage for the supplemental data element. Supplemental data indicates the data is additional information requested to augment the measure information. Risk adjustment factor indicates the data is additional information used to calculate risk adjustment factors when applying a risk model to the measure calculation.
Cardinality | 0..*
Terminology Binding | MeasureDataUsage (Extensible)
Type | CodeableConcept
Measure.supplementalData.description
Element Id | Measure.supplementalData.description
Definition | The human readable description of this supplemental data.
Cardinality | 0..1
Type | string
Measure.supplementalData.criteria
Element Id | Measure.supplementalData.criteria
Definition | The criteria for the supplemental data. This is typically the name of a valid expression defined within a referenced library, but it may also be a path to a specific data element. The criteria defines the data to be returned for this element.
Cardinality | 1..1
Type | Expression

14.14.7 Resource Measure - Mappings

URL: https://hl7.org/fhir/R4/measure-mappings.html

14.14.7 Resource Measure - Mappings

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the measure resource (see Mappings to Other Standards for further information & status).

14.14.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

Measure
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

14.14.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

Measure | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate

effectivePeriod | Definition.effectivePeriod

14.14.7.3 RIM Mapping (http://hl7.org/v3)

Measure
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.14.8 Resource Measure - Extensions & Profiles

URL: https://hl7.org/fhir/R4/measure-profiles.html

14.14.8 Resource Measure - Extensions & Profiles

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the Measure resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.14.8.1 Profiles

Profile | Description | Context
ShareableMeasure | Enforces the minimum information set for the measure metadata required by HL7 and other organizations that share and publish measures | Shareable Measure

14.14.8.2 Extensions

replaces | Measure | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

cqf-measureInfo | measureInfo | for Clinical Reasoning Extensions
workflow-instantiatesCanonical | instantiatesCanonical | for Workflow Pattern HL7 Extensions
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.14.8.3 Search Extensions

No Search Extensions defined for this resource

14.14.9 Resource Measure - Operations

URL: https://hl7.org/fhir/R4/measure-operations.html

14.14.9 Resource Measure - Operations

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Security Category: Anonymous | Compartments: Not linked to any defined compartments

This resource has 5 operations operations associated with it:

$evaluate-measure | Evaluate Measure
$data-requirements | Data Requirements
$submit-data | Submit Data
$collect-data | Collect Data
$care-gaps | Care Gaps

For more information about operations, including how they are invoked, see Operations.

14.14.9.1 Operation $evaluate-measure on Measure

URL: https://hl7.org/fhir/R4/measure-operation-evaluate-measure.html

14.14.9.1 Operation $evaluate-measure on Measure

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The evaluate-measure operation is used to calculate an eMeasure and obtain the results.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-evaluate-measure
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/$evaluate-measure

URL: [base]/Measure/[id]/$evaluate-measure

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period start to be 2014-01-01T00:00:00 inclusive

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

measure | 0..1 | string (reference) | | | The measure to evaluate. This parameter is only required when the operation is invoked on the resource type, it is not used when invoking the operation on a Measure instance

reportType | 0..1 | code | | | The type of measure report: subject, subject-list, or population. If not specified, a default value of subject will be used if the subject parameter is supplied, otherwise, population will be used

subject | 0..1 | string (reference) | | | Subject for which the measure will be calculated. If not specified, the measure will be calculated for all subjects that meet the requirements of the measure. If specified, the measure will only be calculated for the referenced subject(s)

practitioner | 0..1 | string (reference) | | | Practitioner for which the measure will be calculated. If specified, the measure will be calculated only for subjects that have a primary relationship to the identified practitioner

lastReceivedOn | 0..1 | dateTime | | | The date the results of this measure were last received. This parameter is only valid for patient-level reports and is used to indicate when the last time a result for this patient was received. This information can be used to limit the set of resources returned for a patient-level report

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | MeasureReport | | | The results of the measure calculation. See the MeasureReport resource for a complete description of the output of this operation. Note that implementations may choose to return a MeasureReport with a status of pending to indicate that the report is still being generated. In this case, the client can use a polling method to continually request the MeasureReport until the status is updated to complete Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to calculate the measure for the given subject, or all subjects if no subject is supplied, and return the results as a MeasureReport resource of the appropriate type. Note that whether or not this operation affects the state of the server depends on whether the server persists the generated MeasureReport. If the MeasureReport is not persisted, this operation can be invoked with GET

14.14.9.1.1 Examples

Request: Request evaluation of the CMS146 measure for patient 124 over a measurement period of the first 3 months of 2014

```
GET [base]/Measure/CMS146/$evaluate-measure?periodStart=2014&periodEnd=2014

GET [base]/Measure/CMS146/$evaluate-measure?subject=124&periodStart=2014-01&periodend=2014-03
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<MeasureReport xmlns="http://hl7.org/fhir">
  <id value="evaluate-measure-operation-response"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
  <!-- Generated... -->
    </div>
  </text>
  <contained>
    <Organization>
      <id value="reporter"/>
      <name value="Good Health Hospital"/>
    </Organization>
  </contained>
  <status value="complete"/>
  <type value="summary"/>
  <measure value="Measure/CMS146"/>
  <reporter>
    <reference value="#reporter"/>
  </reporter>
  <period>
      <start value="2014-01-01" />
      <end value="2014-03-31" />
  </period>
  <group id="CMS146-group-1">
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="500"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
```

```
  <count value="200"/>
</population>
<population>
  <code><coding><code value="denominator"/></coding></code>
  <count value="500"/>
</population>
<population>
  <code><coding><code value="denominator-exclusion"/></coding></code>
  <count value="100"/>
</population>
<stratifier>
  <code><text value="stratifier-ages-up-to-9"/></code>
  <stratum>
    <value><text value="true"/></value>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
  <stratum>
    <value><text value="false"/></value>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
</stratifier>
<stratifier>
  <code><text value="stratifier-ages-10-plus"/></code>
  <stratum>
    <value><text value="true"/></value>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
  <stratum>
    <value><text value="false"/></value>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
```

```
      </population>
    </stratum>
  </stratifier>
  <stratifier>
    <code><text value="stratifier-gender"/></code>
    <stratum>
      <value><text value="male"/></value>
      <population>
        <code><coding><code value="initial-population"/></coding></code>
        <count value="250"/>
      </population>
      <population>
        <code><coding><code value="numerator"/></coding></code>
        <count value="100"/>
      </population>
      <population>
        <code><coding><code value="denominator"/></coding></code>
        <count value="250"/>
      </population>
      <population>
        <code><coding><code value="denominator-exclusion"/></coding></code>
        <count value="50"/>
      </population>
    </stratum>
    <stratum>
      <value><text value="female"/></value>
      <population>
        <code><coding><code value="initial-population"/></coding></code>
        <count value="250"/>
      </population>
      <population>
        <code><coding><code value="numerator"/></coding></code>
        <count value="100"/>
      </population>
      <population>
        <code><coding><code value="denominator"/></coding></code>
        <count value="250"/>
      </population>
      <population>
        <code><coding><code value="denominator-exclusion"/></coding></code>
        <count value="50"/>
      </population>
    </stratum>
    <stratum>
      <value><text value="other"/></value>
      <population>
        <code><coding><code value="initial-population"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="numerator"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="denominator"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="denominator-exclusion"/></coding></code>
        <count value="0"/>
      </population>
    </stratum>
    <stratum>
      <value><text value="unknown"/></value>
      <population>
        <code><coding><code value="initial-population"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="numerator"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="denominator"/></coding></code>
        <count value="0"/>
      </population>
      <population>
        <code><coding><code value="denominator-exclusion"/></coding></code>
        <count value="0"/>
      </population>
    </stratum>
  </stratifier>
 </group>
</MeasureReport>
```

For more information about operations, including how they are invoked, see Operations.

14.14.9.2 Operation $data-requirements on Measure

URL: https://hl7.org/fhir/R4/measure-operation-data-requirements.html

14.14.9.2 Operation $data-requirements on Measure

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The data-requirements operation aggregates and returns the parameters and data requirements for the measure and all its dependencies as a single module definition.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/[id]/$data-requirements

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period start to be 2014-01-01T00:00:00 inclusive

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering is a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the measure Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the measure. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the libraries referenced by the measures. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements). This operation defines what resources are subsequently referenced in the evaluatedResources element of the MeasureReport when submitting measure data

14.14.9.2.1 Examples

Request: Request data requirements for the CMS146 measure

```
GET [base]/Measure/CMS146/$data-requirements?periodStart=2014&periodEnd=2014
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK
```

```
{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "resource": "Library/library-quick-model-definition"
    }
  ],
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Other Female Reproductive Conditions"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
```

```
        "confirmed"
      ]
    },
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1011"
    }
  ]
},
{
  "type": "Condition",
  "codeFilter": [
    {
      "path": "category",
      "valueCode": [
        "diagnosis"
      ]
    },
    {
      "path": "clinicalStatus",
      "valueCode": [
        "confirmed"
      ]
    },
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1012"
    }
  ]
},
{
  "type": "Encounter",
  "codeFilter": [
    {
      "path": "status",
      "valueCode": [
        "finished"
      ]
    },
    {
      "path": "class",
      "valueCode": [
        "ambulatory"
      ]
    },
    {
      "path": "type",
      "valueSetString": "2.16.840.1.113883.3.464.1003.101.12.1061"
    }
  ]
},
{
  "type": "DiagnosticReport",
  "codeFilter": [
    {
      "path": "diagnosis",
      "valueSetString": "2.16.840.1.113883.3.464.1003.198.12.1012"
    }
  ]
},
{
  "type": "Medication",
  "codeFilter": [
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
  "type": "MedicationRequest",
  "codeFilter": [
    {
      "path": "status",
      "valueCode": [
        "active"
      ]
    },
    {
      "path": "medication.code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
  "type": "MedicationStatement",
  "codeFilter": [
    {
      "path": "status",
```

```
          "valueCode": [
            "completed"
          ]
        },
        {
          "path": "medication.code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    }
  ]
}
```

For more information about operations, including how they are invoked, see Operations.

14.14.9.3 Operation $submit-data on Measure

URL: https://hl7.org/fhir/R4/measure-operation-submit-data.html

14.14.9.3 Operation $submit-data on Measure

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The submit-data operation is used to submit data-of-interest for a measure. There is no expectation that the submitted data represents all the data-of-interest, only that all the data submitted is relevant to the calculation of the measure for a particular subject or population.

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Measure-submit-data`

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/$submit-data

URL: [base]/Measure/[id]/$submit-data

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
measureReport	1..1	MeasureReport			The measure report being submitted
resource	0..*	Resource			The individual resources that make up the data-of-interest being submitted

The effect of invoking this operation is that the submitted data is posted to the receiving system and can be used for subsequent calculation of the relevant quality measure. The data-of-interest for a measure can be determined by examining the measure definition, or by invoking the $data-requirements operation

14.14.9.3.1 Examples

Request: Submit data of interest for CMS146

```
POST [base]/Measure/Measure-mrp/$submit-data
```

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "measureReport",
      "resource": {
      "resourceType": "MeasureReport",
      "status": "complete",
      "type": "data-collection",
      "measure": "Measure/measure-mrp",
      "patient": {
      "reference": "Patient/Practitioner-2178"
      },
      "period": {
      "start": "2017-01-01T00:00:00+00:00",
      "end": "2017-12-31T00:00:00+00:00"
      },
      "evaluatedResource": [
        {
        "reference": "Patient/Practitioner-2178"
        },
        {
        "reference": "Encounter/Encounter-62912"
        },
        {
        "reference": "Procedure/Procedure-89972"
        }
      ]
    }
  },
  {
      "name": "resource",
      "resource": {
        "resourceType": "Patient",
        "id": "Practitioner-2178",
        "meta": {
```

```
      "versionId": "8",
      "lastUpdated": "2018-05-31T07:15:08.000+00:00",
      "profile": [
        "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-patient"
      ]
    },
    "text": {
      "status": "generated",
      "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><div class=\"hapiHeaderText\">Natalie Huff <b>WALTER
</b></div><table
class=\"hapiPropertyTable\"><tbody><tr><td>Identifier</td><td>000002418</td></tr><tr><td>Address</td><td><span>623
SW French Place </span><br/><span>Westland </span><span>MI </span></td></tr><tr><td>Date of birth</td><td><span>26
July 1974</span></td></tr></tbody></table></div>"
    },
    "extension": [
      {
        "url": "http://mihin.org/extension/copyright",
        "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
 permissions and limitations under the License."
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/Race",
              "code": "2106-3",
              "display": "White"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/Ethnicity",
              "code": "2186-5",
              "display": "Not Hispanic or Latino"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-religion",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/ReligiousAffiliation",
              "code": "1020",
              "display": "Hinduism"
            }
          ]
        }
      }
    ],
    "identifier": [
      {
        "use": "official",
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "SB",
              "display": "Social Beneficiary Identifier"
            }
          ],
          "text": "US Social Security Number"
        },
        "system": "http://hl7.org/fhir/sid/us-ssn",
        "value": "000002418"
      }
    ],
    "active": true,
    "name": [
      {
        "family": "Walter",
        "given": [
          "Natalie",
          "Huff"
        ]
      }
    ],
    "telecom": [
```

```
      {
        "system": "phone",
        "value": "248-555-3608",
        "use": "home"
      },
      {
        "system": "phone",
        "value": "248-555-4649",
        "use": "mobile"
      }
    ],
    "gender": "female",
    "birthDate": "1974-07-26",
    "address": [
      {
        "use": "home",
        "type": "postal",
        "line": [
          "623 SW French Place"
        ],
        "city": "Westland",
        "district": "Wayne County",
        "state": "MI",
        "postalCode": "48185"
      }
    ]
  }
},
{
  "name": "resource",
  "resource": {
    "resourceType": "Encounter",
    "id": "Encounter-62912",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2018-05-31T08:21:50.000+00:00",
      "profile": [
        "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-encounter"
      ]
    },
    "extension": [
      {
        "url": "http://mihin.org/extension/copyright",
        "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
 permissions and limitations under the License."
      }
    ],
    "status": "finished",
    "class": {
      "system": "http://hl7.org/fhir/encounter-class",
      "code": "inpatient",
      "display": "Inpatient"
    },
    "type": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "56876005",
            "display": "Drug rehabilitation and detoxification"
          }
        ]
      }
    ],
    "subject": {
      "reference": "Patient/Practitioner-2178",
      "display": "Natalie Huff Walter"
    },
    "period": {
      "start": "2018-05-29T11:00:00-04:00",
      "end": "2018-05-29T11:00:00-04:00"
    },
    "hospitalization": {
      "dischargeDisposition": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/discharge-disposition",
            "code": "home",
            "display": "Home"
          }
        ]
      }
    }
  }
},
```

```
    {
      "name": "resource",
      "resource": {
        "resourceType": "Procedure",
        "id": "Procedure-89972",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2018-05-30T23:56:19.000+00:00",
          "profile": [
            "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-procedure"
          ]
        },
        "extension": [
          {
            "url": "http://mihin.org/extension/copyright",
            "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
    Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
    can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
    or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
    WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
     permissions and limitations under the License."
          }
        ],
        "identifier": [
          {
            "system": "mihin.org/fhir/sid/elementId",
            "value": "ca097928-b18c-406c-a01b-6ad4a321fc28"
          }
        ],
        "status": "completed",
        "code": {
          "coding": [
            {
              "system": "http://www.ama-assn.org/go/cpt",
              "code": "1111F",
              "display": "Medication reconciliation postdischarge"
            }
          ]
        },
        "subject": {
          "reference": "Patient/Practitioner-2178",
          "display": "Natalie Huff Walter"
        },
        "performedPeriod": {
          "start": "2018-06-02T14:00:00-05:00",
          "end": "2018-06-02T14:00:00-05:00"
        }
      }
    }
  ]
}
```


Response: Returned on successful evaluation

```
HTTP/1.1 200 OK
```

For more information about operations, including how they are invoked, see Operations.

14.14.9.4 Operation $collect-data on Measure

URL: https://hl7.org/fhir/R4/measure-operation-collect-data.html

14.14.9.4 Operation $collect-data on Measure

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The collect-data operation is used to collect the data-of-interest for the given measure.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-collect-data
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/$collect-data

URL: [base]/Measure/[id]/$collect-data

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period s

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

measure | 0..1 | string (reference) | | | The measure to evaluate. This parameter is only required when the operation is invoked on the resource type, it is not used when invoking the operation on a Measure instance

subject | 0..1 | string (reference) | | | Subject for which the measure will be collected. If not specified, measure data will be collected for all subjects that meet the requirements of the measure. If specified, the measure will only be calculated for the referenced subject(s)
practitioner | 0..1 | string (reference) | | | Practitioner for which the measure will be collected. If specified, measure data will be collected only for subjects that have a primary relationship to the identified practitioner
lastReceivedOn | 0..1 | dateTime | | | The date the results of this measure were last received. This parameter used to indicate when the last time data for this measure was collected. This information is used to support incremental data collection scenarios
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
measureReport | 1..1 | MeasureReport | | | A MeasureReport of type data-collection detailing the results of the operation
resource | 0..* | Resource | | | The result resources that make up the data-of-interest for the measure

The effect of invoking this operation is to gather the data required to perform an evaluation of the measure. If the lastReceivedOn parameter is supplied, only data that is new or has been changed since the lastReceivedOn date is included in the response. Note that the resulting MeasureReport is a transient resource

14.14.9.4.1 Examples

Request: Request the data-of-interest since 2018-07-21 for Measure/measure-mrp for Patient/Practitioner-2178

```
GET [base]/Measure/measure-mrp/$collect-data?periodStart=2018&periodEnd=2018&subject=Patient/Practitioner-2178

GET [base]/Measure/measure-mrp/$collect-
    data?periodStart=2018&periodEnd=2018&subject=Patient/Practitioner-2178&lastReceivedOn=2018-07-21
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "measureReport",
      "resource": {
      "resourceType": "MeasureReport",
      "status": "complete",
      "type": "data-collection",
      "measure": "Measure/measure-mrp",
      "patient": {
      "reference": "Patient/Practitioner-2178"
      },
      "period": {
      "start": "2017-01-01T00:00:00+00:00",
      "end": "2017-12-31T00:00:00+00:00"
      },
      "evaluatedResource": [
        {
        "reference": "Patient/Practitioner-2178"
        },
        {
        "reference": "Encounter/Encounter-62912"
        },
        {
        "reference": "Procedure/Procedure-89972"
        }
      ]
    }
  },
  {
      "name": "resource",
      "resource": {
        "resourceType": "Patient",
        "id": "Practitioner-2178",
        "meta": {
          "versionId": "8",
          "lastUpdated": "2018-05-31T07:15:08.000+00:00",
          "profile": [
            "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-patient"
          ]
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><div class=\"hapiHeaderText\">Natalie Huff <b>WALTER
    </b></div><table
    class=\"hapiPropertyTable\"><tbody><tr><td>Identifier</td><td>000002418</td></tr><tr><td>Address</td><td><span>623
    SW French Place </span><br/><span>Westland </span><span>MI </span></td></tr><tr><td>Date of birth</td><td><span>26
    July 1974</span></td></tr></tbody></table></div>"
        },
        "extension": [
          {
            "url": "http://mihin.org/extension/copyright",
            "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
    Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
    can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
```

```
or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
 permissions and limitations under the License."
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/Race",
              "code": "2106-3",
              "display": "White"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/Ethnicity",
              "code": "2186-5",
              "display": "Not Hispanic or Latino"
            }
          ]
        }
      },
      {
        "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-religion",
        "valueCodeableConcept": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/ReligiousAffiliation",
              "code": "1020",
              "display": "Hinduism"
            }
          ]
        }
      }
    ],
    "identifier": [
      {
        "use": "official",
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "SB",
              "display": "Social Beneficiary Identifier"
            }
          ],
          "text": "US Social Security Number"
        },
        "system": "http://hl7.org/fhir/sid/us-ssn",
        "value": "000002418"
      }
    ],
    "active": true,
    "name": [
      {
        "family": "Walter",
        "given": [
          "Natalie",
          "Huff"
        ]
      }
    ],
    "telecom": [
      {
        "system": "phone",
        "value": "248-555-3608",
        "use": "home"
      },
      {
        "system": "phone",
        "value": "248-555-4649",
        "use": "mobile"
      }
    ],
    "gender": "female",
    "birthDate": "1974-07-26",
    "address": [
      {
        "use": "home",
        "type": "postal",
        "line": [
          "623 SW French Place"
        ],
```

```
        "city": "Westland",
        "district": "Wayne County",
        "state": "MI",
        "postalCode": "48185"
      }
    ]
  }
},
{
  "name": "resource",
  "resource": {
    "resourceType": "Encounter",
    "id": "Encounter-62912",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2018-05-31T08:21:50.000+00:00",
      "profile": [
        "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-encounter"
      ]
    },
    "extension": [
      {
        "url": "http://mihin.org/extension/copyright",
        "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
 permissions and limitations under the License."
      }
    ],
    "status": "finished",
    "class": {
      "system": "http://hl7.org/fhir/encounter-class",
      "code": "inpatient",
      "display": "Inpatient"
    },
    "type": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "56876005",
            "display": "Drug rehabilitation and detoxification"
          }
        ]
      }
    ],
    "subject": {
      "reference": "Patient/Practitioner-2178",
      "display": "Natalie Huff Walter"
    },
    "period": {
      "start": "2018-05-29T11:00:00-04:00",
      "end": "2018-05-29T11:00:00-04:00"
    },
    "hospitalization": {
      "dischargeDisposition": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/discharge-disposition",
            "code": "home",
            "display": "Home"
          }
        ]
      }
    }
  }
},
{
  "name": "resource",
  "resource": {
    "resourceType": "Procedure",
    "id": "Procedure-89972",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2018-05-30T23:56:19.000+00:00",
      "profile": [
        "http://hl7.org/fhir/us/hedis/StructureDefinition/hedis-procedure"
      ]
    },
    "extension": [
      {
        "url": "http://mihin.org/extension/copyright",
        "valueString": "Copyright 2014-2017 Michigan Health Information Network Shared Services. Licensed under the
Apache License, Version 2.0 (the 'License'); you can not use this file except in compliance with the License. You
can obtain a copy of the License at  http://www.apache.org/licenses/LICENSE-2.0.  Unless required by applicable law
or agreed to in writing, software distributed under the License is distributed on an 'AS IS' BASIS, WITHOUT
WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing
```

```
     permissions and limitations under the License."
        }
      ],
      "identifier": [

        {
          "system": "mihin.org/fhir/sid/elementId",
          "value": "ca097928-b18c-406c-a01b-6ad4a321fc28"
        }
      ],
      "status": "completed",
      "code": {
        "coding": [
          {
            "system": "http://www.ama-assn.org/go/cpt",
            "code": "1111F",
            "display": "Medication reconciliation postdischarge"
          }
        ]
      },
      "subject": {
        "reference": "Patient/Practitioner-2178",
        "display": "Natalie Huff Walter"
      },
      "performedPeriod": {
        "start": "2018-06-02T14:00:00-05:00",
        "end": "2018-06-02T14:00:00-05:00"
      }
    }
  }
 ]
}
```

For more information about operations, including how they are invoked, see Operations.

14.14.9.5 Operation $care-gaps on Measure

URL: https://hl7.org/fhir/R4/measure-operation-care-gaps.html

14.14.9.5 Operation $care-gaps on Measure

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: N/A

The care-gaps operation is used to determine gaps-in-care based on the results of quality measures.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-care-gaps
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/$care-gaps

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period s

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

topic | 1..1 | string | | | The topic to be used to determine which measures are considered for the care gaps report. Any measure with the given topic will be included in the report

subject | 1..1 | string (reference) | | | Subject for which the care gaps report will be produced

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The result of the care gaps report will be returned as a document bundle with a MeasureReport entry for each included measure Note: as this is the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to calculate a set of measures for a particular topic (e.g. Preventive Care and Screening) and return a document describing the results of each measure for the given subject. Note that it is up to the server to determine whether or not the generated care gaps report is persisted. If the server does not persist the results, the operation does not affect state and can be invoked with a GET

14.14.9.5.1 Examples

Request: Request a care gaps report for a patient for all Preventive Care and Screening measures

```
GET [base]/Measure/$care-
    gaps?periodStart=2014&periodEnd=2014&topic=Preventive%20Care%20and%20Screening&patient=Patient/123
```

Response: Returned on successful evaluation

```
HTTP/1.1 200 OK

<Bundle xmlns="http://hl7.org/fhir">
    <type value="document"/>
    <entry>
```

```
    <resource>
        <Composition xmlns="http://hl7.org/fhir">
            <status value="final"/>
            <type>
                <coding>
                    <system value="http://loinc.org"/>
                    <code value="57024-2"/>
                </coding>
            </type>
            <subject>
                <reference value="Patient/Patient-1153"/>
            </subject>
            <date value="2018-07-22"/>
            <author>
      <reference value="Organization/123"/>
            </author>
            <title value="Preventive Care and Screening Care Gap Report"/>
            <section>
                <title value="Breast Cancer Screening Cohort Definition"/>
                <entry>
                    <reference value="MeasureReport/be224ee3-20aa-4058-bf41-52a6be342d81"/>
                </entry>
            </section>
            <section>
              <title value="Cervical Cancer Screening Cohort Definition"/>
              <entry>
                  <reference value="MeasureReport/86fa5aed-8193-4f55-835b-5bbafbd1e7d6"/>
              </entry>
            </section>
        </Composition>
    </resource>
</entry>
<entry>
    <resource>
        <MeasureReport xmlns="http://hl7.org/fhir">
            <id value="be224ee3-20aa-4058-bf41-52a6be342d81"/>
            <contained>
                <Bundle xmlns="http://hl7.org/fhir">
                    <id value="3eb57f94-7b33-48b1-9673-3b18634dca04"/>
                    <type value="collection"/>
                </Bundle>
            </contained>
            <status value="complete"/>
            <type value="individual"/>
            <measure value="Measure/measure-bcs"/>
            <subject>
                <reference value="Patient/Patient-1153"/>
            </subject>
            <period>
                <start value="2017-01-01T00:00:00+00:00"/>
                <end value="2017-12-31T00:00:00+00:00"/>
            </period>
            <group>
                <population>
                    <code>
                        <coding>
                            <code value="initial-population"/>
                        </coding>
                    </code>
                    <count value="1"/>
                </population>
                <population>
                    <code>
                        <coding>
                            <code value="numerator"/>
                        </coding>
                    </code>
                    <count value="0"/>
                </population>
                <population>
                    <code>
                        <coding>
                            <code value="denominator"/>
                        </coding>
                    </code>
                    <count value="1"/>
                </population>
            </group>
            <evaluatedResource>
                <reference value="#3eb57f94-7b33-48b1-9673-3b18634dca04"/>
            </evaluatedResource>
        </MeasureReport>
    </resource>
</entry>
<entry>
  <resource>
    <MeasureReport xmlns="http://hl7.org/fhir">
        <id value="86fa5aed-8193-4f55-835b-5bbafbd1e7d6"/>
        <contained>
```

```
                <Bundle xmlns="http://hl7.org/fhir">
                    <id value="8e6e9224-59b6-4edd-b17f-58981a754e57"/>
                    <type value="collection"/>
                </Bundle>
            </contained>
            <status value="complete"/>
            <type value="individual"/>
            <measure value="Measure/measure-ccs"/>
            <subject>
                <reference value="Patient/Patient-1153"/>
            </subject>
            <period>
                <start value="2017-01-01T00:00:00+00:00"/>
                <end value="2017-12-31T00:00:00+00:00"/>
            </period>
            <group>
                <population>
                    <code>
                        <coding>
                            <code value="initial-population"/>
                        </coding>
                    </code>
                    <count value="1"/>
                </population>
                <population>
                    <code>
                        <coding>
                            <code value="numerator"/>
                        </coding>
                    </code>
                    <count value="1"/>
                </population>
                <population>
                    <code>
                        <coding>
                            <code value="denominator"/>
                        </coding>
                    </code>
                    <count value="1"/>
                </population>
            </group>
            <evaluatedResource>
                <reference value="#8e6e9224-59b6-4edd-b17f-58981a754e57"/>
            </evaluatedResource>
        </MeasureReport>
      </resource>
    </entry>
</Bundle>
```

For more information about operations, including how they are invoked, see Operations.

14.14.10 Resource Measure - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/measure-version-maps.html

No validation errors - all conversions are clean

14.14.10 Resource Measure - R3/R4 Conversions

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Anonymous | Compartments: Not linked to any defined compartments

R3 : R4 Conversion maps for Measure.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (0 errors). (see documentation)

14.14.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/Measure3to4" = "R3 to R4 Conversions for Measure"

uses "http://hl7.org/fhir/3.0/StructureDefinition/Measure" alias MeasureR3 as source
uses "http://hl7.org/fhir/StructureDefinition/Measure" alias Measure as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group Measure(source src : MeasureR3, target tgt : Measure) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
```

```
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.contributor as vs where type = 'author' -> tgt.author as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'editor' -> tgt.editor as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'reviewer' -> tgt.reviewer as vt then Contributor(vs, vt);
  src.contributor as vs where type = 'endorser' -> tgt.endorser as vt then Contributor(vs, vt);
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
  src.disclaimer -> tgt.disclaimer;
  src.scoring -> tgt.scoring;
  src.compositeScoring -> tgt.compositeScoring;
  src.type -> tgt.type;
  src.riskAdjustment -> tgt.riskAdjustment;
  src.rateAggregation -> tgt.rateAggregation;
  src.rationale -> tgt.rationale;
  src.clinicalRecommendationStatement -> tgt.clinicalRecommendationStatement;
  src.improvementNotation as s ->  tgt.improvementNotation as cc,  cc.text = s;
  src.definition -> tgt.definition;
  src.guidance -> tgt.guidance;
  src.set as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.set',
    vt.value = v;
  src.group as s -> tgt.group as t then group(s, t);
  src.supplementalData as s -> tgt.supplementalData as t then supplementalData(s, t);
}

group group(source src, target tgt) extends BackboneElement {
  src.identifier as vs -> tgt.code as vt then Identifier2Codeable(vs, vt);
  src.name as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.group.name',  vt.value = v;
  src.description -> tgt.description;
  src.population as s -> tgt.population as t then population(s, t);
  src.stratifier as s -> tgt.stratifier as t then stratifier(s, t);
}

group population(source src, target tgt) extends BackboneElement {
  src.identifier as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.group.population.identifier',  vt.value = v;
  src.code -> tgt.code;
  src.name as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.group.population.name',  vt.value = v;
  src.description -> tgt.description;
  src.criteria as s ->  tgt.criteria as t,  t.expression = s;
}

group stratifier(source src, target tgt) extends BackboneElement {
  src.identifier as vs -> tgt.code as vt then Identifier2Codeable(vs, vt);
  src.criteria as s ->  tgt.criteria as t,  t.expression = s;
  src.path as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.group.stratifier.path',  vt.value = v;
}

group supplementalData(source src, target tgt) extends BackboneElement {
  src.identifier as vs -> tgt.code as vt then Identifier2Codeable(vs, vt);
  src.usage -> tgt.usage;
  src.description -> tgt.description;
  src.criteria as s ->  tgt.criteria as t,  t.expression = s;
  src.path as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.supplementalData.path',  vt.value = v;
}
```

14.14.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/Measure4to3" = "R4 to R3 Conversion for Measure"

uses "http://hl7.org/fhir/StructureDefinition/Measure" alias Measure as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/Measure" alias MeasureR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group Measure(source src : MeasureR3, target tgt : Measure) extends DomainResource <<type+>> {
  src.url -> tgt.url;
  src.identifier -> tgt.identifier;
  src.version -> tgt.version;
  src.name -> tgt.name;
  src.title -> tgt.title;
  src.status -> tgt.status;
  src.experimental -> tgt.experimental;
  src.date -> tgt.date;
  src.publisher -> tgt.publisher;
  src.contact -> tgt.contact;
  src.description -> tgt.description;
  src.useContext -> tgt.useContext;
  src.jurisdiction -> tgt.jurisdiction;
  src.purpose -> tgt.purpose;
```

```
  src.usage -> tgt.usage;
  src.copyright -> tgt.copyright;
  src.approvalDate -> tgt.approvalDate;
  src.lastReviewDate -> tgt.lastReviewDate;
  src.effectivePeriod -> tgt.effectivePeriod;
  src.topic -> tgt.topic;
  src.author as vs ->  tgt.contributor as vt,  vt.type = 'author' then Contributor(vs, vt);
  src.editor as vs ->  tgt.contributor as vt,  vt.type = 'editor',  vt.contact = vs;
  src.reviewer as vs ->  tgt.contributor as vt,  vt.type = 'reviewer',  vt.contact = vs;
  src.endorser as vs ->  tgt.contributor as vt,  vt.type = 'endorser',  vt.contact = vs;
  src.relatedArtifact -> tgt.relatedArtifact;
  src.library -> tgt.library;
  src.disclaimer -> tgt.disclaimer;
  src.scoring -> tgt.scoring;
  src.compositeScoring -> tgt.compositeScoring;
  src.type -> tgt.type;
  src.riskAdjustment -> tgt.riskAdjustment;
  src.rateAggregation -> tgt.rateAggregation;
  src.rationale -> tgt.rationale;
  src.clinicalRecommendationStatement -> tgt.clinicalRecommendationStatement;
  src.improvementNotation as cc then {
    cc.text -> tgt.improvementNotation;
  };
  src.definition -> tgt.definition;
  src.guidance -> tgt.guidance;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.set' then {
    e.value -> tgt.set;
  };
  src.group as s -> tgt.group as t then group(s, t);
  src.supplementalData as s -> tgt.supplementalData as t then supplementalData(s, t);
}

group group(source src, target tgt) extends BackboneElement {
  src.code as vs -> tgt.identifier as vt then Codeable2Identifier(vs, vt);
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.group.name' then {
    e.value -> tgt.name;
  };
  src.description -> tgt.description;
  src.population as s -> tgt.population as t then population(s, t);
  src.stratifier as s -> tgt.stratifier as t then stratifier(s, t);
}

group population(source src, target tgt) extends BackboneElement {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    Measure.group.population.identifier' then {
    e.value -> tgt.identifier;
  };
  src.code -> tgt.code;
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.group.population.name'
    then {
    e.value -> tgt.name;
  };
  src.description -> tgt.description;
  src.criteria as s then {
    s.expression -> tgt.criteria;
  };
}

group stratifier(source src, target tgt) extends BackboneElement {
  src.code as vs -> tgt.identifier as vt then Codeable2Identifier(vs, vt);
  src.criteria as s then {
    s.expression -> tgt.criteria;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.group.stratifier.path'
    then {
    e.value -> tgt.path;
  };
}

group supplementalData(source src, target tgt) extends BackboneElement {
  src.code as vs -> tgt.identifier as vt then Codeable2Identifier(vs, vt);
  src.usage -> tgt.usage;
  src.description -> tgt.description;
  src.criteria as s then {
    s.expression -> tgt.criteria;
  };
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-Measure.supplementalData.path'
    then {
    e.value -> tgt.path;
  };
}
```

14.14.10.3 R4 Validation Errors

14.14.11 Shareable Measure

URL: https://hl7.org/fhir/R4/measure-shareable.html

Enforces the minimum information set for the measure metadata required by HL7 and other organizations that share and publish measures

14.14.11 Shareable Measure

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative

14.14.11.1 Content

Profiles:
ShareableMeasure | Enforces the minimum information set for the measure metadata required by HL7 and other organizations that share and publish measures

14.14.11.2 StructureDefinition: ShareableMeasure

URL: https://hl7.org/fhir/R4/shareablemeasure.html

14.14.11.2 StructureDefinition: ShareableMeasure

Clinical Quality Information Work Group | Maturity Level: 1 | Trial Use | Use Context: Any

The official URL for this profile is:

`http://hl7.org/fhir/StructureDefinition/shareablemeasure`

Enforces the minimum information set for the measure metadata required by HL7 and other organizations that share and publish measures

This profile was published on [no date] as a draft by HL7.

14.14.11.2.1 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots, and how the XML and JSON presentations work.

- Text Summary
- Differential Table
- Snapshot Table
- XML Template
- JSON Template
- All

This structure is derived from Measure.

Summary

Mandatory: 6 elements Must-Support: 6 elements

This structure is derived from Measure.

Name | Flags | Card. | Type | Description & Constraints
Measure | | 1..1
url | | 1..1 | uri
identifier | S | 0..* | Identifier
version | | 1..1 | string
name | | 1..1 | string
title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Name | Flags | Card. | Type | Description & Constraints
Measure | I | 1..1 | | A quality measure definition
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this measure, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the measure
version | ΣI | 1..1 | string | Business version of the measure
name | ΣI | 1..1 | string | Name for this measure (computer friendly)
title | SΣI | 0..1 | string | Name for this measure (human friendly)

subtitle | I | 0..1 | string | Subordinate title of the measure
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the measure
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this measure is defined
usage | I | 0..1 | string | Describes the clinical usage of the measure
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the measure was approved by publisher
lastReviewDate | I | 0..1 | date | When the measure was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the measure is expected to be used
topic | I | 0..* | CodeableConcept | The category of the measure, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | I | 0..* | canonical(Library) | Logic used by the measure
disclaimer | ΣI | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | ΣI | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort Binding: MeasureScoring (extensible)
compositeScoring | ΣI | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted Binding: CompositeMeasureScoring (extensible)
type | ΣI | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite Binding: MeasureType (extensible)
riskAdjustment | ΣI | 0..1 | string | How risk adjustment is applied for this measure
rateAggregation | ΣI | 0..1 | string | How is rate aggregation performed for this measure
rationale | ΣI | 0..1 | markdown | Detailed description of why the measure exists
clinicalRecommendationStatement | ΣI | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | ΣI | 0..1 | CodeableConcept | increase | decrease Binding: MeasureImprovementNotation (required)
definition | ΣI | 0..* | markdown | Defined terms used in the measure documentation
guidance | ΣI | 0..1 | markdown | Additional guidance for implementers
group | I | 0..* | BackboneElement | Population criteria group
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the group
description | I | 0..1 | string | Summary description
population | I | 0..* | BackboneElement | Population criteria
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation Binding: MeasurePopulationType (extensible)
description | I | 0..1 | string | The human readable description of this population criteria
criteria | I | 1..1 | Expression | The criteria that defines this population
stratifier | I | 0..* | BackboneElement | Stratifier criteria for the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the stratifier
description | I | 0..1 | string | The human readable description of this stratifier
criteria | I | 0..1 | Expression | How the measure should be stratified
component | I | 0..* | BackboneElement | Stratifier criteria component for the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the stratifier component
description | I | 0..1 | string | The human readable description of this stratifier component
criteria | I | 1..1 | Expression | Component of how the measure should be stratified
supplementalData | I | 0..* | BackboneElement | What other data should be reported with the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the supplemental data

usage | I | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor Binding: MeasureDataUsage (extensible)
description | I | 0..1 | string | The human readable description of this supplemental data
criteria | I | 1..1 | Expression | Expression describing additional data to be reported
Documentation for this format

```
<!-- Shareable Measure -->

<Measure xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this measure, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the measure --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the measure -->
 <name value="[string]"/><!--  1..1 Name for this measure (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this measure (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the measure -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device   --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the measure -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for measure (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this measure is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the measure -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the measure was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the measure was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the measure is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the measure, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <library value="[canonical]"/><!--  0..* Logic used by the measure -->
 <disclaimer value="[markdown]"/><!--  0..1 Disclaimer for use of the measure or its referenced content -->
 <scoring><!--  0..1 CodeableConcept proportion | ratio | continuous-variable | cohort   --></scoring>
 <compositeScoring><!--  0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted   --></compositeScoring>
 <type><!--  0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite   --></type>
 <riskAdjustment value="[string]"/><!--  0..1 How risk adjustment is applied for this measure -->
 <rateAggregation value="[string]"/><!--  0..1 How is rate aggregation performed for this measure -->
 <rationale value="[markdown]"/><!--  0..1 Detailed description of why the measure exists -->
 <clinicalRecommendationStatement value="[markdown]"/><!--  0..1 Summary of clinical guidelines -->
 <improvementNotation><!--  0..1 CodeableConcept increase | decrease   --></improvementNotation>
 <definition value="[markdown]"/><!--  0..* Defined terms used in the measure documentation -->
 <guidance value="[markdown]"/><!--  0..1 Additional guidance for implementers -->
 <group>  0..* BackboneElement  <!--  0..* Population criteria group -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..1 CodeableConcept Meaning of the group --></code>
  <description value="[string]"/><!--  0..1 Summary description -->
  <population>  0..* BackboneElement  <!--  0..* Population criteria -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code><!--  0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <description value="[string]"/><!--  0..1 The human readable description of this population criteria -->
   <criteria><!--  1..1 Expression
       The criteria that defines this population --></criteria>
  </population>
  <stratifier>  0..* BackboneElement  <!--  0..* Stratifier criteria for the measure -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code><!--  0..1 CodeableConcept
       Meaning of the stratifier --></code>
   <description value="[string]"/><!--  0..1 The human readable description of this stratifier -->
   <criteria><!--  0..1 Expression
       How the measure should be stratified --></criteria>
   <component>  0..* BackboneElement  <!--  0..* Stratifier criteria component for the measure -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <code><!--  0..1 CodeableConcept
```

```
        Meaning of the stratifier component --></code>
    <description value="[string]"/><!--  0..1
        The human readable description of this stratifier component -->
    <criteria><!--  1..1 Expression
        Component of how the measure should be stratified --></criteria>
   </component>
  </stratifier>
 </group>
 <supplementalData>  0..* BackboneElement  <!--  0..* What other data should be reported with the measure -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..1 CodeableConcept
      Meaning of the supplemental data --></code>
  <usage><!--  0..* CodeableConcept supplemental-data | risk-adjustment-factor   --></usage>
  <description value="[string]"/><!--  0..1 The human readable description of this supplemental data -->
  <criteria><!--  1..1 Expression
      Expression describing additional data to be reported --></criteria>
 </supplementalData>
</Measure>
```

```
{ // Shareable Measure
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this measure, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the measure
    "version" : "<string>", // C? R! Business version of the measure
    "name" : "<string>", // C? R! Name for this measure (computer friendly)
    "title" : "<string>", // C?Name for this measure (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the measure
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device
    "subjectReference" : { Reference(Group) }, // C?E.g. Patient, Practitioner, RelatedPerson, Organization, Location,
    Device
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the measure
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for measure (if applicable)
    "purpose" : "<markdown>", // C?Why this measure is defined
    "usage" : "<string>", // C?Describes the clinical usage of the measure
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the measure was approved by publisher
    "lastReviewDate" : "<date>", // C?When the measure was last reviewed
    "effectivePeriod" : { Period }, // C?When the measure is expected to be used
    "topic" : [{ CodeableConcept }], // C?The category of the measure, such as Education, Treatment, Assessment, etc.
    "author" : [{ ContactDetail }], // C?Who authored the content
    "editor" : [{ ContactDetail }], // C?Who edited the content
    "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
    "endorser" : [{ ContactDetail }], // C?Who endorsed the content
    "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
    "library" : ["<canonical>"], // C?Logic used by the measure
    "disclaimer" : "<markdown>", // C?Disclaimer for use of the measure or its referenced content
    "scoring" : { CodeableConcept }, // C?proportion | ratio | continuous-variable | cohort
    "compositeScoring" : { CodeableConcept }, // C?opportunity | all-or-nothing | linear | weighted
    "type" : [{ CodeableConcept }], // C?process | outcome | structure | patient-reported-outcome | composite
    "riskAdjustment" : "<string>", // C?How risk adjustment is applied for this measure
    "rateAggregation" : "<string>", // C?How is rate aggregation performed for this measure
    "rationale" : "<markdown>", // C?Detailed description of why the measure exists
    "clinicalRecommendationStatement" : "<markdown>", // C?Summary of clinical guidelines
    "improvementNotation" : { CodeableConcept }, // C?increase | decrease
    "definition" : ["<markdown>"], // C?Defined terms used in the measure documentation
    "guidance" : "<markdown>", // C?Additional guidance for implementers
    "group" : [{ BackboneElement }] // C?Population criteria group
    "supplementalData" : [{ BackboneElement }], // C?What other data should be reported with the measure
  }
```

This structure is derived from Measure.

Summary

Mandatory: 6 elements Must-Support: 6 elements

Differential View

This structure is derived from Measure.

Name	Flags	Card.	Type	Description & Constraints
Measure		1..1		
url		1..1	uri	
identifier	S	0..*	Identifier	
version		1..1	string	

name | | 1..1 | string
title | S | 0..1 | string
experimental | | 1..1 | boolean
date | S | 0..1 | dateTime
publisher | | 1..1 | string
contact | S | 0..* | ContactDetail
description | | 1..1 | markdown
useContext | S | 0..* | UsageContext
jurisdiction | S | 0..* | CodeableConcept
Documentation for this format

Snapshot View

Name | Flags | Card. | Type | Description & Constraints
Measure | I | 1..1 | | A quality measure definition
id | Σ | 0..1 | string | Logical id of this artifact
meta | ΣI | 0..1 | Meta | Metadata about the resource
implicitRules | ?!ΣI | 0..1 | uri | A set of rules under which this content was created
language | I | 0..1 | code | Language of the resource content Binding: Common Languages (preferred) Max Binding: All Languages
text | I | 0..1 | Narrative | Text summary of the resource, for human interpretation
contained | | 0..* | Resource | Contained, inline Resources
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!I | 0..* | Extension | Extensions that cannot be ignored
url | ΣI | 1..1 | uri | Canonical identifier for this measure, represented as a URI (globally unique)
identifier | SΣI | 0..* | Identifier | Additional identifier for the measure
version | ΣI | 1..1 | string | Business version of the measure
name | ΣI | 1..1 | string | Name for this measure (computer friendly)
title | SΣI | 0..1 | string | Name for this measure (human friendly)
subtitle | I | 0..1 | string | Subordinate title of the measure
status | ?!ΣI | 1..1 | code | draft | active | retired | unknown Binding: PublicationStatus (required)
experimental | ΣI | 1..1 | boolean | For testing purposes, not real usage
subject[x] | I | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Binding: Subject type (extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(group.html)
date | SΣI | 0..1 | dateTime | Date last changed
publisher | ΣI | 1..1 | string | Name of the publisher (organization or individual)
contact | SΣI | 0..* | ContactDetail | Contact details for the publisher
description | ΣI | 1..1 | markdown | Natural language description of the measure
useContext | SΣI | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | SΣI | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Binding: Jurisdiction (extensible)
purpose | I | 0..1 | markdown | Why this measure is defined
usage | I | 0..1 | string | Describes the clinical usage of the measure
copyright | I | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | I | 0..1 | date | When the measure was approved by publisher
lastReviewDate | I | 0..1 | date | When the measure was last reviewed
effectivePeriod | ΣI | 0..1 | Period | When the measure is expected to be used
topic | I | 0..* | CodeableConcept | The category of the measure, such as Education, Treatment, Assessment, etc. Binding: DefinitionTopic (example)
author | I | 0..* | ContactDetail | Who authored the content
editor | I | 0..* | ContactDetail | Who edited the content
reviewer | I | 0..* | ContactDetail | Who reviewed the content
endorser | I | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | I | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | I | 0..* | canonical(Library) | Logic used by the measure
disclaimer | ΣI | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | ΣI | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort Binding: MeasureScoring (extensible)
compositeScoring | ΣI | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted Binding: CompositeMeasureScoring (extensible)
type | ΣI | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite Binding: MeasureType (extensible)
riskAdjustment | ΣI | 0..1 | string | How risk adjustment is applied for this measure
rateAggregation | ΣI | 0..1 | string | How is rate aggregation performed for this measure
rationale | ΣI | 0..1 | markdown | Detailed description of why the measure exists
clinicalRecommendationStatement | ΣI | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | ΣI | 0..1 | CodeableConcept | increase | decrease Binding: MeasureImprovementNotation (required)
definition | ΣI | 0..* | markdown | Defined terms used in the measure documentation
guidance | ΣI | 0..1 | markdown | Additional guidance for implementers
group | I | 0..* | BackboneElement | Population criteria group
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the group
description | I | 0..1 | string | Summary description
population | I | 0..* | BackboneElement | Population criteria

id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation Binding: MeasurePopulationType (extensible)
description | I | 0..1 | string | The human readable description of this population criteria
criteria | I | 1..1 | Expression | The criteria that defines this population
stratifier | I | 0..* | BackboneElement | Stratifier criteria for the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the stratifier
description | I | 0..1 | string | The human readable description of this stratifier
criteria | I | 0..1 | Expression | How the measure should be stratified
component | I | 0..* | BackboneElement | Stratifier criteria component for the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the stratifier component
description | I | 0..1 | string | The human readable description of this stratifier component
criteria | I | 1..1 | Expression | Component of how the measure should be stratified
supplementalData | I | 0..* | BackboneElement | What other data should be reported with the measure
id | | 0..1 | string | Unique id for inter-element referencing
extension | I | 0..* | Extension | Additional content defined by implementations
modifierExtension | ?!ΣI | 0..* | Extension | Extensions that cannot be ignored even if unrecognized
code | I | 0..1 | CodeableConcept | Meaning of the supplemental data
usage | I | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor Binding: MeasureDataUsage (extensible)
description | I | 0..1 | string | The human readable description of this supplemental data
criteria | I | 1..1 | Expression | Expression describing additional data to be reported
Documentation for this format

XML Template

```
<!-- Shareable Measure -->

<Measure xmlns="http://hl7.org/fhir"
>
  <!-- from Element: extension -->
 <id><!-- 0..1 string Logical id of this artifact --></id>
 <meta><!--  0..1 Meta Metadata about the resource --></meta>
 <implicitRules value="[uri]"/><!--  0..1 A set of rules under which this content was created -->
 <language value="[code]"/><!--  0..1 Language of the resource content   -->
 <text><!--  0..1 Narrative
     Text summary of the resource, for human interpretation --></text>
 <contained><!-- 0..* Resource Contained, inline Resources --></contained>
 <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored  --></modifierExtension>
 <url value="[uri]"/><!--  1..1 Canonical identifier for this measure, represented as a URI (globally unique) -->
 <identifier><!--  0..* Identifier Additional identifier for the measure --></identifier>
 <version value="[string]"/><!--  1..1 Business version of the measure -->
 <name value="[string]"/><!--  1..1 Name for this measure (computer friendly) -->
 <title value="[string]"/><!--  0..1 Name for this measure (human friendly) -->
 <subtitle value="[string]"/><!--  0..1 Subordinate title of the measure -->
 <status value="[code]"/><!--  1..1 draft | active | retired | unknown   -->
 <experimental value="[boolean]"/><!--  1..1 For testing purposes, not real usage -->
 <subject[x]><!--  0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device   --></subject[x]>
 <date value="[dateTime]"/><!--  0..1 Date last changed -->
 <publisher value="[string]"/><!--  1..1 Name of the publisher (organization or individual) -->
 <contact><!--  0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  1..1 Natural language description of the measure -->
 <useContext><!--  0..* UsageContext
     The context that the content is intended to support --></useContext>
 <jurisdiction><!--  0..* CodeableConcept Intended jurisdiction for measure (if applicable)   --></jurisdiction>
 <purpose value="[markdown]"/><!--  0..1 Why this measure is defined -->
 <usage value="[string]"/><!--  0..1 Describes the clinical usage of the measure -->
 <copyright value="[markdown]"/><!--  0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!--  0..1 When the measure was approved by publisher -->
 <lastReviewDate value="[date]"/><!--  0..1 When the measure was last reviewed -->
 <effectivePeriod><!--  0..1 Period
     When the measure is expected to be used --></effectivePeriod>
 <topic><!--  0..* CodeableConcept The category of the measure, such as Education, Treatment, Assessment, etc.
    --></topic>
 <author><!--  0..* ContactDetail Who authored the content --></author>
 <editor><!--  0..* ContactDetail Who edited the content --></editor>
 <reviewer><!--  0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!--  0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!--  0..* RelatedArtifact
     Additional documentation, citations, etc. --></relatedArtifact>
 <library value="[canonical]"/><!--  0..* Logic used by the measure -->
 <disclaimer value="[markdown]"/><!--  0..1 Disclaimer for use of the measure or its referenced content -->
 <scoring><!--  0..1 CodeableConcept proportion | ratio | continuous-variable | cohort   --></scoring>
 <compositeScoring><!--  0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted   --></compositeScoring>
```

```
 <type><!--  0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite   --></type>
 <riskAdjustment value="[string]"/><!--  0..1 How risk adjustment is applied for this measure -->
 <rateAggregation value="[string]"/><!--  0..1 How is rate aggregation performed for this measure -->
 <rationale value="[markdown]"/><!--  0..1 Detailed description of why the measure exists -->
 <clinicalRecommendationStatement value="[markdown]"/><!--  0..1 Summary of clinical guidelines -->
 <improvementNotation><!--  0..1 CodeableConcept increase | decrease   --></improvementNotation>
 <definition value="[markdown]"/><!--  0..* Defined terms used in the measure documentation -->
 <guidance value="[markdown]"/><!--  0..1 Additional guidance for implementers -->
 <group>  0..* BackboneElement  <!--  0..* Population criteria group -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..1 CodeableConcept Meaning of the group --></code>
  <description value="[string]"/><!--  0..1 Summary description -->
  <population>  0..* BackboneElement  <!--  0..* Population criteria -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code><!--  0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <description value="[string]"/><!--  0..1 The human readable description of this population criteria -->
   <criteria><!--  1..1 Expression
       The criteria that defines this population --></criteria>
  </population>
  <stratifier>  0..* BackboneElement  <!--  0..* Stratifier criteria for the measure -->
   <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
   <code><!--  0..1 CodeableConcept
       Meaning of the stratifier --></code>
   <description value="[string]"/><!--  0..1 The human readable description of this stratifier -->
   <criteria><!--  0..1 Expression
       How the measure should be stratified --></criteria>
   <component>  0..* BackboneElement  <!--  0..* Stratifier criteria component for the measure -->
    <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
    <code><!--  0..1 CodeableConcept
        Meaning of the stratifier component --></code>
    <description value="[string]"/><!--  0..1
        The human readable description of this stratifier component -->
    <criteria><!--  1..1 Expression
        Component of how the measure should be stratified --></criteria>
   </component>
  </stratifier>
 </group>
 <supplementalData>  0..* BackboneElement  <!--  0..* What other data should be reported with the measure -->
  <modifierExtension><!--  0..* Extension  Extensions that cannot be ignored even if unrecognized
    --></modifierExtension>
  <code><!--  0..1 CodeableConcept
      Meaning of the supplemental data --></code>
  <usage><!--  0..* CodeableConcept supplemental-data | risk-adjustment-factor   --></usage>
  <description value="[string]"/><!--  0..1 The human readable description of this supplemental data -->
  <criteria><!--  1..1 Expression
      Expression describing additional data to be reported --></criteria>
 </supplementalData>
</Measure>
```

JSON Template

```
{ // Shareable Measure
  // from Element: extension
    "meta" : { Meta }, // C?Metadata about the resource
    "implicitRules" : "<uri>", // C?A set of rules under which this content was created
    "language" : "<code>", // C?Language of the resource content
    "text" : { Narrative }, // C?Text summary of the resource, for human interpretation
    "contained" : [{ Resource }], //Contained, inline Resources
    "modifierExtension" : [{ Extension }], // C?Extensions that cannot be ignored
    "url" : "<uri>", // C? R! Canonical identifier for this measure, represented as a URI (globally unique)
    "identifier" : [{ Identifier }], // C?Additional identifier for the measure
    "version" : "<string>", // C? R! Business version of the measure
    "name" : "<string>", // C? R! Name for this measure (computer friendly)
    "title" : "<string>", // C?Name for this measure (human friendly)
    "subtitle" : "<string>", // C?Subordinate title of the measure
    "status" : "<code>", // C? R! draft | active | retired | unknown
    "experimental" : <boolean>, // C? R! For testing purposes, not real usage
// value[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
    "subjectCodeableConcept" : { CodeableConcept }, // C?E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device
    "subjectReference" : { Reference(Group) }, // C?E.g. Patient, Practitioner, RelatedPerson, Organization, Location,
    Device
    "date" : "<dateTime>", // C?Date last changed
    "publisher" : "<string>", // C? R! Name of the publisher (organization or individual)
    "contact" : [{ ContactDetail }], // C?Contact details for the publisher
    "description" : "<markdown>", // C? R! Natural language description of the measure
    "useContext" : [{ UsageContext }], // C?The context that the content is intended to support
    "jurisdiction" : [{ CodeableConcept }], // C?Intended jurisdiction for measure (if applicable)
    "purpose" : "<markdown>", // C?Why this measure is defined
    "usage" : "<string>", // C?Describes the clinical usage of the measure
    "copyright" : "<markdown>", // C?Use and/or publishing restrictions
    "approvalDate" : "<date>", // C?When the measure was approved by publisher
    "lastReviewDate" : "<date>", // C?When the measure was last reviewed
    "effectivePeriod" : { Period }, // C?When the measure is expected to be used
```

```
  "topic" : [{ CodeableConcept }], // C?The category of the measure, such as Education, Treatment, Assessment, etc.
  "author" : [{ ContactDetail }], // C?Who authored the content
  "editor" : [{ ContactDetail }], // C?Who edited the content
  "reviewer" : [{ ContactDetail }], // C?Who reviewed the content
  "endorser" : [{ ContactDetail }], // C?Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // C?Additional documentation, citations, etc.
  "library" : ["<canonical>"], // C?Logic used by the measure
  "disclaimer" : "<markdown>", // C?Disclaimer for use of the measure or its referenced content
  "scoring" : { CodeableConcept }, // C?proportion | ratio | continuous-variable | cohort
  "compositeScoring" : { CodeableConcept }, // C?opportunity | all-or-nothing | linear | weighted
  "type" : [{ CodeableConcept }], // C?process | outcome | structure | patient-reported-outcome | composite
  "riskAdjustment" : "<string>", // C?How risk adjustment is applied for this measure
  "rateAggregation" : "<string>", // C?How is rate aggregation performed for this measure
  "rationale" : "<markdown>", // C?Detailed description of why the measure exists
  "clinicalRecommendationStatement" : "<markdown>", // C?Summary of clinical guidelines
  "improvementNotation" : { CodeableConcept }, // C?increase | decrease
  "definition" : ["<markdown>"], // C?Defined terms used in the measure documentation
  "guidance" : "<markdown>", // C?Additional guidance for implementers
  "group" : [{ BackboneElement }] // C?Population criteria group
  "supplementalData" : [{ BackboneElement }], // C?What other data should be reported with the measure
}
```

Alternate definitions: Master Definition (XML, JSON), Schematron

14.14.11.2.2 Terminology Bindings

Path	Name	Conformance	ValueSet
Measure.language	Common Languages	preferred	Common Languages
Measure.status	PublicationStatus	required	PublicationStatus
Measure.subject[x]	Subject type	extensible	Subject type
Measure.jurisdiction	Jurisdiction	extensible	Jurisdiction
Measure.topic	DefinitionTopic	example	DefinitionTopic
Measure.scoring	MeasureScoring	extensible	MeasureScoring
Measure.compositeScoring	CompositeMeasureScoring	extensible	CompositeMeasureScoring
Measure.type	MeasureType	extensible	MeasureType
Measure.improvementNotation	MeasureImprovementNotation	required	MeasureImprovementNotation
Measure.group.population.code	MeasurePopulationType	extensible	MeasurePopulationType
Measure.supplementalData.usage	MeasureDataUsage	extensible	MeasureDataUsage

14.14.12 JSON Schema for Measure

URL: https://hl7.org/fhir/R4/measure.schema.json.html

14.14.12 JSON Schema for Measure

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for Measure, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/Measure",
  "$ref": "#/definitions/Measure",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "Measure": {
      "description": "The Measure resource provides the definition of a quality measure.",
      "properties": {
        "resourceType": {
          "description": "This is a Measure resource",
          "const": "Measure"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
```

```
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this measure when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this measure is (or will be)
published. This URL can be the target of a canonical reference. It SHALL remain the same when the measure is stored
on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this measure when it is represented in other
formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the measure when it is referenced in a
 specification, model, design or instance. This is an arbitrary value managed by the measure author and is not
expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not
available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a
version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g.
1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification.
Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the measure. This name should be usable as an identifier
for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the measure.",
```

```
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the measure giving additional information about its
content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this measure. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this measure is authored for testing purposes (or
education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectCodeableConcept": {
      "description": "The intended subjects for the measure. If this element is not provided, a Patient subject is
assumed, but the subject of the measure can be anything.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "The intended subjects for the measure. If this element is not provided, a Patient subject is
assumed, but the subject of the measure can be anything.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the measure was published. The date must change when the
business version changes and it must change if the status code changes. In addition, it should change when the
substantive content of the measure changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the measure.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the measure from a consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate measure instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
```

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    "jurisdiction": {
      "description": "A legal or geographic region in which the measure is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this measure is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description, from a clinical perspective, of how the measure is used.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the measure and/or its contents. Copyright statements are
generally legal restrictions on the use and publishing of the measure.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the measure content was or is planned to be in active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the measure. Topics provide a high-level
categorization grouping types of measures that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
```

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        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "library": {
      "description": "A reference to a Library resource containing the formal logic used by the measure.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "disclaimer": {
      "description": "Notices and disclaimers regarding the use of the measure or related to intellectual property
(such as code systems) referenced by the measure.",
      "$ref": "#/definitions/markdown"
    },
    "_disclaimer": {
      "description": "Extensions for disclaimer",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "scoring": {
      "description": "Indicates how the calculation is performed for the measure, including proportion, ratio,
continuous-variable, and cohort. The value set is extensible, allowing additional measure scoring types to be
represented.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "compositeScoring": {
      "description": "If this is a composite measure, the scoring method used to combine the component measures to
determine the composite score.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "type": {
      "description": "Indicates whether the measure is used to examine a process, an outcome over time, a patient-
reported outcome, or a structure measure such as utilization.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "riskAdjustment": {
      "description": "A description of the risk adjustment factors that may impact the resulting score for the
measure and how they may be accounted for when computing and reporting measure results.",
      "$ref": "#/definitions/string"
    },
    "_riskAdjustment": {
      "description": "Extensions for riskAdjustment",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "rateAggregation": {
      "description": "Describes how to combine the information calculated, based on logic in each of several
populations, into one summarized result.",
      "$ref": "#/definitions/string"
    },
    "_rateAggregation": {
      "description": "Extensions for rateAggregation",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "rationale": {
      "description": "Provides a succinct statement of the need for the measure. Usually includes statements
pertaining to importance criterion: impact, gap in care, and evidence.",
      "$ref": "#/definitions/markdown"
    },
    "_rationale": {
      "description": "Extensions for rationale",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "clinicalRecommendationStatement": {
      "description": "Provides a summary of relevant clinical guidelines or other clinical recommendations
supporting the measure.",
      "$ref": "#/definitions/markdown"
    },
    "_clinicalRecommendationStatement": {
      "description": "Extensions for clinicalRecommendationStatement",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "improvementNotation": {
      "description": "Information on whether an increase or decrease in score is the preferred result (e.g., a
higher score indicates better quality OR a lower score indicates better quality OR quality is within a range).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "definition": {
```

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      "description": "Provides a description of an individual term used within the measure.",
      "items": {
        "$ref": "#/definitions/markdown"
      },
      "type": "array"
    },
    "_definition": {
      "description": "Extensions for definition",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "guidance": {
      "description": "Additional guidance for the measure including how it can be used in a clinical context, and
the intent of the measure.",
      "$ref": "#/definitions/markdown"
    },
    "_guidance": {
      "description": "Extensions for guidance",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "group": {
      "description": "A group of population criteria for the measure.",
      "items": {
        "$ref": "#/definitions/Measure_Group"
      },
      "type": "array"
    },
    "supplementalData": {
      "description": "The supplemental data criteria for the measure report, specified as either the name of a valid
 CQL expression within a referenced library, or a valid FHIR Resource Path.",
      "items": {
        "$ref": "#/definitions/Measure_SupplementalData"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "resourceType"
  ]
},
"Measure_Group": {
  "description": "The Measure resource provides the definition of a quality measure.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Indicates a meaning for the group. This can be as simple as a unique identifier, or it can
establish meaning in a broader context by drawing from a terminology, allowing groups to be correlated across
measures.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "The human readable description of this population group.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
```

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    "population": {
      "description": "A population criteria for the measure.",
      "items": {
        "$ref": "#/definitions/Measure_Population"
      },
      "type": "array"
    },
    "stratifier": {
      "description": "The stratifier criteria for the measure report, specified as either the name of a valid CQL
expression defined within a referenced library or a valid FHIR Resource Path.",
      "items": {
        "$ref": "#/definitions/Measure_Stratifier"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Measure_Population": {
  "description": "The Measure resource provides the definition of a quality measure.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The type of population criteria.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "The human readable description of this population criteria.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "criteria": {
      "description": "An expression that specifies the criteria for the population, typically the name of an
expression in a library.",
      "$ref": "Expression.schema.json#/definitions/Expression"
    }
  },
  "additionalProperties": false,
  "required": [
    "criteria"
  ]
},
"Measure_Stratifier": {
  "description": "The Measure resource provides the definition of a quality measure.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
```

```
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Indicates a meaning for the stratifier. This can be as simple as a unique identifier, or it
can establish meaning in a broader context by drawing from a terminology, allowing stratifiers to be correlated
across measures.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "The human readable description of this stratifier criteria.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "criteria": {
      "description": "An expression that specifies the criteria for the stratifier. This is typically the name of an
 expression defined within a referenced library, but it may also be a path to a stratifier element.",
      "$ref": "Expression.schema.json#/definitions/Expression"
    },
    "component": {
      "description": "A component of the stratifier criteria for the measure report, specified as either the name of
 a valid CQL expression defined within a referenced library or a valid FHIR Resource Path.",
      "items": {
        "$ref": "#/definitions/Measure_Component"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"Measure_Component": {
  "description": "The Measure resource provides the definition of a quality measure.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "Indicates a meaning for the stratifier component. This can be as simple as a unique
identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing stratifiers to
be correlated across measures.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "description": {
      "description": "The human readable description of this stratifier criteria component.",
      "$ref": "#/definitions/string"
    },
    "_description": {
```

```
        "description": "Extensions for description",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "criteria": {
        "description": "An expression that specifies the criteria for this component of the stratifier. This is typically the name of an expression defined within a referenced library, but it may also be a path to a stratifier element.",
        "$ref": "Expression.schema.json#/definitions/Expression"
      }
    },
    "additionalProperties": false,
    "required": [
      "criteria"
    ]
  },
  "Measure_SupplementalData": {
    "description": "The Measure resource provides the definition of a quality measure.",
    "properties": {
      "id": {
        "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
        "$ref": "string.schema.json#/definitions/string"
      },
      "extension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "modifierExtension": {
        "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
        "items": {
          "$ref": "Extension.schema.json#/definitions/Extension"
        },
        "type": "array"
      },
      "code": {
        "description": "Indicates a meaning for the supplemental data. This can be as simple as a unique identifier, or it can establish meaning in a broader context by drawing from a terminology, allowing supplemental data to be correlated across measures.",
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "usage": {
        "description": "An indicator of the intended usage for the supplemental data element. Supplemental data indicates the data is additional information requested to augment the measure information. Risk adjustment factor indicates the data is additional information used to calculate risk adjustment factors when applying a risk model to the measure calculation.",
        "items": {
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "type": "array"
      },
      "description": {
        "description": "The human readable description of this supplemental data.",
        "$ref": "#/definitions/string"
      },
      "_description": {
        "description": "Extensions for description",
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "criteria": {
        "description": "The criteria for the supplemental data. This is typically the name of a valid expression defined within a referenced library, but it may also be a path to a specific data element. The criteria defines the data to be returned for this element.",
        "$ref": "Expression.schema.json#/definitions/Expression"
      }
    },
    "additionalProperties": false,
    "required": [
      "criteria"
    ]
  }
 }
}
```


14.14.13 Resource Measure - Maturity Dependency View

URL: https://hl7.org/fhir/R4/measure-dependencies.html

14.14.13 Resource Measure - Maturity Dependency View

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for Measure FMM level 2

Name | Card. | Type | Dependency Analysis
Measure | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
library | 0..* | canonical(Library)
disclaimer | 0..1 | markdown
scoring | 0..1 | CodeableConcept
compositeScoring | 0..1 | CodeableConcept
type | 0..* | CodeableConcept
riskAdjustment | 0..1 | string
rateAggregation | 0..1 | string
rationale | 0..1 | markdown
clinicalRecommendationStatement | 0..1 | markdown
improvementNotation | 0..1 | CodeableConcept
definition | 0..* | markdown
guidance | 0..1 | markdown
group | 0..* | BackboneElement
code | 0..1 | CodeableConcept
description | 0..1 | string
population | 0..* | BackboneElement
code | 0..1 | CodeableConcept
description | 0..1 | string
criteria | 1..1 | Expression
stratifier | 0..* | BackboneElement
code | 0..1 | CodeableConcept
description | 0..1 | string
criteria | 0..1 | Expression
component | 0..* | BackboneElement
code | 0..1 | CodeableConcept
description | 0..1 | string
criteria | 1..1 | Expression
supplementalData | 0..* | BackboneElement
code | 0..1 | CodeableConcept
usage | 0..* | CodeableConcept
description | 0..1 | string
criteria | 1..1 | Expression

14.15 Resource MeasureReport - Content

URL: https://hl7.org/fhir/R4/measurereport.html
Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Patient

The MeasureReport resource contains the results of the calculation of a measure; and optionally a reference to the resources involved in that calculation.

14.15.1 Scope and Usage

The MeasureReport resource represents the results of calculating a measure for a specific subject or group of subjects. The $evaluate-measure operation of the Measure resource is defined to return a MeasureReport. The resource is capable of representing three different levels of report: individual, subject-list, and summary.

The resource draws requirements from the HL7 CDA R2 Implementation Guide: Quality Reporting Document Architecture - Category 1 (QRDA I) DSTU Release 3 (US Realm) and the HL7 Implementation Guide for CDA Release 2: Quality Reporting Document Architecture - Category III (QRDA III), DSTU Release 1 implementation guides.

For a detailed discussion of how to use the Measure and MeasureReport resources, refer to the Quality Reporting topic.

Note that this resource is a special case of the more general notion of a query evaluation result. However, because the general case requires the ability to represent arbitrary content, this resource uses a simple indicator structure to describe population sizes for each population type defined in the measure. The intent is to be able to represent the more general case as well, either by generalizing this resource, or by making this structure a profile of a more general resource, and we are actively seeking comments about what approaches might be taken to achieve that aim.

14.15.2 Boundaries and Relationships

Although the MeasureReport is conceptually an Observation, there is enough specific information required to support the quality reporting use case to warrant a separate resource.

The resource is differentiated from a general purpose query result because it communicates specific information related to quality measurement evaluation that would be difficult to convey generally without imposing some other structure on top of the general results.

14.15.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
MeasureReport | ITU | | DomainResource | Results of a measure evaluation + Rule: Measure Reports used for data collection SHALL NOT communicate group and score information + Rule: Stratifiers SHALL be either a single criteria or a set of criteria components
Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Additional identifier for the MeasureReport
status | ?!Σ | 1..1 | code | complete | pending | error MeasureReportStatus (Required)
type | ΣI | 1..1 | code | individual | subject-list | summary | data-collection MeasureReportType (Required)
measure | Σ | 1..1 | canonical(Measure) | What measure was calculated
subject | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Location | Device | RelatedPerson | Group) | What individual(s) the report is for
date | Σ | 0..1 | dateTime | When the report was generated
reporter | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Location | Organization) | Who is reporting the data
period | Σ | 1..1 | Period | What period the report covers
improvementNotation | ?!Σ | 0..1 | CodeableConcept | increase | decrease MeasureImprovementNotation (Required)
group | I | 0..* | BackboneElement | Measure results for each group
code | Σ | 0..1 | CodeableConcept | Meaning of the group
population | | 0..* | BackboneElement | The populations in the group
code | Σ | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
subjectResults | | 0..1 | Reference(List) | For subject-list reports, the subject results in this population
measureScore | Σ | 0..1 | Quantity | What score this group achieved
stratifier | | 0..* | BackboneElement | Stratification results
code | | 0..* | CodeableConcept | What stratifier of the group
stratum | | 0..* | BackboneElement | Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier components

value | | 0..1 | CodeableConcept | The stratum value, e.g. male
component | | 0..* | BackboneElement | Stratifier component values
code | | 1..1 | CodeableConcept | What stratifier component of the group
value | | 1..1 | CodeableConcept | The stratum component value, e.g. male
population | | 0..* | BackboneElement | Population results in this stratum
code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
subjectResults | | 0..1 | Reference(List) | For subject-list reports, the subject results in this population
measureScore | | 0..1 | Quantity | What score this stratum achieved
evaluatedResource | | 0..* | Reference(Any) | What data was used to calculate the measure score
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MeasureReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Additional identifier for the MeasureReport --></identifier>
 <status value="[code]"/><!-- 1..1 complete | pending | error -->
 <type value="[code]"/><!--  1..1 individual | subject-list | summary | data-collection -->
 <measure><!-- 1..1 canonical(Measure) What measure was calculated --></measure>
 <subject><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Location|
   Device|RelatedPerson|Group) What individual(s) the report is for --></subject>
 <date value="[dateTime]"/><!-- 0..1 When the report was generated -->
 <reporter><!-- 0..1 Reference(Practitioner|PractitionerRole|Location|
   Organization) Who is reporting the data --></reporter>
 <period><!-- 1..1 Period What period the report covers --></period>
 <improvementNotation><!-- 0..1 CodeableConcept increase | decrease --></improvementNotation>
 <group>  <!--  0..* Measure results for each group -->
  <code><!-- 0..1 CodeableConcept Meaning of the group --></code>
  <population>  <!-- 0..* The populations in the group -->
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <count value="[integer]"/><!-- 0..1 Size of the population -->
   <subjectResults><!-- 0..1 Reference(List) For subject-list reports, the subject results in this population
    --></subjectResults>
  </population>
  <measureScore><!-- 0..1 Quantity What score this group achieved --></measureScore>
  <stratifier>  <!-- 0..* Stratification results -->
   <code><!-- 0..* CodeableConcept What stratifier of the group --></code>
   <stratum>  <!-- 0..* Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier
    components -->
    <value><!-- 0..1 CodeableConcept The stratum value, e.g. male --></value>
    <component>  <!-- 0..* Stratifier component values -->
     <code><!-- 1..1 CodeableConcept What stratifier component of the group --></code>
     <value><!-- 1..1 CodeableConcept The stratum component value, e.g. male --></value>
    </component>
    <population>  <!-- 0..* Population results in this stratum -->
     <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
     <count value="[integer]"/><!-- 0..1 Size of the population -->
     <subjectResults><!-- 0..1 Reference(List) For subject-list reports, the subject results in this population
    --></subjectResults>
    </population>
    <measureScore><!-- 0..1 Quantity What score this stratum achieved --></measureScore>
   </stratum>
  </stratifier>
 </group>
 <evaluatedResource><!-- 0..* Reference(Any) What data was used to calculate the measure score --></evaluatedResource>
</MeasureReport>
```

JSON Template

```
{
  "resourceType" : "MeasureReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Additional identifier for the MeasureReport
  "status" : "<code>", // R!  complete | pending | error
  "type" : "<code>", // C? R!  individual | subject-list | summary | data-collection
  "measure" : { canonical(Measure) }, // R!  What measure was calculated
  "subject" : { Reference(Patient|Practitioner|PractitionerRole|Location|
   Device|RelatedPerson|Group) }, // What individual(s) the report is for
  "date" : "<dateTime>", // When the report was generated
  "reporter" : { Reference(Practitioner|PractitionerRole|Location|
   Organization) }, // Who is reporting the data
  "period" : { Period }, // R!  What period the report covers
  "improvementNotation" : { CodeableConcept }, // increase | decrease
  "group" : [{ // C? Measure results for each group
    "code" : { CodeableConcept }, // Meaning of the group
    "population" : [{ // The populations in the group
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
```

```
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "count" : <integer>, // Size of the population
      "subjectResults" : { Reference(List) } // For subject-list reports, the subject results in this population
    }],
    "measureScore" : { Quantity }, // What score this group achieved
    "stratifier" : [{ // Stratification results
      "code" : [{ CodeableConcept }], // What stratifier of the group
      "stratum" : [{ // Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier
    components
        "value" : { CodeableConcept }, // The stratum value, e.g. male
        "component" : [{ // Stratifier component values
          "code" : { CodeableConcept }, // R!  What stratifier component of the group
          "value" : { CodeableConcept } // R!  The stratum component value, e.g. male
        }],
        "population" : [{ // Population results in this stratum
          "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator |
    denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-
    observation
          "count" : <integer>, // Size of the population
          "subjectResults" : { Reference(List) } // For subject-list reports, the subject results in this population
        }],
        "measureScore" : { Quantity } // What score this stratum achieved
      }]
    }]
  }],
  "evaluatedResource" : [{ Reference(Any) }] // What data was used to calculate the measure score
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MeasureReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MeasureReport.identifier [ Identifier ], ... ; # 0..* Additional identifier for the MeasureReport
  fhir:MeasureReport.status [ code ]; # 1..1 complete | pending | error
  fhir:MeasureReport.type [ code ]; # 1..1 individual | subject-list | summary | data-collection
  fhir:MeasureReport.measure [ canonical(Measure) ]; # 1..1 What measure was calculated
  fhir:MeasureReport.subject [ Reference(Patient|Practitioner|PractitionerRole|Location|Device|RelatedPerson|Group) ]; #
     0..1 What individual(s) the report is for
  fhir:MeasureReport.date [ dateTime ]; # 0..1 When the report was generated
  fhir:MeasureReport.reporter [ Reference(Practitioner|PractitionerRole|Location|Organization) ]; # 0..1 Who is
    reporting the data
  fhir:MeasureReport.period [ Period ]; # 1..1 What period the report covers
  fhir:MeasureReport.improvementNotation [ CodeableConcept ]; # 0..1 increase | decrease
  fhir:MeasureReport.group [ # 0..* Measure results for each group
    fhir:MeasureReport.group.code [ CodeableConcept ]; # 0..1 Meaning of the group
    fhir:MeasureReport.group.population [ # 0..* The populations in the group
      fhir:MeasureReport.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-observation
      fhir:MeasureReport.group.population.count [ integer ]; # 0..1 Size of the population
      fhir:MeasureReport.group.population.subjectResults [ Reference(List) ]; # 0..1 For subject-list reports, the
    subject results in this population
    ], ...;
    fhir:MeasureReport.group.measureScore [ Quantity ]; # 0..1 What score this group achieved
    fhir:MeasureReport.group.stratifier [ # 0..* Stratification results
      fhir:MeasureReport.group.stratifier.code [ CodeableConcept ], ... ; # 0..* What stratifier of the group
      fhir:MeasureReport.group.stratifier.stratum [ # 0..* Stratum results, one for each unique value, or set of values,
     in the stratifier, or stratifier components
        fhir:MeasureReport.group.stratifier.stratum.value [ CodeableConcept ]; # 0..1 The stratum value, e.g. male
        fhir:MeasureReport.group.stratifier.stratum.component [ # 0..* Stratifier component values
          fhir:MeasureReport.group.stratifier.stratum.component.code [ CodeableConcept ]; # 1..1 What stratifier
    component of the group
          fhir:MeasureReport.group.stratifier.stratum.component.value [ CodeableConcept ]; # 1..1 The stratum component
    value, e.g. male
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.population [ # 0..* Population results in this stratum
          fhir:MeasureReport.group.stratifier.stratum.population.code [ CodeableConcept ]; # 0..1 initial-population |
    numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population |
     measure-population-exclusion | measure-observation
          fhir:MeasureReport.group.stratifier.stratum.population.count [ integer ]; # 0..1 Size of the population
          fhir:MeasureReport.group.stratifier.stratum.population.subjectResults [ Reference(List) ]; # 0..1 For subject-
    list reports, the subject results in this population
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.measureScore [ Quantity ]; # 0..1 What score this stratum achieved
      ], ...;
    ], ...;
  ], ...;
  fhir:MeasureReport.evaluatedResource [ Reference(Any) ], ... ; # 0..* What data was used to calculate the measure
    score
]
```

Changes since R3

MeasureReport

MeasureReport.identifier | Max Cardinality changed from 1 to *
MeasureReport.status | Change value set from http://hl7.org/fhir/ValueSet/measure-report-status to http://hl7.org/fhir/ValueSet/measure-report-status|4.0.1
MeasureReport.type | Change value set from http://hl7.org/fhir/ValueSet/measure-report-type to http://hl7.org/fhir/ValueSet/measure-report-type|4.0.1
MeasureReport.measure | Type changed from Reference(Measure) to canonical(Measure)
MeasureReport.subject | Added Element
MeasureReport.reporter | Added Element
MeasureReport.improvementNotation | Added Element
MeasureReport.group.code | Added Element
MeasureReport.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.population.subjectResults | Added Element
MeasureReport.group.measureScore | Type changed from decimal to Quantity
MeasureReport.group.stratifier.code | Added Element
MeasureReport.group.stratifier.stratum.value | Min Cardinality changed from 1 to 0 Type changed from string to CodeableConcept
MeasureReport.group.stratifier.stratum.component | Added Element
MeasureReport.group.stratifier.stratum.component.code | Added Mandatory Element
MeasureReport.group.stratifier.stratum.component.value | Added Mandatory Element
MeasureReport.group.stratifier.stratum.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.stratifier.stratum.population.subjectResults | Added Element
MeasureReport.group.stratifier.stratum.measureScore | Type changed from decimal to Quantity
MeasureReport.evaluatedResource | Added Element
MeasureReport.patient | deleted
MeasureReport.reportingOrganization | deleted
MeasureReport.group.identifier | deleted
MeasureReport.group.population.identifier | deleted
MeasureReport.group.population.patients | deleted
MeasureReport.group.stratifier.identifier | deleted
MeasureReport.group.stratifier.stratum.population.identifier | deleted
MeasureReport.group.stratifier.stratum.population.patients | deleted
MeasureReport.evaluatedResources | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

Structure

Name | Flags | Card. | Type | Description & Constraints
MeasureReport | ITU | | DomainResource | Results of a measure evaluation + Rule: Measure Reports used for data collection SHALL NOT communicate group and score information + Rule: Stratifiers SHALL be either a single criteria or a set of criteria components Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | Σ | 0..* | Identifier | Additional identifier for the MeasureReport
status | ?!Σ | 1..1 | code | complete | pending | error MeasureReportStatus (Required)
type | ΣI | 1..1 | code | individual | subject-list | summary | data-collection MeasureReportType (Required)
measure | Σ | 1..1 | canonical(Measure) | What measure was calculated
subject | Σ | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Location | Device | RelatedPerson | Group) | What individual(s) the report is for
date | Σ | 0..1 | dateTime | When the report was generated
reporter | Σ | 0..1 | Reference(Practitioner | PractitionerRole | Location | Organization) | Who is reporting the data
period | Σ | 1..1 | Period | What period the report covers
improvementNotation | ?!Σ | 0..1 | CodeableConcept | increase | decrease MeasureImprovementNotation (Required)
group | I | 0..* | BackboneElement | Measure results for each group
code | Σ | 0..1 | CodeableConcept | Meaning of the group
population | | 0..* | BackboneElement | The populations in the group
code | Σ | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
subjectResults | | 0..1 | Reference(List) | For subject-list reports, the subject results in this population
measureScore | Σ | 0..1 | Quantity | What score this group achieved
stratifier | | 0..* | BackboneElement | Stratification results
code | | 0..* | CodeableConcept | What stratifier of the group
stratum | | 0..* | BackboneElement | Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier components
value | | 0..1 | CodeableConcept | The stratum value, e.g. male
component | | 0..* | BackboneElement | Stratifier component values
code | | 1..1 | CodeableConcept | What stratifier component of the group
value | | 1..1 | CodeableConcept | The stratum component value, e.g. male
population | | 0..* | BackboneElement | Population results in this stratum

code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)

count | | 0..1 | integer | Size of the population

subjectResults | | 0..1 | Reference(List) | For subject-list reports, the subject results in this population

measureScore | | 0..1 | Quantity | What score this stratum achieved

evaluatedResource | | 0..* | Reference(Any) | What data was used to calculate the measure score

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MeasureReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Additional identifier for the MeasureReport --></identifier>
 <status value="[code]"/><!-- 1..1 complete | pending | error -->
 <type value="[code]"/><!--  1..1 individual | subject-list | summary | data-collection -->
 <measure><!-- 1..1 canonical(Measure) What measure was calculated --></measure>
 <subject><!-- 0..1 Reference(Patient|Practitioner|PractitionerRole|Location|
   Device|RelatedPerson|Group) What individual(s) the report is for --></subject>
 <date value="[dateTime]"/><!-- 0..1 When the report was generated -->
 <reporter><!-- 0..1 Reference(Practitioner|PractitionerRole|Location|
   Organization) Who is reporting the data --></reporter>
 <period><!-- 1..1 Period What period the report covers --></period>
 <improvementNotation><!-- 0..1 CodeableConcept increase | decrease --></improvementNotation>
 <group>  <!--  0..* Measure results for each group -->
  <code><!-- 0..1 CodeableConcept Meaning of the group --></code>
  <population>  <!-- 0..* The populations in the group -->
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <count value="[integer]"/><!-- 0..1 Size of the population -->
   <subjectResults><!-- 0..1 Reference(List) For subject-list reports, the subject results in this population
    --></subjectResults>
  </population>
  <measureScore><!-- 0..1 Quantity What score this group achieved --></measureScore>
  <stratifier>  <!-- 0..* Stratification results -->
   <code><!-- 0..* CodeableConcept What stratifier of the group --></code>
   <stratum>  <!-- 0..* Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier
    components -->
    <value><!-- 0..1 CodeableConcept The stratum value, e.g. male --></value>
    <component>  <!-- 0..* Stratifier component values -->
     <code><!-- 1..1 CodeableConcept What stratifier component of the group --></code>
     <value><!-- 1..1 CodeableConcept The stratum component value, e.g. male --></value>
    </component>
    <population>  <!-- 0..* Population results in this stratum -->
     <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
     <count value="[integer]"/><!-- 0..1 Size of the population -->
     <subjectResults><!-- 0..1 Reference(List) For subject-list reports, the subject results in this population
    --></subjectResults>
    </population>
    <measureScore><!-- 0..1 Quantity What score this stratum achieved --></measureScore>
   </stratum>
  </stratifier>
 </group>
 <evaluatedResource><!-- 0..* Reference(Any) What data was used to calculate the measure score --></evaluatedResource>
</MeasureReport>
```

JSON Template

```
{
  "resourceType" : "MeasureReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Additional identifier for the MeasureReport
  "status" : "<code>", // R!  complete | pending | error
  "type" : "<code>", // C? R!  individual | subject-list | summary | data-collection
  "measure" : { canonical(Measure) }, // R!  What measure was calculated
  "subject" : { Reference(Patient|Practitioner|PractitionerRole|Location|
   Device|RelatedPerson|Group) }, // What individual(s) the report is for
  "date" : "<dateTime>", // When the report was generated
  "reporter" : { Reference(Practitioner|PractitionerRole|Location|
   Organization) }, // Who is reporting the data
  "period" : { Period }, // R!  What period the report covers
  "improvementNotation" : { CodeableConcept }, // increase | decrease
  "group" : [{ // C? Measure results for each group
    "code" : { CodeableConcept }, // Meaning of the group
    "population" : [{ // The populations in the group
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "count" : <integer>, // Size of the population
      "subjectResults" : { Reference(List) } // For subject-list reports, the subject results in this population
    }],
    "measureScore" : { Quantity }, // What score this group achieved
    "stratifier" : [{ // Stratification results
      "code" : [{ CodeableConcept }], // What stratifier of the group
```

```
      "stratum" : [{ // Stratum results, one for each unique value, or set of values, in the stratifier, or stratifier
    components
        "value" : { CodeableConcept }, // The stratum value, e.g. male
        "component" : [{ // Stratifier component values
          "code" : { CodeableConcept }, // R!  What stratifier component of the group
          "value" : { CodeableConcept } // R!  The stratum component value, e.g. male
        }],
        "population" : [{ // Population results in this stratum
          "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator |
    denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-
    observation
          "count" : <integer>, // Size of the population
          "subjectResults" : { Reference(List) } // For subject-list reports, the subject results in this population
        }],
        "measureScore" : { Quantity } // What score this stratum achieved
      }]
    }]
  }],
  "evaluatedResource" : [{ Reference(Any) }] // What data was used to calculate the measure score
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:MeasureReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MeasureReport.identifier [ Identifier ], ... ; # 0..* Additional identifier for the MeasureReport
  fhir:MeasureReport.status [ code ]; # 1..1 complete | pending | error
  fhir:MeasureReport.type [ code ]; # 1..1 individual | subject-list | summary | data-collection
  fhir:MeasureReport.measure [ canonical(Measure) ]; # 1..1 What measure was calculated
  fhir:MeasureReport.subject [ Reference(Patient|Practitioner|PractitionerRole|Location|Device|RelatedPerson|Group) ]; #
     0..1 What individual(s) the report is for
  fhir:MeasureReport.date [ dateTime ]; # 0..1 When the report was generated
  fhir:MeasureReport.reporter [ Reference(Practitioner|PractitionerRole|Location|Organization) ]; # 0..1 Who is
    reporting the data
  fhir:MeasureReport.period [ Period ]; # 1..1 What period the report covers
  fhir:MeasureReport.improvementNotation [ CodeableConcept ]; # 0..1 increase | decrease
  fhir:MeasureReport.group [ # 0..* Measure results for each group
    fhir:MeasureReport.group.code [ CodeableConcept ]; # 0..1 Meaning of the group
    fhir:MeasureReport.group.population [ # 0..* The populations in the group
      fhir:MeasureReport.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-observation
      fhir:MeasureReport.group.population.count [ integer ]; # 0..1 Size of the population
      fhir:MeasureReport.group.population.subjectResults [ Reference(List) ]; # 0..1 For subject-list reports, the
    subject results in this population
    ], ...;
    fhir:MeasureReport.group.measureScore [ Quantity ]; # 0..1 What score this group achieved
    fhir:MeasureReport.group.stratifier [ # 0..* Stratification results
      fhir:MeasureReport.group.stratifier.code [ CodeableConcept ], ... ; # 0..* What stratifier of the group
      fhir:MeasureReport.group.stratifier.stratum [ # 0..* Stratum results, one for each unique value, or set of values,
     in the stratifier, or stratifier components
        fhir:MeasureReport.group.stratifier.stratum.value [ CodeableConcept ]; # 0..1 The stratum value, e.g. male
        fhir:MeasureReport.group.stratifier.stratum.component [ # 0..* Stratifier component values
          fhir:MeasureReport.group.stratifier.stratum.component.code [ CodeableConcept ]; # 1..1 What stratifier
    component of the group
          fhir:MeasureReport.group.stratifier.stratum.component.value [ CodeableConcept ]; # 1..1 The stratum component
    value, e.g. male
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.population [ # 0..* Population results in this stratum
          fhir:MeasureReport.group.stratifier.stratum.population.code [ CodeableConcept ]; # 0..1 initial-population |
    numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population |
     measure-population-exclusion | measure-observation
          fhir:MeasureReport.group.stratifier.stratum.population.count [ integer ]; # 0..1 Size of the population
          fhir:MeasureReport.group.stratifier.stratum.population.subjectResults [ Reference(List) ]; # 0..1 For subject-
    list reports, the subject results in this population
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.measureScore [ Quantity ]; # 0..1 What score this stratum achieved
      ], ...;
    ], ...;
  ], ...;
  fhir:MeasureReport.evaluatedResource [ Reference(Any) ], ... ; # 0..* What data was used to calculate the measure
    score
]
```

Changes since Release 3

MeasureReport

MeasureReport.identifier | Max Cardinality changed from 1 to *

MeasureReport.status | Change value set from http://hl7.org/fhir/ValueSet/measure-report-status to http://hl7.org/fhir/ValueSet/measure-report-status|4.0.1

MeasureReport.type | Change value set from http://hl7.org/fhir/ValueSet/measure-report-type to http://hl7.org/fhir/ValueSet/measure-report-type|4.0.1

MeasureReport.measure | Type changed from Reference(Measure) to canonical(Measure)

MeasureReport.subject | Added Element
MeasureReport.reporter | Added Element
MeasureReport.improvementNotation | Added Element
MeasureReport.group.code | Added Element
MeasureReport.group.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.population.subjectResults | Added Element
MeasureReport.group.measureScore | Type changed from decimal to Quantity
MeasureReport.group.stratifier.code | Added Element
MeasureReport.group.stratifier.stratum.value | Min Cardinality changed from 1 to 0 Type changed from string to CodeableConcept
MeasureReport.group.stratifier.stratum.component | Added Element
MeasureReport.group.stratifier.stratum.component.code | Added Mandatory Element
MeasureReport.group.stratifier.stratum.component.value | Added Mandatory Element
MeasureReport.group.stratifier.stratum.population.code | Change code system for extensibly bound codes from "http://hl7.org/fhir/measure-population" to "http://terminology.hl7.org/CodeSystem/measure-population"
MeasureReport.group.stratifier.stratum.population.subjectResults | Added Element
MeasureReport.group.stratifier.stratum.measureScore | Type changed from decimal to Quantity
MeasureReport.evaluatedResource | Added Element
MeasureReport.patient | deleted
MeasureReport.reportingOrganization | deleted
MeasureReport.group.identifier | deleted
MeasureReport.group.population.identifier | deleted
MeasureReport.group.population.patients | deleted
MeasureReport.group.stratifier.identifier | deleted
MeasureReport.group.stratifier.stratum.population.identifier | deleted
MeasureReport.group.stratifier.stratum.population.patients | deleted
MeasureReport.evaluatedResources | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R3 <--> R4 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors).)

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.15.3.1 Terminology Bindings

Path | Definition | Type | Reference
MeasureReport.status | The status of the measure report. | Required | MeasureReportStatus
MeasureReport.type | The type of the measure report. | Required | MeasureReportType
MeasureReport.improvementNotation | Observation values that indicate what change in a measurement value or score is indicative of an improvement in the measured item or scored issue. | Required | MeasureImprovementNotation
MeasureReport.group.population.code MeasureReport.group.stratifier.stratum.population.code | The type of population (e.g. initial, numerator, denominator, etc.). | Extensible | MeasurePopulationType

14.15.3.2 Constraints

id | Level | Location | Description | Expression
mrp-1 | Rule | (base) | Measure Reports used for data collection SHALL NOT communicate group and score information | (type != 'data-collection') or group.exists().not()
mrp-2 | Rule | (base) | Stratifiers SHALL be either a single criteria or a set of criteria components | group.stratifier.stratum.all(value.exists() xor component.exists())

14.15.4 MeasureReport Search

Although the MeasureReport resource does define search parameters, it is up to the individual measure evaluation service whether or not historical records of measure evaluation requests are preserved. A service might not support searching on MeasureReports at all, or it may support searching only for a pre-defined expiration period. The search parameters are defined to provide consumers with a consistent interface to searching if it is available for a specific service implementation. In any case, services should detail the support they do provide using a CapabilityStatement.

14.15.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | The date of the measure report | MeasureReport.date
evaluated-resource | reference | An evaluated resource referenced by the measure report | MeasureReport.evaluatedResource (Any)
identifier | token | External identifier of the measure report to be returned | MeasureReport.identifier
measure | reference | The measure to return measure report results for | MeasureReport.measure (Measure)
patient | reference | The identity of a patient to search for individual measure report results for | MeasureReport.subject.where(resolve() is Patient) (Patient)

period | date | The period of the measure report | MeasureReport.period
reporter | reference | The reporter to return measure report results for | MeasureReport.reporter (Practitioner, Organization, PractitionerRole, Location)
status | token | The status of the measure report | MeasureReport.status
subject | reference | The identity of a subject to search for individual measure report results for | MeasureReport.subject (Practitioner, Group, Device, Patient, PractitionerRole, RelatedPerson, Location)

14.15.6 Resource MeasureReport - Examples

URL: https://hl7.org/fhir/R4/measurereport-examples.html

These examples illustrate the use of the MeasureReport resource to report individual, subject-listing, and summary-level results for measure calculation.

14.15.6 Resource MeasureReport - Examples

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Patient

Example Name | id | Format
CMS 146 Summary Report | measurereport-cms146-cat3-example | XML | JSON | Turtle
CMS 146 Patient Listing Report | measurereport-cms146-cat2-example | XML | JSON | Turtle
CMS 146 Individual Report | measurereport-cms146-cat1-example | XML | JSON | Turtle
Example MeasureReport for reporting public health data elements | hiv-indicators | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.15.7 Resource MeasureReport - Detailed Descriptions

URL: https://hl7.org/fhir/R4/measurereport-definitions.html

14.15.7 Resource MeasureReport - Detailed Descriptions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Security Category: Business | Compartments: Patient

Detailed Descriptions for the elements in the MeasureReport resource.

MeasureReport
Element Id | MeasureReport
Definition | The MeasureReport resource contains the results of the calculation of a measure; and optionally a reference to the resources involved in that calculation.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element mrp-1 Rule Measure Reports used for data collection SHALL NOT communicate group and score information (type != 'data-collection') or group.exists().not() mrp-2 Rule Stratifiers SHALL be either a single criteria or a set of criteria components group.stratifier.stratum.all(value.exists() xor component.exists())
Defined on this element
mrp-1 | Rule | Measure Reports used for data collection SHALL NOT communicate group and score information | (type != 'data-collection') or group.exists().not()
mrp-2 | Rule | Stratifiers SHALL be either a single criteria or a set of criteria components | group.stratifier.stratum.all(value.exists() xor component.exists())
MeasureReport.identifier
Element Id | MeasureReport.identifier
Definition | A formal identifier that is used to identify this MeasureReport when it is represented in other formats or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II data type - e.g. to identify this {{title}} outside of FHIR, where the logical URL is not possible to use.
MeasureReport.status
Element Id | MeasureReport.status
Definition | The MeasureReport status. No data will be available until the MeasureReport status is complete.
Cardinality | 1..1
Terminology Binding | MeasureReportStatus (Required)
Type | code
Is Modifier | true (Reason: This element is labelled as a modifier because it is a status element that contains status entered-in-error which means that the resource should not be treated as valid)
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MeasureReport.type
Element Id | MeasureReport.type
Definition | The type of measure report. This may be an individual report, which provides the score for the measure for an individual member of the population; a subject-listing, which returns the list of members that meet the various criteria in the

measure; a summary report, which returns a population count for each of the criteria in the measure; or a data-collection, which enables the MeasureReport to be used to exchange the data-of-interest for a quality measure.
Cardinality | 1..1
Terminology Binding | MeasureReportType (Required)
Type | code
Summary | true
Comments | Data-collection reports are used only to communicate data-of-interest for a measure. They do not necessarily include all the data for a particular subject or population, but they may.
Invariants | Affect this element mrp-1 Rule Measure Reports used for data collection SHALL NOT communicate group and score information (type != 'data-collection') or group.exists().not()
Affect this element
mrp-1 | Rule | Measure Reports used for data collection SHALL NOT communicate group and score information | (type != 'data-collection') or group.exists().not()
MeasureReport.measure
Element Id | MeasureReport.measure
Definition | A reference to the Measure that was calculated to produce this report.
Cardinality | 1..1
Type | canonical(Measure)
Summary | true
MeasureReport.subject
Element Id | MeasureReport.subject
Definition | Optional subject identifying the individual or individuals the report is for.
Cardinality | 0..1
Type | Reference(Patient | Practitioner | PractitionerRole | Location | Device | RelatedPerson | Group)
Summary | true
MeasureReport.date
Element Id | MeasureReport.date
Definition | The date this measure report was generated.
Cardinality | 0..1
Type | dateTime
Summary | true
MeasureReport.reporter
Element Id | MeasureReport.reporter
Definition | The individual, location, or organization that is reporting the data.
Cardinality | 0..1
Type | Reference(Practitioner | PractitionerRole | Location | Organization)
Summary | true
MeasureReport.period
Element Id | MeasureReport.period
Definition | The reporting period for which the report was calculated.
Cardinality | 1..1
Type | Period
Summary | true
MeasureReport.improvementNotation
Element Id | MeasureReport.improvementNotation
Definition | Whether improvement in the measure is noted by an increase or decrease in the measure score.
Cardinality | 0..1
Terminology Binding | MeasureImprovementNotation (Required)
Type | CodeableConcept
Is Modifier | true (Reason: Improvement notation determines how to interpret the measure score (i.e. whether an increase is an improvement))
Summary | true
Comments | This element is typically defined by the measure, but reproduced here to ensure the measure score can be interpreted. The element is labeled as a modifier because it changes the interpretation of the reported measure score.
MeasureReport.group
Element Id | MeasureReport.group
Definition | The results of the calculation, one for each population group in the measure.
Cardinality | 0..*
Invariants | Affect this element mrp-1 Rule Measure Reports used for data collection SHALL NOT communicate group and score information (type != 'data-collection') or group.exists().not()
Affect this element
mrp-1 | Rule | Measure Reports used for data collection SHALL NOT communicate group and score information | (type != 'data-collection') or group.exists().not()
MeasureReport.group.code
Element Id | MeasureReport.group.code
Definition | The meaning of the population group as defined in the measure definition.
Cardinality | 0..1
Type | CodeableConcept
Summary | true
MeasureReport.group.population
Element Id | MeasureReport.group.population
Definition | The populations that make up the population group, one for each type of population appropriate for the measure.
Cardinality | 0..*

MeasureReport.group.population.code
Element Id | MeasureReport.group.population.code
Definition | The type of the population.
Cardinality | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
Summary | true
MeasureReport.group.population.count
Element Id | MeasureReport.group.population.count
Definition | The number of members of the population.
Cardinality | 0..1
Type | integer
MeasureReport.group.population.subjectResults
Element Id | MeasureReport.group.population.subjectResults
Definition | This element refers to a List of subject level MeasureReport resources, one for each subject in this population.
Cardinality | 0..1
Type | Reference(List)
MeasureReport.group.measureScore
Element Id | MeasureReport.group.measureScore
Definition | The measure score for this population group, calculated as appropriate for the measure type and scoring method, and based on the contents of the populations defined in the group.
Cardinality | 0..1
Type | Quantity
Summary | true
MeasureReport.group.stratifier
Element Id | MeasureReport.group.stratifier
Definition | When a measure includes multiple stratifiers, there will be a stratifier group for each stratifier defined by the measure.
Cardinality | 0..*
MeasureReport.group.stratifier.code
Element Id | MeasureReport.group.stratifier.code
Definition | The meaning of this stratifier, as defined in the measure definition.
Cardinality | 0..*
Type | CodeableConcept
MeasureReport.group.stratifier.stratum
Element Id | MeasureReport.group.stratifier.stratum
Definition | This element contains the results for a single stratum within the stratifier. For example, when stratifying on administrative gender, there will be four strata, one for each possible gender value.
Cardinality | 0..*
MeasureReport.group.stratifier.stratum.value
Element Id | MeasureReport.group.stratifier.stratum.value
Definition | The value for this stratum, expressed as a CodeableConcept. When defining stratifiers on complex values, the value must be rendered such that the value for each stratum within the stratifier is unique.
Cardinality | 0..1
Type | CodeableConcept
MeasureReport.group.stratifier.stratum.component
Element Id | MeasureReport.group.stratifier.stratum.component
Definition | A stratifier component value.
Cardinality | 0..*
MeasureReport.group.stratifier.stratum.component.code
Element Id | MeasureReport.group.stratifier.stratum.component.code
Definition | The code for the stratum component value.
Cardinality | 1..1
Type | CodeableConcept
MeasureReport.group.stratifier.stratum.component.value
Element Id | MeasureReport.group.stratifier.stratum.component.value
Definition | The stratum component value.
Cardinality | 1..1
Type | CodeableConcept
MeasureReport.group.stratifier.stratum.population
Element Id | MeasureReport.group.stratifier.stratum.population
Definition | The populations that make up the stratum, one for each type of population appropriate to the measure.
Cardinality | 0..*
MeasureReport.group.stratifier.stratum.population.code
Element Id | MeasureReport.group.stratifier.stratum.population.code
Definition | The type of the population.
Cardinality | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
MeasureReport.group.stratifier.stratum.population.count
Element Id | MeasureReport.group.stratifier.stratum.population.count
Definition | The number of members of the population in this stratum.
Cardinality | 0..1

Type | integer
MeasureReport.group.stratifier.stratum.population.subjectResults
Element Id | MeasureReport.group.stratifier.stratum.population.subjectResults
Definition | This element refers to a List of subject level MeasureReport resources, one for each subject in this population in this stratum.
Cardinality | 0..1
Type | Reference(List)
MeasureReport.group.stratifier.stratum.measureScore
Element Id | MeasureReport.group.stratifier.stratum.measureScore
Definition | The measure score for this stratum, calculated as appropriate for the measure type and scoring method, and based on only the members of this stratum.
Cardinality | 0..1
Type | Quantity
MeasureReport.evaluatedResource
Element Id | MeasureReport.evaluatedResource
Definition | A reference to a Bundle containing the Resources that were used in the calculation of this measure.
Cardinality | 0..*
Type | Reference(Any)

14.15.8 Resource MeasureReport - Mappings

URL: https://hl7.org/fhir/R4/measurereport-mappings.html

14.15.8 Resource MeasureReport - Mappings

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Patient

Mappings:

FiveWs Pattern Mapping

Mappings for the measurereport resource (see Mappings to Other Standards for further information & status).

14.15.8.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

MeasureReport
identifier | FiveWs.identifier
status | FiveWs.status
type | FiveWs.class
measure | FiveWs.what[x]
subject | FiveWs.subject[x]
date | FiveWs.recorded
reporter | FiveWs.source

14.15.9 Resource MeasureReport - Extensions & Profiles

URL: https://hl7.org/fhir/R4/measurereport-profiles.html

14.15.9 Resource MeasureReport - Extensions & Profiles

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Patient

This table lists profiles and extensions for the MeasureReport resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.15.9.1 Profiles

No Profiles defined for this resource

14.15.9.2 Extensions

No Extensions defined for this resource
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions

workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.15.9.3 Search Extensions

No Search Extensions defined for this resource

14.15.10 Resource MeasureReport - R3/R4 Conversions

URL: https://hl7.org/fhir/R4/measurereport-version-maps.html

No validation errors - all conversions are clean

14.15.10 Resource MeasureReport - R3/R4 Conversions

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Patient

R3 : R4 Conversion maps for MeasureReport.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and 3 r3 resources are invalid (0 errors). (see documentation)

14.15.10.1 R3 to R4

```
map "http://hl7.org/fhir/StructureMap/MeasureReport3to4" = "R3 to R4 Conversions for MeasureReport"

conceptmap "MeasureReportType" {
  prefix s = "http://hl7.org/fhir/measure-report-type"
  prefix t = "http://hl7.org/fhir/measure-report-type"

  s:"'patient-list'" - t:"'subject-list'"
  s:individual - t:individual
  s:summary - t:summary
}

uses "http://hl7.org/fhir/3.0/StructureDefinition/MeasureReport" alias MeasureReportR3 as source
uses "http://hl7.org/fhir/StructureDefinition/MeasureReport" alias MeasureReport as target

imports "http://hl7.org/fhir/StructureMap/*3to4"

group MeasureReport(source src : MeasureReportR3, target tgt : MeasureReport) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type as v -> tgt.type = translate(v, '#MeasureReportType', 'code');
  src.measure -> tgt.measure;
  src.patient -> tgt.subject;
  src.date -> tgt.date;
  src.reportingOrganization -> tgt.reporter;
  src.period -> tgt.period;
  src.group as s -> tgt.group as t then group(s, t);
  src.evaluatedResources -> tgt.evaluatedResource;
}

group group(source src, target tgt) extends BackboneElement {
  src.identifier as vs -> tgt.code as vt then Identifier2Codeable(vs, vt);
  src.population as s -> tgt.population as t then population(s, t);
  src.measureScore -> tgt.measureScore;
  src.stratifier as s -> tgt.stratifier as t then stratifier(s, t);
}

group population(source src, target tgt) extends BackboneElement {
  src.identifier as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MeasureReport.group.identifier',  vt.value = v;
  src.code -> tgt.code;
  src.count -> tgt.count;
  src.patients -> tgt.subjectResults;
}

group stratifier(source src, target tgt) extends BackboneElement {
  src.identifier as vs -> tgt.code as vt then Identifier2Codeable(vs, vt);
  src.stratum as s -> tgt.stratum as t then stratum(s, t);
}

group stratum(source src, target tgt) extends BackboneElement {
  src.value as s ->  tgt.value as cc,  cc.text = s;
  src.population as s -> tgt.population as t then population2(s, t);
  src.measureScore -> tgt.measureScore;
}

group population2(source src, target tgt) extends BackboneElement {
  src.identifier as v ->  tgt.extension as vt,  vt.url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MeasureReport.group.stratifier.stratum.population.identifier',  vt.value = v;
  src.code -> tgt.code;
  src.count -> tgt.count;
  src.patients -> tgt.subjectResults;
}
```

14.15.10.2 R4 to R3

```
map "http://hl7.org/fhir/StructureMap/MeasureReport4to3" = "R4 to R3 Conversion for MeasureReport"

conceptmap "MeasureReportType" {
  prefix s = "http://hl7.org/fhir/measure-report-type"
  prefix t = "http://hl7.org/fhir/measure-report-type"

  s:"'subject-list'" - t:"'patient-list'"
  s:individual - t:individual
  s:summary - t:summary
}

uses "http://hl7.org/fhir/StructureDefinition/MeasureReport" alias MeasureReport as source
uses "http://hl7.org/fhir/3.0/StructureDefinition/MeasureReport" alias MeasureReportR3 as target

imports "http://hl7.org/fhir/StructureMap/*4to3"

group MeasureReport(source src : MeasureReportR3, target tgt : MeasureReport) extends DomainResource <<type+>> {
  src.identifier -> tgt.identifier;
  src.status -> tgt.status;
  src.type as v -> tgt.type = translate(v, '#MeasureReportType', 'code');
  src.measure -> tgt.measure;
  src.subject -> tgt.patient;
  src.date -> tgt.date;
  src.reporter -> tgt.reportingOrganization;
  src.period -> tgt.period;
  src.group as s -> tgt.group as t then group(s, t);
  src.evaluatedResource -> tgt.evaluatedResources;
}

group group(source src, target tgt) extends BackboneElement {
  src.code as vs -> tgt.identifier as vt then Codeable2Identifier(vs, vt);
  src.population as s -> tgt.population as t then population(s, t);
  src.measureScore -> tgt.measureScore;
  src.stratifier as s -> tgt.stratifier as t then stratifier(s, t);
}

group population(source src, target tgt) extends BackboneElement {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MeasureReport.group.population.identifier' then {
    e.value -> tgt.identifier;
  };
  src.code -> tgt.code;
  src.count -> tgt.count;
  src.subjectResults -> tgt.patients;
}

group stratifier(source src, target tgt) extends BackboneElement {
  src.code as vs -> tgt.identifier as vt then Codeable2Identifier(vs, vt);
  src.stratum as s -> tgt.stratum as t then stratum(s, t);
}

group stratum(source src, target tgt) extends BackboneElement {
  src.value as cc then {
    cc.text -> tgt.value;
  };
  src.population as s -> tgt.population as t then population2(s, t);
  src.measureScore -> tgt.measureScore;
}

group population2(source src, target tgt) extends BackboneElement {
  src.extension as e where url = 'http://hl7.org/fhir/3.0/StructureDefinition/extension-
    MeasureReport.group.stratifier.stratum.population.identifier' then {
    e.value -> tgt.identifier;
  };
  src.code -> tgt.code;
  src.count -> tgt.count;
  src.subjectResults -> tgt.patients;
}
```

14.15.10.3 R4 Validation Errors

14.15.11 JSON Schema for MeasureReport

URL: https://hl7.org/fhir/R4/measurereport.schema.json.html

14.15.11 JSON Schema for MeasureReport

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is a representation of the json schema for MeasureReport, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/MeasureReport",
  "$ref": "#/definitions/MeasureReport",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "MeasureReport": {
      "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
```

```
reference to the resources involved in that calculation.",
  "properties": {
    "resourceType": {
      "description": "This is a MeasureReport resource",
      "const": "MeasureReport"
    },
    "id": {
      "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
 never changes.",
      "$ref": "id.schema.json#/definitions/id"
    },
    "meta": {
      "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
Changes to the content might not always be associated with version changes to the resource.",
      "$ref": "Meta.schema.json#/definitions/Meta"
    },
    "implicitRules": {
      "description": "A reference to a set of rules that were followed when the resource was constructed, and which
must be understood when processing the content. Often, this is a reference to an implementation guide that defines
the special rules along with other profiles etc.",
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this MeasureReport when it is represented in
other formats or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "status": {
      "description": "The MeasureReport status. No data will be available until the MeasureReport status is
complete.",
      "enum": [
        "complete",
        "pending",
        "error"
      ]
```

```
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "The type of measure report. This may be an individual report, which provides the score for the
 measure for an individual member of the population; a subject-listing, which returns the list of members that meet
the various criteria in the measure; a summary report, which returns a population count for each of the criteria in
the measure; or a data-collection, which enables the MeasureReport to be used to exchange the data-of-interest for a
 quality measure.",
      "enum": [
        "individual",
        "subject-list",
        "summary",
        "data-collection"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "measure": {
      "description": "A reference to the Measure that was calculated to produce this report.",
      "$ref": "canonical.schema.json#/definitions/canonical"
    },
    "subject": {
      "description": "Optional subject identifying the individual or individuals the report is for.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date this measure report was generated.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "reporter": {
      "description": "The individual, location, or organization that is reporting the data.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "period": {
      "description": "The reporting period for which the report was calculated.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "improvementNotation": {
      "description": "Whether improvement in the measure is noted by an increase or decrease in the measure score.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "group": {
      "description": "The results of the calculation, one for each population group in the measure.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Group"
      },
      "type": "array"
    },
    "evaluatedResource": {
      "description": "A reference to a Bundle containing the Resources that were used in the calculation of this
measure.",
      "items": {
        "$ref": "Reference.schema.json#/definitions/Reference"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "period",
    "measure",
    "resourceType"
  ]
},
"MeasureReport_Group": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
```

```
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The meaning of the population group as defined in the measure definition.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "population": {
      "description": "The populations that make up the population group, one for each type of population appropriate
 for the measure.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Population"
      },
      "type": "array"
    },
    "measureScore": {
      "description": "The measure score for this population group, calculated as appropriate for the measure type
and scoring method, and based on the contents of the populations defined in the group.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    },
    "stratifier": {
      "description": "When a measure includes multiple stratifiers, there will be a stratifier group for each
stratifier defined by the measure.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Stratifier"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MeasureReport_Population": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The type of the population.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "count": {
      "description": "The number of members of the population.",
      "$ref": "#/definitions/integer"
    },
    "_count": {
      "description": "Extensions for count",
      "$ref": "Element.schema.json#/definitions/Element"
```

```
    },
    "subjectResults": {
      "description": "This element refers to a List of subject level MeasureReport resources, one for each subject
in this population.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    }
  },
  "additionalProperties": false
},
"MeasureReport_Stratifier": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The meaning of this stratifier, as defined in the measure definition.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "stratum": {
      "description": "This element contains the results for a single stratum within the stratifier. For example,
when stratifying on administrative gender, there will be four strata, one for each possible gender value.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Stratum"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"MeasureReport_Stratum": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
```

```
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "value": {
      "description": "The value for this stratum, expressed as a CodeableConcept. When defining stratifiers on
complex values, the value must be rendered such that the value for each stratum within the stratifier is unique.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "component": {
      "description": "A stratifier component value.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Component"
      },
      "type": "array"
    },
    "population": {
      "description": "The populations that make up the stratum, one for each type of population appropriate to the
measure.",
      "items": {
        "$ref": "#/definitions/MeasureReport_Population1"
      },
      "type": "array"
    },
    "measureScore": {
      "description": "The measure score for this stratum, calculated as appropriate for the measure type and scoring
 method, and based on only the members of this stratum.",
      "$ref": "Quantity.schema.json#/definitions/Quantity"
    }
  },
  "additionalProperties": false
},
"MeasureReport_Component": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "code": {
      "description": "The code for the stratum component value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "value": {
      "description": "The stratum component value.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    }
  },
  "additionalProperties": false,
  "required": [
    "code",
    "value"
  ]
},
"MeasureReport_Population1": {
  "description": "The MeasureReport resource contains the results of the calculation of a measure; and optionally a
reference to the resources involved in that calculation.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
```

```
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
    definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the element and that modifies the understanding of the element in which it is contained and/or the understanding of
    the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
    use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
     of the definition of the extension. Applications processing a resource are required to check for modifier
    extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
    (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "code": {
          "description": "The type of the population.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "count": {
          "description": "The number of members of the population in this stratum.",
          "$ref": "#/definitions/integer"
        },
        "_count": {
          "description": "Extensions for count",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "subjectResults": {
          "description": "This element refers to a List of subject level MeasureReport resources, one for each subject
    in this population in this stratum.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false
    }
  }
}
```


14.15.12 Resource MeasureReport - Maturity Dependency View

URL: https://hl7.org/fhir/R4/measurereport-dependencies.html

14.15.12 Resource MeasureReport - Maturity Dependency View

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Dependency Graph for MeasureReport FMM level 2

Name | Card. | Type | Dependency Analysis
MeasureReport | | DomainResource
identifier | 0..* | Identifier
status | 1..1 | code
type | 1..1 | code
measure | 1..1 | canonical(Measure)
subject | 0..1 | Reference(Patient | Practitioner | PractitionerRole | Location | Device | RelatedPerson | Group) | Type Warning: (Group = FMM1-Trial Use vs. Element = FMM2-Trial Use)
date | 0..1 | dateTime
reporter | 0..1 | Reference(Practitioner | PractitionerRole | Location | Organization)
period | 1..1 | Period
improvementNotation | 0..1 | CodeableConcept
group | 0..* | BackboneElement
code | 0..1 | CodeableConcept
population | 0..* | BackboneElement
code | 0..1 | CodeableConcept
count | 0..1 | integer
subjectResults | 0..1 | Reference(List) | Type Warning: (List = FMM1-Trial Use vs. Element = FMM2-Trial Use)
measureScore | 0..1 | Quantity
stratifier | 0..* | BackboneElement
code | 0..* | CodeableConcept
stratum | 0..* | BackboneElement
value | 0..1 | CodeableConcept
component | 0..* | BackboneElement

code | 1..1 | CodeableConcept
value | 1..1 | CodeableConcept
population | 0..* | BackboneElement
code | 0..1 | CodeableConcept
count | 0..1 | integer
subjectResults | 0..1 | Reference(List) | Type Warning: (List = FMM1-Trial Use vs. Element = FMM2-Trial Use)
measureScore | 0..1 | Quantity
evaluatedResource | 0..* | Reference(Any) | Type Warning: (Any = FMM1-Trial Use vs. Element = FMM2-Trial Use)

14.15.13 Value Set Sc-valueset-measure-report-status - Status Code Mapping

URL: https://hl7.org/fhir/R4/sc-valueset-measure-report-status.html

14.15.13 Value Set Sc-valueset-measure-report-status - Status Code Mapping

Clinical Quality Information Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Patient

This is the narrative for the resource. See also the XML, JSON or Turtle format.

14.15.14 MeasureReportStatusCanonicalMap (http://hl7.org/fhir/ConceptMap/sc-measure-report-status)

Mapping from http://hl7.org/fhir/ValueSet/measure-report-status to http://hl7.org/fhir/ValueSet/resource-status

DRAFT. Published on 01/11/2019 9:29:23 AM by HL7 (FHIR Project) (http://hl7.org/fhir , fhir@lists.hl7.org).

Canonical Mapping for "The status of the measure report."

Source Code | Equivalence | Destination Code
pending | equivalent | active
error | equivalent | failed
complete | equivalent | complete

14.16 Resource ResearchDefinition - Content

URL: https://hl7.org/fhir/R4/researchdefinition.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The ResearchDefinition resource describes the conditional state (population and any exposures being compared within the population) and outcome (if specified) that the knowledge (evidence, assertion, recommendation) is about.

14.16.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

TODO

14.16.2 Boundaries and Relationships

TODO

14.16.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchDefinition | ITU | | DomainResource | A research context or question + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this research definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the research definition
version | Σ | 0..1 | string | Business version of the research definition

name | ΣI | 0..1 | string | Name for this research definition (computer friendly)
title | Σ | 0..1 | string | Name for this research definition (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the ResearchDefinition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the research definition
comment | | 0..* | string | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for research definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this research definition is defined
usage | | 0..1 | string | Describes the clinical usage of the ResearchDefinition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the research definition was approved by publisher
lastReviewDate | | 0..1 | date | When the research definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the research definition is expected to be used
topic | | 0..* | CodeableConcept | The category of the ResearchDefinition, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the ResearchDefinition
population | Σ | 1..1 | Reference(ResearchElementDefinition) | What population?
exposure | Σ | 0..1 | Reference(ResearchElementDefinition) | What exposure?
exposureAlternative | Σ | 0..1 | Reference(ResearchElementDefinition) | What alternative exposure state?
outcome | Σ | 0..1 | Reference(ResearchElementDefinition) | What outcome?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this research definition, represented as a URI (globally unique)
     -->
 <identifier><!-- 0..* Identifier Additional identifier for the research definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the research definition -->
 <name value="[string]"/><!--  0..1 Name for this research definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this research definition (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the ResearchDefinition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the research definition -->
 <comment value="[string]"/><!-- 0..* Used for footnotes or explanatory notes -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for research definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this research definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the ResearchDefinition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the research definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the research definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the research definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the ResearchDefinition, such as Education, Treatment, Assessment, etc.
     --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the ResearchDefinition --></library>
 <population><!-- 1..1 Reference(ResearchElementDefinition) What population? --></population>
 <exposure><!-- 0..1 Reference(ResearchElementDefinition) What exposure? --></exposure>
 <exposureAlternative><!-- 0..1 Reference(ResearchElementDefinition) What alternative exposure state?
    --></exposureAlternative>
 <outcome><!-- 0..1 Reference(ResearchElementDefinition) What outcome? --></outcome>
```

```
</ResearchDefinition>
```

JSON Template

```
{
  "resourceType" : "ResearchDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this research definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the research definition
  "version" : "<string>", // Business version of the research definition
  "name" : "<string>", // C? Name for this research definition (computer friendly)
  "title" : "<string>", // Name for this research definition (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the ResearchDefinition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the research definition
  "comment" : ["<string>"], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for research definition (if applicable)
  "purpose" : "<markdown>", // Why this research definition is defined
  "usage" : "<string>", // Describes the clinical usage of the ResearchDefinition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the research definition was approved by publisher
  "lastReviewDate" : "<date>", // When the research definition was last reviewed
  "effectivePeriod" : { Period }, // When the research definition is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the ResearchDefinition, such as Education, Treatment, Assessment,
    etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the ResearchDefinition
  "population" : { Reference(ResearchElementDefinition) }, // R!  What population?
  "exposure" : { Reference(ResearchElementDefinition) }, // What exposure?
  "exposureAlternative" : { Reference(ResearchElementDefinition) }, // What alternative exposure state?
  "outcome" : { Reference(ResearchElementDefinition) } // What outcome?
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchDefinition.url [ uri ]; # 0..1 Canonical identifier for this research definition, represented as a URI
    (globally unique)
  fhir:ResearchDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the research definition
  fhir:ResearchDefinition.version [ string ]; # 0..1 Business version of the research definition
  fhir:ResearchDefinition.name [ string ]; # 0..1 Name for this research definition (computer friendly)
  fhir:ResearchDefinition.title [ string ]; # 0..1 Name for this research definition (human friendly)
  fhir:ResearchDefinition.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:ResearchDefinition.subtitle [ string ]; # 0..1 Subordinate title of the ResearchDefinition
  fhir:ResearchDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ResearchDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # ResearchDefinition.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One
    of these 2
    fhir:ResearchDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ResearchDefinition.subjectReference [ Reference(Group) ]
  fhir:ResearchDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ResearchDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ResearchDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ResearchDefinition.description [ markdown ]; # 0..1 Natural language description of the research definition
  fhir:ResearchDefinition.comment [ string ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:ResearchDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ResearchDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for research definition
    (if applicable)
  fhir:ResearchDefinition.purpose [ markdown ]; # 0..1 Why this research definition is defined
  fhir:ResearchDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the ResearchDefinition
  fhir:ResearchDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ResearchDefinition.approvalDate [ date ]; # 0..1 When the research definition was approved by publisher
  fhir:ResearchDefinition.lastReviewDate [ date ]; # 0..1 When the research definition was last reviewed
  fhir:ResearchDefinition.effectivePeriod [ Period ]; # 0..1 When the research definition is expected to be used
  fhir:ResearchDefinition.topic [ CodeableConcept ], ... ; # 0..* The category of the ResearchDefinition, such as
    Education, Treatment, Assessment, etc.
  fhir:ResearchDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ResearchDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ResearchDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ResearchDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
```

```
  fhir:ResearchDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:ResearchDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the ResearchDefinition
  fhir:ResearchDefinition.population [ Reference(ResearchElementDefinition) ]; # 1..1 What population?
  fhir:ResearchDefinition.exposure [ Reference(ResearchElementDefinition) ]; # 0..1 What exposure?
  fhir:ResearchDefinition.exposureAlternative [ Reference(ResearchElementDefinition) ]; # 0..1 What alternative exposure
     state?
  fhir:ResearchDefinition.outcome [ Reference(ResearchElementDefinition) ]; # 0..1 What outcome?
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchDefinition | ITU | | DomainResource | A research context or question + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this research definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the research definition
version | Σ | 0..1 | string | Business version of the research definition
name | ΣI | 0..1 | string | Name for this research definition (computer friendly)
title | Σ | 0..1 | string | Name for this research definition (human friendly)
shortTitle | | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the ResearchDefinition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the research definition
comment | | 0..* | string | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for research definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this research definition is defined
usage | | 0..1 | string | Describes the clinical usage of the ResearchDefinition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the research definition was approved by publisher
lastReviewDate | | 0..1 | date | When the research definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the research definition is expected to be used
topic | | 0..* | CodeableConcept | The category of the ResearchDefinition, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the ResearchDefinition
population | Σ | 1..1 | Reference(ResearchElementDefinition) | What population?
exposure | Σ | 0..1 | Reference(ResearchElementDefinition) | What exposure?
exposureAlternative | Σ | 0..1 | Reference(ResearchElementDefinition) | What alternative exposure state?
outcome | Σ | 0..1 | Reference(ResearchElementDefinition) | What outcome?
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this research definition, represented as a URI (globally unique)
     -->
 <identifier><!-- 0..* Identifier Additional identifier for the research definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the research definition -->
 <name value="[string]"/><!--  0..1 Name for this research definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this research definition (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the ResearchDefinition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the research definition -->
```

```
 <comment value="[string]"/><!-- 0..* Used for footnotes or explanatory notes -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for research definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this research definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the ResearchDefinition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the research definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the research definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the research definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the ResearchDefinition, such as Education, Treatment, Assessment, etc.
     --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the ResearchDefinition --></library>
 <population><!-- 1..1 Reference(ResearchElementDefinition) What population? --></population>
 <exposure><!-- 0..1 Reference(ResearchElementDefinition) What exposure? --></exposure>
 <exposureAlternative><!-- 0..1 Reference(ResearchElementDefinition) What alternative exposure state?
    --></exposureAlternative>
 <outcome><!-- 0..1 Reference(ResearchElementDefinition) What outcome? --></outcome>
</ResearchDefinition>
```

JSON Template

```
{
  "resourceType" : "ResearchDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this research definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the research definition
  "version" : "<string>", // Business version of the research definition
  "name" : "<string>", // C? Name for this research definition (computer friendly)
  "title" : "<string>", // Name for this research definition (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the ResearchDefinition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the research definition
  "comment" : ["<string>"], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for research definition (if applicable)
  "purpose" : "<markdown>", // Why this research definition is defined
  "usage" : "<string>", // Describes the clinical usage of the ResearchDefinition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the research definition was approved by publisher
  "lastReviewDate" : "<date>", // When the research definition was last reviewed
  "effectivePeriod" : { Period }, // When the research definition is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the ResearchDefinition, such as Education, Treatment, Assessment,
    etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the ResearchDefinition
  "population" : { Reference(ResearchElementDefinition) }, // R!  What population?
  "exposure" : { Reference(ResearchElementDefinition) }, // What exposure?
  "exposureAlternative" : { Reference(ResearchElementDefinition) }, // What alternative exposure state?
  "outcome" : { Reference(ResearchElementDefinition) } // What outcome?
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchDefinition.url [ uri ]; # 0..1 Canonical identifier for this research definition, represented as a URI
    (globally unique)
  fhir:ResearchDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the research definition
  fhir:ResearchDefinition.version [ string ]; # 0..1 Business version of the research definition
  fhir:ResearchDefinition.name [ string ]; # 0..1 Name for this research definition (computer friendly)
  fhir:ResearchDefinition.title [ string ]; # 0..1 Name for this research definition (human friendly)
  fhir:ResearchDefinition.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:ResearchDefinition.subtitle [ string ]; # 0..1 Subordinate title of the ResearchDefinition
  fhir:ResearchDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ResearchDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # ResearchDefinition.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One
```

```
    of these 2
    fhir:ResearchDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ResearchDefinition.subjectReference [ Reference(Group) ]
  fhir:ResearchDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ResearchDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ResearchDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ResearchDefinition.description [ markdown ]; # 0..1 Natural language description of the research definition
  fhir:ResearchDefinition.comment [ string ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:ResearchDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to support
  fhir:ResearchDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for research definition
    (if applicable)
  fhir:ResearchDefinition.purpose [ markdown ]; # 0..1 Why this research definition is defined
  fhir:ResearchDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the ResearchDefinition
  fhir:ResearchDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ResearchDefinition.approvalDate [ date ]; # 0..1 When the research definition was approved by publisher
  fhir:ResearchDefinition.lastReviewDate [ date ]; # 0..1 When the research definition was last reviewed
  fhir:ResearchDefinition.effectivePeriod [ Period ]; # 0..1 When the research definition is expected to be used
  fhir:ResearchDefinition.topic [ CodeableConcept ], ... ; # 0..* The category of the ResearchDefinition, such as
    Education, Treatment, Assessment, etc.
  fhir:ResearchDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ResearchDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ResearchDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ResearchDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:ResearchDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:ResearchDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the ResearchDefinition
  fhir:ResearchDefinition.population [ Reference(ResearchElementDefinition) ]; # 1..1 What population?
  fhir:ResearchDefinition.exposure [ Reference(ResearchElementDefinition) ]; # 0..1 What exposure?
  fhir:ResearchDefinition.exposureAlternative [ Reference(ResearchElementDefinition) ]; # 0..1 What alternative exposure
     state?
  fhir:ResearchDefinition.outcome [ Reference(ResearchElementDefinition) ]; # 0..1 What outcome?
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.16.3.1 Terminology Bindings

Path | Definition | Type | Reference

ResearchDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus

ResearchDefinition.subject[x] | The possible types of subjects for the research (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType

ResearchDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

ResearchDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic

14.16.3.2 Constraints

id | Level | Location | Description | Expression

rsd-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.16.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

composed-of | reference | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='composed-of').resource (Any)

context | token | A use context assigned to the research definition | (ResearchDefinition.useContext.value as CodeableConcept)

context-quantity | quantity | A quantity- or range-valued use context assigned to the research definition | (ResearchDefinition.useContext.value as Quantity) | (ResearchDefinition.useContext.value as Range)

context-type | token | A type of use context assigned to the research definition | ResearchDefinition.useContext.code

context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the research definition | On ResearchDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)

context-type-value | composite | A use context type and value assigned to the research definition | On ResearchDefinition.useContext: context-type: code context: value.as(CodeableConcept)

date | date | The research definition publication date | ResearchDefinition.date

depends-on | reference | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='depends-on').resource | ResearchDefinition.library (Library, Any)

derived-from | reference | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='derived-from').resource (Any)

description | string | The description of the research definition | ResearchDefinition.description

effective | date | The time during which the research definition is intended to be in use | ResearchDefinition.effectivePeriod

identifier | token | External identifier for the research definition | ResearchDefinition.identifier

jurisdiction | token | Intended jurisdiction for the research definition | ResearchDefinition.jurisdiction

name | string | Computationally friendly name of the research definition | ResearchDefinition.name

predecessor | reference | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the research definition | ResearchDefinition.publisher
status | token | The current status of the research definition | ResearchDefinition.status
successor | reference | What resource is being referenced | ResearchDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the research definition | ResearchDefinition.title
topic | token | Topics associated with the ResearchDefinition | ResearchDefinition.topic
url | uri | The uri that identifies the research definition | ResearchDefinition.url
version | token | The business version of the research definition | ResearchDefinition.version

14.16.5 Resource ResearchDefinition - Examples

URL: https://hl7.org/fhir/R4/researchdefinition-examples.html

14.16.5 Resource ResearchDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

TODO

Example Name | id | Format
Example of researchdefinition | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.16.6 Resource ResearchDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/researchdefinition-definitions.html

14.16.6 Resource ResearchDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ResearchDefinition resource.

ResearchDefinition
Element Id | ResearchDefinition
Definition | The ResearchDefinition resource describes the conditional state (population and any exposures being compared within the population) and outcome (if specified) that the knowledge (evidence, assertion, recommendation) is about.
Cardinality | 0..*
Type | DomainResource
Invariants | Defined on this element rsd-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
rsd-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ResearchDefinition.url
Element Id | ResearchDefinition.url
Definition | An absolute URI that is used to identify this research definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this research definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the research definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the research definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ResearchDefinition.identifier
Element Id | ResearchDefinition.identifier
Definition | A formal identifier that is used to identify this research definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this research definition outside of FHIR, where it is not possible to use the logical URI.
ResearchDefinition.version
Element Id | ResearchDefinition.version

Definition | The identifier that is used to identify this version of the research definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the research definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different research definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the research definition with the format [url]|[version].
ResearchDefinition.name
Element Id | ResearchDefinition.name
Definition | A natural language name identifying the research definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ResearchDefinition.title
Element Id | ResearchDefinition.title
Definition | A short, descriptive, user-friendly title for the research definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ResearchDefinition.shortTitle
Element Id | ResearchDefinition.shortTitle
Definition | The short title provides an alternate title for use in informal descriptive contexts where the full, formal title is not necessary.
Cardinality | 0..1
Type | string
Requirements | Need to be able to reference the content by a short description, but still provide a longer, more formal title for the content.
ResearchDefinition.subtitle
Element Id | ResearchDefinition.subtitle
Definition | An explanatory or alternate title for the ResearchDefinition giving additional information about its content.
Cardinality | 0..1
Type | string
ResearchDefinition.status
Element Id | ResearchDefinition.status
Definition | The status of this research definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of research definitions that are appropriate for use versus not.
ResearchDefinition.experimental
Element Id | ResearchDefinition.experimental
Definition | A Boolean value to indicate that this research definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level research definition.
Summary | true
Comments | Allows filtering of research definitions that are appropriate for use versus not.
ResearchDefinition.subject[x]
Element Id | ResearchDefinition.subject[x]
Definition | The intended subjects for the ResearchDefinition. If this element is not provided, a Patient subject is assumed, but the subject of the ResearchDefinition can be anything.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)

Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
Comments | The subject of the ResearchDefinition is critical in interpreting the criteria definitions, as the logic in the ResearchDefinitions is evaluated with respect to a particular subject. This corresponds roughly to the notion of a Compartment in that it limits what content is available based on its relationship to the subject. In CQL, this corresponds to the context declaration.
ResearchDefinition.date
Element Id | ResearchDefinition.date
Definition | The date (and optionally time) when the research definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the research definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the research definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ResearchDefinition.publisher
Element Id | ResearchDefinition.publisher
Definition | The name of the organization or individual that published the research definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the research definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the research definition is the organization or individual primarily responsible for the maintenance and upkeep of the research definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the research definition. This item SHOULD be populated unless the information is available from context.
ResearchDefinition.contact
Element Id | ResearchDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ResearchDefinition.description
Element Id | ResearchDefinition.description
Definition | A free text natural language description of the research definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the research definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the research definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the research definition is presumed to be the predominant language in the place the research definition was created).
ResearchDefinition.comment
Element Id | ResearchDefinition.comment
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | string
ResearchDefinition.useContext
Element Id | ResearchDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate research definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ResearchDefinition.jurisdiction
Element Id | ResearchDefinition.jurisdiction
Definition | A legal or geographic region in which the research definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the research definition to be used in jurisdictions other than those for which it was originally designed or intended.
ResearchDefinition.purpose
Element Id | ResearchDefinition.purpose

Definition | Explanation of why this research definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the research definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this research definition.
ResearchDefinition.usage
Element Id | ResearchDefinition.usage
Definition | A detailed description, from a clinical perspective, of how the ResearchDefinition is used.
Cardinality | 0..1
Type | string
ResearchDefinition.copyright
Element Id | ResearchDefinition.copyright
Definition | A copyright statement relating to the research definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the research definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the research definition and/or its content.
Alternate Names | License; Restrictions
ResearchDefinition.approvalDate
Element Id | ResearchDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
To Do | Consider how this applies to this resource.
ResearchDefinition.lastReviewDate
Element Id | ResearchDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
To Do | Consider how this applies to this resource.
ResearchDefinition.effectivePeriod
Element Id | ResearchDefinition.effectivePeriod
Definition | The period during which the research definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the research definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a research definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
To Do | Consider how this applies to this resource.
ResearchDefinition.topic
Element Id | ResearchDefinition.topic
Definition | Descriptive topics related to the content of the ResearchDefinition. Topics provide a high-level categorization grouping types of ResearchDefinitions that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the ResearchDefinition so that it can be found by topical searches.
To Do | Consider how this applies to this resource.
ResearchDefinition.author
Element Id | ResearchDefinition.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
ResearchDefinition.editor
Element Id | ResearchDefinition.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
ResearchDefinition.reviewer
Element Id | ResearchDefinition.reviewer

Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
ResearchDefinition.endorser
Element Id | ResearchDefinition.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
To Do | Consider how this applies to this resource.
ResearchDefinition.relatedArtifact
Element Id | ResearchDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | ResearchDefinitions must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
To Do | Consider how this applies to this resource.
ResearchDefinition.library
Element Id | ResearchDefinition.library
Definition | A reference to a Library resource containing the formal logic used by the ResearchDefinition.
Cardinality | 0..*
Type | canonical(Library)
To Do | Consider how this applies to this resource.
ResearchDefinition.population
Element Id | ResearchDefinition.population
Definition | A reference to a ResearchElementDefinition resource that defines the population for the research.
Cardinality | 1..1
Type | Reference(ResearchElementDefinition)
Summary | true
ResearchDefinition.exposure
Element Id | ResearchDefinition.exposure
Definition | A reference to a ResearchElementDefinition resource that defines the exposure for the research.
Cardinality | 0..1
Type | Reference(ResearchElementDefinition)
Summary | true
ResearchDefinition.exposureAlternative
Element Id | ResearchDefinition.exposureAlternative
Definition | A reference to a ResearchElementDefinition resource that defines the exposureAlternative for the research.
Cardinality | 0..1
Type | Reference(ResearchElementDefinition)
Summary | true
ResearchDefinition.outcome
Element Id | ResearchDefinition.outcome
Definition | A reference to a ResearchElementDefinition resomece that defines the outcome for the research.
Cardinality | 0..1
Type | Reference(ResearchElementDefinition)
Summary | true

14.16.7 Resource ResearchDefinition - Mappings

URL: https://hl7.org/fhir/R4/researchdefinition-mappings.html

14.16.7 Resource ResearchDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the researchdefinition resource (see Mappings to Other Standards for further information & status).

14.16.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ResearchDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version

status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

14.16.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ResearchDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
subject[x] | Definition.subject
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.16.7.3 RIM Mapping (http://hl7.org/v3)

ResearchDefinition
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.16.8 Resource ResearchDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/researchdefinition-profiles.html

14.16.8 Resource ResearchDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ResearchDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.16.8.1 Profiles

No Profiles defined for this resource

14.16.8.2 Extensions

replaces | ResearchDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.16.8.3 Search Extensions

No Search Extensions defined for this resource

14.16.9 JSON Schema for ResearchDefinition

URL: https://hl7.org/fhir/R4/researchdefinition.schema.json.html

14.16.9 JSON Schema for ResearchDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ResearchDefinition, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ResearchDefinition",
  "$ref": "#/definitions/ResearchDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ResearchDefinition": {
      "description": "The ResearchDefinition resource describes the conditional state (population and any exposures
    being compared within the population) and outcome (if specified) that the knowledge (evidence, assertion,
    recommendation) is about.",
      "properties": {
        "resourceType": {
          "description": "This is a ResearchDefinition resource",
          "const": "ResearchDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
        },
        "_language": {
          "description": "Extensions for language",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "text": {
          "description": "A human-readable narrative that contains a summary of the resource and can be used to
    represent the content of the resource to a human. The narrative need not encode all the structured data, but is
    required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
    Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
          "$ref": "Narrative.schema.json#/definitions/Narrative"
        },
        "contained": {
          "description": "These resources do not have an independent existence apart from the resource that contains
    them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
          "items": {
            "$ref": "ResourceList.schema.json#/definitions/ResourceList"
          },
          "type": "array"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
     definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
    that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of
    the resource and that modifies the understanding of the element that contains it and/or the understanding of the
    containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
    of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
    extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
    met as part of the definition of the extension. Applications processing a resource are required to check for
    modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
    DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
```

```
    },
    "url": {
      "description": "An absolute URI that is used to identify this research definition when it is referenced in a
specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique
and SHOULD be a literal address at which at which an authoritative instance of this research definition is (or will
be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the research
definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this research definition when it is represented
in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the research definition when it is
referenced in a specification, model, design or instance. This is an arbitrary value managed by the research
definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if
 a managed version is not available. There is also no expectation that versions can be placed in a lexicographical
sequence. To provide a version consistent with the Decision Support Service specification, use the format
Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision
Support Service specification. Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the research definition. This name should be usable as an
identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the research definition.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "shortTitle": {
      "description": "The short title provides an alternate title for use in informal descriptive contexts where the
 full, formal title is not necessary.",
      "$ref": "#/definitions/string"
    },
    "_shortTitle": {
      "description": "Extensions for shortTitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the ResearchDefinition giving additional information
about its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this research definition. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this research definition is authored for testing purposes (or
 education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
```

```
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectCodeableConcept": {
      "description": "The intended subjects for the ResearchDefinition. If this element is not provided, a Patient subject is assumed, but the subject of the ResearchDefinition can be anything.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "The intended subjects for the ResearchDefinition. If this element is not provided, a Patient subject is assumed, but the subject of the ResearchDefinition can be anything.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the research definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the research definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the research definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the research definition from a consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_comment": {
      "description": "Extensions for comment",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate research definition instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the research definition is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this research definition is needed and why it has been designed as it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description, from a clinical perspective, of how the ResearchDefinition is used.",
```

```
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the research definition and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the research definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the research definition content was or is planned to be in active
use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the ResearchDefinition. Topics provide a high-
level categorization grouping types of ResearchDefinitions that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "library": {
      "description": "A reference to a Library resource containing the formal logic used by the
ResearchDefinition.",
      "items": {
```

```
            "$ref": "canonical.schema.json#/definitions/canonical"
          },
          "type": "array"
        },
        "population": {
          "description": "A reference to a ResearchElementDefinition resource that defines the population for the
    research.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "exposure": {
          "description": "A reference to a ResearchElementDefinition resource that defines the exposure for the
    research.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "exposureAlternative": {
          "description": "A reference to a ResearchElementDefinition resource that defines the exposureAlternative for
    the research.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        },
        "outcome": {
          "description": "A reference to a ResearchElementDefinition resomece that defines the outcome for the
    research.",
          "$ref": "Reference.schema.json#/definitions/Reference"
        }
      },
      "additionalProperties": false,
      "required": [
        "resourceType",
        "population"
      ]
    }
  }
}
```

14.16.10 Resource ResearchDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/researchdefinition-dependencies.html

14.16.10 Resource ResearchDefinition - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ResearchDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
ResearchDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
shortTitle | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
comment | 0..* | string
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
library | 0..* | canonical(Library)

population | 1..1 | Reference(ResearchElementDefinition)
exposure | 0..1 | Reference(ResearchElementDefinition)
exposureAlternative | 0..1 | Reference(ResearchElementDefinition)
outcome | 0..1 | Reference(ResearchElementDefinition)

14.17 Resource ResearchElementDefinition - Content

URL: https://hl7.org/fhir/R4/researchelementdefinition.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The ResearchElementDefinition resource describes a "PICO" element that knowledge (evidence, assertion, recommendation) is about.

14.17.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

TODO

14.17.2 Boundaries and Relationships

TODO

This resource is referenced by ResearchDefinition

14.17.3 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchElementDefinition | ITU | | DomainResource | A population, intervention, or exposure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this research element definition, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the research element definition
version | Σ | 0..1 | string | Business version of the research element definition
name | ΣI | 0..1 | string | Name for this research element definition (computer friendly)
title | Σ | 0..1 | string | Name for this research element definition (human friendly)
shortTitle | Σ | 0..1 | string | Title for use in informal contexts
subtitle | | 0..1 | string | Subordinate title of the ResearchElementDefinition
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | Σ | 0..1 | boolean | For testing purposes, not real usage
subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)
subjectCodeableConcept | | | CodeableConcept
subjectReference | | | Reference(Group)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the research element definition
comment | | 0..* | string | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for research element definition (if applicable) Jurisdiction (Extensible)
purpose | | 0..1 | markdown | Why this research element definition is defined
usage | | 0..1 | string | Describes the clinical usage of the ResearchElementDefinition
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the research element definition was approved by publisher
lastReviewDate | | 0..1 | date | When the research element definition was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the research element definition is expected to be used
topic | | 0..* | CodeableConcept | The category of the ResearchElementDefinition, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content

relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
library | | 0..* | canonical(Library) | Logic used by the ResearchElementDefinition
type | Σ | 1..1 | code | population | exposure | outcome ResearchElementType (Required)
variableType | | 0..1 | code | dichotomous | continuous | descriptive VariableType (Required)
characteristic | Σ | 1..* | BackboneElement | What defines the members of the research element
definition[x] | Σ | 1..1 | | What code or expression defines members?
definitionCodeableConcept | | | CodeableConcept
definitionCanonical | | | canonical(ValueSet)
definitionExpression | | | Expression
definitionDataRequirement | | | DataRequirement
usageContext | | 0..* | UsageContext | What code/value pairs define members?
exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members
unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)
studyEffectiveDescription | | 0..1 | string | What time period does the study cover
studyEffective[x] | | 0..1 | | What time period does the study cover
studyEffectiveDateTime | | | dateTime
studyEffectivePeriod | | | Period
studyEffectiveDuration | | | Duration
studyEffectiveTiming | | | Timing
studyEffectiveTimeFromStart | | 0..1 | Duration | Observation time from study start
studyEffectiveGroupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median GroupMeasure (Required)
participantEffectiveDescription | | 0..1 | string | What time period do participants cover
participantEffective[x] | | 0..1 | | What time period do participants cover
participantEffectiveDateTime | | | dateTime
participantEffectivePeriod | | | Period
participantEffectiveDuration | | | Duration
participantEffectiveTiming | | | Timing
participantEffectiveTimeFromStart | | 0..1 | Duration | Observation time from study start
participantEffectiveGroupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median GroupMeasure (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchElementDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this research element definition, represented as a URI (globally
     unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the research element definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the research element definition -->
 <name value="[string]"/><!--  0..1 Name for this research element definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this research element definition (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the ResearchElementDefinition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the research element definition -->
 <comment value="[string]"/><!-- 0..* Used for footnotes or explanatory notes -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for research element definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this research element definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the ResearchElementDefinition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the research element definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the research element definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the research element definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the ResearchElementDefinition, such as Education, Treatment,
    Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the ResearchElementDefinition --></library>
 <type value="[code]"/><!-- 1..1 population | exposure | outcome -->
 <variableType value="[code]"/><!-- 0..1 dichotomous | continuous | descriptive -->
 <characteristic>  <!-- 1..* What defines the members of the research element -->
  <definition[x]><!-- 1..1 CodeableConcept|canonical(ValueSet)|Expression|
    DataRequirement What code or expression defines members? --></definition[x]>
  <usageContext><!-- 0..* UsageContext What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!-- 0..1 Whether the characteristic includes or excludes members -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <studyEffectiveDescription value="[string]"/><!-- 0..1 What time period does the study cover -->
```

```
  <studyEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period does the study cover
    --></studyEffective[x]>
  <studyEffectiveTimeFromStart><!-- 0..1 Duration Observation time from study start --></studyEffectiveTimeFromStart>
  <studyEffectiveGroupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean |
     median-of-median -->
  <participantEffectiveDescription value="[string]"/><!-- 0..1 What time period do participants cover -->
  <participantEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period do participants cover
    --></participantEffective[x]>
  <participantEffectiveTimeFromStart><!-- 0..1 Duration Observation time from study start
    --></participantEffectiveTimeFromStart>
  <participantEffectiveGroupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-
    mean | median-of-median -->
 </characteristic>
</ResearchElementDefinition>
```

JSON Template

```
{
  "resourceType" : "ResearchElementDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this research element definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the research element definition
  "version" : "<string>", // Business version of the research element definition
  "name" : "<string>", // C? Name for this research element definition (computer friendly)
  "title" : "<string>", // Name for this research element definition (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the ResearchElementDefinition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the research element definition
  "comment" : ["<string>"], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for research element definition (if applicable)
  "purpose" : "<markdown>", // Why this research element definition is defined
  "usage" : "<string>", // Describes the clinical usage of the ResearchElementDefinition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the research element definition was approved by publisher
  "lastReviewDate" : "<date>", // When the research element definition was last reviewed
  "effectivePeriod" : { Period }, // When the research element definition is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the ResearchElementDefinition, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the ResearchElementDefinition
  "type" : "<code>", // R!  population | exposure | outcome
  "variableType" : "<code>", // dichotomous | continuous | descriptive
  "characteristic" : [{ // R!  What defines the members of the research element
    // definition[x]: What code or expression defines members?. One of these 4:
    "definitionCodeableConcept" : { CodeableConcept },
    "definitionCanonical" : { canonical(ValueSet) },
    "definitionExpression" : { Expression },
    "definitionDataRequirement" : { DataRequirement },
    "usageContext" : [{ UsageContext }], // What code/value pairs define members?
    "exclude" : <boolean>, // Whether the characteristic includes or excludes members
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "studyEffectiveDescription" : "<string>", // What time period does the study cover
    // studyEffective[x]: What time period does the study cover. One of these 4:
    "studyEffectiveDateTime" : "<dateTime>",
    "studyEffectivePeriod" : { Period },
    "studyEffectiveDuration" : { Duration },
    "studyEffectiveTiming" : { Timing },
    "studyEffectiveTimeFromStart" : { Duration }, // Observation time from study start
    "studyEffectiveGroupMeasure" : "<code>", // mean | median | mean-of-mean | mean-of-median | median-of-mean | median-
    of-median
    "participantEffectiveDescription" : "<string>", // What time period do participants cover
    // participantEffective[x]: What time period do participants cover. One of these 4:
    "participantEffectiveDateTime" : "<dateTime>",
    "participantEffectivePeriod" : { Period },
    "participantEffectiveDuration" : { Duration },
    "participantEffectiveTiming" : { Timing },
    "participantEffectiveTimeFromStart" : { Duration }, // Observation time from study start
    "participantEffectiveGroupMeasure" : "<code>" // mean | median | mean-of-mean | mean-of-median | median-of-mean |
    median-of-median
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:ResearchElementDefinition;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchElementDefinition.url [ uri ]; # 0..1 Canonical identifier for this research element definition,
    represented as a URI (globally unique)
  fhir:ResearchElementDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the research element
    definition
  fhir:ResearchElementDefinition.version [ string ]; # 0..1 Business version of the research element definition
  fhir:ResearchElementDefinition.name [ string ]; # 0..1 Name for this research element definition (computer friendly)
  fhir:ResearchElementDefinition.title [ string ]; # 0..1 Name for this research element definition (human friendly)
  fhir:ResearchElementDefinition.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:ResearchElementDefinition.subtitle [ string ]; # 0..1 Subordinate title of the ResearchElementDefinition
  fhir:ResearchElementDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ResearchElementDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # ResearchElementDefinition.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location,
    Device. One of these 2
    fhir:ResearchElementDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ResearchElementDefinition.subjectReference [ Reference(Group) ]
  fhir:ResearchElementDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ResearchElementDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ResearchElementDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ResearchElementDefinition.description [ markdown ]; # 0..1 Natural language description of the research element
    definition
  fhir:ResearchElementDefinition.comment [ string ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:ResearchElementDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:ResearchElementDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for research
    element definition (if applicable)
  fhir:ResearchElementDefinition.purpose [ markdown ]; # 0..1 Why this research element definition is defined
  fhir:ResearchElementDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the ResearchElementDefinition
  fhir:ResearchElementDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ResearchElementDefinition.approvalDate [ date ]; # 0..1 When the research element definition was approved by
    publisher
  fhir:ResearchElementDefinition.lastReviewDate [ date ]; # 0..1 When the research element definition was last reviewed
  fhir:ResearchElementDefinition.effectivePeriod [ Period ]; # 0..1 When the research element definition is expected to
    be used
  fhir:ResearchElementDefinition.topic [ CodeableConcept ], ... ; # 0..* The category of the ResearchElementDefinition,
    such as Education, Treatment, Assessment, etc.
  fhir:ResearchElementDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ResearchElementDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ResearchElementDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ResearchElementDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:ResearchElementDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations,
    etc.
  fhir:ResearchElementDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the
    ResearchElementDefinition
  fhir:ResearchElementDefinition.type [ code ]; # 1..1 population | exposure | outcome
  fhir:ResearchElementDefinition.variableType [ code ]; # 0..1 dichotomous | continuous | descriptive
  fhir:ResearchElementDefinition.characteristic [ # 1..* What defines the members of the research element
    # ResearchElementDefinition.characteristic.definition[x] : 1..1 What code or expression defines members?. One of
    these 4
      fhir:ResearchElementDefinition.characteristic.definitionCodeableConcept [ CodeableConcept ]
      fhir:ResearchElementDefinition.characteristic.definitionCanonical [ canonical(ValueSet) ]
      fhir:ResearchElementDefinition.characteristic.definitionExpression [ Expression ]
      fhir:ResearchElementDefinition.characteristic.definitionDataRequirement [ DataRequirement ]
    fhir:ResearchElementDefinition.characteristic.usageContext [ UsageContext ], ... ; # 0..* What code/value pairs
    define members?
    fhir:ResearchElementDefinition.characteristic.exclude [ boolean ]; # 0..1 Whether the characteristic includes or
    excludes members
    fhir:ResearchElementDefinition.characteristic.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome
    described in?
    fhir:ResearchElementDefinition.characteristic.studyEffectiveDescription [ string ]; # 0..1 What time period does the
     study cover
    # ResearchElementDefinition.characteristic.studyEffective[x] : 0..1 What time period does the study cover. One of
    these 4
      fhir:ResearchElementDefinition.characteristic.studyEffectiveDateTime [ dateTime ]
      fhir:ResearchElementDefinition.characteristic.studyEffectivePeriod [ Period ]
      fhir:ResearchElementDefinition.characteristic.studyEffectiveDuration [ Duration ]
      fhir:ResearchElementDefinition.characteristic.studyEffectiveTiming [ Timing ]
    fhir:ResearchElementDefinition.characteristic.studyEffectiveTimeFromStart [ Duration ]; # 0..1 Observation time from
     study start
    fhir:ResearchElementDefinition.characteristic.studyEffectiveGroupMeasure [ code ]; # 0..1 mean | median | mean-of-
    mean | mean-of-median | median-of-mean | median-of-median
    fhir:ResearchElementDefinition.characteristic.participantEffectiveDescription [ string ]; # 0..1 What time period do
     participants cover
    # ResearchElementDefinition.characteristic.participantEffective[x] : 0..1 What time period do participants cover.
    One of these 4
      fhir:ResearchElementDefinition.characteristic.participantEffectiveDateTime [ dateTime ]
      fhir:ResearchElementDefinition.characteristic.participantEffectivePeriod [ Period ]
      fhir:ResearchElementDefinition.characteristic.participantEffectiveDuration [ Duration ]
      fhir:ResearchElementDefinition.characteristic.participantEffectiveTiming [ Timing ]
    fhir:ResearchElementDefinition.characteristic.participantEffectiveTimeFromStart [ Duration ]; # 0..1 Observation
    time from study start
    fhir:ResearchElementDefinition.characteristic.participantEffectiveGroupMeasure [ code ]; # 0..1 mean | median |
    mean-of-mean | mean-of-median | median-of-mean | median-of-median
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ResearchElementDefinition | ITU | | DomainResource | A population, intervention, or exposure definition + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | Σ | 0..1 | uri | Canonical identifier for this research element definition, represented as a URI (globally unique)

identifier | Σ | 0..* | Identifier | Additional identifier for the research element definition

version | Σ | 0..1 | string | Business version of the research element definition

name | ΣI | 0..1 | string | Name for this research element definition (computer friendly)

title | Σ | 0..1 | string | Name for this research element definition (human friendly)

shortTitle | Σ | 0..1 | string | Title for use in informal contexts

subtitle | | 0..1 | string | Subordinate title of the ResearchElementDefinition

status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | Σ | 0..1 | boolean | For testing purposes, not real usage

subject[x] | | 0..1 | | E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device Subject type (Extensible)

subjectCodeableConcept | | | CodeableConcept

subjectReference | | | Reference(Group)

date | Σ | 0..1 | dateTime | Date last changed

publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)

contact | Σ | 0..* | ContactDetail | Contact details for the publisher

description | Σ | 0..1 | markdown | Natural language description of the research element definition

comment | | 0..* | string | Used for footnotes or explanatory notes

useContext | Σ | 0..* | UsageContext | The context that the content is intended to support

jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for research element definition (if applicable) Jurisdiction (Extensible)

purpose | | 0..1 | markdown | Why this research element definition is defined

usage | | 0..1 | string | Describes the clinical usage of the ResearchElementDefinition

copyright | | 0..1 | markdown | Use and/or publishing restrictions

approvalDate | | 0..1 | date | When the research element definition was approved by publisher

lastReviewDate | | 0..1 | date | When the research element definition was last reviewed

effectivePeriod | Σ | 0..1 | Period | When the research element definition is expected to be used

topic | | 0..* | CodeableConcept | The category of the ResearchElementDefinition, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)

author | | 0..* | ContactDetail | Who authored the content

editor | | 0..* | ContactDetail | Who edited the content

reviewer | | 0..* | ContactDetail | Who reviewed the content

endorser | | 0..* | ContactDetail | Who endorsed the content

relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.

library | | 0..* | canonical(Library) | Logic used by the ResearchElementDefinition

type | Σ | 1..1 | code | population | exposure | outcome ResearchElementType (Required)

variableType | | 0..1 | code | dichotomous | continuous | descriptive VariableType (Required)

characteristic | Σ | 1..* | BackboneElement | What defines the members of the research element

definition[x] | Σ | 1..1 | | What code or expression defines members?

definitionCodeableConcept | | | CodeableConcept

definitionCanonical | | | canonical(ValueSet)

definitionExpression | | | Expression

definitionDataRequirement | | | DataRequirement

usageContext | | 0..* | UsageContext | What code/value pairs define members?

exclude | | 0..1 | boolean | Whether the characteristic includes or excludes members

unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)

studyEffectiveDescription | | 0..1 | string | What time period does the study cover

studyEffective[x] | | 0..1 | | What time period does the study cover

studyEffectiveDateTime | | | dateTime

studyEffectivePeriod | | | Period

studyEffectiveDuration | | | Duration

studyEffectiveTiming | | | Timing

studyEffectiveTimeFromStart | | 0..1 | Duration | Observation time from study start

studyEffectiveGroupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median GroupMeasure (Required)

participantEffectiveDescription | | 0..1 | string | What time period do participants cover

participantEffective[x] | | 0..1 | | What time period do participants cover

participantEffectiveDateTime | | | dateTime

participantEffectivePeriod | | | Period

participantEffectiveDuration | | | Duration

participantEffectiveTiming | | | Timing

participantEffectiveTimeFromStart | | 0..1 | Duration | Observation time from study start

participantEffectiveGroupMeasure | | 0..1 | code | mean | median | mean-of-mean | mean-of-median | median-of-mean | median-of-median GroupMeasure (Required)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchElementDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this research element definition, represented as a URI (globally
     unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the research element definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the research element definition -->
 <name value="[string]"/><!--  0..1 Name for this research element definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this research element definition (human friendly) -->
 <shortTitle value="[string]"/><!-- 0..1 Title for use in informal contexts -->
 <subtitle value="[string]"/><!-- 0..1 Subordinate title of the ResearchElementDefinition -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <subject[x]><!-- 0..1 CodeableConcept|Reference(Group) E.g. Patient, Practitioner, RelatedPerson, Organization,
    Location, Device --></subject[x]>
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the research element definition -->
 <comment value="[string]"/><!-- 0..* Used for footnotes or explanatory notes -->
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for research element definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this research element definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the ResearchElementDefinition -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the research element definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the research element definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the research element definition is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the ResearchElementDefinition, such as Education, Treatment,
    Assessment, etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <library><!-- 0..* canonical(Library) Logic used by the ResearchElementDefinition --></library>
 <type value="[code]"/><!-- 1..1 population | exposure | outcome -->
 <variableType value="[code]"/><!-- 0..1 dichotomous | continuous | descriptive -->
 <characteristic>  <!-- 1..* What defines the members of the research element -->
  <definition[x]><!-- 1..1 CodeableConcept|canonical(ValueSet)|Expression|
    DataRequirement What code or expression defines members? --></definition[x]>
  <usageContext><!-- 0..* UsageContext What code/value pairs define members? --></usageContext>
  <exclude value="[boolean]"/><!-- 0..1 Whether the characteristic includes or excludes members -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <studyEffectiveDescription value="[string]"/><!-- 0..1 What time period does the study cover -->
  <studyEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period does the study cover
    --></studyEffective[x]>
  <studyEffectiveTimeFromStart><!-- 0..1 Duration Observation time from study start --></studyEffectiveTimeFromStart>
  <studyEffectiveGroupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-mean |
     median-of-median -->
  <participantEffectiveDescription value="[string]"/><!-- 0..1 What time period do participants cover -->
  <participantEffective[x]><!-- 0..1 dateTime|Period|Duration|Timing What time period do participants cover
    --></participantEffective[x]>
  <participantEffectiveTimeFromStart><!-- 0..1 Duration Observation time from study start
    --></participantEffectiveTimeFromStart>
  <participantEffectiveGroupMeasure value="[code]"/><!-- 0..1 mean | median | mean-of-mean | mean-of-median | median-of-
    mean | median-of-median -->
 </characteristic>
</ResearchElementDefinition>
```

JSON Template

```
{
  "resourceType" : "ResearchElementDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this research element definition, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the research element definition
  "version" : "<string>", // Business version of the research element definition
  "name" : "<string>", // C? Name for this research element definition (computer friendly)
  "title" : "<string>", // Name for this research element definition (human friendly)
  "shortTitle" : "<string>", // Title for use in informal contexts
  "subtitle" : "<string>", // Subordinate title of the ResearchElementDefinition
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  // subject[x]: E.g. Patient, Practitioner, RelatedPerson, Organization, Location, Device. One of these 2:
  "subjectCodeableConcept" : { CodeableConcept },
  "subjectReference" : { Reference(Group) },
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the research element definition
  "comment" : ["<string>"], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
```

```
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for research element definition (if applicable)
  "purpose" : "<markdown>", // Why this research element definition is defined
  "usage" : "<string>", // Describes the clinical usage of the ResearchElementDefinition
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the research element definition was approved by publisher
  "lastReviewDate" : "<date>", // When the research element definition was last reviewed
  "effectivePeriod" : { Period }, // When the research element definition is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the ResearchElementDefinition, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "library" : [{ canonical(Library) }], // Logic used by the ResearchElementDefinition
  "type" : "<code>", // R!  population | exposure | outcome
  "variableType" : "<code>", // dichotomous | continuous | descriptive
  "characteristic" : [{ // R!  What defines the members of the research element
    // definition[x]: What code or expression defines members?. One of these 4:
    "definitionCodeableConcept" : { CodeableConcept },
    "definitionCanonical" : { canonical(ValueSet) },
    "definitionExpression" : { Expression },
    "definitionDataRequirement" : { DataRequirement },
    "usageContext" : [{ UsageContext }], // What code/value pairs define members?
    "exclude" : <boolean>, // Whether the characteristic includes or excludes members
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "studyEffectiveDescription" : "<string>", // What time period does the study cover
    // studyEffective[x]: What time period does the study cover. One of these 4:
    "studyEffectiveDateTime" : "<dateTime>",
    "studyEffectivePeriod" : { Period },
    "studyEffectiveDuration" : { Duration },
    "studyEffectiveTiming" : { Timing },
    "studyEffectiveTimeFromStart" : { Duration }, // Observation time from study start
    "studyEffectiveGroupMeasure" : "<code>", // mean | median | mean-of-mean | mean-of-median | median-of-mean | median-
    of-median
    "participantEffectiveDescription" : "<string>", // What time period do participants cover
    // participantEffective[x]: What time period do participants cover. One of these 4:
    "participantEffectiveDateTime" : "<dateTime>",
    "participantEffectivePeriod" : { Period },
    "participantEffectiveDuration" : { Duration },
    "participantEffectiveTiming" : { Timing },
    "participantEffectiveTimeFromStart" : { Duration }, // Observation time from study start
    "participantEffectiveGroupMeasure" : "<code>" // mean | median | mean-of-mean | mean-of-median | median-of-mean |
    median-of-median
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchElementDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchElementDefinition.url [ uri ]; # 0..1 Canonical identifier for this research element definition,
    represented as a URI (globally unique)
  fhir:ResearchElementDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the research element
    definition
  fhir:ResearchElementDefinition.version [ string ]; # 0..1 Business version of the research element definition
  fhir:ResearchElementDefinition.name [ string ]; # 0..1 Name for this research element definition (computer friendly)
  fhir:ResearchElementDefinition.title [ string ]; # 0..1 Name for this research element definition (human friendly)
  fhir:ResearchElementDefinition.shortTitle [ string ]; # 0..1 Title for use in informal contexts
  fhir:ResearchElementDefinition.subtitle [ string ]; # 0..1 Subordinate title of the ResearchElementDefinition
  fhir:ResearchElementDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ResearchElementDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  # ResearchElementDefinition.subject[x] : 0..1 E.g. Patient, Practitioner, RelatedPerson, Organization, Location,
    Device. One of these 2
    fhir:ResearchElementDefinition.subjectCodeableConcept [ CodeableConcept ]
    fhir:ResearchElementDefinition.subjectReference [ Reference(Group) ]
  fhir:ResearchElementDefinition.date [ dateTime ]; # 0..1 Date last changed
  fhir:ResearchElementDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ResearchElementDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ResearchElementDefinition.description [ markdown ]; # 0..1 Natural language description of the research element
    definition
  fhir:ResearchElementDefinition.comment [ string ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:ResearchElementDefinition.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:ResearchElementDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for research
    element definition (if applicable)
  fhir:ResearchElementDefinition.purpose [ markdown ]; # 0..1 Why this research element definition is defined
  fhir:ResearchElementDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the ResearchElementDefinition
  fhir:ResearchElementDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ResearchElementDefinition.approvalDate [ date ]; # 0..1 When the research element definition was approved by
    publisher
  fhir:ResearchElementDefinition.lastReviewDate [ date ]; # 0..1 When the research element definition was last reviewed
  fhir:ResearchElementDefinition.effectivePeriod [ Period ]; # 0..1 When the research element definition is expected to
    be used
```

```
  fhir:ResearchElementDefinition.topic [ CodeableConcept ], ... ; # 0..* The category of the ResearchElementDefinition,
    such as Education, Treatment, Assessment, etc.
  fhir:ResearchElementDefinition.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:ResearchElementDefinition.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:ResearchElementDefinition.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:ResearchElementDefinition.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:ResearchElementDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations,
    etc.
  fhir:ResearchElementDefinition.library [ canonical(Library) ], ... ; # 0..* Logic used by the
    ResearchElementDefinition
  fhir:ResearchElementDefinition.type [ code ]; # 1..1 population | exposure | outcome
  fhir:ResearchElementDefinition.variableType [ code ]; # 0..1 dichotomous | continuous | descriptive
  fhir:ResearchElementDefinition.characteristic [ # 1..* What defines the members of the research element
    # ResearchElementDefinition.characteristic.definition[x] : 1..1 What code or expression defines members?. One of
    these 4
      fhir:ResearchElementDefinition.characteristic.definitionCodeableConcept [ CodeableConcept ]
      fhir:ResearchElementDefinition.characteristic.definitionCanonical [ canonical(ValueSet) ]
      fhir:ResearchElementDefinition.characteristic.definitionExpression [ Expression ]
      fhir:ResearchElementDefinition.characteristic.definitionDataRequirement [ DataRequirement ]
    fhir:ResearchElementDefinition.characteristic.usageContext [ UsageContext ], ... ; # 0..* What code/value pairs
    define members?
    fhir:ResearchElementDefinition.characteristic.exclude [ boolean ]; # 0..1 Whether the characteristic includes or
    excludes members
    fhir:ResearchElementDefinition.characteristic.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome
    described in?
    fhir:ResearchElementDefinition.characteristic.studyEffectiveDescription [ string ]; # 0..1 What time period does the
     study cover
    # ResearchElementDefinition.characteristic.studyEffective[x] : 0..1 What time period does the study cover. One of
    these 4
      fhir:ResearchElementDefinition.characteristic.studyEffectiveDateTime [ dateTime ]
      fhir:ResearchElementDefinition.characteristic.studyEffectivePeriod [ Period ]
      fhir:ResearchElementDefinition.characteristic.studyEffectiveDuration [ Duration ]
      fhir:ResearchElementDefinition.characteristic.studyEffectiveTiming [ Timing ]
    fhir:ResearchElementDefinition.characteristic.studyEffectiveTimeFromStart [ Duration ]; # 0..1 Observation time from
     study start
    fhir:ResearchElementDefinition.characteristic.studyEffectiveGroupMeasure [ code ]; # 0..1 mean | median | mean-of-
    mean | mean-of-median | median-of-mean | median-of-median
    fhir:ResearchElementDefinition.characteristic.participantEffectiveDescription [ string ]; # 0..1 What time period do
     participants cover
    # ResearchElementDefinition.characteristic.participantEffective[x] : 0..1 What time period do participants cover.
    One of these 4
      fhir:ResearchElementDefinition.characteristic.participantEffectiveDateTime [ dateTime ]
      fhir:ResearchElementDefinition.characteristic.participantEffectivePeriod [ Period ]
      fhir:ResearchElementDefinition.characteristic.participantEffectiveDuration [ Duration ]
      fhir:ResearchElementDefinition.characteristic.participantEffectiveTiming [ Timing ]
    fhir:ResearchElementDefinition.characteristic.participantEffectiveTimeFromStart [ Duration ]; # 0..1 Observation
    time from study start
    fhir:ResearchElementDefinition.characteristic.participantEffectiveGroupMeasure [ code ]; # 0..1 mean | median |
    mean-of-mean | mean-of-median | median-of-mean | median-of-median
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.17.3.1 Terminology Bindings

Path | Definition | Type | Reference

ResearchElementDefinition.status | The lifecycle status of an artifact. | Required | PublicationStatus

ResearchElementDefinition.subject[x] | The possible types of subjects for a measure (E.g. Patient, Practitioner, Organization, Location, etc.). | Extensible | SubjectType

ResearchElementDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet

ResearchElementDefinition.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic

ResearchElementDefinition.type | The possible types of research elements (E.g. Population, Exposure, Outcome). | Required | ResearchElementType

ResearchElementDefinition.variableType | The possible types of variables for exposures or outcomes (E.g. Dichotomous, Continuous, Descriptive). | Required | VariableType

ResearchElementDefinition.characteristic.unitOfMeasure | Unified Code for Units of Measure (UCUM). | Required | UCUMCodes

ResearchElementDefinition.characteristic.studyEffectiveGroupMeasure

ResearchElementDefinition.characteristic.participantEffectiveGroupMeasure | Possible group measure aggregates (E.g. Mean, Median). | Required | GroupMeasure

14.17.3.2 Constraints

id | Level | Location | Description | Expression

red-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.17.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='composed-of').resource (Any)
context | token | A use context assigned to the research element definition | (ResearchElementDefinition.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the research element definition | (ResearchElementDefinition.useContext.value as Quantity) | (ResearchElementDefinition.useContext.value as Range)
context-type | token | A type of use context assigned to the research element definition | ResearchElementDefinition.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the research element definition | On ResearchElementDefinition.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the research element definition | On ResearchElementDefinition.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The research element definition publication date | ResearchElementDefinition.date
depends-on | reference | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='depends-on').resource | ResearchElementDefinition.library (Library, Any)
derived-from | reference | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the research element definition | ResearchElementDefinition.description
effective | date | The time during which the research element definition is intended to be in use | ResearchElementDefinition.effectivePeriod
identifier | token | External identifier for the research element definition | ResearchElementDefinition.identifier
jurisdiction | token | Intended jurisdiction for the research element definition | ResearchElementDefinition.jurisdiction
name | string | Computationally friendly name of the research element definition | ResearchElementDefinition.name
predecessor | reference | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the research element definition | ResearchElementDefinition.publisher
status | token | The current status of the research element definition | ResearchElementDefinition.status
successor | reference | What resource is being referenced | ResearchElementDefinition.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the research element definition | ResearchElementDefinition.title
topic | token | Topics associated with the ResearchElementDefinition | ResearchElementDefinition.topic
url | uri | The uri that identifies the research element definition | ResearchElementDefinition.url
version | token | The business version of the research element definition | ResearchElementDefinition.version

14.17.5 Resource ResearchElementDefinition - Examples

URL: https://hl7.org/fhir/R4/researchelementdefinition-examples.html

TODO

14.17.5 Resource ResearchElementDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Example Name | id | Format
Example of researchelementdefinition | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.17.6 Resource ResearchElementDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/R4/researchelementdefinition-definitions.html

14.17.6 Resource ResearchElementDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ResearchElementDefinition resource.

ResearchElementDefinition
Element Id | ResearchElementDefinition
Definition | The ResearchElementDefinition resource describes a "PICO" element that knowledge (evidence, assertion, recommendation) is about.
Cardinality | 0..*
Type | DomainResource
Requirements | Need to be able to define and reuse the definition of individual elements of a research question.
Comments | PICO stands for Population (the population within which exposures are being compared), Intervention (the conditional state or exposure state being described for its effect on outcomes), Comparison (the alternative conditional state or alternative

exposure state being compared against), and Outcome (the result or effect of the intervention in the population).
Invariants | Defined on this element red-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Defined on this element
red-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ResearchElementDefinition.url
Element Id | ResearchElementDefinition.url
Definition | An absolute URI that is used to identify this research element definition when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this research element definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the research element definition is stored on different servers.
Cardinality | 0..1
Type | uri
Requirements | Allows the research element definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.
ResearchElementDefinition.identifier
Element Id | ResearchElementDefinition.identifier
Definition | A formal identifier that is used to identify this research element definition when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifier, not a resource identifier (see discussion)
Cardinality | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this research element definition outside of FHIR, where it is not possible to use the logical URI.
ResearchElementDefinition.version
Element Id | ResearchElementDefinition.version
Definition | The identifier that is used to identify this version of the research element definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the research element definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Cardinality | 0..1
Type | string
Summary | true
Comments | There may be different research element definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the research element definition with the format [url]|[version].
ResearchElementDefinition.name
Element Id | ResearchElementDefinition.name
Definition | A natural language name identifying the research element definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Cardinality | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.
Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
Affect this element
inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
ResearchElementDefinition.title
Element Id | ResearchElementDefinition.title
Definition | A short, descriptive, user-friendly title for the research element definition.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ResearchElementDefinition.shortTitle
Element Id | ResearchElementDefinition.shortTitle
Definition | The short title provides an alternate title for use in informal descriptive contexts where the full, formal title is not necessary.

Cardinality | 0..1
Type | string
Requirements | Need to be able to reference the content by a short description, but still provide a longer, more formal title for the content.
Summary | true
ResearchElementDefinition.subtitle
Element Id | ResearchElementDefinition.subtitle
Definition | An explanatory or alternate title for the ResearchElementDefinition giving additional information about its content.
Cardinality | 0..1
Type | string
ResearchElementDefinition.status
Element Id | ResearchElementDefinition.status
Definition | The status of this research element definition. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of research element definitions that are appropriate for use versus not.
ResearchElementDefinition.experimental
Element Id | ResearchElementDefinition.experimental
Definition | A Boolean value to indicate that this research element definition is authored for testing purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.
Cardinality | 0..1
Type | boolean
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level research element definition.
Summary | true
Comments | Allows filtering of research element definitions that are appropriate for use versus not.
ResearchElementDefinition.subject[x]
Element Id | ResearchElementDefinition.subject[x]
Definition | The intended subjects for the ResearchElementDefinition. If this element is not provided, a Patient subject is assumed, but the subject of the ResearchElementDefinition can be anything.
Cardinality | 0..1
Terminology Binding | Subject type (Extensible)
Type | CodeableConcept|Reference(Group)
[x] Note | See Choice of Data Types for further information about how to use [x]
Meaning if Missing | Patient
Comments | The subject of the ResearchElementDefinition is critical in interpreting the criteria definitions, as the logic in the ResearchElementDefinitions is evaluated with respect to a particular subject. This corresponds roughly to the notion of a Compartment in that it limits what content is available based on its relationship to the subject. In CQL, this corresponds to the context declaration.
ResearchElementDefinition.date
Element Id | ResearchElementDefinition.date
Definition | The date (and optionally time) when the research element definition was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the research element definition changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the research element definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
ResearchElementDefinition.publisher
Element Id | ResearchElementDefinition.publisher
Definition | The name of the organization or individual that published the research element definition.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the research element definition. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the research element definition is the organization or individual primarily responsible for the maintenance and upkeep of the research element definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the research element definition. This item SHOULD be populated unless the information is available from context.
ResearchElementDefinition.contact
Element Id | ResearchElementDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.

ResearchElementDefinition.description
Element Id | ResearchElementDefinition.description
Definition | A free text natural language description of the research element definition from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the research element definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the research element definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the research element definition is presumed to be the predominant language in the place the research element definition was created).
ResearchElementDefinition.comment
Element Id | ResearchElementDefinition.comment
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | string
ResearchElementDefinition.useContext
Element Id | ResearchElementDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to assist with indexing and searching for appropriate research element definition instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
ResearchElementDefinition.jurisdiction
Element Id | ResearchElementDefinition.jurisdiction
Definition | A legal or geographic region in which the research element definition is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the research element definition to be used in jurisdictions other than those for which it was originally designed or intended.
ResearchElementDefinition.purpose
Element Id | ResearchElementDefinition.purpose
Definition | Explanation of why this research element definition is needed and why it has been designed as it has.
Cardinality | 0..1
Type | markdown
Comments | This element does not describe the usage of the research element definition. Instead, it provides traceability of ''why'' the resource is either needed or ''why'' it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this research element definition.
ResearchElementDefinition.usage
Element Id | ResearchElementDefinition.usage
Definition | A detailed description, from a clinical perspective, of how the ResearchElementDefinition is used.
Cardinality | 0..1
Type | string
ResearchElementDefinition.copyright
Element Id | ResearchElementDefinition.copyright
Definition | A copyright statement relating to the research element definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the research element definition.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the research element definition and/or its content.
Alternate Names | License; Restrictions
ResearchElementDefinition.approvalDate
Element Id | ResearchElementDefinition.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
To Do | Consider how this applies to this resource.
ResearchElementDefinition.lastReviewDate
Element Id | ResearchElementDefinition.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.

Comments | If specified, this date follows the original approval date.
To Do | Consider how this applies to this resource.
ResearchElementDefinition.effectivePeriod
Element Id | ResearchElementDefinition.effectivePeriod
Definition | The period during which the research element definition content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the research element definition are or are expected to be used instead.
Summary | true
Comments | The effective period for a research element definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
To Do | Consider how this applies to this resource.
ResearchElementDefinition.topic
Element Id | ResearchElementDefinition.topic
Definition | Descriptive topics related to the content of the ResearchElementDefinition. Topics provide a high-level categorization grouping types of ResearchElementDefinitions that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the ResearchElementDefinition so that it can be found by topical searches.
To Do | Consider how this applies to this resource.
ResearchElementDefinition.author
Element Id | ResearchElementDefinition.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
ResearchElementDefinition.editor
Element Id | ResearchElementDefinition.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*
Type | ContactDetail
ResearchElementDefinition.reviewer
Element Id | ResearchElementDefinition.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
ResearchElementDefinition.endorser
Element Id | ResearchElementDefinition.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
ResearchElementDefinition.relatedArtifact
Element Id | ResearchElementDefinition.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | ResearchElementDefinitions must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
To Do | Consider how this applies to this resource.
ResearchElementDefinition.library
Element Id | ResearchElementDefinition.library
Definition | A reference to a Library resource containing the formal logic used by the ResearchElementDefinition.
Cardinality | 0..*
Type | canonical(Library)
To Do | Consider how this applies to this resource.
ResearchElementDefinition.type
Element Id | ResearchElementDefinition.type
Definition | The type of research element, a population, an exposure, or an outcome.
Cardinality | 1..1
Terminology Binding | ResearchElementType (Required)
Type | code
Summary | true
ResearchElementDefinition.variableType
Element Id | ResearchElementDefinition.variableType
Definition | The type of the outcome (e.g. Dichotomous, Continuous, or Descriptive).
Cardinality | 0..1
Terminology Binding | VariableType (Required)
Type | code

ResearchElementDefinition.characteristic
Element Id | ResearchElementDefinition.characteristic
Definition | A characteristic that defines the members of the research element. Multiple characteristics are applied with "and" semantics.
Cardinality | 1..*
Summary | true
Comments | Characteristics can be defined flexibly to accommodate different use cases for membership criteria, ranging from simple codes, all the way to using an expression language to express the criteria.
ResearchElementDefinition.characteristic.definition[x]
Element Id | ResearchElementDefinition.characteristic.definition[x]
Definition | Define members of the research element using Codes (such as condition, medication, or observation), Expressions (using an expression language such as FHIRPath or CQL) or DataRequirements (such as Diabetes diagnosis onset in the last year).
Cardinality | 1..1
Type | CodeableConcept|canonical(ValueSet)|Expression|DataRequirement
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | Need to be able to define members in simple codes when the membership aligns well with terminology, with common criteria such as observations in a value set or lab tests within a year, or with expression language to support criteria that do not fit in the above.
Summary | true
ResearchElementDefinition.characteristic.usageContext
Element Id | ResearchElementDefinition.characteristic.usageContext
Definition | Use UsageContext to define the members of the population, such as Age Ranges, Genders, Settings.
Cardinality | 0..*
Type | UsageContext
Requirements | Need to be able to define members more structurally when more information can be communicated such as age range.
To Do | Define semantics when multiple definition types are present, or disallow mixtures.
ResearchElementDefinition.characteristic.exclude
Element Id | ResearchElementDefinition.characteristic.exclude
Definition | When true, members with this characteristic are excluded from the element.
Cardinality | 0..1
Type | boolean
Meaning if Missing | False
ResearchElementDefinition.characteristic.unitOfMeasure
Element Id | ResearchElementDefinition.characteristic.unitOfMeasure
Definition | Specifies the UCUM unit for the outcome.
Cardinality | 0..1
Terminology Binding | UCUM Codes (Required)
Type | CodeableConcept
ResearchElementDefinition.characteristic.studyEffectiveDescription
Element Id | ResearchElementDefinition.characteristic.studyEffectiveDescription
Definition | A narrative description of the time period the study covers.
Cardinality | 0..1
Type | string
ResearchElementDefinition.characteristic.studyEffective[x]
Element Id | ResearchElementDefinition.characteristic.studyEffective[x]
Definition | Indicates what effective period the study covers.
Cardinality | 0..1
Type | dateTime|Period|Duration|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
ResearchElementDefinition.characteristic.studyEffectiveTimeFromStart
Element Id | ResearchElementDefinition.characteristic.studyEffectiveTimeFromStart
Definition | Indicates duration from the study initiation.
Cardinality | 0..1
Type | Duration
ResearchElementDefinition.characteristic.studyEffectiveGroupMeasure
Element Id | ResearchElementDefinition.characteristic.studyEffectiveGroupMeasure
Definition | Indicates how elements are aggregated within the study effective period.
Cardinality | 0..1
Terminology Binding | GroupMeasure (Required)
Type | code
ResearchElementDefinition.characteristic.participantEffectiveDescription
Element Id | ResearchElementDefinition.characteristic.participantEffectiveDescription
Definition | A narrative description of the time period the study covers.
Cardinality | 0..1
Type | string
ResearchElementDefinition.characteristic.participantEffective[x]
Element Id | ResearchElementDefinition.characteristic.participantEffective[x]
Definition | Indicates what effective period the study covers.
Cardinality | 0..1
Type | dateTime|Period|Duration|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
ResearchElementDefinition.characteristic.participantEffectiveTimeFromStart

Element Id | ResearchElementDefinition.characteristic.participantEffectiveTimeFromStart
Definition | Indicates duration from the participant's study entry.
Cardinality | 0..1
Type | Duration
ResearchElementDefinition.characteristic.participantEffectiveGroupMeasure
Element Id | ResearchElementDefinition.characteristic.participantEffectiveGroupMeasure
Definition | Indicates how elements are aggregated within the study effective period.
Cardinality | 0..1
Terminology Binding | GroupMeasure (Required)
Type | code

14.17.7 Resource ResearchElementDefinition - Mappings

URL: https://hl7.org/fhir/R4/researchelementdefinition-mappings.html

14.17.7 Resource ResearchElementDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the researchelementdefinition resource (see Mappings to Other Standards for further information & status).

14.17.7.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

ResearchElementDefinition
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
experimental | FiveWs.class
date | FiveWs.recorded
publisher | FiveWs.witness
purpose | FiveWs.why[x]

14.17.7.2 Workflow Pattern (http://hl7.org/fhir/workflow)

ResearchElementDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
experimental | Definition.experimental
subject[x] | Definition.subject
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.17.7.3 RIM Mapping (http://hl7.org/v3)

ResearchElementDefinition
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}
endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.17.8 Resource ResearchElementDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/R4/researchelementdefinition-profiles.html

14.17.8 Resource ResearchElementDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the ResearchElementDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.17.8.1 Profiles

No Profiles defined for this resource

14.17.8.2 Extensions

replaces | ResearchElementDefinition | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.17.8.3 Search Extensions

No Search Extensions defined for this resource

14.17.9 JSON Schema for ResearchElementDefinition

URL: https://hl7.org/fhir/R4/researchelementdefinition.schema.json.html

14.17.9 JSON Schema for ResearchElementDefinition

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for ResearchElementDefinition, which is just a part of the full JSON Schema.

```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/ResearchElementDefinition",
  "$ref": "#/definitions/ResearchElementDefinition",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "ResearchElementDefinition": {
      "description": "The ResearchElementDefinition resource describes a \"PICO\" element that knowledge (evidence,
    assertion, recommendation) is about.",
      "properties": {
        "resourceType": {
          "description": "This is a ResearchElementDefinition resource",
          "const": "ResearchElementDefinition"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
          "$ref": "#/definitions/uri"
        },
        "_implicitRules": {
          "description": "Extensions for implicitRules",
          "$ref": "Element.schema.json#/definitions/Element"
        },
        "language": {
          "description": "The base language in which the resource is written.",
          "$ref": "#/definitions/code"
```

```
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to
represent the content of the resource to a human. The narrative need not encode all the structured data, but is
required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative.
Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains
them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
 definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the resource and that modifies the understanding of the element that contains it and/or the understanding of the
containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use
of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be
met as part of the definition of the extension. Applications processing a resource are required to check for
modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or
DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this research element definition when it is
referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be
globally unique and SHOULD be a literal address at which at which an authoritative instance of this research element
 definition is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same
 when the research element definition is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this research element definition when it is
represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the research element definition when
it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the research
element definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g.
yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a
lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the
format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision
 Support Service specification. Note that a version is required for non-experimental active artifacts.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the research element definition. This name should be
usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the research element definition.",
```

```
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "shortTitle": {
      "description": "The short title provides an alternate title for use in informal descriptive contexts where the
 full, formal title is not necessary.",
      "$ref": "#/definitions/string"
    },
    "_shortTitle": {
      "description": "Extensions for shortTitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subtitle": {
      "description": "An explanatory or alternate title for the ResearchElementDefinition giving additional
information about its content.",
      "$ref": "#/definitions/string"
    },
    "_subtitle": {
      "description": "Extensions for subtitle",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this research element definition. Enables tracking the life-cycle of the
content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "experimental": {
      "description": "A Boolean value to indicate that this research element definition is authored for testing
purposes (or education/evaluation/marketing) and is not intended to be used for genuine usage.",
      "$ref": "#/definitions/boolean"
    },
    "_experimental": {
      "description": "Extensions for experimental",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "subjectCodeableConcept": {
      "description": "The intended subjects for the ResearchElementDefinition. If this element is not provided, a
Patient subject is assumed, but the subject of the ResearchElementDefinition can be anything.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "subjectReference": {
      "description": "The intended subjects for the ResearchElementDefinition. If this element is not provided, a
Patient subject is assumed, but the subject of the ResearchElementDefinition can be anything.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "date": {
      "description": "The date  (and optionally time) when the research element definition was published. The date
must change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the research element definition changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the research element definition.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the research element definition from a
consumer\u0027s perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
```

```
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "comment": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "#/definitions/string"
      },
      "type": "array"
    },
    "_comment": {
      "description": "Extensions for comment",
      "items": {
        "$ref": "Element.schema.json#/definitions/Element"
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate research element
definition instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the research element definition is intended to be
used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "purpose": {
      "description": "Explanation of why this research element definition is needed and why it has been designed as
it has.",
      "$ref": "#/definitions/markdown"
    },
    "_purpose": {
      "description": "Extensions for purpose",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "usage": {
      "description": "A detailed description, from a clinical perspective, of how the ResearchElementDefinition is
used.",
      "$ref": "#/definitions/string"
    },
    "_usage": {
      "description": "Extensions for usage",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "copyright": {
      "description": "A copyright statement relating to the research element definition and/or its contents.
Copyright statements are generally legal restrictions on the use and publishing of the research element
definition.",
      "$ref": "#/definitions/markdown"
    },
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the research element definition content was or is planned to be in
active use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the ResearchElementDefinition. Topics provide a
high-level categorization grouping types of ResearchElementDefinitions that can be useful for filtering and
searching.",
```

```
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "library": {
      "description": "A reference to a Library resource containing the formal logic used by the
ResearchElementDefinition.",
      "items": {
        "$ref": "canonical.schema.json#/definitions/canonical"
      },
      "type": "array"
    },
    "type": {
      "description": "The type of research element, a population, an exposure, or an outcome.",
      "enum": [
        "population",
        "exposure",
        "outcome"
      ]
    },
    "_type": {
      "description": "Extensions for type",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "variableType": {
      "description": "The type of the outcome (e.g. Dichotomous, Continuous, or Descriptive).",
      "enum": [
        "dichotomous",
        "continuous",
        "descriptive"
      ]
    },
    "_variableType": {
      "description": "Extensions for variableType",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "characteristic": {
      "description": "A characteristic that defines the members of the research element. Multiple characteristics
are applied with \"and\" semantics.",
      "items": {
        "$ref": "#/definitions/ResearchElementDefinition_Characteristic"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "characteristic",
    "resourceType"
```

```
  ]
},
"ResearchElementDefinition_Characteristic": {
  "description": "The ResearchElementDefinition resource describes a \"PICO\" element that knowledge (evidence,
assertion, recommendation) is about.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "definitionCodeableConcept": {
      "description": "Define members of the research element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "definitionCanonical": {
      "description": "Define members of the research element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "pattern": "^\\S*$",
      "type": "string"
    },
    "_definitionCanonical": {
      "description": "Extensions for definitionCanonical",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "definitionExpression": {
      "description": "Define members of the research element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "Expression.schema.json#/definitions/Expression"
    },
    "definitionDataRequirement": {
      "description": "Define members of the research element using Codes (such as condition, medication, or
observation), Expressions ( using an expression language such as FHIRPath or CQL) or DataRequirements (such as
Diabetes diagnosis onset in the last year).",
      "$ref": "DataRequirement.schema.json#/definitions/DataRequirement"
    },
    "usageContext": {
      "description": "Use UsageContext to define the members of the population, such as Age Ranges, Genders,
Settings.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "exclude": {
      "description": "When true, members with this characteristic are excluded from the element.",
      "$ref": "#/definitions/boolean"
    },
    "_exclude": {
      "description": "Extensions for exclude",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "unitOfMeasure": {
      "description": "Specifies the UCUM unit for the outcome.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "studyEffectiveDescription": {
      "description": "A narrative description of the time period the study covers.",
      "$ref": "#/definitions/string"
    },
    "_studyEffectiveDescription": {
```

```
      "description": "Extensions for studyEffectiveDescription",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "studyEffectiveDateTime": {
      "description": "Indicates what effective period the study covers.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_studyEffectiveDateTime": {
      "description": "Extensions for studyEffectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "studyEffectivePeriod": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "studyEffectiveDuration": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "studyEffectiveTiming": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "studyEffectiveTimeFromStart": {
      "description": "Indicates duration from the study initiation.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "studyEffectiveGroupMeasure": {
      "description": "Indicates how elements are aggregated within the study effective period.",
      "enum": [
        "mean",
        "median",
        "mean-of-mean",
        "mean-of-median",
        "median-of-mean",
        "median-of-median"
      ]
    },
    "_studyEffectiveGroupMeasure": {
      "description": "Extensions for studyEffectiveGroupMeasure",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participantEffectiveDescription": {
      "description": "A narrative description of the time period the study covers.",
      "$ref": "#/definitions/string"
    },
    "_participantEffectiveDescription": {
      "description": "Extensions for participantEffectiveDescription",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participantEffectiveDateTime": {
      "description": "Indicates what effective period the study covers.",
      "pattern": "^([0-9]([0-9]([0-9][1-9]|[1-9]0)|[1-9]00)|[1-9]000)(-(0[1-9]|1[0-2])(-(0[1-9]|[1-2][0-9]|3[0-
1])(T([01][0-9]|2[0-3]):[0-5][0-9]:([0-5][0-9]|60)(\\.[0-9]+)?(Z|(\\+|-)((0[0-9]|1[0-3]):[0-5][0-9]|14:00)))?)?)?$",
      "type": "string"
    },
    "_participantEffectiveDateTime": {
      "description": "Extensions for participantEffectiveDateTime",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "participantEffectivePeriod": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "participantEffectiveDuration": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "participantEffectiveTiming": {
      "description": "Indicates what effective period the study covers.",
      "$ref": "Timing.schema.json#/definitions/Timing"
    },
    "participantEffectiveTimeFromStart": {
      "description": "Indicates duration from the participant\u0027s study entry.",
      "$ref": "Duration.schema.json#/definitions/Duration"
    },
    "participantEffectiveGroupMeasure": {
      "description": "Indicates how elements are aggregated within the study effective period.",
      "enum": [
        "mean",
        "median",
        "mean-of-mean",
        "mean-of-median",
        "median-of-mean",
        "median-of-median"
      ]
    },
```

```
        "_participantEffectiveGroupMeasure": {
          "description": "Extensions for participantEffectiveGroupMeasure",
          "$ref": "Element.schema.json#/definitions/Element"
        }
      },
      "additionalProperties": false
    }
  }
}
```

14.17.10 Resource ResearchElementDefinition - Maturity Dependency View

URL: https://hl7.org/fhir/R4/researchelementdefinition-dependencies.html

14.17.10 Resource ResearchElementDefinition - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for ResearchElementDefinition FMM level 0

Name | Card. | Type | Dependency Analysis
ResearchElementDefinition | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
shortTitle | 0..1 | string
subtitle | 0..1 | string
status | 1..1 | code
experimental | 0..1 | boolean
subject[x] | 0..1
subjectCodeableConcept | | CodeableConcept
subjectReference | | Reference(Group
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
comment | 0..* | string
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
purpose | 0..1 | markdown
usage | 0..1 | string
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
library | 0..* | canonical(Library)
type | 1..1 | code
variableType | 0..1 | code
characteristic | 1..* | BackboneElement
definition[x] | 1..1
definitionCodeableConcept | | CodeableConcept
definitionCanonical | | canonical
definitionExpression | | Expression
definitionDataRequirement | | DataRequirement
usageContext | 0..* | UsageContext
exclude | 0..1 | boolean
unitOfMeasure | 0..1 | CodeableConcept
studyEffectiveDescription | 0..1 | string
studyEffective[x] | 0..1
studyEffectiveDateTime | | dateTime
studyEffectivePeriod | | Period
studyEffectiveDuration | | Duration
studyEffectiveTiming | | Timing
studyEffectiveTimeFromStart | 0..1 | Duration
studyEffectiveGroupMeasure | 0..1 | code
participantEffectiveDescription | 0..1 | string

participantEffective[x] | 0..1
participantEffectiveDateTime | | dateTime
participantEffectivePeriod | | Period
participantEffectiveDuration | | Duration
participantEffectiveTiming | | Timing
participantEffectiveTimeFromStart | 0..1 | Duration
participantEffectiveGroupMeasure | 0..1 | code

14.18 Resource RiskEvidenceSynthesis - Content

URL: https://hl7.org/fhir/R4/riskevidencesynthesis.html
Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.

14.18.1 Scope and Usage

The scope of the RiskEvidenceSynthesis resource is to describe the likelihood of an outcome in a population with an exposure where the risk estimate is derived from the combination of research studies. Risk estimates are not effects and do not represent a difference between exposure states. Expressing risk estimates is done throughout reporting of biomedical research, systematic reviews, and clinical reference across all disciplines. Risk estimates as a "synthesis" of research studies is a different concept for risk estimation than risk estimates for an individual person.

This resource is referenced by EffectEvidenceSynthesis

14.18.2 Resource Content

- Structure
- UML
- XML
- JSON
- Turtle
- R3 Diff
- All

Structure

Name | Flags | Card. | Type | Description & Constraints
RiskEvidenceSynthesis | ITU | | DomainResource | A quantified estimate of risk based on a body of evidence + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | Σ | 0..1 | uri | Canonical identifier for this risk evidence synthesis, represented as a URI (globally unique)
identifier | Σ | 0..* | Identifier | Additional identifier for the risk evidence synthesis
version | Σ | 0..1 | string | Business version of the risk evidence synthesis
name | ΣI | 0..1 | string | Name for this risk evidence synthesis (computer friendly)
title | Σ | 0..1 | string | Name for this risk evidence synthesis (human friendly)
status | ?!Σ | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
date | Σ | 0..1 | dateTime | Date last changed
publisher | Σ | 0..1 | string | Name of the publisher (organization or individual)
contact | Σ | 0..* | ContactDetail | Contact details for the publisher
description | Σ | 0..1 | markdown | Natural language description of the risk evidence synthesis
note | | 0..* | Annotation | Used for footnotes or explanatory notes
useContext | Σ | 0..* | UsageContext | The context that the content is intended to support
jurisdiction | Σ | 0..* | CodeableConcept | Intended jurisdiction for risk evidence synthesis (if applicable) Jurisdiction (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the risk evidence synthesis was approved by publisher
lastReviewDate | | 0..1 | date | When the risk evidence synthesis was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the risk evidence synthesis is expected to be used
topic | | 0..* | CodeableConcept | The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
synthesisType | | 0..1 | CodeableConcept | Type of synthesis SynthesisType (Extensible)
studyType | | 0..1 | CodeableConcept | Type of study StudyType (Extensible)
population | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposure | Σ | 0..1 | Reference(EvidenceVariable) | What exposure?
outcome | Σ | 1..1 | Reference(EvidenceVariable) | What outcome?

sampleSize | | 0..1 | BackboneElement | What sample size was involved?
description | | 0..1 | string | Description of sample size
numberOfStudies | | 0..1 | integer | How many studies?
numberOfParticipants | | 0..1 | integer | How many participants?
riskEstimate | Σ | 0..1 | BackboneElement | What was the estimated risk
description | | 0..1 | string | Description of risk estimate
type | | 0..1 | CodeableConcept | Type of risk estimate RiskEstimateType (Extensible)
value | | 0..1 | decimal | Point estimate
unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)
denominatorCount | | 0..1 | integer | Sample size for group measured
numeratorCount | | 0..1 | integer | Number with the outcome
precisionEstimate | | 0..* | BackboneElement | How precise the estimate is
type | | 0..1 | CodeableConcept | Type of precision estimate PrecisionEstimateType (Extensible)
level | | 0..1 | decimal | Level of confidence interval
from | | 0..1 | decimal | Lower bound
to | | 0..1 | decimal | Upper bound
certainty | | 0..* | BackboneElement | How certain is the risk
rating | | 0..* | CodeableConcept | Certainty rating QualityOfEvidenceRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
certaintySubcomponent | | 0..* | BackboneElement | A component that contributes to the overall certainty
type | | 0..1 | CodeableConcept | Type of subcomponent of certainty rating CertaintySubcomponentType (Extensible)
rating | | 0..* | CodeableConcept | Subcomponent certainty rating CertaintySubcomponentRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskEvidenceSynthesis xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this risk evidence synthesis, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the risk evidence synthesis --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the risk evidence synthesis -->
 <name value="[string]"/><!--  0..1 Name for this risk evidence synthesis (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this risk evidence synthesis (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the risk evidence synthesis -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for risk evidence synthesis (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the risk evidence synthesis was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the risk evidence synthesis was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the risk evidence synthesis is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment,
     etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <synthesisType><!-- 0..1 CodeableConcept Type of synthesis --></synthesisType>
 <studyType><!-- 0..1 CodeableConcept Type of study --></studyType>
 <population><!-- 1..1 Reference(EvidenceVariable) What population? --></population>
 <exposure><!-- 0..1 Reference(EvidenceVariable) What exposure? --></exposure>
 <outcome><!-- 1..1 Reference(EvidenceVariable) What outcome? --></outcome>
 <sampleSize>  <!-- 0..1 What sample size was involved? -->
  <description value="[string]"/><!-- 0..1 Description of sample size -->
  <numberOfStudies value="[integer]"/><!-- 0..1 How many studies? -->
  <numberOfParticipants value="[integer]"/><!-- 0..1 How many participants? -->
 </sampleSize>
 <riskEstimate>  <!-- 0..1 What was the estimated risk -->
  <description value="[string]"/><!-- 0..1 Description of risk estimate -->
  <type><!-- 0..1 CodeableConcept Type of risk estimate --></type>
  <value value="[decimal]"/><!-- 0..1 Point estimate -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <denominatorCount value="[integer]"/><!-- 0..1 Sample size for group measured -->
  <numeratorCount value="[integer]"/><!-- 0..1 Number with the outcome -->
  <precisionEstimate>  <!-- 0..* How precise the estimate is -->
   <type><!-- 0..1 CodeableConcept Type of precision estimate --></type>
   <level value="[decimal]"/><!-- 0..1 Level of confidence interval -->
   <from value="[decimal]"/><!-- 0..1 Lower bound -->
   <to value="[decimal]"/><!-- 0..1 Upper bound -->
  </precisionEstimate>
 </riskEstimate>
 <certainty>  <!-- 0..* How certain is the risk -->
  <rating><!-- 0..* CodeableConcept Certainty rating --></rating>
  <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
```

```
  <certaintySubcomponent>  <!-- 0..* A component that contributes to the overall certainty -->
   <type><!-- 0..1 CodeableConcept Type of subcomponent of certainty rating --></type>
   <rating><!-- 0..* CodeableConcept Subcomponent certainty rating --></rating>
   <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  </certaintySubcomponent>
 </certainty>
</RiskEvidenceSynthesis>
```

JSON Template

```
{
  "resourceType" : "RiskEvidenceSynthesis",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this risk evidence synthesis, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the risk evidence synthesis
  "version" : "<string>", // Business version of the risk evidence synthesis
  "name" : "<string>", // C? Name for this risk evidence synthesis (computer friendly)
  "title" : "<string>", // Name for this risk evidence synthesis (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the risk evidence synthesis
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for risk evidence synthesis (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the risk evidence synthesis was approved by publisher
  "lastReviewDate" : "<date>", // When the risk evidence synthesis was last reviewed
  "effectivePeriod" : { Period }, // When the risk evidence synthesis is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EffectEvidenceSynthesis, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "synthesisType" : { CodeableConcept }, // Type of synthesis
  "studyType" : { CodeableConcept }, // Type of study
  "population" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposure" : { Reference(EvidenceVariable) }, // What exposure?
  "outcome" : { Reference(EvidenceVariable) }, // R!  What outcome?
  "sampleSize" : { // What sample size was involved?
    "description" : "<string>", // Description of sample size
    "numberOfStudies" : <integer>, // How many studies?
    "numberOfParticipants" : <integer> // How many participants?
  },
  "riskEstimate" : { // What was the estimated risk
    "description" : "<string>", // Description of risk estimate
    "type" : { CodeableConcept }, // Type of risk estimate
    "value" : <decimal>, // Point estimate
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "denominatorCount" : <integer>, // Sample size for group measured
    "numeratorCount" : <integer>, // Number with the outcome
    "precisionEstimate" : [{ // How precise the estimate is
      "type" : { CodeableConcept }, // Type of precision estimate
      "level" : <decimal>, // Level of confidence interval
      "from" : <decimal>, // Lower bound
      "to" : <decimal> // Upper bound
    }]
  },
  "certainty" : [{ // How certain is the risk
    "rating" : [{ CodeableConcept }], // Certainty rating
    "note" : [{ Annotation }], // Used for footnotes or explanatory notes
    "certaintySubcomponent" : [{ // A component that contributes to the overall certainty
      "type" : { CodeableConcept }, // Type of subcomponent of certainty rating
      "rating" : [{ CodeableConcept }], // Subcomponent certainty rating
      "note" : [{ Annotation }] // Used for footnotes or explanatory notes
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskEvidenceSynthesis;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskEvidenceSynthesis.url [ uri ]; # 0..1 Canonical identifier for this risk evidence synthesis, represented as a
     URI (globally unique)
  fhir:RiskEvidenceSynthesis.identifier [ Identifier ], ... ; # 0..* Additional identifier for the risk evidence
    synthesis
  fhir:RiskEvidenceSynthesis.version [ string ]; # 0..1 Business version of the risk evidence synthesis
  fhir:RiskEvidenceSynthesis.name [ string ]; # 0..1 Name for this risk evidence synthesis (computer friendly)
  fhir:RiskEvidenceSynthesis.title [ string ]; # 0..1 Name for this risk evidence synthesis (human friendly)
  fhir:RiskEvidenceSynthesis.status [ code ]; # 1..1 draft | active | retired | unknown
```

```
  fhir:RiskEvidenceSynthesis.date [ dateTime ]; # 0..1 Date last changed
  fhir:RiskEvidenceSynthesis.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:RiskEvidenceSynthesis.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:RiskEvidenceSynthesis.description [ markdown ]; # 0..1 Natural language description of the risk evidence
    synthesis
  fhir:RiskEvidenceSynthesis.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:RiskEvidenceSynthesis.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:RiskEvidenceSynthesis.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for risk evidence
    synthesis (if applicable)
  fhir:RiskEvidenceSynthesis.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:RiskEvidenceSynthesis.approvalDate [ date ]; # 0..1 When the risk evidence synthesis was approved by publisher
  fhir:RiskEvidenceSynthesis.lastReviewDate [ date ]; # 0..1 When the risk evidence synthesis was last reviewed
  fhir:RiskEvidenceSynthesis.effectivePeriod [ Period ]; # 0..1 When the risk evidence synthesis is expected to be used
  fhir:RiskEvidenceSynthesis.topic [ CodeableConcept ], ... ; # 0..* The category of the EffectEvidenceSynthesis, such
    as Education, Treatment, Assessment, etc.
  fhir:RiskEvidenceSynthesis.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:RiskEvidenceSynthesis.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:RiskEvidenceSynthesis.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:RiskEvidenceSynthesis.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:RiskEvidenceSynthesis.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:RiskEvidenceSynthesis.synthesisType [ CodeableConcept ]; # 0..1 Type of synthesis
  fhir:RiskEvidenceSynthesis.studyType [ CodeableConcept ]; # 0..1 Type of study
  fhir:RiskEvidenceSynthesis.population [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:RiskEvidenceSynthesis.exposure [ Reference(EvidenceVariable) ]; # 0..1 What exposure?
  fhir:RiskEvidenceSynthesis.outcome [ Reference(EvidenceVariable) ]; # 1..1 What outcome?
  fhir:RiskEvidenceSynthesis.sampleSize [ # 0..1 What sample size was involved?
    fhir:RiskEvidenceSynthesis.sampleSize.description [ string ]; # 0..1 Description of sample size
    fhir:RiskEvidenceSynthesis.sampleSize.numberOfStudies [ integer ]; # 0..1 How many studies?
    fhir:RiskEvidenceSynthesis.sampleSize.numberOfParticipants [ integer ]; # 0..1 How many participants?
  ];
  fhir:RiskEvidenceSynthesis.riskEstimate [ # 0..1 What was the estimated risk
    fhir:RiskEvidenceSynthesis.riskEstimate.description [ string ]; # 0..1 Description of risk estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.type [ CodeableConcept ]; # 0..1 Type of risk estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.value [ decimal ]; # 0..1 Point estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome described
     in?
    fhir:RiskEvidenceSynthesis.riskEstimate.denominatorCount [ integer ]; # 0..1 Sample size for group measured
    fhir:RiskEvidenceSynthesis.riskEstimate.numeratorCount [ integer ]; # 0..1 Number with the outcome
    fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate [ # 0..* How precise the estimate is
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.type [ CodeableConcept ]; # 0..1 Type of precision
    estimate
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.level [ decimal ]; # 0..1 Level of confidence interval
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.from [ decimal ]; # 0..1 Lower bound
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.to [ decimal ]; # 0..1 Upper bound
    ], ...;
  ];
  fhir:RiskEvidenceSynthesis.certainty [ # 0..* How certain is the risk
    fhir:RiskEvidenceSynthesis.certainty.rating [ CodeableConcept ], ... ; # 0..* Certainty rating
    fhir:RiskEvidenceSynthesis.certainty.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
    fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent [ # 0..* A component that contributes to the overall
    certainty
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.type [ CodeableConcept ]; # 0..1 Type of subcomponent
    of certainty rating
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.rating [ CodeableConcept ], ... ; # 0..* Subcomponent
    certainty rating
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.note [ Annotation ], ... ; # 0..* Used for footnotes or
     explanatory notes
    ], ...;
  ], ...;
]
```

Changes since R3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
RiskEvidenceSynthesis	ITU		DomainResource	A quantified estimate of risk based on a body of evidence + Warning: Name should be usable as an identifier for the module by machine processing applications such as code generation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url	Σ	0..1	uri	Canonical identifier for this risk evidence synthesis, represented as a URI (globally unique)
identifier	Σ	0..*	Identifier	Additional identifier for the risk evidence synthesis
version	Σ	0..1	string	Business version of the risk evidence synthesis
name	ΣI	0..1	string	Name for this risk evidence synthesis (computer friendly)
title	Σ	0..1	string	Name for this risk evidence synthesis (human friendly)
status	?!Σ	1..1	code	draft | active | retired | unknown PublicationStatus (Required)
date	Σ	0..1	dateTime	Date last changed
publisher	Σ	0..1	string	Name of the publisher (organization or individual)
contact	Σ	0..*	ContactDetail	Contact details for the publisher
description	Σ	0..1	markdown	Natural language description of the risk evidence synthesis
note		0..*	Annotation	Used for footnotes or explanatory notes
useContext	Σ	0..*	UsageContext	The context that the content is intended to support
jurisdiction	Σ	0..*	CodeableConcept	Intended jurisdiction for risk evidence synthesis (if applicable) Jurisdiction (Extensible)

copyright | | 0..1 | markdown | Use and/or publishing restrictions
approvalDate | | 0..1 | date | When the risk evidence synthesis was approved by publisher
lastReviewDate | | 0..1 | date | When the risk evidence synthesis was last reviewed
effectivePeriod | Σ | 0..1 | Period | When the risk evidence synthesis is expected to be used
topic | | 0..* | CodeableConcept | The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment, etc. DefinitionTopic (Example)
author | | 0..* | ContactDetail | Who authored the content
editor | | 0..* | ContactDetail | Who edited the content
reviewer | | 0..* | ContactDetail | Who reviewed the content
endorser | | 0..* | ContactDetail | Who endorsed the content
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
synthesisType | | 0..1 | CodeableConcept | Type of synthesis SynthesisType (Extensible)
studyType | | 0..1 | CodeableConcept | Type of study StudyType (Extensible)
population | Σ | 1..1 | Reference(EvidenceVariable) | What population?
exposure | Σ | 0..1 | Reference(EvidenceVariable) | What exposure?
outcome | Σ | 1..1 | Reference(EvidenceVariable) | What outcome?
sampleSize | | 0..1 | BackboneElement | What sample size was involved?
description | | 0..1 | string | Description of sample size
numberOfStudies | | 0..1 | integer | How many studies?
numberOfParticipants | | 0..1 | integer | How many participants?
riskEstimate | Σ | 0..1 | BackboneElement | What was the estimated risk
description | | 0..1 | string | Description of risk estimate
type | | 0..1 | CodeableConcept | Type of risk estimate RiskEstimateType (Extensible)
value | | 0..1 | decimal | Point estimate
unitOfMeasure | | 0..1 | CodeableConcept | What unit is the outcome described in? UCUM Codes (Required)
denominatorCount | | 0..1 | integer | Sample size for group measured
numeratorCount | | 0..1 | integer | Number with the outcome
precisionEstimate | | 0..* | BackboneElement | How precise the estimate is
type | | 0..1 | CodeableConcept | Type of precision estimate PrecisionEstimateType (Extensible)
level | | 0..1 | decimal | Level of confidence interval
from | | 0..1 | decimal | Lower bound
to | | 0..1 | decimal | Upper bound
certainty | | 0..* | BackboneElement | How certain is the risk
rating | | 0..* | CodeableConcept | Certainty rating QualityOfEvidenceRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
certaintySubcomponent | | 0..* | BackboneElement | A component that contributes to the overall certainty
type | | 0..1 | CodeableConcept | Type of subcomponent of certainty rating CertaintySubcomponentType (Extensible)
rating | | 0..* | CodeableConcept | Subcomponent certainty rating CertaintySubcomponentRating (Extensible)
note | | 0..* | Annotation | Used for footnotes or explanatory notes
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RiskEvidenceSynthesis xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Canonical identifier for this risk evidence synthesis, represented as a URI (globally
    unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the risk evidence synthesis --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the risk evidence synthesis -->
 <name value="[string]"/><!--  0..1 Name for this risk evidence synthesis (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this risk evidence synthesis (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <date value="[dateTime]"/><!-- 0..1 Date last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the risk evidence synthesis -->
 <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
 <useContext><!-- 0..* UsageContext The context that the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for risk evidence synthesis (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <approvalDate value="[date]"/><!-- 0..1 When the risk evidence synthesis was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the risk evidence synthesis was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the risk evidence synthesis is expected to be used --></effectivePeriod>
 <topic><!-- 0..* CodeableConcept The category of the EffectEvidenceSynthesis, such as Education, Treatment, Assessment,
     etc. --></topic>
 <author><!-- 0..* ContactDetail Who authored the content --></author>
 <editor><!-- 0..* ContactDetail Who edited the content --></editor>
 <reviewer><!-- 0..* ContactDetail Who reviewed the content --></reviewer>
 <endorser><!-- 0..* ContactDetail Who endorsed the content --></endorser>
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <synthesisType><!-- 0..1 CodeableConcept Type of synthesis --></synthesisType>
 <studyType><!-- 0..1 CodeableConcept Type of study --></studyType>
 <population><!-- 1..1 Reference(EvidenceVariable) What population? --></population>
 <exposure><!-- 0..1 Reference(EvidenceVariable) What exposure? --></exposure>
 <outcome><!-- 1..1 Reference(EvidenceVariable) What outcome? --></outcome>
 <sampleSize>  <!-- 0..1 What sample size was involved? -->
  <description value="[string]"/><!-- 0..1 Description of sample size -->
```

```
  <numberOfStudies value="[integer]"/><!-- 0..1 How many studies? -->
  <numberOfParticipants value="[integer]"/><!-- 0..1 How many participants? -->
 </sampleSize>
 <riskEstimate>  <!-- 0..1 What was the estimated risk -->
  <description value="[string]"/><!-- 0..1 Description of risk estimate -->
  <type><!-- 0..1 CodeableConcept Type of risk estimate --></type>
  <value value="[decimal]"/><!-- 0..1 Point estimate -->
  <unitOfMeasure><!-- 0..1 CodeableConcept What unit is the outcome described in? --></unitOfMeasure>
  <denominatorCount value="[integer]"/><!-- 0..1 Sample size for group measured -->
  <numeratorCount value="[integer]"/><!-- 0..1 Number with the outcome -->
  <precisionEstimate>  <!-- 0..* How precise the estimate is -->
   <type><!-- 0..1 CodeableConcept Type of precision estimate --></type>
   <level value="[decimal]"/><!-- 0..1 Level of confidence interval -->
   <from value="[decimal]"/><!-- 0..1 Lower bound -->
   <to value="[decimal]"/><!-- 0..1 Upper bound -->
  </precisionEstimate>
 </riskEstimate>
 <certainty>  <!-- 0..* How certain is the risk -->
  <rating><!-- 0..* CodeableConcept Certainty rating --></rating>
  <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  <certaintySubcomponent>  <!-- 0..* A component that contributes to the overall certainty -->
   <type><!-- 0..1 CodeableConcept Type of subcomponent of certainty rating --></type>
   <rating><!-- 0..* CodeableConcept Subcomponent certainty rating --></rating>
   <note><!-- 0..* Annotation Used for footnotes or explanatory notes --></note>
  </certaintySubcomponent>
 </certainty>
</RiskEvidenceSynthesis>
```

JSON Template

```
{
  "resourceType" : "RiskEvidenceSynthesis",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Canonical identifier for this risk evidence synthesis, represented as a URI (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the risk evidence synthesis
  "version" : "<string>", // Business version of the risk evidence synthesis
  "name" : "<string>", // C? Name for this risk evidence synthesis (computer friendly)
  "title" : "<string>", // Name for this risk evidence synthesis (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "date" : "<dateTime>", // Date last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the risk evidence synthesis
  "note" : [{ Annotation }], // Used for footnotes or explanatory notes
  "useContext" : [{ UsageContext }], // The context that the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for risk evidence synthesis (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "approvalDate" : "<date>", // When the risk evidence synthesis was approved by publisher
  "lastReviewDate" : "<date>", // When the risk evidence synthesis was last reviewed
  "effectivePeriod" : { Period }, // When the risk evidence synthesis is expected to be used
  "topic" : [{ CodeableConcept }], // The category of the EffectEvidenceSynthesis, such as Education, Treatment,
    Assessment, etc.
  "author" : [{ ContactDetail }], // Who authored the content
  "editor" : [{ ContactDetail }], // Who edited the content
  "reviewer" : [{ ContactDetail }], // Who reviewed the content
  "endorser" : [{ ContactDetail }], // Who endorsed the content
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "synthesisType" : { CodeableConcept }, // Type of synthesis
  "studyType" : { CodeableConcept }, // Type of study
  "population" : { Reference(EvidenceVariable) }, // R!  What population?
  "exposure" : { Reference(EvidenceVariable) }, // What exposure?
  "outcome" : { Reference(EvidenceVariable) }, // R!  What outcome?
  "sampleSize" : { // What sample size was involved?
    "description" : "<string>", // Description of sample size
    "numberOfStudies" : <integer>, // How many studies?
    "numberOfParticipants" : <integer> // How many participants?
  },
  "riskEstimate" : { // What was the estimated risk
    "description" : "<string>", // Description of risk estimate
    "type" : { CodeableConcept }, // Type of risk estimate
    "value" : <decimal>, // Point estimate
    "unitOfMeasure" : { CodeableConcept }, // What unit is the outcome described in?
    "denominatorCount" : <integer>, // Sample size for group measured
    "numeratorCount" : <integer>, // Number with the outcome
    "precisionEstimate" : [{ // How precise the estimate is
      "type" : { CodeableConcept }, // Type of precision estimate
      "level" : <decimal>, // Level of confidence interval
      "from" : <decimal>, // Lower bound
      "to" : <decimal> // Upper bound
    }]
  },
  "certainty" : [{ // How certain is the risk
    "rating" : [{ CodeableConcept }], // Certainty rating
    "note" : [{ Annotation }], // Used for footnotes or explanatory notes
    "certaintySubcomponent" : [{ // A component that contributes to the overall certainty
      "type" : { CodeableConcept }, // Type of subcomponent of certainty rating
      "rating" : [{ CodeableConcept }], // Subcomponent certainty rating
      "note" : [{ Annotation }] // Used for footnotes or explanatory notes
    }]
```

```
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RiskEvidenceSynthesis;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RiskEvidenceSynthesis.url [ uri ]; # 0..1 Canonical identifier for this risk evidence synthesis, represented as a
     URI (globally unique)
  fhir:RiskEvidenceSynthesis.identifier [ Identifier ], ... ; # 0..* Additional identifier for the risk evidence
    synthesis
  fhir:RiskEvidenceSynthesis.version [ string ]; # 0..1 Business version of the risk evidence synthesis
  fhir:RiskEvidenceSynthesis.name [ string ]; # 0..1 Name for this risk evidence synthesis (computer friendly)
  fhir:RiskEvidenceSynthesis.title [ string ]; # 0..1 Name for this risk evidence synthesis (human friendly)
  fhir:RiskEvidenceSynthesis.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:RiskEvidenceSynthesis.date [ dateTime ]; # 0..1 Date last changed
  fhir:RiskEvidenceSynthesis.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:RiskEvidenceSynthesis.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:RiskEvidenceSynthesis.description [ markdown ]; # 0..1 Natural language description of the risk evidence
    synthesis
  fhir:RiskEvidenceSynthesis.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
  fhir:RiskEvidenceSynthesis.useContext [ UsageContext ], ... ; # 0..* The context that the content is intended to
    support
  fhir:RiskEvidenceSynthesis.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for risk evidence
    synthesis (if applicable)
  fhir:RiskEvidenceSynthesis.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:RiskEvidenceSynthesis.approvalDate [ date ]; # 0..1 When the risk evidence synthesis was approved by publisher
  fhir:RiskEvidenceSynthesis.lastReviewDate [ date ]; # 0..1 When the risk evidence synthesis was last reviewed
  fhir:RiskEvidenceSynthesis.effectivePeriod [ Period ]; # 0..1 When the risk evidence synthesis is expected to be used
  fhir:RiskEvidenceSynthesis.topic [ CodeableConcept ], ... ; # 0..* The category of the EffectEvidenceSynthesis, such
    as Education, Treatment, Assessment, etc.
  fhir:RiskEvidenceSynthesis.author [ ContactDetail ], ... ; # 0..* Who authored the content
  fhir:RiskEvidenceSynthesis.editor [ ContactDetail ], ... ; # 0..* Who edited the content
  fhir:RiskEvidenceSynthesis.reviewer [ ContactDetail ], ... ; # 0..* Who reviewed the content
  fhir:RiskEvidenceSynthesis.endorser [ ContactDetail ], ... ; # 0..* Who endorsed the content
  fhir:RiskEvidenceSynthesis.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:RiskEvidenceSynthesis.synthesisType [ CodeableConcept ]; # 0..1 Type of synthesis
  fhir:RiskEvidenceSynthesis.studyType [ CodeableConcept ]; # 0..1 Type of study
  fhir:RiskEvidenceSynthesis.population [ Reference(EvidenceVariable) ]; # 1..1 What population?
  fhir:RiskEvidenceSynthesis.exposure [ Reference(EvidenceVariable) ]; # 0..1 What exposure?
  fhir:RiskEvidenceSynthesis.outcome [ Reference(EvidenceVariable) ]; # 1..1 What outcome?
  fhir:RiskEvidenceSynthesis.sampleSize [ # 0..1 What sample size was involved?
    fhir:RiskEvidenceSynthesis.sampleSize.description [ string ]; # 0..1 Description of sample size
    fhir:RiskEvidenceSynthesis.sampleSize.numberOfStudies [ integer ]; # 0..1 How many studies?
    fhir:RiskEvidenceSynthesis.sampleSize.numberOfParticipants [ integer ]; # 0..1 How many participants?
  ];
  fhir:RiskEvidenceSynthesis.riskEstimate [ # 0..1 What was the estimated risk
    fhir:RiskEvidenceSynthesis.riskEstimate.description [ string ]; # 0..1 Description of risk estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.type [ CodeableConcept ]; # 0..1 Type of risk estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.value [ decimal ]; # 0..1 Point estimate
    fhir:RiskEvidenceSynthesis.riskEstimate.unitOfMeasure [ CodeableConcept ]; # 0..1 What unit is the outcome described
     in?
    fhir:RiskEvidenceSynthesis.riskEstimate.denominatorCount [ integer ]; # 0..1 Sample size for group measured
    fhir:RiskEvidenceSynthesis.riskEstimate.numeratorCount [ integer ]; # 0..1 Number with the outcome
    fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate [ # 0..* How precise the estimate is
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.type [ CodeableConcept ]; # 0..1 Type of precision
    estimate
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.level [ decimal ]; # 0..1 Level of confidence interval
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.from [ decimal ]; # 0..1 Lower bound
      fhir:RiskEvidenceSynthesis.riskEstimate.precisionEstimate.to [ decimal ]; # 0..1 Upper bound
    ], ...;
  ];
  fhir:RiskEvidenceSynthesis.certainty [ # 0..* How certain is the risk
    fhir:RiskEvidenceSynthesis.certainty.rating [ CodeableConcept ], ... ; # 0..* Certainty rating
    fhir:RiskEvidenceSynthesis.certainty.note [ Annotation ], ... ; # 0..* Used for footnotes or explanatory notes
    fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent [ # 0..* A component that contributes to the overall
    certainty
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.type [ CodeableConcept ]; # 0..1 Type of subcomponent
    of certainty rating
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.rating [ CodeableConcept ], ... ; # 0..* Subcomponent
    certainty rating
      fhir:RiskEvidenceSynthesis.certainty.certaintySubcomponent.note [ Annotation ], ... ; # 0..* Used for footnotes or
     explanatory notes
    ], ...;
  ], ...;
]
```

Changes since Release 3

This resource did not exist in Release 2

This analysis is available as XML or JSON.

See the Profiles & Extensions and the alternate definitions: Master Definition XML + JSON, XML Schema/Schematron + JSON Schema, ShEx (for Turtle) + see the extensions & the dependency analysis

14.18.2.1 Terminology Bindings

Path | Definition | Type | Reference
RiskEvidenceSynthesis.status | The lifecycle status of an artifact. | Required | PublicationStatus
RiskEvidenceSynthesis.jurisdiction | Countries and regions within which this artifact is targeted for use. | Extensible | Jurisdiction ValueSet
RiskEvidenceSynthesis.topic | High-level categorization of the definition, used for searching, sorting, and filtering. | Example | DefinitionTopic
RiskEvidenceSynthesis.synthesisType | Types of combining results from a body of evidence (eg. summary data meta-analysis). | Extensible | SynthesisType
RiskEvidenceSynthesis.studyType | Types of research studies (types of research methods). | Extensible | StudyType
RiskEvidenceSynthesis.riskEstimate.type | Whether the risk estimate is dichotomous, continuous or qualitative and the specific type of risk estimate (eg proportion or median). | Extensible | RiskEstimateType
RiskEvidenceSynthesis.riskEstimate.unitOfMeasure | Unified Code for Units of Measure (UCUM). | Required | UCUMCodes
RiskEvidenceSynthesis.riskEstimate.precisionEstimate.type | Method of reporting variability of estimates, such as confidence intervals, interquartile range or standard deviation. | Extensible | PrecisionEstimateType
RiskEvidenceSynthesis.certainty.rating | The quality of the evidence described. The code system used specifies the quality scale used to grade this evidence source while the code specifies the actual quality score (represented as a coded value) associated with the evidence. | Extensible | QualityOfEvidenceRating
RiskEvidenceSynthesis.certainty.certaintySubcomponent.type | The subcomponent classification of quality of evidence rating systems. | Extensible | CertaintySubcomponentType
RiskEvidenceSynthesis.certainty.certaintySubcomponent.rating | The quality rating of the subcomponent of a quality of evidence rating. | Extensible | CertaintySubcomponentRating

14.18.2.2 Constraints

id | Level | Location | Description | Expression
rvs-0 | Warning | (base) | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

14.18.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
context | token | A use context assigned to the risk evidence synthesis | (RiskEvidenceSynthesis.useContext.value as CodeableConcept)
context-quantity | quantity | A quantity- or range-valued use context assigned to the risk evidence synthesis | (RiskEvidenceSynthesis.useContext.value as Quantity) | (RiskEvidenceSynthesis.useContext.value as Range)
context-type | token | A type of use context assigned to the risk evidence synthesis | RiskEvidenceSynthesis.useContext.code
context-type-quantity | composite | A use context type and quantity- or range-based value assigned to the risk evidence synthesis | On RiskEvidenceSynthesis.useContext: context-type: code context-quantity: value.as(Quantity) | value.as(Range)
context-type-value | composite | A use context type and value assigned to the risk evidence synthesis | On RiskEvidenceSynthesis.useContext: context-type: code context: value.as(CodeableConcept)
date | date | The risk evidence synthesis publication date | RiskEvidenceSynthesis.date
description | string | The description of the risk evidence synthesis | RiskEvidenceSynthesis.description
effective | date | The time during which the risk evidence synthesis is intended to be in use | RiskEvidenceSynthesis.effectivePeriod
identifier | token | External identifier for the risk evidence synthesis | RiskEvidenceSynthesis.identifier
jurisdiction | token | Intended jurisdiction for the risk evidence synthesis | RiskEvidenceSynthesis.jurisdiction
name | string | Computationally friendly name of the risk evidence synthesis | RiskEvidenceSynthesis.name
publisher | string | Name of the publisher of the risk evidence synthesis | RiskEvidenceSynthesis.publisher
status | token | The current status of the risk evidence synthesis | RiskEvidenceSynthesis.status
title | string | The human-friendly name of the risk evidence synthesis | RiskEvidenceSynthesis.title
url | uri | The uri that identifies the risk evidence synthesis | RiskEvidenceSynthesis.url
version | token | The business version of the risk evidence synthesis | RiskEvidenceSynthesis.version

14.18.4 Resource RiskEvidenceSynthesis - Examples

URL: https://hl7.org/fhir/R4/riskevidencesynthesis-examples.html

14.18.4 Resource RiskEvidenceSynthesis - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

TODO

Example Name | id | Format
Example of riskevidencesynthesis | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.18.5 Resource RiskEvidenceSynthesis - Detailed Descriptions

URL: https://hl7.org/fhir/R4/riskevidencesynthesis-definitions.html

14.18.5 Resource RiskEvidenceSynthesis - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Security Category: Business | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the RiskEvidenceSynthesis resource.

RiskEvidenceSynthesis

Element Id | RiskEvidenceSynthesis

Definition | The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.

Cardinality | 0..*

Type | DomainResource

Invariants | Defined on this element rvs-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

Defined on this element

rvs-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

RiskEvidenceSynthesis.url

Element Id | RiskEvidenceSynthesis.url

Definition | An absolute URI that is used to identify this risk evidence synthesis when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this risk evidence synthesis is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the risk evidence synthesis is stored on different servers.

Cardinality | 0..1

Type | uri

Requirements | Allows the risk evidence synthesis to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid: but real http: addresses are preferred. Multiple instances may share the same URL if they have a distinct version. The determination of when to create a new version of a resource (same url, new version) vs. defining a new artifact is up to the author. Considerations for making this decision are found in Technical and Business Versions. In some cases, the resource can no longer be found at the stated url, but the url itself cannot change. Implementations can use the meta.source element to indicate where the current master source of the resource can be found.

RiskEvidenceSynthesis.identifier

Element Id | RiskEvidenceSynthesis.identifier

Definition | A formal identifier that is used to identify this risk evidence synthesis when it is represented in other formats, or referenced in a specification, model, design or an instance.

Note | This is a business identifier, not a resource identifier (see discussion)

Cardinality | 0..*

Type | Identifier

Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.

Summary | true

Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, and can then identify this risk evidence synthesis outside of FHIR, where it is not possible to use the logical URI.

RiskEvidenceSynthesis.version

Element Id | RiskEvidenceSynthesis.version

Definition | The identifier that is used to identify this version of the risk evidence synthesis when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the risk evidence synthesis author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Cardinality | 0..1

Type | string

Summary | true

Comments | There may be different risk evidence synthesis instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a reference to a particular business version of the risk evidence synthesis with the format [url]|[version].

RiskEvidenceSynthesis.name

Element Id | RiskEvidenceSynthesis.name

Definition | A natural language name identifying the risk evidence synthesis. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Cardinality | 0..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alphanumeric type name to ensure that it is machine-processing friendly.

Invariants | Affect this element inv-0 Warning Name should be usable as an identifier for the module by machine processing applications such as code generation name.matches('[A-Z]([A-Za-z0-9_]){0,254}')

Affect this element

inv-0 | Warning | Name should be usable as an identifier for the module by machine processing applications such as code generation | name.matches('[A-Z]([A-Za-z0-9_]){0,254}')
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
RiskEvidenceSynthesis.title
Element Id | RiskEvidenceSynthesis.title
Definition | A short, descriptive, user-friendly title for the risk evidence synthesis.
Cardinality | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
RiskEvidenceSynthesis.status
Element Id | RiskEvidenceSynthesis.status
Definition | The status of this risk evidence synthesis. Enables tracking the life-cycle of the content.
Cardinality | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true (Reason: This is labeled as "Is Modifier" because applications should not use a retired {{title}} without due consideration)
Summary | true
Comments | Allows filtering of risk evidence synthesiss that are appropriate for use versus not.
RiskEvidenceSynthesis.date
Element Id | RiskEvidenceSynthesis.date
Definition | The date (and optionally time) when the risk evidence synthesis was published. The date must change when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the risk evidence synthesis changes.
Cardinality | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the risk evidence synthesis. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
RiskEvidenceSynthesis.publisher
Element Id | RiskEvidenceSynthesis.publisher
Definition | The name of the organization or individual that published the risk evidence synthesis.
Cardinality | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the risk evidence synthesis. May also allow for contact.
Summary | true
Comments | Usually an organization but may be an individual. The publisher (or steward) of the risk evidence synthesis is the organization or individual primarily responsible for the maintenance and upkeep of the risk evidence synthesis. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the risk evidence synthesis. This item SHOULD be populated unless the information is available from context.
RiskEvidenceSynthesis.contact
Element Id | RiskEvidenceSynthesis.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Cardinality | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
RiskEvidenceSynthesis.description
Element Id | RiskEvidenceSynthesis.description
Definition | A free text natural language description of the risk evidence synthesis from a consumer's perspective.
Cardinality | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the risk evidence synthesis was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the risk evidence synthesis as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the risk evidence synthesis is presumed to be the predominant language in the place the risk evidence synthesis was created).
RiskEvidenceSynthesis.note
Element Id | RiskEvidenceSynthesis.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation
RiskEvidenceSynthesis.useContext
Element Id | RiskEvidenceSynthesis.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance plans, studies, ...) and may be used to

assist with indexing and searching for appropriate risk evidence synthesis instances.
Cardinality | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation that all or any of the contexts apply.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
RiskEvidenceSynthesis.jurisdiction
Element Id | RiskEvidenceSynthesis.jurisdiction
Definition | A legal or geographic region in which the risk evidence synthesis is intended to be used.
Cardinality | 0..*
Terminology Binding | Jurisdiction (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the risk evidence synthesis to be used in jurisdictions other than those for which it was originally designed or intended.
To Do | Note: This was added back in because this is a metadata resource and it's a required element for metadata resource searching.
RiskEvidenceSynthesis.copyright
Element Id | RiskEvidenceSynthesis.copyright
Definition | A copyright statement relating to the risk evidence synthesis and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the risk evidence synthesis.
Cardinality | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the risk evidence synthesis and/or its content.
Alternate Names | License; Restrictions
RiskEvidenceSynthesis.approvalDate
Element Id | RiskEvidenceSynthesis.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Cardinality | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes or editorial corrections.
RiskEvidenceSynthesis.lastReviewDate
Element Id | RiskEvidenceSynthesis.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval but does not change the original approval date.
Cardinality | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this date follows the original approval date.
RiskEvidenceSynthesis.effectivePeriod
Element Id | RiskEvidenceSynthesis.effectivePeriod
Definition | The period during which the risk evidence synthesis content was or is planned to be in active use.
Cardinality | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of the risk evidence synthesis are or are expected to be used instead.
Summary | true
Comments | The effective period for a risk evidence synthesis determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
RiskEvidenceSynthesis.topic
Element Id | RiskEvidenceSynthesis.topic
Definition | Descriptive topics related to the content of the RiskEvidenceSynthesis. Topics provide a high-level categorization grouping types of EffectEvidenceSynthesiss that can be useful for filtering and searching.
Cardinality | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the RiskEvidenceSynthesis so that it can be found by topical searches.
RiskEvidenceSynthesis.author
Element Id | RiskEvidenceSynthesis.author
Definition | An individiual or organization primarily involved in the creation and maintenance of the content.
Cardinality | 0..*
Type | ContactDetail
RiskEvidenceSynthesis.editor
Element Id | RiskEvidenceSynthesis.editor
Definition | An individual or organization primarily responsible for internal coherence of the content.
Cardinality | 0..*

Type | ContactDetail
RiskEvidenceSynthesis.reviewer
Element Id | RiskEvidenceSynthesis.reviewer
Definition | An individual or organization primarily responsible for review of some aspect of the content.
Cardinality | 0..*
Type | ContactDetail
RiskEvidenceSynthesis.endorser
Element Id | RiskEvidenceSynthesis.endorser
Definition | An individual or organization responsible for officially endorsing the content for use in some setting.
Cardinality | 0..*
Type | ContactDetail
RiskEvidenceSynthesis.relatedArtifact
Element Id | RiskEvidenceSynthesis.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Cardinality | 0..*
Type | RelatedArtifact
Requirements | RiskEvidenceSynthesiss must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
RiskEvidenceSynthesis.synthesisType
Element Id | RiskEvidenceSynthesis.synthesisType
Definition | Type of synthesis eg meta-analysis.
Cardinality | 0..1
Terminology Binding | SynthesisType (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.studyType
Element Id | RiskEvidenceSynthesis.studyType
Definition | Type of study eg randomized trial.
Cardinality | 0..1
Terminology Binding | StudyType (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.population
Element Id | RiskEvidenceSynthesis.population
Definition | A reference to a EvidenceVariable resource that defines the population for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
RiskEvidenceSynthesis.exposure
Element Id | RiskEvidenceSynthesis.exposure
Definition | A reference to a EvidenceVariable resource that defines the exposure for the research.
Cardinality | 0..1
Type | Reference(EvidenceVariable)
Summary | true
RiskEvidenceSynthesis.outcome
Element Id | RiskEvidenceSynthesis.outcome
Definition | A reference to a EvidenceVariable resomece that defines the outcome for the research.
Cardinality | 1..1
Type | Reference(EvidenceVariable)
Summary | true
RiskEvidenceSynthesis.sampleSize
Element Id | RiskEvidenceSynthesis.sampleSize
Definition | A description of the size of the sample involved in the synthesis.
Cardinality | 0..1
RiskEvidenceSynthesis.sampleSize.description
Element Id | RiskEvidenceSynthesis.sampleSize.description
Definition | Human-readable summary of sample size.
Cardinality | 0..1
Type | string
RiskEvidenceSynthesis.sampleSize.numberOfStudies
Element Id | RiskEvidenceSynthesis.sampleSize.numberOfStudies
Definition | Number of studies included in this evidence synthesis.
Cardinality | 0..1
Type | integer
RiskEvidenceSynthesis.sampleSize.numberOfParticipants
Element Id | RiskEvidenceSynthesis.sampleSize.numberOfParticipants
Definition | Number of participants included in this evidence synthesis.
Cardinality | 0..1
Type | integer
RiskEvidenceSynthesis.riskEstimate
Element Id | RiskEvidenceSynthesis.riskEstimate
Definition | The estimated risk of the outcome.

Cardinality | 0..1
Summary | true
RiskEvidenceSynthesis.riskEstimate.description
Element Id | RiskEvidenceSynthesis.riskEstimate.description
Definition | Human-readable summary of risk estimate.
Cardinality | 0..1
Type | string
RiskEvidenceSynthesis.riskEstimate.type
Element Id | RiskEvidenceSynthesis.riskEstimate.type
Definition | Examples include proportion and mean.
Cardinality | 0..1
Terminology Binding | RiskEstimateType (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.riskEstimate.value
Element Id | RiskEvidenceSynthesis.riskEstimate.value
Definition | The point estimate of the risk estimate.
Cardinality | 0..1
Type | decimal
RiskEvidenceSynthesis.riskEstimate.unitOfMeasure
Element Id | RiskEvidenceSynthesis.riskEstimate.unitOfMeasure
Definition | Specifies the UCUM unit for the outcome.
Cardinality | 0..1
Terminology Binding | UCUM Codes (Required)
Type | CodeableConcept
RiskEvidenceSynthesis.riskEstimate.denominatorCount
Element Id | RiskEvidenceSynthesis.riskEstimate.denominatorCount
Definition | The sample size for the group that was measured for this risk estimate.
Cardinality | 0..1
Type | integer
RiskEvidenceSynthesis.riskEstimate.numeratorCount
Element Id | RiskEvidenceSynthesis.riskEstimate.numeratorCount
Definition | The number of group members with the outcome of interest.
Cardinality | 0..1
Type | integer
RiskEvidenceSynthesis.riskEstimate.precisionEstimate
Element Id | RiskEvidenceSynthesis.riskEstimate.precisionEstimate
Definition | A description of the precision of the estimate for the effect.
Cardinality | 0..*
RiskEvidenceSynthesis.riskEstimate.precisionEstimate.type
Element Id | RiskEvidenceSynthesis.riskEstimate.precisionEstimate.type
Definition | Examples include confidence interval and interquartile range.
Cardinality | 0..1
Terminology Binding | PrecisionEstimateType (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.riskEstimate.precisionEstimate.level
Element Id | RiskEvidenceSynthesis.riskEstimate.precisionEstimate.level
Definition | Use 95 for a 95% confidence interval.
Cardinality | 0..1
Type | decimal
RiskEvidenceSynthesis.riskEstimate.precisionEstimate.from
Element Id | RiskEvidenceSynthesis.riskEstimate.precisionEstimate.from
Definition | Lower bound of confidence interval.
Cardinality | 0..1
Type | decimal
RiskEvidenceSynthesis.riskEstimate.precisionEstimate.to
Element Id | RiskEvidenceSynthesis.riskEstimate.precisionEstimate.to
Definition | Upper bound of confidence interval.
Cardinality | 0..1
Type | decimal
RiskEvidenceSynthesis.certainty
Element Id | RiskEvidenceSynthesis.certainty
Definition | A description of the certainty of the risk estimate.
Cardinality | 0..*
RiskEvidenceSynthesis.certainty.rating
Element Id | RiskEvidenceSynthesis.certainty.rating
Definition | A rating of the certainty of the effect estimate.
Cardinality | 0..*
Terminology Binding | QualityOfEvidenceRating (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.certainty.note
Element Id | RiskEvidenceSynthesis.certainty.note
Definition | A human-readable string to clarify or explain concepts about the resource.

Cardinality | 0..*
Type | Annotation
RiskEvidenceSynthesis.certainty.certaintySubcomponent
Element Id | RiskEvidenceSynthesis.certainty.certaintySubcomponent
Definition | A description of a component of the overall certainty.
Cardinality | 0..*
RiskEvidenceSynthesis.certainty.certaintySubcomponent.type
Element Id | RiskEvidenceSynthesis.certainty.certaintySubcomponent.type
Definition | Type of subcomponent of certainty rating.
Cardinality | 0..1
Terminology Binding | CertaintySubcomponentType (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.certainty.certaintySubcomponent.rating
Element Id | RiskEvidenceSynthesis.certainty.certaintySubcomponent.rating
Definition | A rating of a subcomponent of rating certainty.
Cardinality | 0..*
Terminology Binding | CertaintySubcomponentRating (Extensible)
Type | CodeableConcept
RiskEvidenceSynthesis.certainty.certaintySubcomponent.note
Element Id | RiskEvidenceSynthesis.certainty.certaintySubcomponent.note
Definition | A human-readable string to clarify or explain concepts about the resource.
Cardinality | 0..*
Type | Annotation

14.18.6 Resource RiskEvidenceSynthesis - Mappings

URL: https://hl7.org/fhir/R4/riskevidencesynthesis-mappings.html

14.18.6 Resource RiskEvidenceSynthesis - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

Mappings:

FiveWs Pattern Mapping

Workflow Pattern

RIM Mapping

Mappings for the riskevidencesynthesis resource (see Mappings to Other Standards for further information & status).

14.18.6.1 FiveWs Pattern Mapping (http://hl7.org/fhir/fivews)

RiskEvidenceSynthesis
url | FiveWs.identifier
identifier | FiveWs.identifier
version | FiveWs.version
status | FiveWs.status
date | FiveWs.recorded
publisher | FiveWs.witness

14.18.6.2 Workflow Pattern (http://hl7.org/fhir/workflow)

RiskEvidenceSynthesis | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
title | Definition.title
status | Definition.status
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
copyright | Definition.copyright
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod

14.18.6.3 RIM Mapping (http://hl7.org/v3)

RiskEvidenceSynthesis
author | .participation[typeCode=AUT]
reviewer | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

endorser | .participation[typeCode=VRF] {not clear whether VRF best corresponds to reviewer or endorser}

14.18.7 Resource RiskEvidenceSynthesis - Extensions & Profiles

URL: https://hl7.org/fhir/R4/riskevidencesynthesis-profiles.html

14.18.7 Resource RiskEvidenceSynthesis - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Security Category: Business | Compartments: Not linked to any defined compartments

This table lists profiles and extensions for the RiskEvidenceSynthesis resource. For background information, see Profiling Resources and Extensibility. Additional profiles and extensions may be found in published Implementation Guides , or in the Conformance resource registry .

14.18.7.1 Profiles

No Profiles defined for this resource

14.18.7.2 Extensions

replaces | RiskEvidenceSynthesis | replaces | for General Extensions for use by FHIR Implementers
Extensions for all resources or elements
resource-pertainsToGoal | pertainsToGoal | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource
Extensions that refer to Any resource
cqf-relativeDateTime | relativeDateTime | for Clinical Reasoning Extensions
event-basedOn | basedOn | for Event Pattern HL7 Extensions
event-partOf | partOf | for Event Pattern HL7 Extensions
flag-detail | detail | for Flag HL7 Extensions
relative-date | Relative Date Criteria | for General Extensions for use by FHIR Implementers
replaces | replaces | for General Extensions for use by FHIR Implementers
request-replaces | replaces | for Request Pattern HL7 Extensions
workflow-supportingInfo | supportingInfo | for Workflow Pattern HL7 Extensions

14.18.7.3 Search Extensions

No Search Extensions defined for this resource

14.18.8 JSON Schema for RiskEvidenceSynthesis

URL: https://hl7.org/fhir/R4/riskevidencesynthesis.schema.json.html

14.18.8 JSON Schema for RiskEvidenceSynthesis

Clinical Decision Support Work Group | Maturity Level: N/A | Standards Status: Informative | Compartments: Not linked to any defined compartments

This is a representation of the json schema for RiskEvidenceSynthesis, which is just a part of the full JSON Schema.


```
{
  "$schema": "http://json-schema.org/draft-06/schema#",
  "id": "http://hl7.org/fhir/json-schema/RiskEvidenceSynthesis",
  "$ref": "#/definitions/RiskEvidenceSynthesis",
  "description": "see http://hl7.org/fhir/json.html#schema for information about the FHIR Json Schemas",
  "definitions": {
    "RiskEvidenceSynthesis": {
      "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus
    exposure state where the risk estimate is derived from a combination of research studies.",
      "properties": {
        "resourceType": {
          "description": "This is a RiskEvidenceSynthesis resource",
          "const": "RiskEvidenceSynthesis"
        },
        "id": {
          "description": "The logical id of the resource, as used in the URL for the resource. Once assigned, this value
     never changes.",
          "$ref": "id.schema.json#/definitions/id"
        },
        "meta": {
          "description": "The metadata about the resource. This is content that is maintained by the infrastructure.
    Changes to the content might not always be associated with version changes to the resource.",
          "$ref": "Meta.schema.json#/definitions/Meta"
        },
        "implicitRules": {
          "description": "A reference to a set of rules that were followed when the resource was constructed, and which
    must be understood when processing the content. Often, this is a reference to an implementation guide that defines
    the special rules along with other profiles etc.",
```

```
      "$ref": "#/definitions/uri"
    },
    "_implicitRules": {
      "description": "Extensions for implicitRules",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "language": {
      "description": "The base language in which the resource is written.",
      "$ref": "#/definitions/code"
    },
    "_language": {
      "description": "Extensions for language",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "text": {
      "description": "A human-readable narrative that contains a summary of the resource and can be used to represent the content of the resource to a human. The narrative need not encode all the structured data, but is required to contain sufficient detail to make it \"clinically safe\" for a human to just read the narrative. Resource definitions may define what content should be represented in the narrative to ensure clinical safety.",
      "$ref": "Narrative.schema.json#/definitions/Narrative"
    },
    "contained": {
      "description": "These resources do not have an independent existence apart from the resource that contains them - they cannot be identified independently, and nor can they have their own independent transaction scope.",
      "items": {
        "$ref": "ResourceList.schema.json#/definitions/ResourceList"
      },
      "type": "array"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the resource and that modifies the understanding of the element that contains it and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer is allowed to define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "url": {
      "description": "An absolute URI that is used to identify this risk evidence synthesis when it is referenced in a specification, model, design or an instance; also called its canonical identifier. This SHOULD be globally unique and SHOULD be a literal address at which at which an authoritative instance of this risk evidence synthesis is (or will be) published. This URL can be the target of a canonical reference. It SHALL remain the same when the risk evidence synthesis is stored on different servers.",
      "$ref": "#/definitions/uri"
    },
    "_url": {
      "description": "Extensions for url",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "identifier": {
      "description": "A formal identifier that is used to identify this risk evidence synthesis when it is represented in other formats, or referenced in a specification, model, design or an instance.",
      "items": {
        "$ref": "Identifier.schema.json#/definitions/Identifier"
      },
      "type": "array"
    },
    "version": {
      "description": "The identifier that is used to identify this version of the risk evidence synthesis when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the risk evidence synthesis author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.",
      "$ref": "#/definitions/string"
    },
    "_version": {
      "description": "Extensions for version",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "name": {
      "description": "A natural language name identifying the risk evidence synthesis. This name should be usable as an identifier for the module by machine processing applications such as code generation.",
      "$ref": "#/definitions/string"
```

```
    },
    "_name": {
      "description": "Extensions for name",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "title": {
      "description": "A short, descriptive, user-friendly title for the risk evidence synthesis.",
      "$ref": "#/definitions/string"
    },
    "_title": {
      "description": "Extensions for title",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "status": {
      "description": "The status of this risk evidence synthesis. Enables tracking the life-cycle of the content.",
      "enum": [
        "draft",
        "active",
        "retired",
        "unknown"
      ]
    },
    "_status": {
      "description": "Extensions for status",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "date": {
      "description": "The date  (and optionally time) when the risk evidence synthesis was published. The date must
change when the business version changes and it must change if the status code changes. In addition, it should
change when the substantive content of the risk evidence synthesis changes.",
      "$ref": "#/definitions/dateTime"
    },
    "_date": {
      "description": "Extensions for date",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "publisher": {
      "description": "The name of the organization or individual that published the risk evidence synthesis.",
      "$ref": "#/definitions/string"
    },
    "_publisher": {
      "description": "Extensions for publisher",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "contact": {
      "description": "Contact details to assist a user in finding and communicating with the publisher.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "description": {
      "description": "A free text natural language description of the risk evidence synthesis from a consumer\u0027s
 perspective.",
      "$ref": "#/definitions/markdown"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "note": {
      "description": "A human-readable string to clarify or explain concepts about the resource.",
      "items": {
        "$ref": "Annotation.schema.json#/definitions/Annotation"
      },
      "type": "array"
    },
    "useContext": {
      "description": "The content was developed with a focus and intent of supporting the contexts that are listed.
These contexts may be general categories (gender, age, ...) or may be references to specific programs (insurance
plans, studies, ...) and may be used to assist with indexing and searching for appropriate risk evidence synthesis
instances.",
      "items": {
        "$ref": "UsageContext.schema.json#/definitions/UsageContext"
      },
      "type": "array"
    },
    "jurisdiction": {
      "description": "A legal or geographic region in which the risk evidence synthesis is intended to be used.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "copyright": {
      "description": "A copyright statement relating to the risk evidence synthesis and/or its contents. Copyright
statements are generally legal restrictions on the use and publishing of the risk evidence synthesis.",
      "$ref": "#/definitions/markdown"
    },
```

```
    "_copyright": {
      "description": "Extensions for copyright",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "approvalDate": {
      "description": "The date on which the resource content was approved by the publisher. Approval happens once
when the content is officially approved for usage.",
      "$ref": "#/definitions/date"
    },
    "_approvalDate": {
      "description": "Extensions for approvalDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "lastReviewDate": {
      "description": "The date on which the resource content was last reviewed. Review happens periodically after
approval but does not change the original approval date.",
      "$ref": "#/definitions/date"
    },
    "_lastReviewDate": {
      "description": "Extensions for lastReviewDate",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "effectivePeriod": {
      "description": "The period during which the risk evidence synthesis content was or is planned to be in active
use.",
      "$ref": "Period.schema.json#/definitions/Period"
    },
    "topic": {
      "description": "Descriptive topics related to the content of the RiskEvidenceSynthesis. Topics provide a high-
level categorization grouping types of EffectEvidenceSynthesiss that can be useful for filtering and searching.",
      "items": {
        "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
      },
      "type": "array"
    },
    "author": {
      "description": "An individiual or organization primarily involved in the creation and maintenance of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "editor": {
      "description": "An individual or organization primarily responsible for internal coherence of the content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "reviewer": {
      "description": "An individual or organization primarily responsible for review of some aspect of the
content.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "endorser": {
      "description": "An individual or organization responsible for officially endorsing the content for use in some
 setting.",
      "items": {
        "$ref": "ContactDetail.schema.json#/definitions/ContactDetail"
      },
      "type": "array"
    },
    "relatedArtifact": {
      "description": "Related artifacts such as additional documentation, justification, or bibliographic
references.",
      "items": {
        "$ref": "RelatedArtifact.schema.json#/definitions/RelatedArtifact"
      },
      "type": "array"
    },
    "synthesisType": {
      "description": "Type of synthesis eg meta-analysis.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "studyType": {
      "description": "Type of study eg randomized trial.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "population": {
      "description": "A reference to a EvidenceVariable resource that defines the population for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "exposure": {
      "description": "A reference to a EvidenceVariable resource that defines the exposure for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
```

```
    },
    "outcome": {
      "description": "A reference to a EvidenceVariable resomece that defines the outcome for the research.",
      "$ref": "Reference.schema.json#/definitions/Reference"
    },
    "sampleSize": {
      "description": "A description of the size of the sample involved in the synthesis.",
      "$ref": "#/definitions/RiskEvidenceSynthesis_SampleSize"
    },
    "riskEstimate": {
      "description": "The estimated risk of the outcome.",
      "$ref": "#/definitions/RiskEvidenceSynthesis_RiskEstimate"
    },
    "certainty": {
      "description": "A description of the certainty of the risk estimate.",
      "items": {
        "$ref": "#/definitions/RiskEvidenceSynthesis_Certainty"
      },
      "type": "array"
    }
  },
  "additionalProperties": false,
  "required": [
    "outcome",
    "resourceType",
    "population"
  ]
},
"RiskEvidenceSynthesis_SampleSize": {
  "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus
exposure state where the risk estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string
value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the
definition and use of extensions. Though any implementer can define an extension, there is a set of requirements
that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of
the element and that modifies the understanding of the element in which it is contained and/or the understanding of
the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the
use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of
extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part
 of the definition of the extension. Applications processing a resource are required to check for modifier
extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource
(including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable summary of sample size.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfStudies": {
      "description": "Number of studies included in this evidence synthesis.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfStudies": {
      "description": "Extensions for numberOfStudies",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numberOfParticipants": {
      "description": "Number of participants included in this evidence synthesis.",
      "$ref": "#/definitions/integer"
    },
    "_numberOfParticipants": {
      "description": "Extensions for numberOfParticipants",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"RiskEvidenceSynthesis_RiskEstimate": {
```

```
  "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "description": {
      "description": "Human-readable summary of risk estimate.",
      "$ref": "#/definitions/string"
    },
    "_description": {
      "description": "Extensions for description",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "type": {
      "description": "Examples include proportion and mean.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "value": {
      "description": "The point estimate of the risk estimate.",
      "$ref": "#/definitions/decimal"
    },
    "_value": {
      "description": "Extensions for value",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "unitOfMeasure": {
      "description": "Specifies the UCUM unit for the outcome.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "denominatorCount": {
      "description": "The sample size for the group that was measured for this risk estimate.",
      "$ref": "#/definitions/integer"
    },
    "_denominatorCount": {
      "description": "Extensions for denominatorCount",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "numeratorCount": {
      "description": "The number of group members with the outcome of interest.",
      "$ref": "#/definitions/integer"
    },
    "_numeratorCount": {
      "description": "Extensions for numeratorCount",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "precisionEstimate": {
      "description": "A description of the precision of the estimate for the effect.",
      "items": {
        "$ref": "#/definitions/RiskEvidenceSynthesis_PrecisionEstimate"
      },
      "type": "array"
    }
  },
  "additionalProperties": false
},
"RiskEvidenceSynthesis_PrecisionEstimate": {
  "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
```

```
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "type": {
      "description": "Examples include confidence interval and interquartile range.",
      "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
    },
    "level": {
      "description": "Use 95 for a 95% confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_level": {
      "description": "Extensions for level",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "from": {
      "description": "Lower bound of confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_from": {
      "description": "Extensions for from",
      "$ref": "Element.schema.json#/definitions/Element"
    },
    "to": {
      "description": "Upper bound of confidence interval.",
      "$ref": "#/definitions/decimal"
    },
    "_to": {
      "description": "Extensions for to",
      "$ref": "Element.schema.json#/definitions/Element"
    }
  },
  "additionalProperties": false
},
"RiskEvidenceSynthesis_Certainty": {
  "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.",
  "properties": {
    "id": {
      "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
      "$ref": "string.schema.json#/definitions/string"
    },
    "extension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
    },
    "modifierExtension": {
      "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
      "items": {
        "$ref": "Extension.schema.json#/definitions/Extension"
      },
      "type": "array"
```

```
        },
        "rating": {
          "description": "A rating of the certainty of the effect estimate.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "note": {
          "description": "A human-readable string to clarify or explain concepts about the resource.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        },
        "certaintySubcomponent": {
          "description": "A description of a component of the overall certainty.",
          "items": {
            "$ref": "#/definitions/RiskEvidenceSynthesis_CertaintySubcomponent"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    },
    "RiskEvidenceSynthesis_CertaintySubcomponent": {
      "description": "The RiskEvidenceSynthesis resource describes the likelihood of an outcome in a population plus exposure state where the risk estimate is derived from a combination of research studies.",
      "properties": {
        "id": {
          "description": "Unique id for the element within a resource (for internal references). This may be any string value that does not contain spaces.",
          "$ref": "string.schema.json#/definitions/string"
        },
        "extension": {
          "description": "May be used to represent additional information that is not part of the basic definition of the element. To make the use of extensions safe and manageable, there is a strict set of governance  applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension.",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "modifierExtension": {
          "description": "May be used to represent additional information that is not part of the basic definition of the element and that modifies the understanding of the element in which it is contained and/or the understanding of the containing element\u0027s descendants. Usually modifier elements provide negation or qualification. To make the use of extensions safe and manageable, there is a strict set of governance applied to the definition and use of extensions. Though any implementer can define an extension, there is a set of requirements that SHALL be met as part of the definition of the extension. Applications processing a resource are required to check for modifier extensions.\n\nModifier extensions SHALL NOT change the meaning of any elements on Resource or DomainResource (including cannot change the meaning of modifierExtension itself).",
          "items": {
            "$ref": "Extension.schema.json#/definitions/Extension"
          },
          "type": "array"
        },
        "type": {
          "description": "Type of subcomponent of certainty rating.",
          "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
        },
        "rating": {
          "description": "A rating of a subcomponent of rating certainty.",
          "items": {
            "$ref": "CodeableConcept.schema.json#/definitions/CodeableConcept"
          },
          "type": "array"
        },
        "note": {
          "description": "A human-readable string to clarify or explain concepts about the resource.",
          "items": {
            "$ref": "Annotation.schema.json#/definitions/Annotation"
          },
          "type": "array"
        }
      },
      "additionalProperties": false
    }
  }
}
```


14.18.9 Resource RiskEvidenceSynthesis - Maturity Dependency View

URL: https://hl7.org/fhir/R4/riskevidencesynthesis-dependencies.html

14.18.9 Resource RiskEvidenceSynthesis - Maturity Dependency View

Clinical Decision Support Work Group | Maturity Level: 0 | Trial Use | Compartments: Not linked to any defined compartments

Dependency Graph for RiskEvidenceSynthesis FMM level 0

Name | Card. | Type | Dependency Analysis
RiskEvidenceSynthesis | | DomainResource
url | 0..1 | uri
identifier | 0..* | Identifier
version | 0..1 | string
name | 0..1 | string
title | 0..1 | string
status | 1..1 | code
date | 0..1 | dateTime
publisher | 0..1 | string
contact | 0..* | ContactDetail
description | 0..1 | markdown
note | 0..* | Annotation
useContext | 0..* | UsageContext
jurisdiction | 0..* | CodeableConcept
copyright | 0..1 | markdown
approvalDate | 0..1 | date
lastReviewDate | 0..1 | date
effectivePeriod | 0..1 | Period
topic | 0..* | CodeableConcept
author | 0..* | ContactDetail
editor | 0..* | ContactDetail
reviewer | 0..* | ContactDetail
endorser | 0..* | ContactDetail
relatedArtifact | 0..* | RelatedArtifact
synthesisType | 0..1 | CodeableConcept
studyType | 0..1 | CodeableConcept
population | 1..1 | Reference(EvidenceVariable)
exposure | 0..1 | Reference(EvidenceVariable)
outcome | 1..1 | Reference(EvidenceVariable)
sampleSize | 0..1 | BackboneElement
description | 0..1 | string
numberOfStudies | 0..1 | integer
numberOfParticipants | 0..1 | integer
riskEstimate | 0..1 | BackboneElement
description | 0..1 | string
type | 0..1 | CodeableConcept
value | 0..1 | decimal
unitOfMeasure | 0..1 | CodeableConcept
denominatorCount | 0..1 | integer
numeratorCount | 0..1 | integer
precisionEstimate | 0..* | BackboneElement
type | 0..1 | CodeableConcept
level | 0..1 | decimal
from | 0..1 | decimal
to | 0..1 | decimal
certainty | 0..* | BackboneElement
rating | 0..* | CodeableConcept
note | 0..* | Annotation
certaintySubcomponent | 0..* | BackboneElement
type | 0..1 | CodeableConcept
rating | 0..* | CodeableConcept
note | 0..* | Annotation